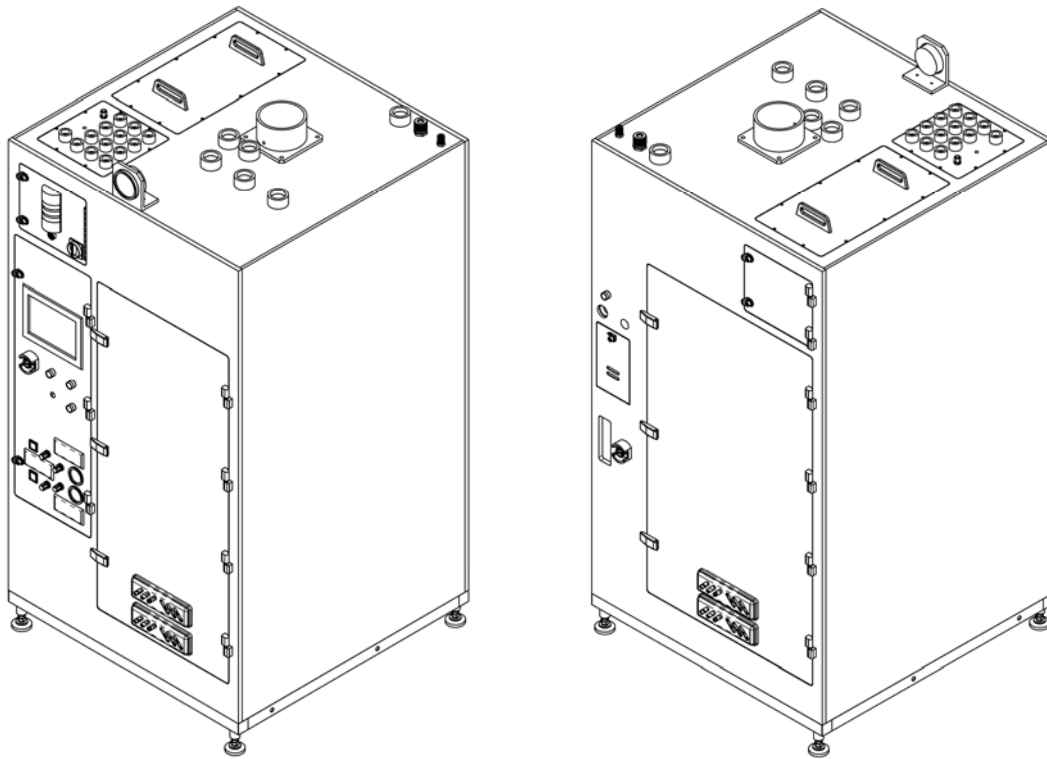

Advanced Equipment Systems Division

Inline Blender, with Integrated Daytanks

Operation and Maintenance Manual

Publication Number 928880-100 Revision A October 2015



October 2015

Release A of this publication applies to the Inline Blender, with Integrated Daytanks, and to all subsequent revisions until otherwise indicated in new editions.

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If you have comments about this publication, you may address them to:

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Warranty and Limitation of Liability

ADVANCED EQUIPMENT SYSTEMS DIVISION

DOMESTIC WARRANTY

DOES NOT INCLUDE AIR LIQUIDE ON-SITE SUPPORT

For Equipment Delivered and Installed Within the Continental U.S.

TERMS

Air Liquide Electronics U.S. LP, (hereinafter referred to as Air Liquide) warrants new Advanced Equipment Systems (AES) Division equipment manufactured by Air Liquide to and only to the original purchaser ("Customer") and at the original point of installation to be free from defects in materials and workmanship for 12 months from startup or acceptance by the customer, when used for its approved purpose.

The startup and commissioning must be performed by Air Liquide personnel. Any change from the specified chemistry for the system to a different chemistry voids the warranty, unless prior written approval is obtained from an authorized representative of the Air Liquide Advanced Equipment Systems Division. This warranty is automatically voided if the equipment is used with Shipley's MEGAPOSIT® MF-701 chemical. In addition, if the Customer fails or refuses to disclose the name or contents of all processing materials being used in the product/equipment to Air Liquide personnel who are servicing, maintaining, or repairing such items, then this warranty shall be null and void. Air Liquide only warrants the use of Shipley MEGAPOSIT® MF-501 with 450HP Teflon®. The warranty period shall begin upon startup or acceptance. However, in no event will it begin later than 12 weeks after shipment from Air Liquide.

Any exceptions to the above must be confirmed in writing by Air Liquide Advanced Equipment Systems Division Manager of Business/Contracts within the above 12-week startup period.

PARTS WARRANTY

Air Liquide's warranty obligation is limited to supplying and/or replacing non-consumable parts that fail or are determined to be defective during the warranty period. This warranty shall not apply in the case of equipment/parts that have been damaged due to Customer or a third party's neglect, abuse, misuse, misapplication, unauthorized alteration, failure to repair or service, or improper repair. Air Liquide does not accept liability for indirect, special, incidental, punitive or consequential damages whatsoever. Consumable parts are not covered by this warranty and are listed later in this document.

CUSTOMER REQUIREMENTS

The warranty requires that the Customer purchase and maintain (and as appropriate, install) the Air Liquide recommended spare parts and that all service operations and preventative maintenance functions (listed below) be performed by the Customer's technicians that have been trained by Air Liquide.

The Customer's responsibilities include service operations and routine preventative maintenance of the equipment. This includes providing labor for all repairs covered in the Air Liquide training and certification program. Additional Customer responsibilities are described later in this document. If the Customer is responsible for preventative maintenance, documentation must be maintained regarding such maintenance and be presented to Air Liquide or its representatives at their request to ensure the validity of the warranty. If equipment or a part fails because the Customer has not performed routine preventative maintenance, the downtime will be charged to the Customer. Additionally, the Customer will be liable for any parts or labor required to fix the failure.

In cases where the equipment or parts have moved and been reinstalled within the Customer's factory without Air Liquide's authorized assistance or is removed or transshipped outside the Continental United States, this warranty shall be null and void.

In addition, the Customer has certain service responsibilities during the warranty period. When the conditions identified below have been met by the Customer, as determined by Air Liquide, the warranty will include all Air Liquide service labor, repair parts (excluding consumables), parts shipment and associated labor/travel costs.

Where the customer has not met these obligations, a purchase order from the Customer to pay for such services will be required before Air Liquide support would commence and the Customer will be charged for all services performed at Air Liquide's applicable rates.

Service Operations

The Customer is solely responsible for maintaining spare parts and operating any Advanced Equipment Systems Division equipment. The equipment operations typically consist of:

- Monitoring equipment and system status
- Evaluating and maintaining system performance
- Restoring equipment to the automatic mode from manual mode and fault states resulting from point of use (POU) faults
- Recovering from events that interrupt production (ITP's)
- Responding to calls for assistance from the fab
- Maintaining an informational database and record of all equipment Preventative Maintenance (PM's) performed
- Purchasing and maintaining the Air Liquide recommended spare parts

Routine Preventative Maintenance

The preventative maintenance activities for which the Customer is responsible include:

- Weekly, monthly, quarterly, semi-annual, and annual PM's for all Dispense Units, valve boxes, sample stations and support equipment
- Spare parts tracking and reordering
- Unscheduled repairs to restore equipment to operation
- Scheduled repairs not covered by PM's
- Pump rebuilds integral to Air Liquide equipment
- Consumable parts tracking and ordering

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For a comprehensive list and detailed explanation of the preventative maintenance activities for which the customer is responsible, please consult the equipment Operations and Maintenance Manual.

A failed component will usually result in a shutdown. However, other shutdowns may occur from conditional faults generated by the control logic. The Customer is responsible for troubleshooting the system and attempting to return the equipment to the Auto mode. Customers are also expected to provide the labor to:

- Replace failed, damaged, or defective components in all fluid flowpaths
- Adjust or replace all failed, damaged, or defective pressure, optical, capacitive, or mechanical sensors and switches
- Replace electronic components in the power distribution and control systems at the direction of the Customer Service Representative.

WARRANTY CLAIMS

When making a warranty claim the Customer is required to furnish the following: a valid purchase order (P.O.), equipment serial number, part number, quantity required, chemical service, and a description of the damage or failure. Air Liquide may require return of the defective part. The P.O. will be used to cover the cost of unreturned defective parts or non-warranty damage as well as tracking the claim. A Returned Material Authorization (RMA) number must be obtained from Air Liquide Technical Service before returning any parts to Air Liquide.

The Customer will be charged at Air Liquide's applicable rates for non-warranty Air Liquide activities. The Air Liquide Service Manager for the Advanced Equipment Systems Division in consultation with the Business/Contracts Manager will resolve any disputed claims.

RESPONSE TIME

Air Liquide Technical Service will make an oral response to a request for service within one hour when received by 4:00 p.m. Central Time Monday through Friday, excluding holidays. When on-site warranty service is required, Air Liquide service personnel will be on site within 24 to 48 hours upon mutual agreement that a service call is necessary. This excludes weekends and Holidays.

In-stock emergency parts will be shipped from Chanhassen, Minnesota or the local Regional Service Office the same day for overnight delivery when calls are received by Air Liquide's Technical Service Department before 2:00 p.m. Central Time during normal business days, excluding weekends and holidays.

TRANS-SHIPMENT OR EQUIPMENT RELOCATION OUTSIDE U.S.A.

In cases where equipment is moved and reinstalled within the Customer's factory, removed or transshipped outside the Continental U.S.A. without Air Liquide's written approval, this warranty shall be null and void.

EQUIPMENT MODIFICATION BY CUSTOMER

Equipment modifications made by Customer shall void the warranty unless prior written approval is obtained in writing from an authorized representative of Air Liquide's Advanced Equipment Systems Division.

LIMITATION OF LIABILITY

In no event shall Air Liquide's liability, whether based on an action or a claim in contract, tort, (including without limitation negligence and to the extent permitted by law, strict liability) or otherwise, to Customer arising out of or related to the equipment exceed the aggregate price of the product purchased by such buyer as of the date such action or claim was filed.

IN NO EVENT SHALL AIR LIQUIDE BE LIABLE FOR SPECIAL, INCIDENTAL, INDIRECT, CONSEQUENTIAL OR PUNITIVE DAMAGES OF ANY KIND, WHETHER FOR BREACH OF ANY WARRANTY, BREACH OR REPUDIATION OF ANY OTHER TERM OR CONDITION OF THE SALE, NEGLIGENCE, ON THE BASIS OF STRICT LIABILITY, OR FOR ANY OTHER REASON.

LIMITATION OF WARRANTY

THERE ARE NO TERMS, CONDITIONS, OR WARRANTIES EXPRESSED, IMPLIED OR STATUTORY, OF MERCHANTABILITY, FITNESS FOR THE PURPOSE INTENDED, CAPACITY OR OTHERWISE, OF THE GOODS ORDERED OTHER THAN, OR DIFFERENT FROM, THIS WARRANTY, AND THIS WARRANTY IS EXPRESSLY IN LIEU OF ANY OTHER WARRANTY, EXPRESSED OR IMPLIED.

Air Liquide reserves the right to void the warranty on equipment that has been (i) serviced by untrained personnel, (ii) installed, started up or commissioned by non Air Liquide authorized personnel or (iii) when the recommended spare parts have not been purchased, installed or maintained.

Air Liquide neither assumes nor authorizes any other party to assume for it any liability in connection with said equipment except as stated herein.

NOTE: Merchandise supplied by Air Liquide is manufactured in compliance with the Fair Labor Standards Act of 1938, as amended.

CONSUMABLE LIST

- Filters
 - Water
 - Chemical
 - Nitrogen
 - Polyester air filters
- O-rings
- Filter housings, diaphragms, and pumps
- Pump Valve Kit
- Rebuild kits for pumps
- All fuses
- Light bulbs for Reset button
- All gaskets and seals
- All nuts, bolts, washers, etc.

ADVANCED EQUIPMENT SYSTEMS DIVISION

DOMESTIC WARRANTY ADDENDUM

PROVIDING FOR AIR LIQUIDE ON-SITE SUPPORT

For Equipment Delivered and Installed
Within the Continental U.S.

NOTE: This addendum should be read in conjunction with Air Liquide's domestic warranty and applies only if the customer has separately purchased these services.

CUSTOMER REQUIREMENTS

The Customer has certain service responsibilities in addition to Air Liquide's responsibilities during the warranty period. When the conditions identified below have been met by the Customer, as determined by Air Liquide, the warranty will include all Air Liquide service labor, repair parts (excluding consumables), parts shipment and associated labor/travel costs. The Customer will provide the Air Liquide Technical Service Representative with a dedicated office in the vendor office area large enough for two individuals to work and store parts and tools.

Where the customer has not met these obligations, a purchase order from the Customer to pay for such services will be required before Air Liquide support would commence and the Customer will be charged for all services performed at Air Liquide's applicable rates.

Service Operations

Customer – The Customer is responsible for maintaining spare parts and operating the Advanced Equipment Systems Division equipment. The Advanced Equipment Systems Division equipment operations typically consist of:

- Monitoring equipment and system status
- Evaluating and maintaining system performance
- Restoring equipment to the automatic mode from manual and fault states resulting from point of use (POU) faults
- Recovering from events that interrupt production (ITP's)
- Responding to calls for assistance from the fab
- Maintaining an informational database and a record of all equipment Preventative Maintenance (PM's) performed
- Purchasing and maintaining the Air Liquide recommended spare parts

Air Liquide – An Air Liquide Technical Service Representative will:

- Assist with developing schedules and criteria for routine system surveillance and hardware inspections
- Assist with evaluating system capacity and measuring performance
- Review all incidents resulting in downtime and assist in developing action plans to eliminate the source of interruption or minimize the duration of the outage
- Review technical and operational issues that have the potential to affect equipment performance and advise the customer on appropriate actions

Routine Preventative Maintenance

Customer – The preventative maintenance activities for which the Customer is responsible include:

- Weekly, monthly, quarterly, semi-annual, and annual PM's for all Dispense Units, valve boxes, sample stations and support equipment
- Spare parts tracking and reordering, and returning of failed parts to Air Liquide (via the RMA process)
- Unscheduled repairs to restore equipment to operation
- Scheduled repairs not covered by PM's
- Pump rebuilds integral to Air Liquide equipment
- Consumable parts tracking and ordering

A failed component will usually result in a shutdown. However, other shutdowns may occur from conditional faults generated by the control logic. The Customer is responsible for troubleshooting the system and attempting to return the equipment to the Auto mode. The Customer is also expected to provide the labor to:

- Replace failed, damaged, or defective components in all fluid flow paths
- Adjust or replace all failed, damaged, or defective pressure, optical, capacitive, or mechanical sensors and switches
- Replace electronic components in the power distribution and control systems at the direction of the Technical Service Representative
- **Air Liquide** – The Air Liquide Technical Service Representative is to be called if the Customer is unable to recover from a shutdown or when the repairs necessary to restore operations were not included in the Advanced Equipment Systems Division equipment training and certification program. These activities are generally restricted to advanced maintenance of the control system. In addition, the Technical Service Representative will provide the following support:
 - Review the PM records and periodically inspect the units for compliance with Air Liquide specifications
 - Ensure that all product upgrades and safety bulletins are considered by the customer
 - Serve as a technical resource for the maintenance group
 - Ensure that the customer receives all warranty replacement parts

RESPONSE TIME

- Subject to the terms of this Warranty, an assigned Technical Service Representative will provide the customer on-site support, Monday through Friday, excluding holidays, 8:00 a.m. to 5:00 p.m., or during normal (day shift) business hours not to exceed 8 working hours per day. Support during other hours is not provided under this Warranty. (Customers may, however, purchase such coverage through a Site Supplement Contract with Air Liquide.)

In-stock emergency parts will be shipped from Chanhassen or the local Regional Office the same day for overnight delivery when calls are received by Air Liquide before 2:00 p.m. Central Time during normal business days, excluding Holidays.

ADVANCED EQUIPMENT SYSTEMS DIVISION

DOMESTIC WARRANTY ADDENDUM

**PROVIDING FOR
LOCAL REGIONAL OFFICE SUPPORT
IF WITHIN 60 MILES**

**For Equipment Delivered and Installed Within the
Continental U.S.**

**NOTE: This addendum should be read in conjunction
with Air Liquide's domestic warranty.**

RESPONSIBILITIES

The Customer has certain service responsibilities in addition to Air Liquide's responsibilities during the warranty period. When the conditions identified below have been met by the Customer, as determined by Air Liquide, the warranty will be extended to include all Air Liquide service labor, repair parts (excluding consumables), parts shipment and associated labor/travel costs.

Where the customer has not met these obligations, a purchase order from the Customer to pay for such services will be required before Air Liquide support would commence and the Customer will be charged for all services performed at Air Liquide's applicable rates.

Service Operations

Customer — The Customer is solely responsible for maintaining spare parts and operating the Advanced Equipment Systems Division equipment. The Advanced Equipment Systems Division equipment operations typically consist of:

- Monitoring equipment and system status
- Evaluating and maintaining system performance
- Restoring equipment to the automatic mode from manual and fault states resulting from point of use (POU) faults
- Recovering from events that interrupt production (ITP's)
- Responding to calls for assistance from the FAB
- Maintaining an informational database and a record of all equipment Preventative Maintenance (PM's) performed
- Purchasing and maintaining the Air Liquide recommended spare parts

Air Liquide — An Air Liquide Technical Service Representative from the Regional Office will provide some or all of the following support operations when purchased as part of the warranty contract:

- Assist with developing schedules and criteria for routine system surveillance and hardware inspections
- Assist with evaluating system capacity and measuring performance
- Review all incidents resulting in downtime and assist in developing action plans to eliminate the source of interruption or minimize the duration of the outage
- Review technical and operational issues that have the potential to affect equipment performance and advise the Customer on appropriate actions

Routine Preventative Maintenance

Customer — The preventative maintenance activities for which the Customer is responsible include:

- Weekly, monthly, quarterly, semi-annual, and annual PM's for all Dispense Units, valve boxes, sample stations and support equipment
- Spare parts tracking and reordering
- Unscheduled repairs to restore equipment to operation
- Scheduled repairs not covered by PM's
- Pump rebuilds integral to Air Liquide equipment
- Consumable parts tracking and ordering

A failed component will usually result in a shutdown. However, other shutdowns may occur from conditional faults generated by the control logic. The Customer is responsible for troubleshooting the system and attempting to return the equipment to the Auto mode. The Customer is also expected to provide the labor to:

- Replace failed, damaged, or defective components in all fluid flow paths
- Adjust or replace all failed, damaged, or defective pressure, optical, capacitive, or mechanical sensors and switches
- Replace electronic components in the power distribution and control systems at the direction of the Technical Service Representative

Air Liquide — When all or some of the support below is purchased as part of the warranty, Service personnel from the local Air Liquide Regional Office are to be called if the Customer is unable to recover from a shutdown or when the repairs necessary to restore operations were not included in the Advanced Equipment Systems Division training and certification program. These activities are generally restricted to advanced maintenance of the control system. In addition, the Technical Service Representative will provide the following support:

- Review the PM records and periodically inspect the units for compliance with Air Liquide specifications
- Ensure that all product upgrades and safety bulletins are considered by the Customer
- Serve as a technical resource for the maintenance group
- Ensure that the Customer receives all warranty replacement parts

RESPONSE TIME

Subject to the terms of this Warranty, an assigned Technical Service Representative from a local Regional Office will make an oral response to calls for assistance within one hour of a message or page, Monday through Fridays, excluding holidays, between 8:00 a.m. and 5:00 p.m. or during normal (day shift) business hours at no additional cost not to exceed 8 working hours per day. If the problem cannot be resolved with phone support, a Technical Service Representative will travel to the site within 24 hours of the last phone conversation with the customer contact.

Support during other hours is not provided under this Warranty. Customers may, however, purchase such coverage through a Site Supplement Contract with Air Liquide. In-stock emergency parts will be shipped from Chanhassen or the local Regional Office the same day for overnight delivery when calls are received by Air Liquide before 2:00 p.m. Central Time during normal business days, excluding Holidays.

ADVANCED EQUIPMENT SYSTEMS DIVISION

INTERNATIONAL WARRANTY

DOES NOT INCLUDE AIR LIQUIDE ON-SITE SUPPORT

For Equipment Delivered and Installed Outside of the
Continental U.S.

TERMS

Air Liquide Electronics U.S. LP, (hereinafter referred to as Air Liquide) warrants new Advanced Equipment Systems Division equipment manufactured by Air Liquide to and only to the original purchaser ("Customer") and at the original point of installation to be free from defects in materials and workmanship for 12 months from startup or acceptance by the Customer when used for its approved purpose.

The startup and commissioning must be performed by Air Liquide personnel. Any change from the specified chemistry for the system to a different chemistry voids the warranty, unless prior written approval is obtained from an authorized representative of the Air Liquide Advanced Equipment Systems Division. This warranty is automatically voided if the equipment is used with Shipley's MEGAPOSIT® MF-701 chemical. In addition, if the Customer fails or refuses to disclose the name and contents of all processing materials being used in the product/equipment to Air Liquide personnel who are servicing, maintaining, or repairing such items, then this warranty shall be null and void. Air Liquide only warrants the use of Shipley MEGAPOSIT® MF-501 with 450HP Teflon®. The warranty period shall begin upon startup or acceptance. However, in no event will it begin later than 12 weeks after shipment from Air Liquide.

Any exceptions to the above must be confirmed in writing by Air Liquide Advanced Equipment Systems Division Manager of Business/Contracts within the above 12-week startup period.

PARTS WARRANTY

Air Liquide's warranty obligation is limited to supplying and/or replacing non-consumable parts that fail or are determined to be defective during the warranty period. This warranty shall not apply in the case of equipment/parts that have been damaged due to Customer or a third party's neglect, abuse, misuse, misapplication, unauthorized alteration, failure to repair or service, or improper repair. Air Liquide does not accept liability for indirect, special, incidental, punitive or consequential damages whatsoever. Consumable parts are not covered by this warranty and are listed later in this document.

CUSTOMER REQUIREMENTS

The warranty requires that the Customer purchase and maintain (and as appropriate, install) the Air Liquide recommended spare parts and that all service operations and preventative maintenance functions (listed below) be performed by the Customer's technicians that have been trained by Air Liquide.

The Customer's responsibilities include service operations and routine preventative maintenance of the system. This includes providing labor for all repairs covered in the Air Liquide training and certification program. Additional Customer responsibilities are described later in this document. If the Customer is responsible for preventative maintenance, documentation must be maintained regarding such maintenance presented to Air Liquide or its representatives at their request to ensure the validity of the warranty. If an equipment or part fails because the Customer has not performed routine preventative maintenance, the downtime will be charged against the Customer of the machine. Additionally, the Customer will be liable for any parts or labor required to fix the failure.

In cases where the equipment or parts have moved and been reinstalled within the Customer's factory without Air Liquide's authorized assistance or is removed or transshipped outside the Continental United States, this warranty shall be null and void.

In addition, the Customer has certain service responsibilities during the warranty period. When the conditions identified below have been met by the Customer, as determined by Air Liquide, the warranty will include all Air Liquide service labor, repair parts (excluding consumables), parts shipment and associated labor/travel costs.

Where the customer has not met these obligations, a purchase order from the Customer to pay for such services will be required before Air Liquide support would commence and the Customer will be charged for all services performed at Air Liquide's applicable rates.

Service Operations

The Customer is solely responsible for maintaining spare parts and operating any Advanced Equipment Systems Division equipment. The Advanced Equipment Systems Division equipment operations typically consist of:

- Monitoring equipment and system status
- Evaluating and maintaining system performance
- Restoring equipment to the automatic mode from manual mode and fault states resulting from point of use (POU) faults
- Recovering from events that interrupt production (ITP's)
- Responding to calls for assistance from the fab
- Maintaining an informational database and record of all equipment Preventative Maintenance (PM's) performed
- Purchasing and maintaining the Air Liquide-recommended spare parts

Routine Preventative Maintenance

The preventative maintenance activities for which the Customer is responsible include:

- Weekly, monthly, quarterly, semi-annual, and annual PM's for all Dispense Units, valve boxes, sample stations and support equipment
- Spare parts tracking and reordering
- Unscheduled repairs to restore equipment to operation
- Scheduled repairs not covered by PM's
- Pump rebuilds integral to Air Liquide equipment
- Consumable parts tracking and ordering

For a comprehensive list and detailed explanation of the preventative maintenance activities for which the customer is responsible please consult the Equipment Operations and Maintenance Manual.

A failed component will usually result in a shutdown. However, other shutdowns may occur from conditional faults generated by the control logic. The Customer is responsible for troubleshooting the system and attempting to return the equipment to the Auto mode. They are also expected to provide the labor to:

- Replace failed, damaged, or defective components in all fluid flow paths
- Adjust or replace all failed, damaged, or defective pressure, optical, capacitive, or mechanical sensors and switches
- Replace electronic components in the power distribution and control systems at the direction of the Technical Service Representative.

WARRANTY CLAIMS

When making a warranty claim the Customer is required to furnish the following: a valid purchase order (P.O.), equipment serial number, part number, quantity required, chemical service, and a description of the damage or failure. Air Liquide may require return of the defective part. The P.O. will be used to cover the cost of unreturned defective parts or non-warranty damage as well as tracking the claim. A Returned Material Authorization (RMA) number must be obtained from Air Liquide Technical Service before returning any parts to Air Liquide.

The Customer will be charged at Air Liquide's applicable rates for non-warranty Air Liquide activities. The Air Liquide Regional Service Manager in consultation with the Business/Contracts Manager will resolve any disputed claims.

RESPONSE TIME

Air Liquide Technical Service will make an oral response to a request for service within one hour when received by 4:00 p.m. Central Time Monday through Friday, excluding holidays.

TRANS-SHIPMENT OR EQUIPMENT RELOCATION

In cases where equipment is moved and reinstalled within or outside the Customer's factory, removed or trans-shipped without Air Liquide's written authorization, this warranty shall be null and void.

EQUIPMENT MODIFICATION BY CUSTOMER

Equipment modifications made by Customer shall void the warranty unless prior written approval is obtained in writing from an authorized representative of Air Liquide's Advanced Equipment Systems Division.

LIMITATION OF LIABILITY

In no event shall Air Liquide's liability, whether based on an action or a claim in contract, tort, (including without limitation negligence and to the extent permitted by law, strict liability) or otherwise, to Customer arising out of or related to the equipment exceed the aggregate price of the product purchased by such buyer as of the date such action or claim was filed.

IN NO EVENT SHALL AIR LIQUIDE BE LIABLE FOR DIRECT, SPECIAL, INCIDENTAL, INDIRECT, CONSEQUENTIAL OR PUNITIVE DAMAGES OF ANY KIND, WHETHER FOR BREACH OF ANY WARRANTY, BREACH OR REPUDIATION OF ANY OTHER TERM OR CONDITION OF THE SALE, NEGLIGENCE, ON THE BASIS OF STRICT LIABILITY, OR FOR ANY OTHER REASON.

LIMITATION OF WARRANTY

THERE ARE NO TERMS, CONDITIONS, OR WARRANTIES EXPRESSED, IMPLIED OR STATUTORY, OF MERCHANTABILITY, FITNESS FOR THE PURPOSE INTENDED, CAPACITY OR OTHERWISE, OF THE GOODS ORDERED OTHER THAN, OR DIFFERENT FROM, THIS WARRANTY. AND THIS WARRANTY IS EXPRESSLY IN LIEU OF ANY OTHER WARRANTY, EXPRESSED OR IMPLIED.

Air Liquide reserves the right to void the warranty on equipment that has been (i) serviced by untrained personnel, (ii) installed, started up or commissioned by non Air Liquide authorized personnel, or (iii) when the recommended spare parts have not been purchased and maintained.

Air Liquide neither assumes nor authorizes any other party to assume for it any liability in connection with said equipment except as hereinbefore stated.

NOTE: Merchandise supplied by Air Liquide is manufactured in compliance with the Fair Labor Standards Act of 1938, as amended.

CONSUMABLE LIST

- Filters
 - Water
 - Chemical
 - Nitrogen
 - Polyester air filters
- O-rings
- Filter housings, diaphragms, and pumps
- Pump Valve Kit
- Rebuild kits for pumps
- All fuses
- Light bulbs for Reset button
- All gaskets and seals
- All nuts, bolts, washers, etc.

ADVANCED EQUIPMENT SYSTEMS DIVISION
INTERNATIONAL WARRANTY ADDENDUM

PROVIDING FOR
AIR LIQUIDE ON-SITE SUPPORT

For Equipment Delivered and Installed Outside of the Continental U.S.

NOTE: This addendum should be read in conjunction with Air Liquide Electronics' international warranty and applies only if the customer has separately purchased these additional services.

RESPONSIBILITIES

The Customer has certain service responsibilities in addition to Air Liquide's responsibilities during the warranty period. When the conditions identified below have been met by the Customer, as determined by Air Liquide, the warranty will be extended to include all Air Liquide service labor, repair parts (excluding consumables), parts shipment and associated labor/travel costs. The Customer will provide the Air Liquide Technical Service Representative with a dedicated office in the vendor office area large enough for two individuals to work and store parts and tools.

Where the operational conditions have NOT been met, a purchase order from the Customer to pay for such services will be required before Air Liquide support would commence and the Customer will be charged for all services performed at Air Liquide's applicable rates.

Service Operations

Customer — The Customer is responsible for maintaining spare parts and operating the Advanced Equipment Systems Division equipment. The Advanced Equipment Systems Division equipment operations typically consist of:

- Monitoring equipment and system status
- Evaluating and maintaining system performance
- Restoring equipment to the automatic mode from manual and fault states resulting from point of use (POU) faults
- Recovering from events that interrupt production (ITP's)
- Responding to calls for assistance from the FAB
- Maintaining an informational database and a record of all equipment Preventative Maintenance (PM's) performed
- Purchasing and maintaining the Air Liquide recommended spare parts

AIR LIQUIDE — An Air Liquide Technical Service Representative will:

- Assist with developing schedules and criteria for routine system surveillance and hardware inspections
- Assist with evaluating system capacity and measuring performance
- Review all incidents resulting in downtime and assist in developing action plans to eliminate the source of interruption or minimize the duration of the outage
- Review technical and operational issues that have the potential to affect equipment performance and advise the Customer on appropriate actions

Routine Preventative Maintenance

Customer — The preventative maintenance activities for which the Customer is responsible include:

- Weekly, monthly, quarterly, semi-annual, and annual PM's for all Dispense Units, valve boxes, sample stations and support equipment
- Spare parts tracking and reordering, and returning of failed parts to Air Liquide (via the RMA process)
- Unscheduled repairs to restore equipment to operation
- Scheduled repairs not covered by PM's
- Pump rebuilds integral to Air Liquide equipment
- Consumable parts tracking and ordering

A failed component will usually result in a shutdown. However, other shutdowns may occur from conditional faults generated by the control logic. The Customer is responsible for troubleshooting the system and attempting to return the equipment to the Auto mode. The Customer is also expected to provide the labor to:

- Replace failed, damaged, or defective components in all fluid flow paths
- Adjust or replace all failed, damaged, or defective pressure, optical, capacitive, or mechanical sensors and switches
- Replace electronic components in the power distribution and control systems at the direction of the Technical service Representative.

AIR LIQUIDE — The Air Liquide Technical Service representative is to be called if the Customer is unable to recover from a shutdown or when the repairs necessary to restore operations were not included in the Advanced Equipment Systems Division equipment training and certification program. These activities are generally restricted to advanced maintenance of the control system. In addition, the Technical Service Representative will provide the following support:

- Review the PM records and periodically inspect the units for compliance with Air Liquide specifications
- Ensure that all product upgrades and safety bulletins are considered by the Customer
- Serve as a technical resource for the maintenance group
- Ensure that the Customer receives all warranty replacement parts

RESPONSE TIME

Subject to the terms of this Warranty, an assigned Technical Service Representative will provide the customer on-site support, Monday through Friday, excluding holidays, 8:00 a.m. to 5:00 p.m. or during normal (day shift) business hours not to exceed 8 working hours per day. Support during other hours is not provided under this Warranty. Customers may, however, purchase such coverage through a Site Supplement Contract with Air Liquide.

In-stock emergency parts will be shipped from Chanhassen or the local Regional Office the same day for overnight delivery when calls are received by Air Liquide before 2:00 p.m. Central Time during normal business days, excluding Holidays.

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Technical Support Contact Information

Customers of Air Liquide Electronics can contact Air Liquide Electronics (AES) Technical Support 24-hours/7-days a week. This includes customers with equipment produced by the Hollister, CA Operations (formerly Applied Chemical Solutions), Systems Chemistry, FSI International, and BOC Edwards.

To reach Air Liquide Electronics Technical Support via Phone

To reach Air Liquide Electronics Technical Support, dial:

1-877-90ALEUS (1-877-902-5387)

1-952-917-5200

A Technical Support Specialist will answer and provide immediate assistance. If other customers are being assisted at the same time, leave a voice message or call again later.

After normal business hours, you can call the Technical Support line at the same number and extension as above. These connections automatically roll over to the on-call technician's cellular telephone, and the technician will respond as soon as possible.

To reach Air Liquide Electronics Technical Support via Email

Customer with questions that do not require an immediate answer, but that need attention, are encouraged to send an email to:

Airliquide_AES_TechSupport@airliquide.com

A Technical Support Representative will respond to your question as soon as possible.

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Introduction

The following preface and summary of changes provides an introduction to this publication.

This manual was originally written in English.

About This Manual

This manual is designed to provide reference and technical information for operating and maintaining the Inline Blender, with Integrated Daytanks.

Who Should Use This Manual

This manual is intended for use by trained Inline Blender, with Integrated Daytanks, operators and technicians. The manual user must be familiar with the chemical unloading, blend and dispense functions of the Inline Blender, with Integrated Daytanks, and its service and maintenance procedures. Procedures that require manufacturer's service are not included in this manual.

How This Manual Is Organized

This manual is organized in the following manner:

- [Safety](#) explains the safety message conventions and equipment safety features used in this manual.
- [Chapter 1, General Description](#), contains a basic description of the Inline Blender, with Integrated Daytanks, and how it operates.
- [Chapter 2, Installation](#), discusses site requirements, uncrating, and installation of the Inline Blender, with Integrated Daytanks.
- [Chapter 3, Component Description](#), illustrates cabinets and compartments and lists components alphabetically.
- [Chapter 4, Operations](#), describes the HMI and provides procedures for the software-controlled operations that the Inline Blender, with Integrated Daytanks, performs.

- [Chapter 5, Startup and Shutdown](#), contains the steps necessary to start up the Inline Blender, with Integrated Daytanks, upon initial installation or after a long-term shutdown, and the steps necessary to shut down the unit (short-term shutdown or long-term shutdown).
- [Chapter 6, Preventive Maintenance](#), describes the frequency of periodic maintenance tasks that required for optimum uptime of the Inline Blender, with Integrated Daytanks.
- [Chapter 7, Alarms and Troubleshooting](#), contains information about Inline Blender, with Integrated Daytanks, alarms and their probable causes and recommended corrective actions.
- [Chapter 8, Component Procedures and Adjustments](#), provides information on various procedures, adjustments, tests, and replacement of components.
- [Appendix A, Maintenance Supplies](#), provides information about manufacturers' tools and test equipment that apply to the Inline Blender, with Integrated Daytanks.
- [Appendix B, System Specifications](#), contains Inline Blender, with Integrated Daytanks, specifications.
- [Appendix C, Input and Output Lists](#), describes I/O status and modules, including timers, inputs and outputs. Like the HMI, they are presented by object (tank, dispense, cabinet, and so on), and by modules communicating with the PLC.
- [Appendix D, System Settings](#), lists the settings for blender configuration, timers, inputs and outputs.
- The [Glossary](#) appendix defines various terms used in this manual that might not be familiar to the reader, followed by a list of acronyms used in component designations and in engineering prints in this manual.
- The [Prints](#) section lists the engineering drawings that are useful in installing, operating, and maintaining the Inline Blender, with Integrated Daytanks.



BEFORE PROCEEDING, READ AND UNDERSTAND THE SAFETY INFORMATION CONTAINED WITHIN THE [SAFETY](#) CHAPTER OF THIS MANUAL.

Summary of Changes

Following is a history of changes to this publication, beginning with the first revision up to the current revision.

Revision A

October 2015. Initial release of this manual, per ECO C14-4704.

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S Safety



**BEFORE PROCEEDING, READ AND UNDERSTAND THE SAFETY INFORMATION
CONTAINED WITHIN THE [SAFETY](#) CHAPTER OF THIS MANUAL.**

This chapter describes the safety-related information that is important for safe equipment operation. Included is a listing of message conventions used in this manual.

The equipment described in this manual uses hazardous chemicals and produces waste products that can be dangerous. Only personnel trained in the procedures and safety messages outlined in this manual should install, operate, or maintain this equipment. When using this equipment, be sure to follow the safety procedures outlined by your facility. These safety procedures should cover the two primary types of hazard training: (1) equipment hazards, and (2) facility-related hazards.

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Safety Message Conventions

Safety messages contained in this manual, **DANGERS**, **WARNINGS**, **CAUTIONS** and **ESD IMPORTANTS** are highlighted in bold for quick identification.

Danger

A Danger message indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury. Messages identified by the word **DANGER** are used sparingly and only for those situations presenting the most serious hazards.

Following is a typical example of a Danger message as it could appear in a manual:



HAZARDOUS CHEMICALS—RISK OF SERIOUS INJURY OR DEATH

Hazardous process chemicals (for example, HF, HCL) are used within the equipment. Prior to working on the unit, such as when replacing any wetted components, complete all the purge and flush procedures.

Warning

A Warning message indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.

Following is a typical example of a Warning message as it could appear in a manual.



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY.

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Wear proper safety gear when performing this procedure.

Caution

A Caution message indicates a potentially hazardous situation which, if not avoided, could result in minor or moderate injury. It may also be used to alert against unsafe practices.

Following is a typical example of a Caution message as it could appear in a manual.



CAUTION

COVER CAN FALL IF IMPROPERLY SECURED, CAUSING PERSONAL INJURY OR EQUIPMENT DAMAGE.

Secure the cover in the open position by placing the notches at the bottom of both arms on the front lip of the cabinet.

IMPORTANT

An Important message alerts the user to actions that could potentially lead to damage to the product, equipment or facilities, if instructions are not followed properly.

An ESD Important message warns of electrostatic discharge that could result in damage to equipment. Following is an example of an ESD Important message as it appears in a publication.



IMPORTANT

Improper handling of electronic components can result in electrostatic discharge, which can cause component failure.

To avoid damage from electrostatic discharge, follow these basic rules:

- **Handle all static-sensitive components at a designated static- safeguarded area or on a portable ESD mat.**
- **Transport all static-sensitive components (for example, circuit boards) in static-shielded containers or packages.**
- **Monitor and test static-protection products and materials to ensure that they are working properly.**

NOTE

A Note message provides tips, shortcuts, and other information that help the user perform a particular action.

Safety Overview

Basic Safety Rules

These safety rules can be used as precautions to apply to ensure the safety of persons involved in maintenance and operations of this equipment; they are not intended as complete safety information for handling chemicals:

- All operations on hazardous chemical systems should be performed **only** by skilled and trained operators.
- Follow the safety practices and procedures establish by the facility.
- Systematically refer to the SDSs, which include first aids measures, risk pictograms and accidental release measures.
- Wear appropriate personal protective equipment (PPE) as specified by facility procedures and SDS.
- Allow the cabinet to exhaust to clear any gas from the cabinet before beginning work.
- While performing in-line work on these systems, two people must be present at all times.
- Ensure that the efficiency of the exhaust system has been checked before starting to work.

Process Material (Chemicals)



THE BLENDER IS DESIGNED TO BLEND ONE CHEMICAL AND ULTRA-PURE WATER.

The chemical to be used is customer- and recipe-specific. In order to keep the blend process generic, this manual does not mention a specific chemical.



FOR SAFETY, ALWAYS REFER TO THE SDS FOR THE CHEMICAL.
Before any operation, refer to the Safety Data Sheet for the chemical that is used. Be aware of the chemical's properties and the safety measures that are needed.

Installation and Setup Safety

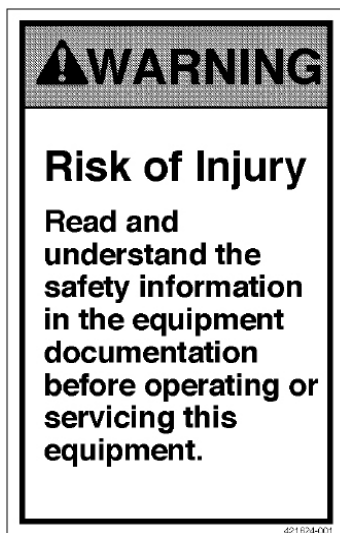
In addition to site requirements and procedures for installation and setup, follow these recommended minimum precautions to help guarantee a safe system startup.

- Install the equipment according to the P&IDs and facilities diagrams in the [Prints](#) section of this manual.
- For safety and correct equipment operation, refer to the Facilities diagram in the [Prints](#) section for site requirements for exhaust pressure and flow.
- No obstacles can be located in front of the Inline Blender, with Integrated Daytanks, cabinet.
- The blender must be located in a secured room where access is restricted. Operating procedures must be in place to ensure that personnel are wearing personal protective equipment (including breathing apparatus, safety glasses, and safety gloves) before being permitted access to these rooms.
- Install and test all the piping according to best industry practices.
- Make sure that the piping network is made of PFA with clear PVC.
- Make sure that vent lines run directly to the appropriate effluent treatment system as approved by the local regulatory agency for the particular chemical species.
- Install the unit on flat ground that supports a minimum weight of 1000 kg/m².
- Perform pressure testing on the piping system in an “as installed” state.
- Perform leak tests on chemical systems before any functional tests.
- Conduct system tests in the facility plant.

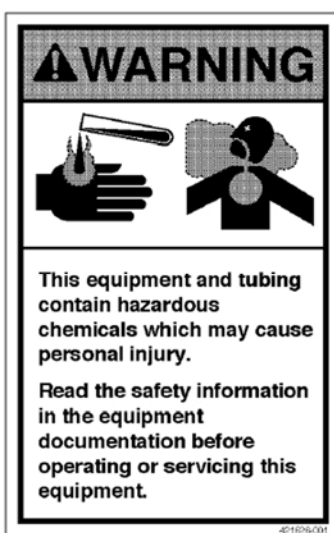
Safety Labels and Emergency Cabinet Switches

The safety labels and emergency cabinet switches on the Inline Blender, with Integrated Daytanks, are shown below. Additional safety labels are placed inside the Wetted and Controls compartments.

Figure S-1. Safety Labels



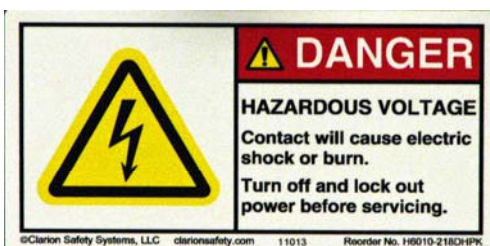
Label P/N 421624-001



Label P/N 421626-001



Label P/N 421489-001



Label P/N 433948-001



Label P/N 314305-001



Label P/N 317043-001

Equipment Safety

Electrical

The electrical cabinet of the Inline Blender, with Integrated Daytanks requires 115-230 VAC, single-phase current, at 50-60 Hz. Use extreme caution when working with electricity or during lighting installation.

The electrical power in the electrical cabinet is switched off by the lockable **MAIN POWER DISCONNECT SWITCH** on the front and back of the Inline Blender, with Integrated Daytanks, and each circuit is protected by a circuit breaker.

Levitronix BPS-600 centrifugal pump P21 requires 115-230 VAC. Two electrical devices, a **BOOSTER PUMP ENABLE PUSHBUTTON** and a **BOOSTER PUMP DISCONNECT SWITCH**, are used by the operator to enable/disable the Levitronix centrifugal pump P21.

The 24 VDC, 10A from DC power supply PWS215 is used to provide power to the modules in the Programmable Logic Controller (PLC), the touch screen (HMI), level sensors, and other electrical components in the system.

Only qualified electricians having a good understanding of the blender's electrical devices and wiring diagram should perform work on electrical systems. The Electrical Schematic listed in the [Prints](#) section shall always be updated in case of any modification.

The facility's electrical safety procedures must be followed.



LINE VOLTAGE MAY BE PRESENT WITH MACHINE POWER OFF.

Contact will cause severe injury or death. Disconnect and lock out power at the main panel before performing maintenance on the blender.

Safety Interlocks

Some of the safety devices (door interlocks) are hard-wired through a relay on a contact placed in the safety interlock line. If any of these relays opens its corresponding contact, the relay on the safety interlock line opens the contact to stop the blender.

IMPORTANT

When one of the safety interlocks is activated, the sensors remain energized to perform data/alarm logging and error recovery functions.

Fire Hazard and Safety

Environmental conditions and uncontrollable events might result in the potential for electrical fire or chemical combustion and should not be discounted despite proactive and precautionary measures for the prevention of fire.

All operators and service technicians must be trained in remedial and corrective action related to safety, fire prevention, control and the proper use of fire extinguisher or the summoning of assistance from professional fire-fighting personnel.

Consult the applicable safety data sheets (SDSs) for hazardous chemical fire prevention and control.

Combustibles

Plastic in components, such as tubing and electrical cables, and the blender enclosure, are combustible.

The material for the valves and tubing is PFA, and the material for the enclosure is polypropylene. These materials are not fire-retardant. These materials handle chemicals without degrading.

Ignition Sources

Electrical components and lights are the only potential ignition sources within the cabinet.

The use of properly rated electrical components and fault-tolerant control systems decreases the risk of fire.

Emergency Stop (EMO)

The **EMO** (Emergency Machine OFF) pushbuttons/switches are located on the front and back of the cabinet. An EMO requires only one momentary push to immediately shut down the whole blender and stop chemical delivery and production. All the solenoid valves close and bring the pneumatic valves to their normal, non-energized position.

IMPORTANT

The PLC, touch screen, and all the sensors remain energized when the EMO is activated to perform data/alarm logging and error recovery functions.

External EMO (single contact)

An external EMO can be added in the EMO circuit of the blender. Two connections are provided to connect an external EMO device. Refer to the electrical schematic listed in the *Prints* section of this manual.



ELECTRICAL HAZARD—RISK OF ELECTRICAL SHOCK AND/OR DAMAGE TO THE MODULE.

External switches connected to the external interlock must be a “dry contact” free of voltage.

Never connect a power source to the safety interlock circuit.

The external EMO must be connected to a dry contact NC in normal conditions.

Emissions

Exhaust

Exhaust ventilation is designed as supplemental control because intermittent activities (such as maintenance or leakage) could present potentially hazardous chemical exposures to personnel. An exhaust pressure sensor detects exhaust pressure and can trigger an alarm if pressure exceeds or goes below normal operating conditions (refer to the Facility & Installation diagram in the [Prints](#) section of this manual). An exhaust pressure indicator is located on the top of the blender cabinet.



FACILITY REQUIREMENTS FOR EXHAUST PRESSURE MUST BE MAINTAINED TO PREVENT POSSIBLE EXPOSURE OF PERSONNEL TO CHEMICAL VAPORS.

Because chemical vapors could be contained in exhaust as a result of leakage or maintenance operations, connect the exhaust to a system that washes the vapors of chemical before ejecting the air into the atmosphere.

Uncontained Chemical

In case of multiple, major failures or unexpected events, there could be a spill of chemical. Major leaks or spills will be contained inside the cabinet up to the overload coupling. Large leaks will flow through this connection to the facility drain.



CONSIDER ALL LIQUID IN THE BLENDER AS HAZARDOUS CHEMICAL.

Always wear PPE, including gloves, safety glasses, and boots, before dealing with any liquid near any part of the blender.

Nitrogen

Nitrogen is used to provide inert blankets on the two interior daytanks.

Drains for Liquids

The Inline Blender, with Integrated Daytanks, drains are connected to facility drains. During normal or maintenance operations, chemical and UPW are sent to the drain.

- A free flow drain, which collects liquid during a maintenance operation or process.
- In the event of a leak, the sump cup is evacuated to this same gravity-drain column.



HAZARDOUS CHEMICALS—RISK OF SERIOUS INJURY OR DEATH
Each drain must be connected separately to a main drain network.

Only compatible chemicals should be sent to the same main drain network.

Solid Hazardous Waste



COMPONENTS THAT HAVE BEEN IN CONTACT WITH CHEMICAL MIGHT CONTAIN HAZARDOUS CHEMICALS OR CHEMICAL RESIDUE.

Always wear proper PPE, take proper safety precautions, and follow site procedures for proper disposal of contaminated equipment and materials used to handle them.

The solid hazardous waste of operation, maintenance and servicing include all the components that have been in contact with any chemical. This includes valves, pump membranes, and filters, plus all equipment used to handle them, such as gloves, PPE, bags, and sponges.

Hazardous Energies & Lockout Information

Refer to the following table for lockout and hazardous energies information. Use the information in [Table S-1](#) for assistance in writing lockout/tagout procedures for this equipment.

For more specific instructions, refer to the Fab's lockout/tagout requirements and procedures.

Table S-1. Inline Blender, with Integrated Daytanks, Lockout/Tagout Information

Equipment Name:	Inline Blender, with Integrated Daytanks
Energy Type:	<u>Chemical</u> Hazard: Exposure to chemical. Magnitude: Variable. Control Method: Primary and secondary containment. <u>Electrical</u> Hazard: Electrical energy in the Controls Compartment. Magnitude: 115-230 VAC, single-phase current. Control Method: Disconnect & lock out MAIN POWER DISCONNECT SWITCH before servicing the blender. <u>Pneumatic</u> Hazard: Mechanical pinch points, pressure-related injuries, noise. Magnitude: Nitrogen >50 psi. Noise <75 dB(A). Control Method: Air and nitrogen line isolation valves. <u>Magnetic Field of Centrifugal Pumps</u> Hazard: Possible adverse effects to people with electronic medical assistance devices such as pacemakers. Also, mechanical pinch points, pressure-related injuries when handling internal pump parts. Magnitude: Not known. Control Method: Warning in Manual.
Shutdown Procedure:	1. Select Operating Mode from the main menu. 2. Press SAFETY and then press STOP for each circuit. 3. Turn the MAIN POWER DISCONNECT SWITCH to the OFF position. 4. Perform the remaining procedures in this table.
Machine and Equipment Isolation:	After the Shutdown Procedure is performed, the PLC inputs and outputs are deactivated and deenergized, and chemical is isolated by primary and secondary containment.
Affixing Lockout or Tagout Devices to the Isolating Devices:	1. Obtain a lockout or tagout device. 2. Secure lockout or tagout device through one of the round holes in the MAIN POWER DISCONNECT SWITCH (lock the padlock or fasten the tag). 3. Make sure that lockout/tagout devices have been secured to the facility's chemical, N₂, and ultra-pure water (UPW) ports. (The procedures for the lockout/tagout of facility ports are determined by facility safety policies and practices, and are not included in this manual.) 4. Verify that the MAIN POWER DISCONNECT SWITCH cannot be turned to the ON position.
Relieving Stored Energies:	After the Shutdown Procedure is performed, chemical in the equipment is not under pressure and electrical energy has been dissipated.

S-14 Safety

Verifying Machine Isolation and De-energization	1. Visually verify that the touch screen panel is blank. 2. Open the electronics compartment door and visually verify that: • LEDs on the PLC components are not illuminated. • LEDs on the solenoid valves are not illuminated. • Screens on the Hach controllers are not illuminated.
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IMPORTANT

Tagout/lockout procedures for the facilities chemical, N₂, and UPW ports and electrical power panel are to be developed by the customer in accordance with site policies and procedures.

1 General Description



WARNING



BEFORE PROCEEDING, READ AND UNDERSTAND THE SAFETY INFORMATION CONTAINED WITHIN THE [SAFETY](#) CHAPTER OF THIS MANUAL.

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BEFORE PROCEEDING, READ AND UNDERSTAND THE SAFETY INFORMATION CONTAINED WITHIN THE [SAFETY](#) CHAPTER OF THIS MANUAL.

Description of the Inline Blender

The Inline Blender, with Integrated Daytanks, is designed to create a chemical blend in accordance with customer-specified recipe requirements, by using a single chemical constituent (Chem1 from a pressurized chemical supply in the Fab) and Ultra-Pure Water (UPW). Advanced Equipment Systems Division does not support the use of the Inline Blender, with Integrated Daytanks, with any other chemical blend, other than by using the customer-specified chemical and UPW.

The blender utilizes inline metrology and instrumentation to control the chemistry and/or UPW flow, and final assay to the tank(s). The blender controls according to the configurable recipe and settings. Chemical flows to the tank(s) if it is within these specifications. Chemical flows to the drain or shuts down if it is **not** within specifications.

The blended chemistry is recirculated within the tank(s) by an onboard pump circuit with inline filter/metrology. The blender controls the outlet valve of the blended chemistry tank(s), allowing downstream units to draw from the tank(s) only if all criteria is in specifications.

The Inline Blender can be installed /operated as a single unit, or as an integrated, redundant pair. When the Inline Blender operates as a pair of units, the units will alternate as available **online** tank(s), but are still capable of operating as a single unit, if the alternate is not available. For details, see the blender pair system drawing listed in the [Prints](#) section of this manual.

The Stages of the Blending Process

The blend is created in two stages; recirculated in a third stage; and distributed in a fourth stage:

Stage 1A: UPW flows from a Fab source past a pressure regulator and a UPW flow controller to an injection port in the Wetted Compartment plumbing.

Stage 1B: At nearly the same time that Stage 1A occurs, chemical Chem1 is supplied to the Inline Blender from a pressurized source in the Fab. Chem1 flows through a manual valve, check valve, AOVs, pressure regulating valve, centrifugal booster pump, and chemical flow controller to an injection port in the Wetted Compartment plumbing.

Stage 2: Chem1 + UPW flow through a static mixer, an accumulator, and past a metrology suite (redundant temperature transmitters and conductivity transmitters), before flowing into integrated daytanks 1A and 1B.

Stage 3: The blended, qualified chemical in daytanks 1A and 1B is recirculated back to daytanks 1A and 1B by means of a centrifugal recirculation pump.

Stage 4: The blended, qualified chemical is distributed from daytanks 1A and 1B, at the same time, to either the T1 Chem Out 1 or T1 Chem Out 2 ports.

See the [Theory of Operations](#) section on page 1-5 for a detailed description of the entire sequence of operation for the Inline Blender, with Integrated Daytanks.

[Table 1-1](#) lists the metrology—conductivity, temperature, flow and pressure monitoring—used at each stage in the process to achieve the required blend recipe.

Table 1-1. Chem1 and Chem 2 Metrology Components

Feature	Components
Pressurized Chem1 and flow controller	• Chemical flow controller J1 (flow 0-2500 milliliters per minute), or similar range. Can vary based in accordance with blending specifications.
UPW and flow controller	• UPW flow controller J99 (flow 0 to 20 liters per minute).
Blended Chem1 + UPW and metrology	• Redundant temperature transmitters. • Redundant conductivity transmitters. • One pressure transducer.
Daytank 1A & daytank 1B recirculation and metrology	• One conductivity probe to Hach analyzer. • One temperature probe to Hart transmitter. • One flowmeter. • One pressure transducer.

Theory of Operations

This section elaborates on the information presented in the [Description of the Blender](#) section. Refer to the P&ID drawing listed in the [Prints](#) section for details.

Stage 1A: UPW Flows to an Injection Port

UPW, from a Fab supply source, flows under 40-60 psi pressure into the Inline Blender. MV.J99.110 must be open; the blender's PLC opens AOV.J99.110. The UPW flows past PRV.J99.110, which can be adjusted to regulate the flow pressure.

The UPW enters flow controller J99, which monitors the flow rate. The UPW exits J99 and flows to an injection port in the plumbing.

Stage 1B: Pressurized Chemical Flows to an Injection Port

At almost the same time as Stage 1A is occurring, pressurized chemical Chem1 from the Fab flows into the Online Blender through a port at the rear top of the cabinet. MV.J1.110 must be open; the blender's PLC opens AOV.J1.110. The chemical flows past PRV.J1.110, which can be adjusted to regulate the flow pressure.

Levitronix BPS-i100 centrifugal pump P1 boosts the flow of Chem1 to and through flow controller J1.

Chem1 enters flow controller J1, which monitors the flow rate. Chem1 exits J1 and flows to an injection port in the plumbing.

Stage 2: Blending the UPW and Chemical

The separate flows of UPW and Chem1 converge at the injection port and flow together through static mixer SM.Q1.910. The helical mixing element inside the static mixer divides, re-directs, and re-orientes the flow of the liquids.

From the static mixer the blended UPW + Chem1 flows through accumulator GC.Q1.910, which stabilizes the blend before it passes through a metrology suite. The suite (consisting of temperature transducer TT.M1.910A, conductivity analyzers CT.M1.910A and CT.M1.910B, and temperature transducer TT.M1.910B) measures, monitors, and controls the health of the blended chemical before it flows past pressure transmitter PT.Q1.910 (which monitors the pressure of the flow after the metrology suite) into daytank 1A (if MV.T1.111A is open) and into daytank 1B (if MV.T1.111B is open).

Stage 3: Recirculating the Blended Chemical From & Back to the Daytanks

Daytanks 1A and 1B use a Levitronix BPS 600 centrifugal pump as their recirculation pump (P21). The two daytanks function as a single container. Chemical is recirculated from pump P21 to both daytanks at the same time. When a Chem Demand signal is received, blended chemical is draw from both daytanks at the same time to transfer to the Fab.

The capacitive level sensors that are located outside the daytanks monitor the Low-Low, Low, High, High-High, and Overfill conditions of blended, qualified chemical in the daytanks. (See the description of each level sensor in the [Daytanks](#) section, starting on page [1-9](#)).

To keep the batch homogenized and within blend specifications, pump P21 recirculates the blend from and back to the daytanks. The blended chemical flows through chemical filter F21 and a metrology suite dedicated to both daytanks. The metrology suite (CT.M21.910, TT.M21.910 and a flowmeter with PT.M21.910 and FT.M21.910) monitors the conductivity, temperature, flow, and pressure of the recirculating blend in the tank. Sample #2 can be taken of the blend that is stored temporarily inside each tank, after the blend recirculates through filter F21.

As required, the recirculating blend can be sent to drain (when the PLC opens AOV.P21.820).

If pump P21 needs to be primed with chemical, the PLC opens AOV.P9.510 (after the operator uses the applicable screen and touch key at the HMI panel). The blended chemical flows through the pump, filter, metrology suite, flowmeter, check valve, and several AOVs to AOV.P9.510. AOV.P9.510 opens and the blend flows through a plumbing line to aspirator P9. A vacuum is created by the DIW that flows through P9. This vacuum draws the chemical through P9 to the main drain pipe and out to the facilities waste drain.

Stage 4: Transferring the Blend from the Daytanks to the Fab

When the chemical is within blending specifications, blended chemical from daytanks 1A and 1B flows through MV.T1.910 (if it is open) and through AOV.T1.910 to either one of two outlet flow paths:

- If MV.T1.991 is open, the blended chemical flows to T1 Chem Out 1.
- If MV.T1.992 is open, the blended chemical flows to T1 Chem Out 2.

The T1 Chem Out 1 and T1 Chem Out 2 ports are located at the bottom, left-hand side of the Inline Blender cabinet. See the Facility & Installation drawing listed in the [Prints](#) section for details.

Future Options

The following options for future use (i.e., they are not currently available) have been built-into the unit. Although they are indicated on the P&ID, their outlets are currently capped. They are described below.

IMPORTANT

IMPLEMENTATION OF THE FOLLOWING OPTIONS MUST BE REVIEWED WITH, AND APPROVED BY, AIR LIQUIDE.

- **AUX OUT 3:** After safety inputs and interconnects of an external unit are integrated with the Inline Blender and when an external unit requests chemical, the blended chemical will flow through AOV.Q1.910 and MV.Q1.910 to an external unit. The port for Aux Out 3 is located at the top of the Inline Blender cabinet. See the Facility & Installation drawing listed in the [Prints](#) section for details.
- **AUX FILL 4:** After safety inputs and interconnects of an external unit are integrated with the Inline Blender and when an external unit requests chemical, the Inline Blender transfers chemical from the daytanks by using centrifugal pump P21. An external unit for the blended chemical might be a Tote Fill Station or the daytank of an M15xx Chemical Distribution Module. The port for Aux Fill 4 is located at the top of the Inline Blender cabinet. See the Facility & Installation drawing listed in the [Prints](#) section for details.
- **BLEND OUT 2:** After safety inputs and interconnects of an external unit are integrated with the Inline Blender and when an external unit requests chemical, the blended chemical will flow through AOV.T2.110 and MV. T2.110 to an external unit. The port for Blend Out 2 is located at the top of the Inline Blender cabinet. See the Facility & Installation drawing listed in the [Prints](#) section for details.

Common Features and Components

Bottle Sample Area

The Bottle Sample Area is used primarily for taking bottle samples of chemical from the Sample #1 and Sample #2 slipstreams in the Inline Blender. The Bottle Sample Area also contains a removable plug, for connecting an optional, customer-provided, external particle counter. The Bottle Sample Area has a magnetic door interlock, which alarms and prevents chemical flow, if the door is opened while a bottle sample is in-progress.

A Sample Compartment Rinse service option is available (Air Liquide P/N 928318-001). The option allows UPW water to be drizzled throughout the Bottle Sample Area, by means of a spray nozzle at the top interior of the compartment. When the option is activated, the PLC opens AOV.L1.210 and UPW water enters the bottle sample area through the Service L1 line.

Chemical Filter

The Inline Blender, with Integrated Daytanks, contains chemical filter F21, whose 20-inch filter element is positioned inside a filter bowl and filter housing. The replaceable filter element is used to trap small particles and provide cleaner chemical for the recirculating qualified blend. Filter F21 is located between the outlet of pump P21 and the metrology suite for the recirculating chemical in the daytank.

Chemical Supply Source

The Inline Blender, with Integrated Daytanks, receives Chem1 from a pressurized chemical supply source in the Fab.

Daytanks

Each daytank (1A and 1B) holds 200 liters of blended chemical.

Daytanks 1A and 1B function as a single container:

- Blended, qualified chemical flows into both daytanks (when the PLC opens AOV.T1.110, and if MV.T1.110, MV.T1.111A, and MV.T1.111B have been manually opened) at the same time.
- Blended chemical is recirculated by pump P21 to and from daytanks 1A and 1B at the same time, through the respective plumbing to each daytank and through the applicable open AOVs and manual valves before and after pump P21 (see the P&ID for details).
- Blended chemical is pulled from both daytanks at the same time through AOV.T1.910 (which is open) to the T1 Chem Out 1 and/or T1 Chem Out 2 ports. Manual isolation valves MV.T1.991 and/or MV.T1.992 must be open.

Four capacitive chemical level sensors are installed on outside rails of daytank 1A and/or daytank 1B. These sensors are:

- Overfill level sensors: CS.T1.58A (daytank 1A) and CS.T1.58B (daytank 1B). For safety reasons, each daytank has its own Overfill level sensor.
- High-High level sensors: CS.T1.18A (daytank 1A) and CS.T1.18B (daytank 1B). For safety reasons, each daytank has its own High-High level sensor.
- High level sensor: CS.T1.17A, which serves daytanks 1A and 1B.
- Low level sensor: CS.T1.12A, which serves daytanks 1A and 1 B.
- Low-Low level sensor: CS.T1.11, which serves daytanks 1A and 1B.

Door Interlocks

The doors of the Inline Blender, with Integrated Daytanks, cabinet are equipped with magnetic interlock switches that determine whether a door is open or closed. In addition, the Bottle Sample Area has a door interlock.

When the applicable door is open, the magnetic interlocks signal the PLC and stops the flow of chemical from the sample tube to the sample bottle. The PLC detects the open interlock.

Electric Keyswitch (Tank 1 Out Override)

Electric keyswitch EKS.T1.910, located in the Controls Compartment, is a component of the Tank Outlet Valve Override Kit. The kit allows the user to pneumatically override AOV.T1.910, independently of the PLC software. Whenever AOV.T1.910 is opened, chemical flows from tandem daytanks T1A/T1B to either T1 Chem Out 1 or T1 Chem Out 2.

Exhaust Pressure Switch and Pressure Indicator

Exhaust pressure switch PS.Z1.491 monitors the exhaust duct pressure for the cabinet and activates an alarm whenever the exhaust low-pressure reading declines below its setpoint. The exhaust alarm signal activates after a user-set time delay, to prevent frequent occurrences of the alarm whenever a compartment door of the Inline Blender is opened momentarily.

Magnehelic exhaust pressure indicator PI.Z1.491 provides a visual display of the exhaust pressure reading.

Float Switches

The sump in the cabinet of the Inline Blender, with Integrated Daytanks, contains two float-type leak detection switches: Warning switch FS.Z5.8 (which sends a Warning signal to the system PLC), and Shutdown switch FS.Z5.9 (which sends a shutdown signal to the system PLC when the level of leak chemical continues to rise).

Flow Accumulator

Flow accumulator GC.Q1.910 reduces the turbulence in the mixed chemical and UPW after they flow through static mixer SM.Q1.910, but before they flow through temperature transmitter TT.M1.910A, conductivity analyzers CT.M1.910A and CT.M1.910B, and temperature transmitter TT.M1.910B.

Flow Controllers

The two flow controllers monitor and control the volume of liquid (in liters/minute) at specific flow pressures.

- Flow controller J1 controls the flow of pressurized chemical from the Fab.
- Flow controller J99 controls the flow of UPW after it flows from the Fab.

Flow Meter

Nitrogen flows through nitrogen filter F.N1.950 and flowmeter FM.N1.953, which filters the nitrogen used as nitrogen blankets at the interior top of daytanks 1A and 1B.

Operator Controls – Front of Cabinet

The main operator panel for the Controls Compartment is located on the front of the Inline Blender, with Integrated Daytanks, cabinet. The operator panel contains the following switches and monitoring components:

- **ALARM** horn: The alarm horn sounds when a system alarm occurs.
- **ALARM SILENCER** button: When the alarm horn sounds, this button can be pressed and held-in to silence the alarm for five minutes.
- **EMO** button/switch: When pressed, immediately stops electrical power to the PLC outputs.
- **MAIN POWER DISCONNECT SWITCH**: This switch turns ON or turns OFF the 115VAC-230 VAC power coming from a power source to the Inline Blender. When the switch is in the ON position, power flows to the unit's power distribution block and to the power supply, where the AC voltage is transformed (stepped-down) to 24 VDC for use by the unit's electrical components.
- **EMO RESET** button: After power has been shut off by pressing the EMO switch, this button is used to restore power.
- **PUMP DISCONNECT SWITCH**: Used to stop the operation of the Levitronix BPS-600 centrifugal recirculation pump.
- **PUMP ENABLE SWITCH**: Used to start the operation of the Levitronix BPS-600 centrifugal recirculation pump.

Operator Controls – Rear of Cabinet

The following operator controls are located at the rear of the Inline Blender cabinet:

- **EMO** button/switch: Immediately stops electrical power to the PLC outputs.
- **SAMPLE PUSH & HOLD** button: To start a bottle sample, push and hold this button.
- **SAMPLE FLOW VALVE** (optional): Used to adjust the flow of chemical for a bottle sample.

PLC/HMI

The Inline Blender, with Integrated Daytanks, uses Allen-Bradley CompactLogix 1769-series digital and analog I/O PLC modules for subsystem control, input/output monitoring, data management, and alarm indications. A 12-inch General Electric QuickPanel HMI (human machine interface) panel provides the user interface with the PLC. The HMI has text and a graphic depiction of the plumbing, valves, and major components, which change color when active. Touch-sensitive softkeys on the screen provide screen-to-screen navigation and parameter entry.

Pneumatic Keyswitch

Pneumatic keyswitch PKS.T1.910, located in the Controls Compartment, is a component of the Tank Outlet Valve Override Kit. PKS.T1.910 receives an electrical signal from EKS.T1.910. The electrical signal activates the pneumatic solenoid in PKS.T1.910, which sends a pneumatic signal to POV.T1.910 (see below).

Pneumatic “OR” Valve

Pneumatic “OR” valve POV.T1.910, located in the Controls Compartment, is a component of the Tank Outlet Valve Override Kit. It is connected to the pilot line of AOV.T1.910. When it is activated, POV.T1.910 sends a pneumatic signal to AOV.T1.910 to open and allow chemical to flow to either T1 Chem Out 1 (providing that MV.T1.991 has been manually opened and is plumbed to the outlet port) or T1 Chem Out 2 (providing that MV.T1.992 has been manually opened and has been plumbed to the outlet port).

Pressure Controls

The following indicating pressure switches, pressure regulating valves, and pressure indicators are located on the front Operator's Panel of the cabinet:

- **SOL VLV PRESSURE:** Pressure switch IPS.A1.951 indicates the current N₂ pressure for the solenoid valves. Pressure regulating valve PRV.A1.951 allows the operator to adjust the pressure setting.
- **N2:** Pressure switch IPS.N1.950 indicates the current N₂ pressure from the Fab to the blender. Pressure regulating valve PRV.N1.950 allows the operator to adjust the facility N₂ pressure setting.
- **UPW BLEND PRESSURE:** Pressure regulating valve PRV.J99.110 is used by the operator to adjust the UPW water pressure setting. The pressure setting is displayed at pressure indicator PI.J99.301.
- **CHEM BLEND PRESSURE:** Pressure regulating valve PRV.J1.110 is used by the operator to adjust the Chem1 pressure setting. The pressure setting is displayed at pressure indicator PI.J1.301.

Pressure Transducers

- Analog pressure transducer PT.Q1.910 is located after Sample #1 AOV.Q1.810 and before PRV.J99.110. PT.Q1.910 monitors the pressure of the blended chemical + UPW after they flow through the metrology suite (TT.M1.910A, CT.M1.910A, CT.M1.910B, TT.M1.910B).
- Analog pressure transducer PT.M21.910 is inside the flowmeter that is located upstream of filter F21 and CT.M21.910 and TT.M21.910. PT.M21.910 monitors the pressure of the recirculating, blended chemical.

Pump

The Inline Blender, with Integrated Daytanks, uses a Levitronix i100 centrifugal pump (P1), which functions as a booster pump for the pressurized Chem 1 from the Fab. A Levitronix BPS-600 centrifugal pump (P21) functions as the blender's recirculating/dispensing pump for the two daytanks.

Static Mixer

A static mixer (SM.Q1.910) contains a helical mixing element that divides, re-directs, and re-orientes the flow of Chem1, after Chem1 + UPW flow from the injection port through the mixer's tube housing.

Status Light Assembly

The status light assembly on the front of the Inline Blender, with Integrated Daytanks, cabinet contains green, amber/yellow, and red lights for visual alarm indication.

Sump Aspirator

The primary function of P9 is as an aspirator of excess liquid in the cabinet sump. It is activated when leak Warning float switch FS.Z5.8 detects liquid in the cabinet sump. The PLC opens AOV.P9.110. The vacuum created by the flow of DI water through P9 draws the liquid from the sump, through the aspirator, and to the unit's drain pipe.

The secondary function of P9 involves priming centrifugal pump P21 with chemical, whenever the PLC determines that chemical is not flowing or recirculating through P21. The PLC opens AOV.P9.510. Chemical is drawn through the pump, filter F21, and the metrology suite to AOV.P9.510, thereby priming P21 with chemical. From AOV.P9.510 the chemical flows to and through P9 to the main drain pipe. The pump-priming function continues until the PLC is satisfied that chemical is flowing through pump P21.

Temperature Transmitters

The temperature transmitters (TT.M1.910A and TT.M1.910B are located upstream of accumulator GC.Q1.910; TT.M21.910 is located after pump P21 and filter F21 for the recirculating blend) consist of a display and temperature sensor. The temperature transmitters either (a) ensure that the liquids the required recipe temperatures, or (b) provide a redundant means of metrology compensation. The blender can be used with either one or both transmitters enabled at each blend point. The temperature transmitters are pre-calibrated to standards. No configuration or calibration is required throughout its service period.

Chemical Containment System

Cabinet Compartments

The cabinet and its compartments are constructed of polypropylene. The cabinet contains different compartments, each with a different function.

Chemical Lines

All pressurized lines and components that contain hazardous process chemical are in compartments that contain door interlock switches or interconnecting double containment piping. If the door to one of these compartments is opened, the associated system is disabled and the plumbing lines within that compartment are depressurized.

Controls Compartment

The Controls Compartment contains the PLC, HMI, circuit breakers, terminal strips, indicating pressure switches, pressure relief valves, and pressure indicators.

Wetted Area Compartments

These compartments contain all the wetted components, such as the pumps, chemical filter, and valves. These compartments have enough containment to hold 150% of the liquid volume potentially contained within the unit components, not including the daytanks. An overflow drain provides additional protection. These compartments are outfitted with lighting to help operators view components.

These compartments also include two-level leak detection, low-exhaust pressure detection, and door-open detection.

Parameters of the Inline Blender, with Integrated Daytanks

Category	Specification
Manufacturer Name and Address	Air Liquide Advanced Equipment Systems Division 18761 Lake Drive East Chanhassen, Minnesota 55317-9384, U.S.A.
Model	Inline Blender, with Integrated Daytanks
Applications	Acids and bases
Cabinet Materials	White Polypropylene
Wetted Components	Teflon PFA tubing/valves/fittings, Teflon PTFE tanks and pumps
Pollution	Degree 1 (no pollution or only dry, nonconductive pollution)
Weight	Refer to the Facility drawing listed in the Prints section
Maximum Pneumatic Working Pressure	125 psi (861 KPa)
Noise levels	68 - 72dB(A); see the Facility drawing listed in the Prints section
Rated Input Voltage	120/230 VAC
Rated Current	Refer the Facility drawing listed in the Prints section
Phase	Refer the Facility drawing listed in the Prints section
Over-current protection	Provided at the facility supply
MTBA	>2500 hours
MTBF	> 4500 hours
MTTR	< 120 minutes

System Connections

The Inline Blender, with Integrated Daytanks, is connected to the following facility ports:

- **Process N₂ supply:** Nitrogen serves as a blanket for daytanks 1A and 1B. It also operates the solenoid valves (which operate the air-operated valves), and provides air pressure in the Controls Compartment to reduce heat and contamination. The Inline Blender does not have N₂ purge functionality.
- **Process UPW:** Ultra-Pure Water is used as a constituent in the blending process. UPW is also used to flush components of excess chemical. UPW is used by the spray guns to clean the interior of the Wetted Compartments of spilled chemical.
- **Facilities Waste Drain:** Used for draining liquid from the Inline Blender, with Integrated Daytanks.
- **Facility exhaust:** Used for venting chemical fumes from the Inline Blender, with Integrated Daytanks, cabinet.
- **Cabinet overflow waste drain.**

Operating Modes

The basic functions and features of the Inline Blender, with Integrated Daytanks, that occur automatically when the modules are in Auto mode are listed in [Table 1-2](#). The table also includes chemical sampling operations available in Auto mode.

Table 1-2. Features and Auto Mode Sequences, Listed by “Module”

“Module”	Automatic and Sampling Operations in AUTO Mode
Pressurized Chemical Flow	• Flow to Inline Blender • Mixing with pressurized chemical in the static mixer
UPW Flow	• Flow to Inline Blender • Mixing with UPW in the static mixer
Inline Blender	• Blending with conductivity regulation • Sampling per an operator request • Dual conductivity check with average and discrepancy check • Temperature compensation of the conductivity • Recipe management • Water rinsing after a blend • Water rinsing before a recipe change
Daytanks 1A and 1B	• Nitrogen blanketing • Leak check • Conductivity monitoring • Temperature compensation of the conductivity • Sampling per an operator request
Cabinet	• Blend Compartment door interlock detection and lights • Wetted Compartment door interlock detection and lights • Bottle Sample Area door interlock detection • Exhaust pressure monitoring • Lights in Controls Compartment

Utilities

Because the presence and quality of the utility supplies are not monitored in the blender, the facility must monitor them:

- Nitrogen must dry and particle free, although the nitrogen blanketing lines in the blender do have filters.
- UPW needs to be particle free (<1000 particles at 0.1 µm per liter) and resistivity $\geq 18,200 \Omega$.

Services

- Flush components and filters with UPW: UPW flows to dilute chemical in the pump and filters and associated plumbing.
- Flush: Flush the blend path at configurable flowrates.

Alarms

The system contains three (3) levels of alarms that indicate improper operation. An understanding of the meaning of each of these alarms is important when operating and servicing the unit.

Each alarm level designates a different degree of importance and requirement for operator response to a machine condition. A description of each alarm level follows. Alarms are coded as follows:

SHUTDOWN LEVEL: Unit has been shutdown. Shutdown alarms cause the PLC to immediately shut down all activity in the unit. When active, all unit outputs are placed in their normal (safe) state. Auto mode operation is not allowed until all shutdown alarms have been corrected.

CRITICAL LEVEL: Suspends operation of the unit and alerts the operator. The unit or sub-unit might be approaching a condition that, unless the operator intervenes, could result in a Shutdown alarm that will shut down the unit.

WARNING LEVEL: Indicates that one of the operations, operating systems, or the chemical supply source from the Inline Blender, with Integrated Daytanks, has been shut down. Warning Level is also used to notify the operator that a batch that was created that mode does not meet the concentration specification and alerts the operator to a condition that might require operator assistance.

Refer to [Chapter 7, Troubleshooting](#), for a list of each individual alarm and its meaning.

Status Lamp/Horn

The system has a status light with green, yellow, and red lights for visual alarm indication. The unit also has an audible horn. The horn activates when an alarm is created and stays on until the condition is removed or the alarm is acknowledged. If the alarm is acknowledged and the condition is present, the alarm re-activates after a user-configurable amount of time. See [Chapter 7, Troubleshooting](#), for additional information.

Security Level Access

A password is required to access service, operation, and engineering functions on the HMI. See [Chapter 4, *System Operation and Process Procedures*](#), for more details.

System Communications

The Inline Blender, with Integrated Daytanks, communicates with other equipment using discrete electrical interface signals and using Ethernet connections.

The unit sends these discrete output signals:

- Chem1 Enable (No Shutdown)
- Chem Ready
- Leak Status
- Chem1 Demand

The unit accepts these discrete input signals:

- Chem1 Ready
- Remote Shutdown from Auxiliary 3
- Remote Demand from Auxiliary 3
- Leak Signal from Auxiliary 3
- Chem Ready for Auxiliary 3
- Chem Ready for Auxiliary 4
- Remote Demand from Auxiliary 4
- Leak Signal from Auxiliary 4
- Remote Shutdown 1
- Remote Shutdown 2

The Inline Blender, with Integrated Daytanks, has an Ethernet interface. This interface provides key alarm information that can be monitored by a Configurable Control System (CSS) or Flexible Factory Control (FFC) system.

Review the Electrical Schematic listed in the [Prints](#) section for I/O details.

2 Installation



WARNING



BEFORE PROCEEDING, READ AND UNDERSTAND THE SAFETY INFORMATION CONTAINED WITHIN THE INSTALLATION AND SETUP SECTION OF THE [SAFETY](#) CHAPTER OF THIS MANUAL.



WARNING



RISK OF EXPLOSION AND PERSONAL INJURY.

Units in plastic enclosures cannot be used in a Class 1, Division 3 environment.

This chapter describes the general procedures for installing the Inline Blender, with Integrated Daytanks. Because each site has its own site-specific practices and requirements, use these procedures only as a general reference guide.

Perform the installation procedures in the order in which they are presented.

Chapter Contents

Installation Safety	2-2
Site Requirements	2-3
Uncrating	2-3
Visual Inspection.....	2-6
Leak Inspection.....	2-7
Leveling	2-7
Facility Connections.....	2-8

Installation Safety



PAYING STRICT ATTENTION TO THE SAFETY INSTRUCTIONS IN THIS MANUAL CAN PREVENT SERIOUS PERSONAL INJURY AND DAMAGE TO THE EQUIPMENT.

Obey the safety instructions given below and in the [SAFETY](#) chapter and take note of appropriate precautions. If you do not, you can cause injury to people and damage to equipment.

- A suitably trained and supervised technician must install/maintain the equipment.
- The installation technician must be familiar with and comply with the safety procedures that relate to the chemical to be used, pressured gases, and electricity. Although not usually considered a dangerous substance, nitrogen is classified as a simple asphyxiate.
- Read applicable Safety Data Sheets (SDSs). Make sure they are current and observe the safety procedures outlined.
- Follow site requirements for use of protective garments, gloves, and eyewear and breathing apparatus.
- Work with at least one qualified co-worker. Never service the equipment unless someone capable of providing aid is present.
- Follow site lockout/tagout procedures when installing and maintaining the equipment. Ensure that all appropriate precautions are taken.
- Use appropriate tools.
- The equipment should be disconnected from any electrical supply during mechanical installation so that it cannot be operated accidentally.
- Leak-test the equipment after installation and repair any leaks found prior to introducing fluids.
- Safely dispose of spilled hazardous substances, contaminated clothing, and contaminated clean-up materials in a facility-approved manner.

Site Requirements

Ensure the facility and the operating area meet all of the requirements listed in [Appendix B, System Specifications](#), and the facilities and installation drawing in the [Prints](#) section.

Ensure that the required site-specific fittings, connectors, special tools, test equipment, and drawings needed for the installation are available.

Ensure that the equipment is in the correct location and that it is positioned at the correct inlets and outlets for power, exhaust, chemical, UPW, CDA, nitrogen, and drains.

Use a pallet jack or a forklift to lift and move the equipment shipping crate.

Uncrating

Purpose

The purpose of this procedure is to uncrate the Inline Blender, with Integrated Daytanks in preparation for installation.

Special Tools Required

- A crowbar or similar tool is required to disassemble the crate.
- A forklift is required to lift the Inline Blender, with Integrated Daytanks off the crate platform.

Precautions



MOVING THE CRATE OR EQUIPMENT WITHOUT A PALLET JACK OR FORKLIFT COULD CAUSE PERSONAL INJURY.

Wear eye protection and industrial gloves for this procedure. Sharp nails or wood splints could cause personal injury.



EXPOSED NAILS AND WOOD SPLINTS COULD CAUSE PERSONAL INJURY.

Wear eye protection and industrial gloves for this procedure. Sharp nails or wood splints could cause personal injury.

IMPORTANT

Refer to the lifting point labels on the system to make sure that the forklift or overhead hoist is positioned at the correct lifting points. Do not lift the system at positions indicated by the DO NOT LIFT FROM THIS SIDE labels on the unit.

Procedure

1. Examine the Shockwatch and Tiltwatch indicators on the crate. If one or both of the indicators are red, make a note of this on the carrier's bill of lading and notify your supervisor. The system may have been damaged during shipment.

NOTE

Two people are required to perform steps 2 and 3.

2. Use crowbars to pry apart the top panel from the side panels. Carefully lift the top panel from the shipping crate.
3. Perform one of the following steps to remove the side panels:
 - a. Use a crowbar to pry apart the side panels at each corner. Remove all exposed nails.
 - b. Use a power saw to cut vertically through the side panel wood and nails.
4. Cut and remove the steel band that secures the unit to the pallet.
5. Use a socket wrench to remove the bolts at the hold-down brackets.
6. Use a forklift capable of lifting 2,500 lb (1136 kg), dry weight, to lift the system off of the wood pallet and into position on the fab floor.

Visual Inspection

Purpose

The purpose of this procedure is to visually look for any signs of shipment damage.

Procedure

1. Examine the cabinet edge welds for cracks, breaks, or fissures.
2. Check the side panels for dents, cracks, large scratches, or holes.
3. Ensure that all interlock panel and door sensors are securely fastened and that sensor wires are not torn or loose.
4. Examine the safety labels. Make sure that they are legible and not torn or missing. Replace any damaged or missing labels.
5. On the front of the cabinet, make sure that the chemical-type identifier/safety label has been affixed and that it is legible.
6. Check the operator interface panel for cracks, chips, or gouges.
7. Examine the lens on all of the pressure indicators and regulators on the front of the cabinet. Look for cracks, chips, or gouges.
8. In the controls compartment, verify that all pilot solenoid valves are securely fastened. Correct any loose or disconnected wiring.
9. In the controls compartment, verify that the cables and wires from the PLC to the operator interface panel are securely fastened.
10. Check that all other wiring is securely connected.
11. Check that equipment ports are not damaged.
12. Verify that all tube nuts are tight and that all tubes are securely fastened to their fittings.

Leak Inspection

Purpose

The purpose of this procedure is to ensure that the bottoms of all wetted compartments were not cracked during shipment.

Procedure

1. Open the door of the blender cabinet.
2. Pour enough UPW or suitable inert liquid in the bottom of the cabinet to cover the surface. Let stand for an appropriate length of time.
3. Check to ensure that no water leaks from the cabinet to the facility floor.
4. Use an industrial vacuum or dry wipes to remove the water from the cabinet floor.

Leveling

Purpose

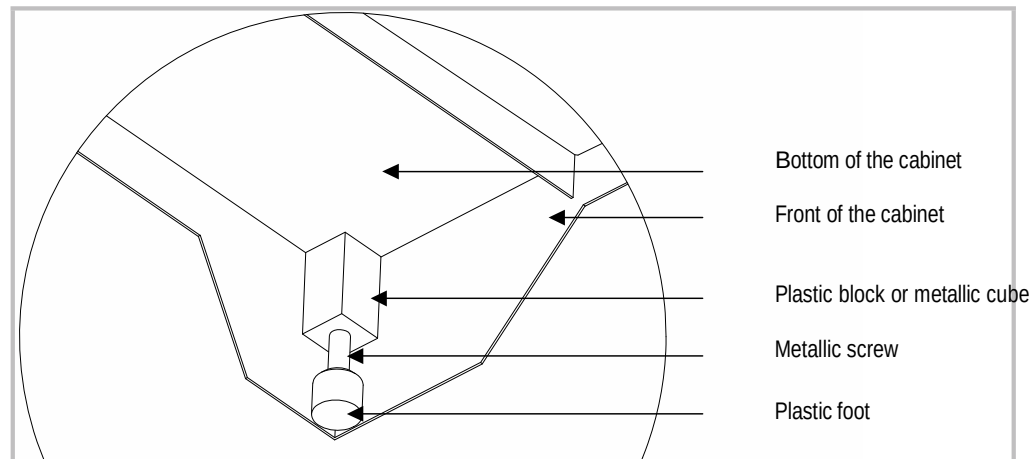
The purpose of this procedure is to level the system before making any connections. Using the adjustable feet, level it from side to side and front to back.

Figure 2-1 shows the design of one of the cabinet feet. The cabinet feet must be adjusted to level the cabinet during installation.

NOTE

Installing the unit on a flat surface that is able to support a minimum weight on the facilities drawing is recommended.

Figure 2-1. Detail of Cabinet Foot



Special Tools Required

An 18-inch or 24-inch carpenter level is required for this leveling procedure.

Procedure

NOTE

When selecting the cabinet operating location, consider safety and maintenance access in addition to other criteria.

1. Place the cabinet in its operating location and position.
2. Individually turn out (turn clockwise) the cabinet levelers (cabinet feet).
3. Place the level on the top front of the cabinet. If it is not level, turn out the applicable cabinet leveler to level the front of the cabinet.
4. Place the level on the top rear of the cabinet. If it is not level, turn out the applicable cabinet leveler to level the rear of the cabinet.
5. Place the level on the top right side of the cabinet. If it is not level, turn out the applicable cabinet leveler to level the right side of the cabinet.
6. Place the level on the top left side of the cabinet. If it is not level, turn out the applicable cabinet leveler to level the right side of the cabinet.
7. Recheck the level of the front, back, and both sides. Make additional adjustments as necessary.
8. Check the fit and operation of all cabinet doors.
9. Turn the leveler locknuts until they are tight against the cabinet frame.
10. Install seismic clamps or bolt the cabinet to the building structure.

Facility Connections

Make the facility connections using the applicable Facilities drawing listed in the [Prints](#) section.

3 Component Description and Location



BEFORE PROCEEDING, READ AND UNDERSTAND THE SAFETY INFORMATION CONTAINED WITHIN THE [SAFETY](#) CHAPTER OF THIS MANUAL.

This chapter describes the cabinet, compartments and components of the Inline Blender, with Integrated Daytanks:

- [Figure 3-1](#) illustrates the cabinet compartments and operator interfaces.
- [Figure 3-2](#) illustrates the pressure regulators by name and component designators.
- [Table 3-1](#) lists components by compartment location and by their location on P&ID 514510-001, which is provided in the [Prints](#) section.

3-2 Component Description and Location

Figure 3-1. Cabinet Components and Operator Interfaces

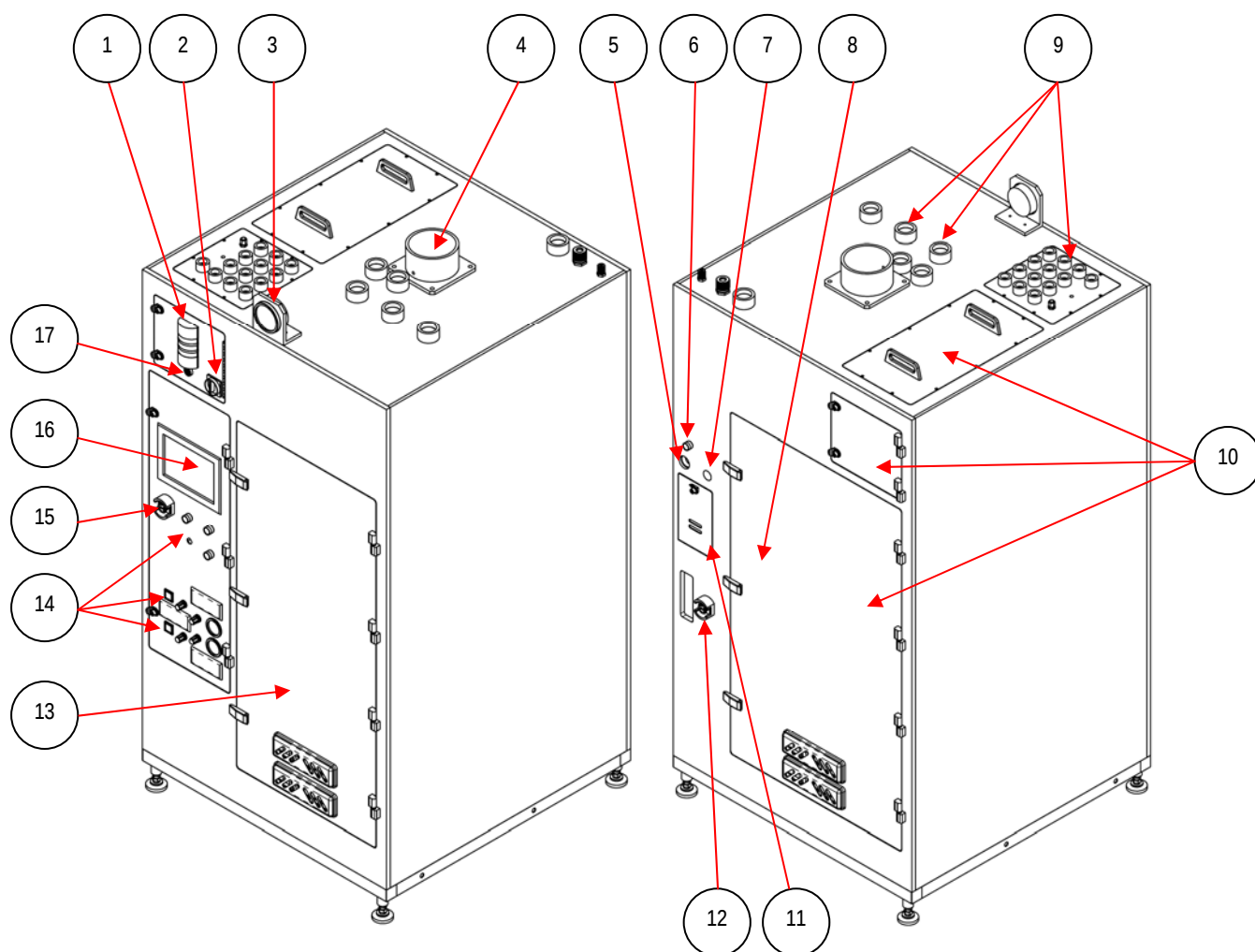


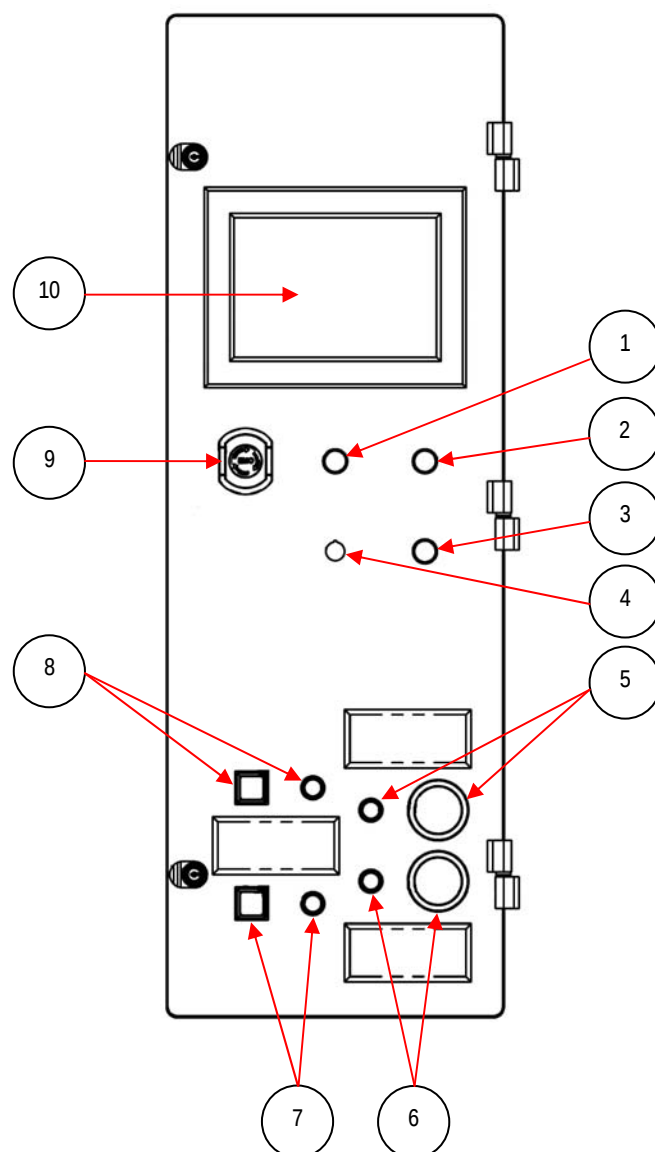
Table 3-1. Inline Blender, with Integrated Daytanks – Front and Back of Cabinet

Call Out	Component Description
1	Status lamp.
2	MAIN POWER DISCONNECT switch.
3	Magnehelic exhaust pressure indicator.
4	Exhaust port.
5	Port for an external particle counter.
6	SAMPLE PUSH AND HOLD switch.
7	Sample flow valve (optional).
8	EMO pushbutton/switch (rear of cabinet).
9	Facilities ports (see the Facility & Installation Drawing, listed in the Prints section, for details).

Table 3-1. Inline Blender, with Integrated Daytanks – Front and Back of Cabinet

Call Out	Component Description
10	Access panels or doors.
11	Bottle Sample area.
12	UPW spray gun compartment.
13	Access door to the Wetted Compartment.
14	Operator panels (see Figure 3-2 for details).
15	EMO pushbutton switch (front of cabinet).
16	GE QuickPanel+ HMI panel.
17	Alarm horn.

Figure 3-2. Inline Blender – Operator Panel Pressure Regulators, Indicators, Switches, and EMO



3-4 **Component Description and Location**

Table 3-2. Inline Blender – Pressure Regulators and Switches – Front Operator Panel

Call Out	Component Description
1	EMO RESET pushbutton.
2	ALARM SILENCER pushbutton.
3	PUMP ENABLE switch.
4	PUMP DISCONNECT switch.
5	UPW BLEND PRESSURE, pressure regulating valve (PRV.J99.110) and pressure indicator (PI.J99.301).
6	CHEM BLEND PRESSURE, pressure regulating valve (PRV.J1.110) and pressure indicator (PI.J1.301).
7	N2 PRESSURE, pressure regulating valve (PRV.N1.950) and indicating pressure switch (IPS.N1.950).
8	SOL VLV PRESSURE, pressure regulating valve (PRV.A1.951) and indicating pressure switch (IPS.A1.951).
9	EMO pushbutton/switch.
10	GE QuickPanel+ HMI panel.

Table 3-3. Inline Blender, with Integrated Daytanks – Components by Type, Tag ID, Description, and Location

Type	Tag ID	Component Description	Component Location
Accumulator	GC.Q1.910	Accumulator, post-static mixer	Wetted Compartment
Air-Operated Valves	AOV.F21.710	Filter F21 to drain	Wetted Compartment
	AOV.J1.110	Pressurized chemical inlet to flow controller J1	Wetted Compartment
	AOV.J1.210	Services to flow controller J1	Wetted Compartment
	AOV.J99.110	UPW inlet to flow controller J99	Wetted Compartment
	AOV.J99.210	Services to flow controller J99	Wetted Compartment
	AOV.L1.210	Services to Bottle Sample Area	Wetted Compartment
	AOV.P9.110	Cabinet sump to aspirator P9	Wetted Compartment
	AOV.P21.110	Blended chemical recirc inlet to pump P21	Wetted Compartment
	AOV.P21.210	Services to pump P21	Wetted Compartment
	AOV.P21.510	Pump P21 priming (to aspirator P9)	Wetted Compartment
	AOV.P21.810	Post-pump P21 to Sample #2	Wetted Compartment
	AOV.P21.820	Post-pump P21 to drain	Wetted Compartment
	AOV.Q1.810	Sample #1 valve to Bottle Sample Area	Wetted Compartment
	AOV.Q1.820	Blended chemical to drain pipe	Wetted Compartment
	AOV.Q1.910	Blended chemical to AUX OUT 3 [future option]	Wetted Compartment
	AOV.S9.110A	Post-pump P21 valve to either recirculation or to AOV.S9.110B	Wetted Compartment
	AOV.S9.110B	Blended chemical to future option AUX FILL 4	Wetted Compartment
	AOV.T1.111	Pump P21 recirc to daytanks T1A or T1B	Wetted Compartment
	AOV.T1.910	Daytank T1 Chem Out 1 or T1 Chem Out 2	Wetted Compartment
	AOV.T1.110	Blended chemical to daytank T1a or T1B	Wetted Compartment
	AOV.T2.110	Blended chemical to future outlet [capped]	Wetted Compartment
	AOV.U1.950	UPW to services	Wetted Compartment
	AOV.U1.960	UPW to Aspirator P9 Enable	Wetted Compartment
Aspirator	P9	Cabinet sump aspirator	Wetted Compartment
Capacitive Sensors	CS.T1.11	Low-Low level sensor for daytanks 1A <u>and</u> 1B	Wetted Compartment
	CS.T1.12A	Low level sensor for daytank 1A <u>and</u> 1B	Wetted Compartment
	CS.T1.17A	High level sensor for daytank 1A <u>and</u> 1B	Wetted Compartment
	CS.T1.18A	High-High level sensor for daytank 1A	Wetted Compartment
	CS.T1.18B	High-High level sensor for daytank 1B	Wetted Compartment
	CS.T1.58A	Overfill sensor for daytank 1A	Wetted Compartment
	CS.T1.58B	Overfill sensor for daytank 1B	Wetted Compartment

3-6 Component Description and Location

Table 3-3. Inline Blender, with Integrated Daytanks – Components by Type, Tag ID, Description, and Location

Type	Tag ID	Component Description	Component Location
Capacitive Sensors (continued)	CS.Z1.17	Liquid in drain sensor	Wetted Compartment
	CS.Z1.58	Drain pipe overflow sensor	Wetted Compartment
Check Valves	CV.A1.110	Facilities N ₂ inlet backflow	Controls Compartment
	CV.J1.110	Pressurized chemical from Fab-to-unit backflow	Wetted Compartment
	CV.J99.110	Facilities UPW supply to flow controller J99 backflow	Wetted Compartment
	CV.L1.820	Sample Bottle Area-to-drain backflow	Wetted Compartment
	CV.M21.910	Post-recirculation flowmeter backflow	Wetted Compartment
	CV.T1.110	Blended chemical to daytanks backflow	Wetted Compartment
	CV.T1.111	Recirculation to daytanks backflow	Wetted Compartment
	CV.T1.910	Daytank 1A or 1B outlet to Chem Out backflow	Wetted Compartment
	CV.U1.950	Facilities UPW return flow backflow	Wetted Compartment
Conductivity Transducers	CT.M1.910A	Post-accumulator initial conductivity transmitter	Wetted Compartment
	CT.M1.910B	Post-accumulator secondary conductivity transmitter	Wetted Compartment
	CT.M21.910	Post-filter F21, pre-flowmeter conductivity transmitter	Wetted Compartment
Electrical Keyswitch	EKS.T1.910	2-position electrical key switch	Controls Compartment
Filter (chemical)	F21	Post-pump P21 chemical filter	Wetted Compartment
Filter (nitrogen)	F.N1.953	Filter for incoming facilities N ₂	Controls Compartment
Float Switches	FS.Z5.8	Cabinet sump warning float switch	Wetted Compartment
	FS.Z5.9	Cabinet sump shutdown float switch	Wetted Compartment
Flow Controllers	J1	Flow controller for pressurized chemical from the Fab	Wetted Compartment
	J99	Flow controller for UPW from the Fab	Wetted Compartment
Flow Meters	FM.N1.953	Facilities N ₂ flow meter	Controls Compartment
Flow Orifices	FO.P21.810	Sample #2 to Bottle Sample Area	Wetted Compartment
	FO.Q1.810	Sample #1 to Bottle Sample Area	Wetted Compartment
	FO.U1.960	UPW return flow to the Fab	Wetted Compartment
Flow Transmitter	FT.M21.910	Post-filter F21 flow transmitter in flowmeter	Wetted Compartment
Indicating Pressure Switches	IPS.A1.951	Pressure switch for facilities N ₂ to solenoids	Controls Compartment
	IPS.N1.950	Pressure switch for facilities N ₂ to unit	Controls Compartment

Table 3-3. Inline Blender, with Integrated Daytanks – Components by Type, Tag ID, Description, and Location

Type	Tag ID	Component Description	Component Location
Interlock Switches	ZS.Z4.5	Bottle Sample Area interlock switch	Bottle Sample Area
	ZS.Z5.5A	Front access door, Wetted Compartment, interlock switch	Wetted Compartment
	ZS.Z5.5B	Rear access door, Wetted Compartment, interlock switch	Wetted Compartment
Lights (cabinet)	LT.Z2.1	Controls Compartment light	Controls Compartment
	LT.Z4.2	Bottle Sample Area light	Bottle Sample Area
	LT.Z5.1A	Upper light, front side	Wetted Compartment
	LT.Z5.1B	Lower light, front side	Wetted Compartment
	LT.Z5.2A	Upper light, rear side	Wetted Compartment
	LT.Z5.2B	Lower light, rear side	Wetted Compartment
Manual Valves	MV.J99.110	Manual isolation valve for pressurized chemical from the Fab	Wetted Compartment
	MV.J99.801	Post-flow controller J99 to drain manual isolation valve	Wetted Compartment
	MV.P21.801	Manual isolation inlet to pump P21 or to the pre-pump drain	Wetted Compartment
	MV.Q1.910	Manual isolation valve for future option AUX OUT 3 [capped]	Wetted Compartment
	MV.S9.110	Manual isolation valve to future option AUX FILL 4 [capped]	Wetted Compartment
	MV.T1.110	Blended chemical to daytanks 1A or 1B	Wetted Compartment
	MV.T1.111A	Manual isolation inlet valve for daytank 1A	Wetted Compartment
	MV.T1.111B	Manual isolation inlet valve for daytank 1B	Wetted Compartment
	MV.T1.910	Daytanks 1A or 1B drain manual isolation valve	Wetted Compartment
	MV.T1.991	Daytanks 1A or 1B manual isolation valve to T1 Chem Out 1 [capped]	Wetted Compartment
	MV.T1.992	Daytanks 1A or 1B manual isolation valve to T1 Chem Out 2 [capped]	Wetted Compartment
	MV.T2.110	Manual isolation valve for future option BLEND OUT 2 [capped]	Wetted Compartment
Pneumatic Keyswitch	PKS.T1.910	Pneumatic keyswitch	Controls Compartment
Pneumatic "OR" Valve	POV.T1.910	Pneumatic "OR" valve	Controls Compartment
Pressure Indicators	PI.J1.301	Pressure regulator for PNEU 31 (Chem1 blend pressure)	Controls Compartment
	PI.J99.301	Pressure regulator for PNEU 30 (UPW blend pressure)	Controls Compartment

3-8 Component Description and Location

Table 3-3. Inline Blender, with Integrated Daytanks – Components by Type, Tag ID, Description, and Location

Type	Tag ID	Component Description	Component Location
Pressure Indicators (continued)	PI.Z1.491	Exhaust low draw pressure indicator (Magnehelic)	Controls Compartment
Pressure Regulating Valves	PRV.A1.951	Facilities N ₂ to solenoids	Controls Compartment
	PRV.J1.110	Pressure valve for pump P1 and flow controller J1	Wetted Compartment
	PRV.J1.301	Pressure valve for PNEU 31 (Chem1 blend pressure)	Controls Compartment
	PRV.J99.110	Pressure valve for flow controller J99	Wetted Compartment
	PRV.J99.301	Pressure valve for PNEU 30 (UPW blend pressure)	Controls Compartment
	PRV.N1.950	Facilities N ₂ inlet	Controls Compartment
	PRV.Q1.910	Post-PT.Q1.910, blended chemical pressure valve	Wetted Compartment
Pressure Safety Valve	PSV.N1.950	Pressure relief valve for facilities N ₂	Wetted Compartment
Pressure Switch	PS.Z1.491	Exhaust low-draw pressure switch	Controls Compartment
Pressure Transmitters	PT.M21.910	Post-filter F21 pressure transmitter in flowmeter	Wetted Compartment
	PT.Q1.910	Post-Sample #1, pre-daytank pressure transmitter	Wetted Compartment
Pump	P1	Levitronix BPS-i100 booster pump	Wetted Compartment
	P21	Levitronix BPS-600 recirculation pump	Wetted Compartment
Spray Guns	SG.U1.950	UPW spray gun	Spray Gun Area
	SG.U1.951	UPW spray gun	Wetted Compartment
Static Mixer	SM.Q1.910	Pressurize chemical + UPW static mixer	Wetted Compartment
Temperature Transmitters	TT.M1.910A	Post-accumulator temperature initial transmitter	Wetted Compartment
	TT.M1.910B	Post-accumulator temperature secondary transmitter	Wetted Compartment
	TT.M21.910	Post-filter F21, pre-flowmeter temperature transmitter	Wetted Compartment
Y-Strainer	YS.P9.110	Cabinet sump-to-aspirator strainer	Bottom of cabinet

4 System Operation and Process Procedures



WARNING

BEFORE PROCEEDING, READ AND UNDERSTAND THE SAFETY INFORMATION CONTAINED WITHIN THE [SAFETY](#) CHAPTER OF THIS MANUAL.

IMPORTANT

Only personnel who are trained and certified by Air Liquide shall service and perform maintenance on this equipment.

IMPORTANT

Settings and configuration functions entered using the HMI screens are found in [Appendix D, System Settings](#).

4 System Operation and Process Procedures4-1

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4-2 ***System Operation and Process Procedures***

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HMI Panel Screens and Functions

This section describes the format, features, and functions of the HMI panel and its screens, screen Objects and Actions, and sub-systems accessed from the HMI panel.

HMI Screen Structure

The HMI screens are part of a dynamic structure that starts at the *System Startup Screen*. A traditional, nested screen hierarchy is not used with this HMI panel. Depending on the Object, icon/component, or plumbing segment that is pressed, and the Action that the user wants to perform, specific pop-up screens, sub-screens, blend recipes, or an alphanumeric keypad appear.

Hundreds of screens, sub-screens, and pop-ups are available in the HMI screen structure. Accessing a screen that is above or below the currently displayed screen depends on the Action and function to be performed.

Security/User Levels

[Table 4-1](#) provides an overview of the types of User Levels used with the system. Access to a password is limited by technical knowledge of the system's electrical, mechanical, and software functions and a need-to-know.

Table 4-1. User Levels

User Level	Definitions
None	View Access to the following screens: - Chemical Dispense, Sample, Service, Drain, Trends, I/O Status, Alarm List. Operate Sump pumps, Filter Vent. Ability to change Modes: Automatic and Manual. Alarm Silence and Alarm Reset buttons are active.
Operator	All the access rights of user level "None", with the addition of access to functions/screens of System Config, Configure Analog Input/Output, Pump Setup/Maintenance/Services, Tank Status/Setup, Sump Pump Timer Setpoints and Filter Vent Timer setpoints. Isotainer supply setpoint modification.
Service	All the access rights of user level "Operator", with the addition of access to functions/screens, set/reset pump/AOV counts, Force AOVs, and Force Pumps.
Technician	All privileges of an Operator plus the following. Pump setup, supply setup, force I/O points, adjust system configuration setpoints (except escalation setpoints), and modify sump setpoints.
Engineer	All privileges of a Technician plus the following: Force valves from the popup screens.
Master	All privileges of an Engineer plus the following: Exit application from the System Admin screen, Hardware Configuration Screen selections, and adjust escalation setpoints.

Description of Operation

The Air Liquide integrated inline blender is a flow-based blender of UPW and/or chemicals to a local PFA tank(s), integrated within the same cabinet enclosure. Its application is ideal for relatively “small consumption” chemistries. The blender utilizes inline metrology and instrumentation to control the chemistry and /or UPW flow, and final assay to the tank(s). The blender controls according to the configurable recipe and settings. Chemical flows to the tank(s) if it is within these specifications. Chemical flows to the drain or shuts down if it is NOT within specifications.

The blended chemistry is recirculated within the tank(s) by an onboard pump circuit with inline filter / metrology. The blender controls the outlet valve of the blended chemistry tank(s), allowing downstream units to draw from the tank(s) only if all criteria are in spec.

This unit can be installed /operated as a single unit or as an integrated, redundant pair of units.

When the system is operated as a pair of units, the units will alternate as available ONLINE tank(s), but are still capable of operating as a single unit, if the alternate is not available.

Additional blend outlet valves are provided for FUTURE Design requirements / applications.

A future AUX FILL 4 OPTION is designed to provide a secondary means of providing chemical to a downstream destination. When this option is installed and configured, the unit will transfer chemical from the tank(s) to a downstream destination utilizing the onboard pump circuit. This application is ideal for tote filling or daytank filling, and requires safety integration signals for operation.

Main Features of the Inline Blender

This inline blender provides user interfaces / features that are typical of most Air Liquide blenders:

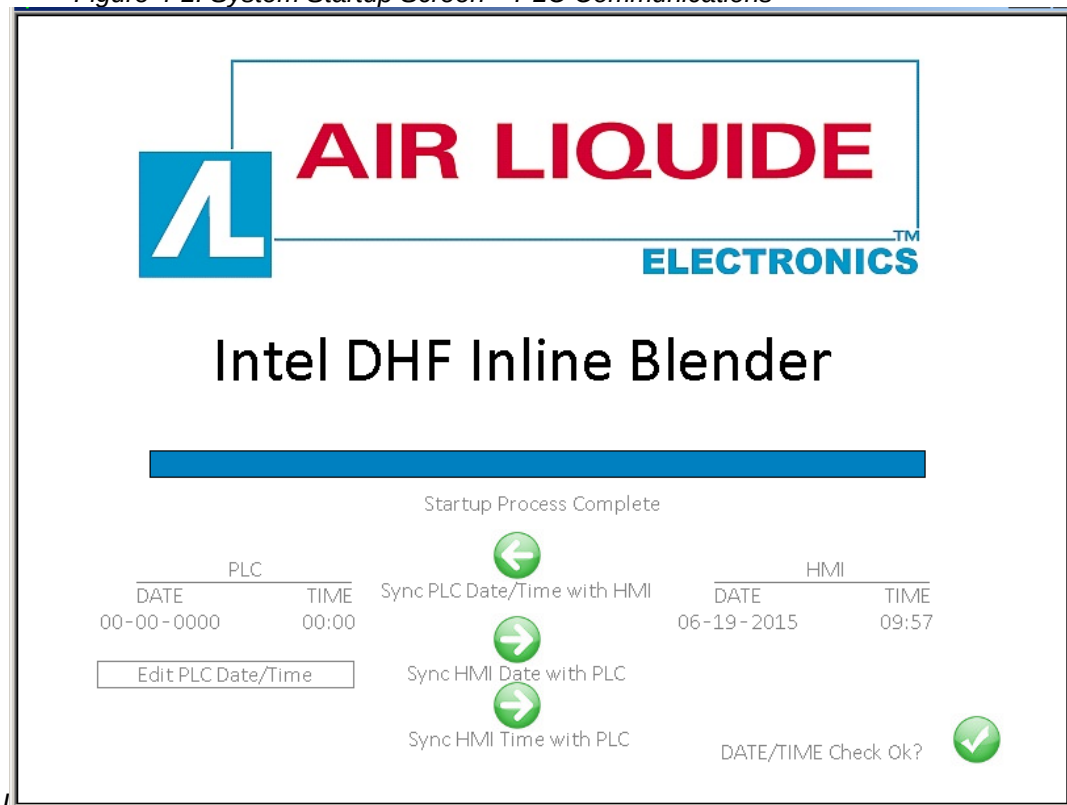
- Onboard Sample Box
- Multiple chemistry sampling locations
- Multiple sampling modes
- Multiple Blend Modes
- Automated services
- Configurable Recipes
- Configurable Alarms / Warnings
- Redundant Blend Metrology with corresponding Alarms / Warnings
- HMI trending visibility
- HMI Alarm / Event Log
- Multiple user login levels
- Facility integration points for safety signals and SCADA package integration (if available).
- Component Cycle Monitoring
- Internal UPW spray gun(s)
- Enclosure Dual Leak detection
- Enclosure Exhaust detection
- Enclosure Door Interlocks

System Screens

Startup Screens

The first screen that displays after the system is powered-up is shown in [Figure 4-1](#).

Figure 4-1. System Startup Screen – PLC Communications



- *System Startup Screen – PLC Communications*: This is the initial screen that the user sees when the system is booted-up.
- To proceed to other screens, review the displayed date / time for the PLC / HMI. Edit it as needed, then accept the status by pressing the green Check icon in the lower right corner of the screen.

Figure 4-2. System Startup Screen – PLC Communications

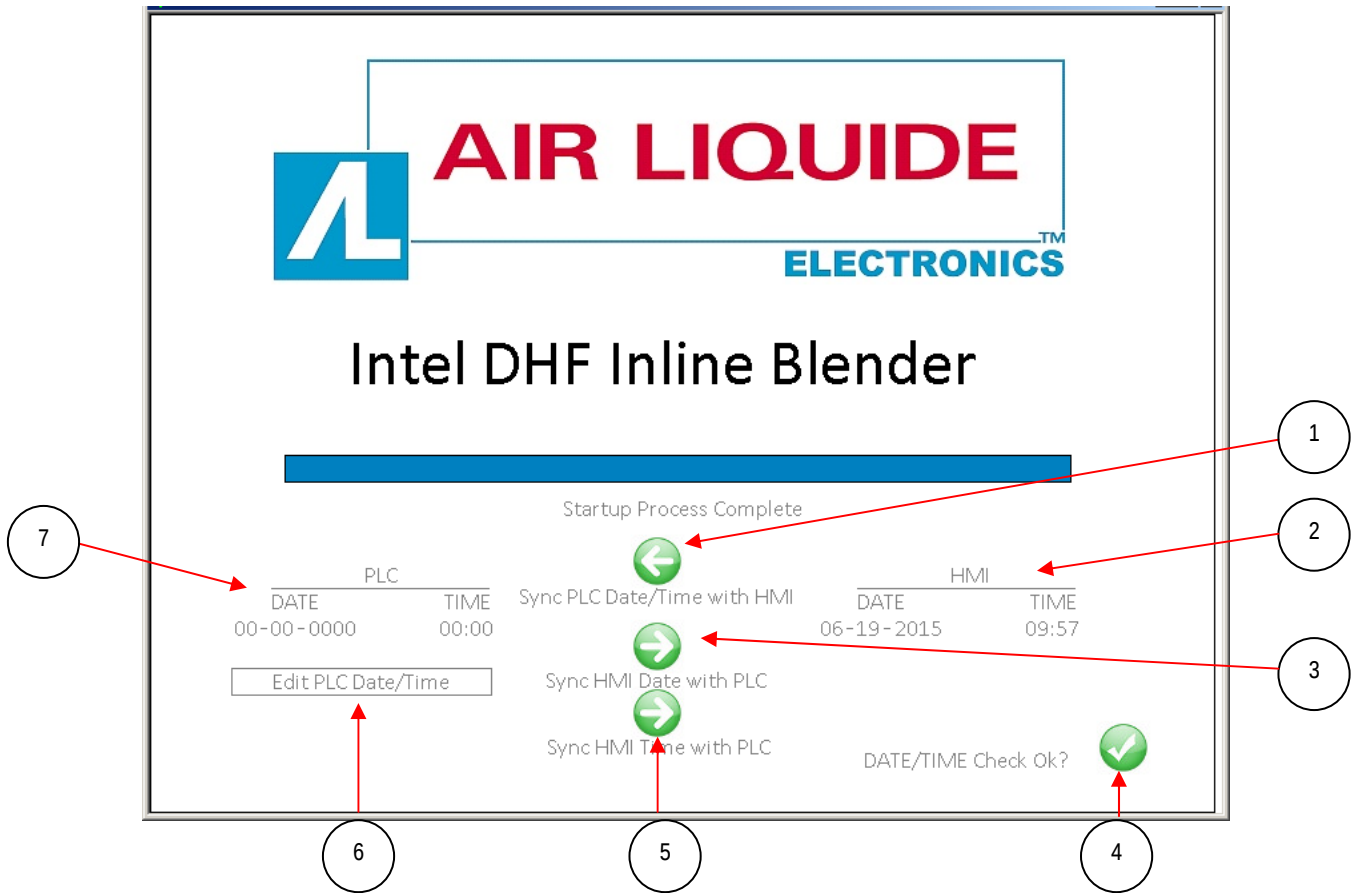


Table 4-2. Home – System Startup Screen – PLC Communications

Callout	Description
1	Update the PLC date <u>and</u> time with the date and time from the HMI panel.
2	Displays the current HMI panel date and time.
3	Update the HMI panel date with the date from the PLC.
4	Accept date and/or time changes and move to the Home Screen.
5	Update the HMI panel time with the time from the PLC.
6	Edit the PLC date and/or time.
7	Displays the current PLC Controller date and time.

Home Screen

The Objects (sub-systems or major components) displayed at the Home Screen, P&ID screens, and other screens show process-flow diagrams that are dynamic and highly editable. Subsystems are displayed on this one screen.

By clicking on an Object or icon the user is presented with pop-up menus. These menus contain the current sub-system state as well as actions, modes, services, and conditions available. The pop-up menu actions are customized based on the machine mode, the sub-system's current state, and available options.

Each of the Objects displayed in *Figure 4-4* can be selected to display a menu of options. Carefully review the choices at each menu. Select and highlight the applicable row on the menu, (for example, change system mode, perform a service, edit a status, etc.).

Figure 4-3. Representative screen only. HOME screen

IMPORTANT Each Object can be in one of four modes: Auto, Manual, Service, or Offline. When the mode is changed, the pop-up menus and sub-screens are displayed for the current mode only.

IMPORTANT The color of a sub-system shows the current mode or state for that sub-system.

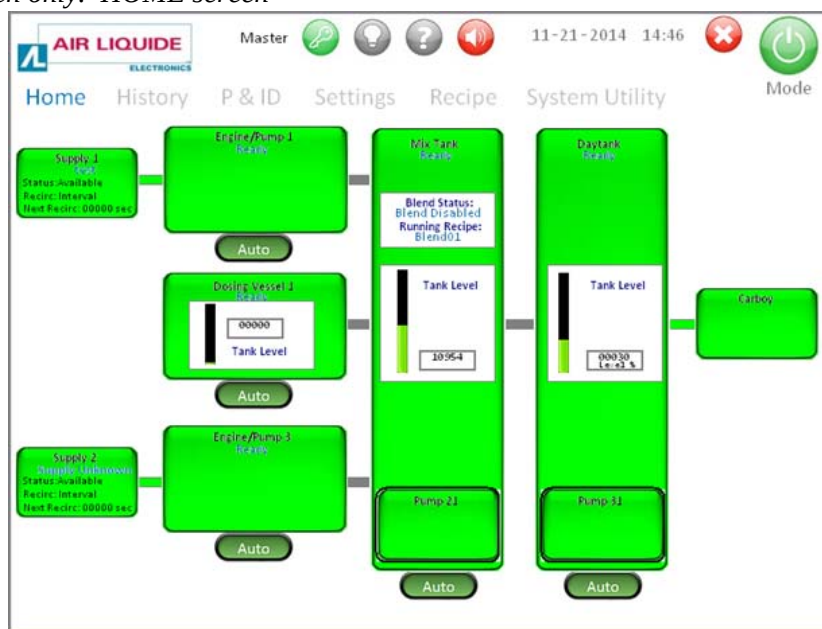


Table 4-3. Mode or Status Colors for Objects on All Screens

Mode or Status	Color of Object on the Screen
Supply Empty or Engine Disabled	Red
System in Auto mode	Green
System in Manual mode	Blue
System in Manual Service mode	Yellow
System in Offline mode	Gray
System in Auto or Manual mode with system not ready, or in an alarm state	White

Figure 4-4. Home Screen

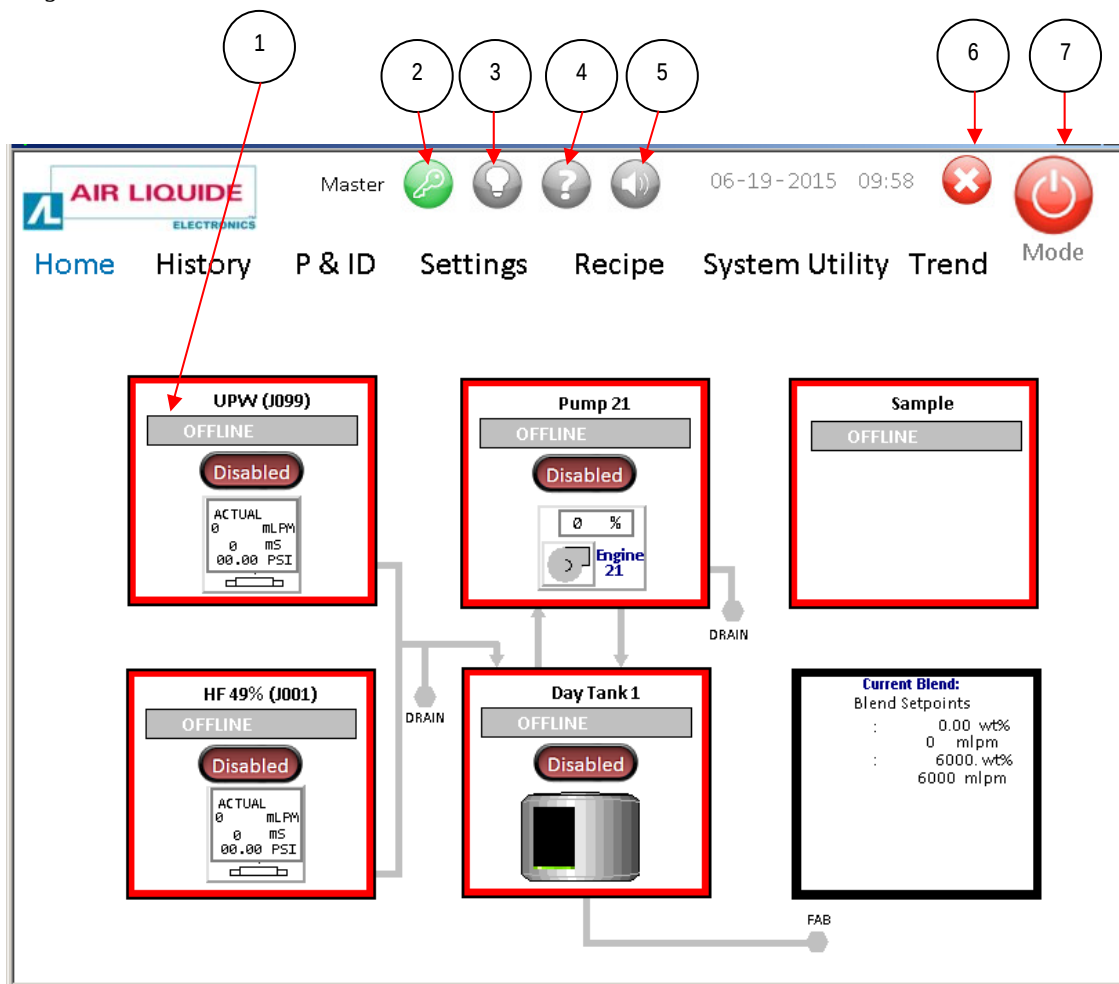
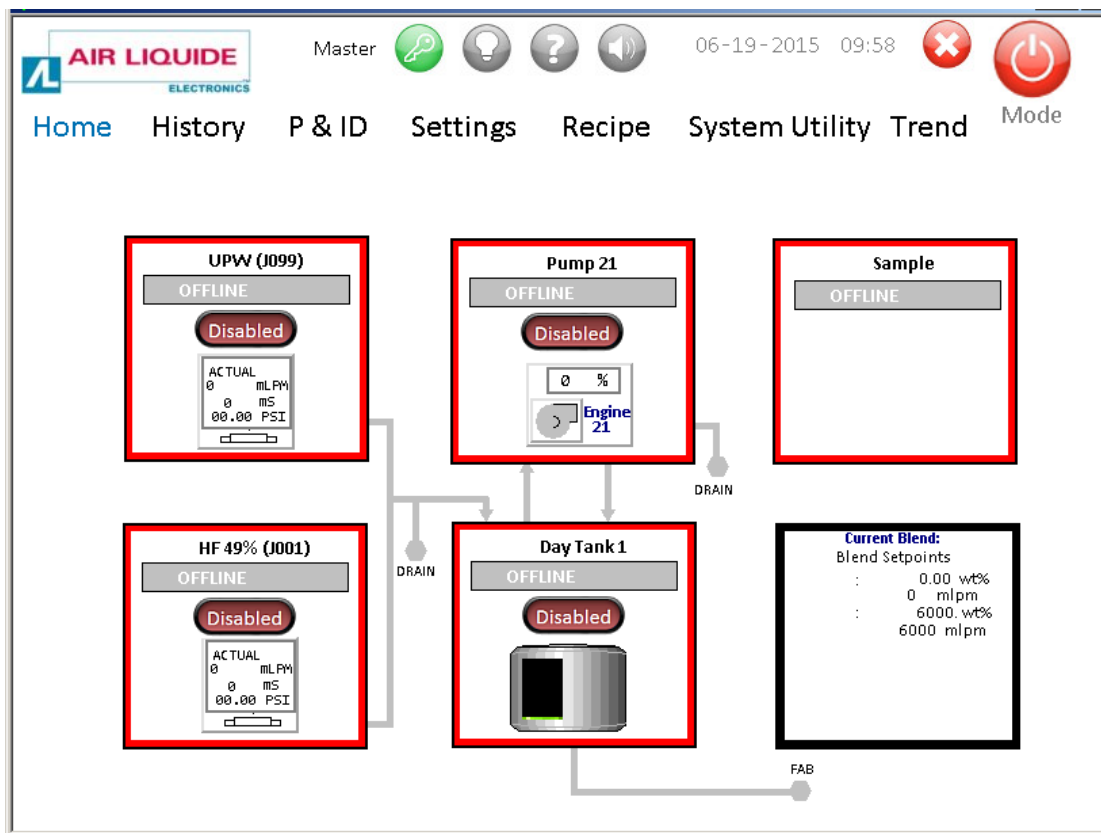


Table 4-4. Home – Main Screen

Callout	Description
1	Press touch key to access This subsystem status and user options.
2	User Log In/Log Out touch key. (The green color of this icon indicates that an active User has logged-on and is using the system).
3	Cabinet Lights touch key. (Gray icon color indicates that the icon has not been pressed and that the cabinet's lights are off. When pressed, the icon color is green, and the lights are on).
4	About touch key. (Press this touch key to view the system's About page).
5	Active Alarms touch key. (A red color indicates that the system has an active alarm or alarms. Press the touch key to view an alarm list).
6	Exit System touch key. (Press this key to exit the runtime to the Windows operating system).
7	Mode Status touch key. Press this touch key to change the mode of the system: - Blue-colored icon = System Manual mode. From Manual mode the user may only change to Auto mode. - Green-colored icon = System Auto mode. From Auto mode the user may only change to Manual mode. - Red-colored icon = System is in Shutdown mode. A system shutdown alarm has occurred. From this state the user may select to put the machine in Manual mode only.

Figure 4-5. Home Screen (continued)



IMPORTANT

Each Object can be in one of four modes: Auto, Manual, Service, or Offline.
When the mode is changed, the pop-up menus and sub-screens are displayed for the current mode only.

Table 4-5. Home Screen (continued)

Note	Description
1	PIPE ANIMATION: GRAY = not used; GREEN = PRESSURIZED CHEM; BLUE = PRESSURIZED UPW
2	SUBSYSTEM BORDER COLOR: RED = DISABLED; BLUE = MANUAL MODE; YELLOW = SERVICE; GREEN = AUTO

Alarms Screens

Figure 4-6. Current Alarms Screen

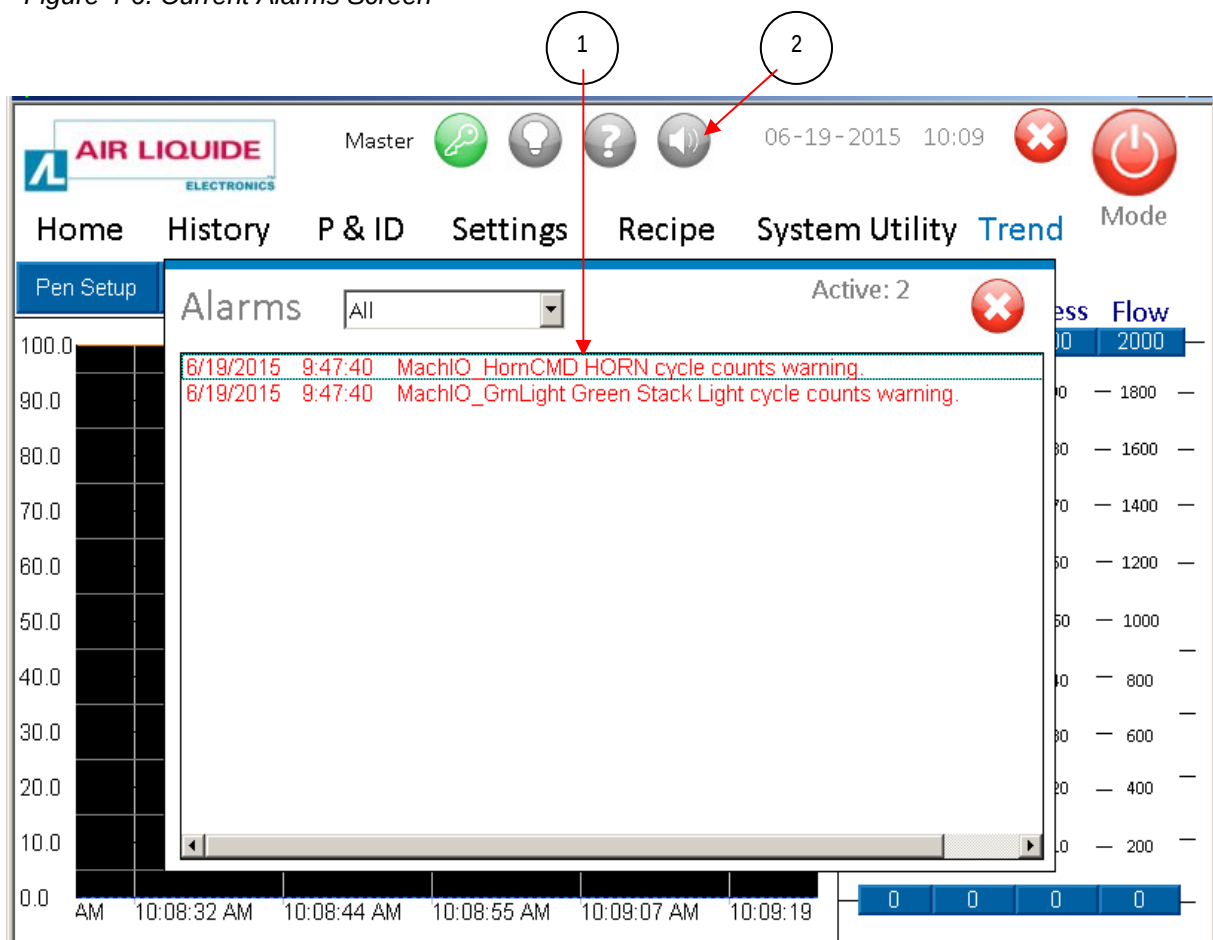


Table 4-6. Current Alarms Screen

Callout	Description
1	Displays a listing of the current alarms: date, time, alarm level, and a brief description of each alarm.
2	ALARMS BUTTON to view the ALARMS SCREEN POPUP

Figure 4-7. Events Screen

**NOTE: REPRESENTATIVE EVENTS SHOWN BELOW.
REVIEW CHAPTER 7 FOR SPECIFIC ALARMS.**

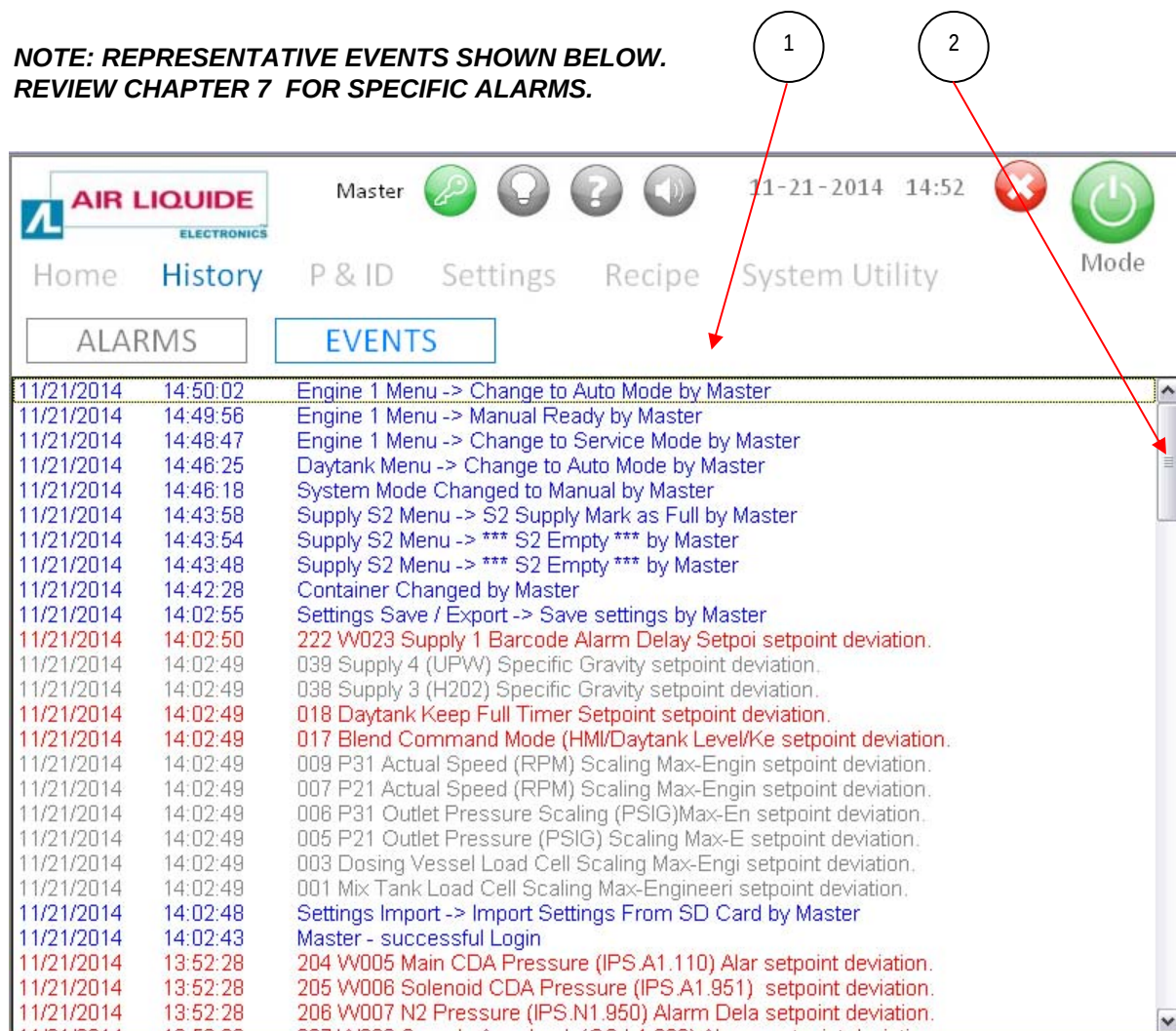


Table 4-7. Events Screen

Callout	Description
1	Displays a listing of the latest, current events: date, time, event, and a brief description of each event. Blue text = A current informational event. Red text = A current critical event. Gray text = An event that is no longer active.
2	Use the elevator bar to scroll down to view older events.

Settings Screen

Figure 4-8. Settings Screen

NOTE: REPRESENTATIVE SCREEN SHOWN.

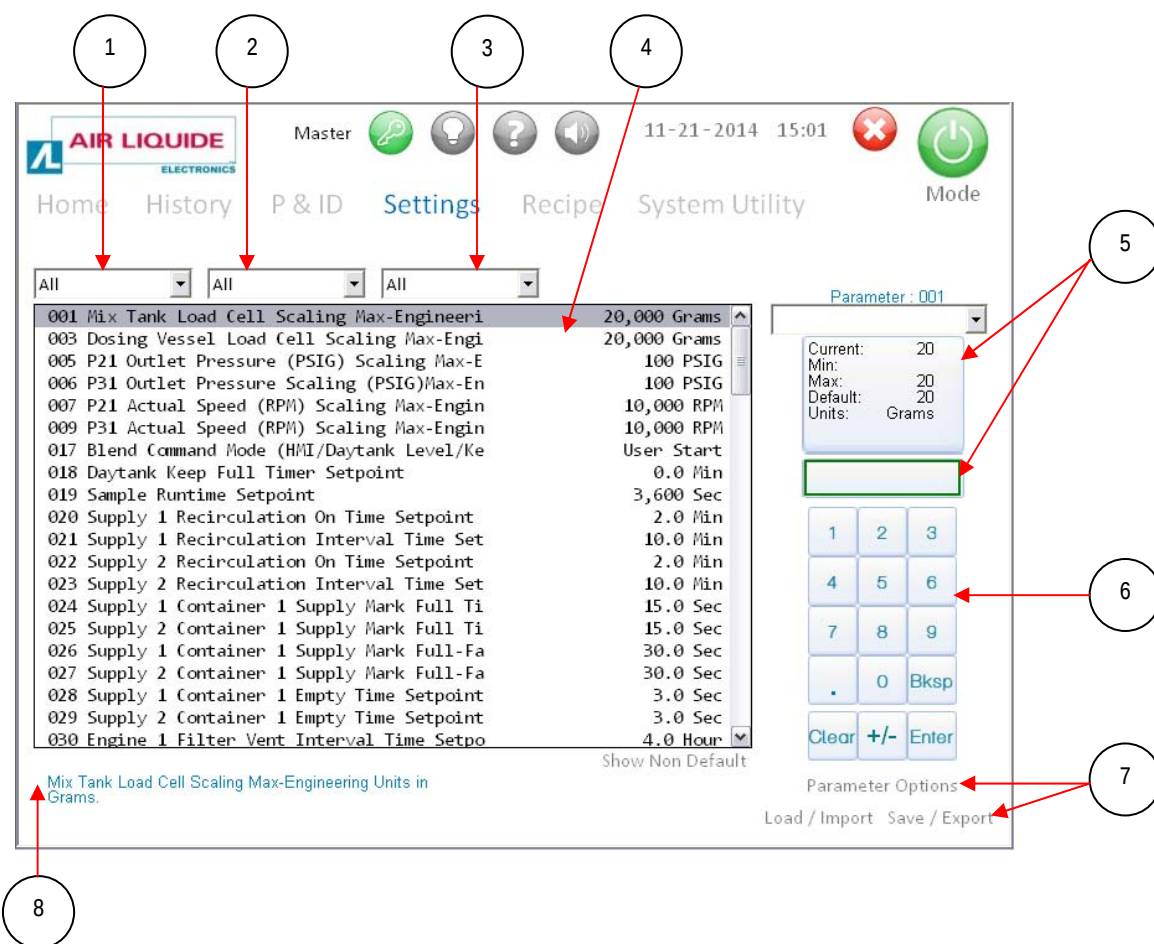


Table 4-8. Settings Screen

Callout	Description
1	Use the pull-down to select (and filter the choices) by sub-system.
2	Use the pull-down to select (and filter the choices) by sub-system object.
3	Use the pull-down to select (and filter the choices) by data type (I.E. Timer, Weight, Volume).
4	Displays the list of parameters based on the filter requirements.
5	Displays current data for the selected parameter.
6	Use the keypad to change default parameter values.
7	Load/Import = Load parameter values from the storage device. Save/Export = Save parameter values to the storage device.
8	Provides Help information for the selected parameter.

Settings (information valid from version 1.00.03 PLC / HMI)

HMI PARAMETERS / SETTINGS						
The values below are for information / training only. The equipment should ONLY be setup according to customer FINGERPRINTED settings. Those settings are not captured in this document.						
Param No	Description	Default	Units / Option 3 Text	Can Change Level	Can See Level	Help
25	DT001 Auto Recirc On Time Setpoint	1440 Min		300		0 Day Tank 1 Recirculation On Time Setpoint
26	DT001 Auto Recirc Off Time Setpoint	1 Min		300		0 Day Tank 1 Recirculation Off Time Setpoint

HMI PARAMETERS / SETTINGS						
The values below are for information / training only. The equipment should ONLY be setup according to customer FINGERPRINTED settings. Those settings are not captured in this document.						
Param No	Description	Default	Units / Option 3 Text	Can Change Level	Can See Level	Help
98	Manual ReZero Flow Controller Timer Setpoint		10 Sec	600		0 Manual ReZero Flow Controller Timer Setpoint
99	P021 Recirculation Speed		6000 RPM	600		0 P021 Recirculation Speed
100	P021 Pump To Drain Speed		3000 RPM	600		0 P021 Pump To Drain Speed
This value will change the way that the equipment samples. 0 = push and hold sample button for sampling. Other values used in this parameter will change the mode to operate as a PUSH to start and PUSH to STOP mode. The sampling will continue within those commands for the time value entered. 9999 ENTRY = PARTICLE COUNTING MODE. When a sample is started in this mode, THERE IS NO PUSH BUTTON INTERACTION. LINES MUST BE INSTALLED AND CORRECTLY ROUTED PRIOR TO STARTING IN THIS MODE.						
101	Sample Timer Setpoint		0 sec	600		0 Sample Timer Setpoint
102	Day Tank Overflow Tank Percent		100%	900		0 Day Tank Overflow Tank Percent
103	Day Tank High High Tank Percent		95%	900		0 Day Tank High High Tank Percent
104	Day Tank High Level Tank Percent		80%	900		0 Day Tank High Level Tank Percent
105	Day Tank Low Level Tank Percent		10%	900		0 Day Tank Low Level Tank Percent
106	Day Tank Low Low Level Tank Percent		0%	900		0 Day Tank Low Low Level Tank Percent
107	Day Tank Volume		400 L	900		0 Day Tank Volume
108	P001 High Pressure Setpoint		50 PSI	600		0 P001 High Pressure Setpoint
109	P001 Low Pressure Setpoint		25 PSI	600		0 P001 Low Pressure Setpoint
110	P099 High Pressure Setpoint		50 PSI	600		0 P099 High Pressure Setpoint
111	P099 Low Pressure Setpoint		25 PSI	600		0 P099 Low Pressure Setpoint
112	P021 High Pressure Setpoint		30 PSI	600		0 P021 High Pressure Setpoint
113	P021 Low Pressure Setpoint		5 PSI	600		0 P021 Low Pressure Setpoint
114	M001 Temperature Sensor Max Differential		0.5 deg C	900		0 M001 Temperature Sensor Max Differential
115	M001 Conductivity Sensor Max Differential		2 mS/cm	900		0 M001 Conductivity Sensor Max Differential
117	P021 High Flow Setpoint		18000 ml/PM	0		0 P021 High Flow Setpoint
118	P021 Low RPM Setpoint		2000 RPM	0		0 P021 Low RPM Setpoint
119	P021 High RPM Setpoint		8900 RPM	0		0 P021 High RPM Setpoint
120	Alarm Horn Silence Timer		3 Sec	0		0 Alarm Horn Silence Timer
121	Alarm Latch Timer		3 Sec	0		0 Alarm Latch Timer
122	Max Number of Blend Retries Exceeded Setpoint		3 Retries	0		0 Max Number of Blend Retries Exceeded Setpoint
125	S001: Dispense Cabinet EMO - PB or external Alarm Delay Setpoint		0 Sec	999		300 S001: Dispense Cabinet EMO - PB or external Alarm Delay Setpoint
126	S002: CDA Pressure IPS A1.951 Alarm Delay Setpoint		1 Sec	999		300 S002: CDA Pressure IPS A1.951 Alarm Delay Setpoint
127	S003: N2 Facility IPS N1.950 Alarm Delay Setpoint		1 Sec	999		300 S003: N2 Facility IPS N1.950 Alarm Delay Setpoint
128	S004: Door Interlock Z5.25.5 Alarm Delay Setpoint		0 Sec	999		300 S004: Door Interlock Z5.25.5 Alarm Delay Setpoint
129	S005: Drain Overflow CS.21.58 Alarm Delay Setpoint		0 Sec	999		300 S005: Drain Overflow CS.21.58 Alarm Delay Setpoint
130	S006: Dispense Cabinet Exhaust PS.21.491 Alarm Delay Setpoint		1 Sec	999		300 S006: Dispense Cabinet Exhaust PS.21.491 Alarm Delay Setpoint
131	S007: Cabinet Drain High Level Leak FS.25.9 Alarm Delay Setpoint		0 Sec	999		300 S007: Cabinet Drain High Level Leak FS.25.9 Alarm Delay Setpoint
132	S008: Cabinet Drain Low Level Leak FS.25.8 Alarm Delay Setpoint		0 Sec	999		300 S008: Cabinet Drain Low Level Leak FS.25.8 Alarm Delay Setpoint
133	S009: Blend To Drain Too Long Alarm Delay Setpoint		0 Sec	999		300 S009: Blend To Drain Too Long Alarm Delay Setpoint
140	S019: Day Tank Over Full Alarm		0 Sec	999		300 S019: Day Tank Over Full Alarm
96	Time Since Batch Finished Alarm Setpoint		1440 min	300		0 Time Since Batch Finished Alarm Setpoint
97	Maintenance Bypass Alarm Delay Setpoint		30 min	900		0 Maintenance Bypass Alarm Delay Setpoint

4-16 System Operation and Process Procedures

HMI PARAMETERS / SETTINGS									
The values below are for information / training only. The equipment should ONLY be setup according to customer FINGERPRINTED settings. Those settings are not captured in this document.									
Param No	Description	Default	Units / Option 3 Text	Can Change Level	Can See Level	Help	NOTES		
150	CO01: Supply J001 Not Ready Alarm Delay Setpoint		5 Sec	600	0	CO01: Supply J001 Not Ready Alarm Delay Setpoint			
151	CO02: Supply J099 Not Ready Alarm Delay Setpoint		5 Sec	600	0	CO02: Supply J099 Not Ready Alarm Delay Setpoint			
152	CO03: Pump 21 Not Ready Alarm Delay Setpoint		5 Sec	600	0	CO03: Pump 21 Not Ready Alarm Delay Setpoint			
153	CO04: M001 Conductivity A Out Of Spec Alarm Delay Setpoint		5 Sec	600	0	CO04: M001 Conductivity A Out Of Spec Alarm Delay Setpoint			
154	CO05: M001 Conductivity B Out Of Spec Alarm Delay Setpoint	6 (60)	Sec	600	0	CO05: M001 Conductivity B Out Of Spec Alarm Delay Setpoint			
155	CO06: M001 Temperature A Out Of Spec Alarm Delay Setpoint		5 Sec	600	0	CO06: M001 Temperature A Out Of Spec Alarm Delay Setpoint			
156	CO07: M001 Temperature B Out Of Spec Alarm Delay Setpoint		5 Sec	600	0	CO07: M001 Temperature B Out Of Spec Alarm Delay Setpoint			
157	CO08: J001 Flow Out Of Spec Alarm Delay Setpoint		5 Sec	600	0	CO08: J001 Flow Out Of Spec Alarm Delay Setpoint			
158	CO09: J099 Flow Out Of Spec Alarm Delay Setpoint		5 Sec	600	0	CO09: J099 Flow Out Of Spec Alarm Delay Setpoint			
159	CO10: J001 Pressure Out Of Spec Alarm Delay Setpoint		5 Sec	600	0	CO10: J001 Pressure Out Of Spec Alarm Delay Setpoint			
160	CO11: J099 Pressure Out Of Spec Alarm Delay Setpoint		5 Sec	600	0	CO11: J099 Pressure Out Of Spec Alarm Delay Setpoint			
161	CO12: J001 High Pressure Alarm Delay Setpoint		5 Sec	600	0	CO12: J001 High Pressure Alarm Delay Setpoint			
162	CO13: J001 Low Pressure Alarm Delay Setpoint		5 Sec	600	0	CO13: J001 Low Pressure Alarm Delay Setpoint			
163	CO14: J099 High Pressure Alarm Delay Setpoint		5 Sec	600	0	CO14: J099 High Pressure Alarm Delay Setpoint			
164	CO15: J099 Low Pressure Alarm Delay Setpoint		5 Sec	600	0	CO15: J099 Low Pressure Alarm Delay Setpoint			
165	CO16: M001 High Conductivity Alarm Delay Setpoint		5 Sec	600	0	CO16: M001 High Conductivity Alarm Delay Setpoint			
166	CO17: M001 Low Conductivity Alarm Delay Setpoint		5 Sec	600	0	CO17: M001 Low Conductivity Alarm Delay Setpoint			
167	CO18: M001 High Temperature Alarm Delay Setpoint		5 Sec	600	0	CO18: M001 High Temperature Alarm Delay Setpoint			
168	CO19: M001 Low Temperature Alarm Delay Setpoint		5 Sec	600	0	CO19: M001 Low Temperature Alarm Delay Setpoint			
169	CO20: T001A High Level Alarm Delay Setpoint		5 Sec	600	0	CO20: T001A High Level Alarm Delay Setpoint			
170	CO21: T001 Low Level Alarm Delay Setpoint		5 Sec	600	0	CO21: T001 Low Level Alarm Delay Setpoint			
171	CO22: T001B High Level Alarm Delay Setpoint		5 Sec	600	0	CO22: T001B High Level Alarm Delay Setpoint			
172	CO23: T001 Fill Timeout Alarm Delay Setpoint	3600 Sec		600	0	CO23: T001 Fill Timeout Alarm Delay Setpoint			
173	CO24: T001 End Of Life Exceeded Alarm Delay Setpoint		600 Sec	600	0	CO24: T001 End Of Life Exceeded Alarm Delay Setpoint			
174	CO25: N2 Facility IPS N1.950 Alarm Delay Setpoint		5 Sec	600	0	CO25: N2 Facility IPS N1.950 Alarm Delay Setpoint			
175	CO26: Remote Shutdown For Fab 1 Alarm Delay Setpoint		0 Sec	600	0	CO26: Remote Shutdown For Fab 1 Alarm Delay Setpoint			
176	CO27: Remote Shutdown For Fab 2 Alarm Delay Setpoint		0 Sec	600	0	CO27: Remote Shutdown For Fab 2 Alarm Delay Setpoint			
177	CO28: Remote Shutdown For Fab 3 Alarm Delay Setpoint		0 Sec	600	0	CO28: Remote Shutdown For Fab 3 Alarm Delay Setpoint			
178	CO29: Remote Shutdown For Fab 4 Alarm Delay Setpoint		0 Sec	600	0	CO29: Remote Shutdown For Fab 4 Alarm Delay Setpoint			
179	CO30: No Conductivity Transmitters Enabled Alarm Delay Setpoint		2 Sec	600	0	CO30: No Conductivity Transmitters Enabled Alarm Delay Setpoint			
180	CO31: No Temperature Transmitters Enabled Alarm Delay Setpoint		2 Sec	600	0	CO31: No Temperature Transmitters Enabled Alarm Delay Setpoint			
181	CO32: M021 Conductivity Out Of Spec		5 Sec	600	0	CO32: M021 Conductivity Out Of Spec			
182	CO33: M021 Temperature Out Of Spec		5 Sec	600	0	CO33: M021 Temperature Out Of Spec			
183	CO34: M021 High Pressure		5 Sec	600	0	CO34: M021 High Pressure			
184	CO35: M021 Low Pressure		5 Sec	600	0	CO35: M021 Low Pressure			
185	CO36: M021 High Conductivity		5 Sec	600	0	CO36: M021 High Conductivity			
186	CO37: M021 Low Conductivity		5 Sec	600	0	CO37: M021 Low Conductivity			
187	CO38: M021 High Temperature		5 Sec	600	0	CO38: M021 High Temperature			
188	CO39: M021 Low Temperature		5 Sec	600	0	CO39: M021 Low Temperature			
189	CO40: Blend To Drain Override Timeout		600 Sec	600	0	CO40: Blend To Drain Override Timeout			

HMI PARAMETERS / SETTINGS				The equipment should ONLY be setup according to customer FINGERPRINTED settings. Those settings are not captured in this document.			
The values below are for information / training only.							
Param No	Description	Default	Units / Option 3 Text	Can Change Level	Can See Level	Help	NOTES
190	CO41: Blend To Drain Too Long		150 Sec	600		0 CO41: Blend To Drain Too Long	
191	CO42: All Tank Outlet Valves Closed		1 Sec	600		0 CO42: All Tank Outlet Valves Closed	
192	CO43: Blend Retries Exceeded Alarm Delay Setpoint		1 Sec	600		0 CO43: Blend Retries Exceeded Alarm Delay Setpoint	
193	CO44: Liquid In Drain CS.21.17 Alarm Delay Setpoint		10 Sec	600		0 CO44: Liquid In Drain CS.21.17 Alarm Delay Setpoint	
194	CO45: J001 Pressure Not Reached In Expected Time Alarm Delay Setpoint		5 Sec	600		0 CO45: J001 Pressure Not Reached In Expected Time Alarm Delay Setpoint	
195	CO46: J099 Pressure Not Reached In Expected Time Alarm Delay Setpoint		5 Sec	600		0 CO46: J099 Pressure Not Reached In Expected Time Alarm Delay Setpoint	
196	CO47: J001 Pressure Low Alarm Delay Setpoint		5 Sec	600		0 CO47: J001 Pressure Low Alarm Delay Setpoint	
197	CO48: J099 Pressure Low Alarm Delay Setpoint		5 Sec	600		0 CO48: J099 Pressure Low Alarm Delay Setpoint	
200	W001: Dispense Cabinet Exhaust Low PS.21.491 Alarm Delay Setpoint		1 Sec	600		0 W001: Dispense Cabinet Exhaust Low PS.21.491 Alarm Delay Setpoint	
201	W002: Sample Door Interlock -25.24.5 Alarm Delay Setpoint		1 Sec	600		0 W002: Sample Door Interlock -25.24.5 Alarm Delay Setpoint	
202	W003: Liquid In Drain CS.21.17 Alarm Delay Setpoint		3 Sec	600		0 W003: Liquid In Drain CS.21.17 Alarm Delay Setpoint	
203	W004: M001 Temp Sensor Exceeded Max Difference Alarm Delay Setpoint		3 Sec	600		0 W004: M001 Temp Sensor Exceeded Max Difference Alarm Delay Setpoint	
204	W005: M001 Conductivity Sensor Exceeding Max Diff Alarm Delay Setpoint		15 Sec	600		0 W005: M001 Conductivity Sensor Exceeding Max Diff Alarm Delay Setpoint	
205	W006: Day Tank 1 Sensor Error Alarm Delay Setpoint		1 Sec	600		0 W006: Day Tank 1 Sensor Error Alarm Delay Setpoint	
206	W007: Day Tank 1A High High Alarm Delay Setpoint		0 Sec	600		0 W007: Day Tank 1A High High Alarm Delay Setpoint	
207	W008: Day Tank 1B High High Alarm Delay Setpoint		0 Sec	600		0 W008: Day Tank 1B High High Alarm Delay Setpoint	
208	W009: Day Tank Low Low Alarm Delay Setpoint		0 Sec	600		0 W009: Day Tank Low Low Alarm Delay Setpoint	
209	W010: Cabinet Low Leak FS.25.8 Alarm Delay Setpoint		1 Sec	600		0 W010: Cabinet Low Leak FS.25.8 Alarm Delay Setpoint	
210	W011: Cabinet High Leak FS.25.9 Alarm Delay Setpoint		0 Sec	600		0 W011: Cabinet High Leak FS.25.9 Alarm Delay Setpoint	
211	W012: HMI Forcing Active Alarm Delay Setpoint		3 Sec	600		0 W012: HMI Forcing Active Alarm Delay Setpoint	
212	W013: PLC Battery Warning Alarm Delay Setpoint		3 Sec	600		0 W013: PLC Battery Warning Alarm Delay Setpoint	
213	W014: J001 Not In Auto Mode Alarm Delay Setpoint		3 Sec	600		0 W014: J001 Not In Auto Mode Alarm Delay Setpoint	
214	W015: J099 Not In Auto Mode Alarm Delay Setpoint		3 Sec	600		0 W015: J099 Not In Auto Mode Alarm Delay Setpoint	
215	W016: P021 Not In Auto Mode Alarm Delay Setpoint		3 Sec	600		0 W016: P021 Not In Auto Mode Alarm Delay Setpoint	
216	W017: Day Tank 1 Not In Auto Mode Alarm Delay Setpoint		3 Sec	600		0 W017: Day Tank 1 Not In Auto Mode Alarm Delay Setpoint	
217	W018: Sample Area Not In Auto Mode Alarm Delay Setpoint		3 Sec	600		0 W018: Sample Area Not In Auto Mode Alarm Delay Setpoint	
218	W019: Machine Not In Auto Mode Alarm Delay Setpoint		3 Sec	600		0 W019: Machine Not In Auto Mode Alarm Delay Setpoint	
219	W020: P021 High Pressure Alarm Delay Setpoint		3 Sec	600		0 W020: P021 High Pressure Alarm Delay Setpoint	
220	W021: P021 Low Pressure Alarm Delay Setpoint		3 Sec	600		0 W021: P021 Low Pressure Alarm Delay Setpoint	
221	W022: P021 Low Flow Alarm Delay Setpoint		3 Sec	600		0 W022: P021 Low Flow Alarm Delay Setpoint	
222	W023: P021 High Flow Alarm Delay Setpoint		3 Sec	600		0 W023: P021 High Flow Alarm Delay Setpoint	
223	W024: P021 Levitronix High RPM Alarm Delay Setpoint		3 Sec	600		0 W024: P021 Levitronix High RPM Alarm Delay Setpoint	
224	W025: P021 Levitronix Low RPM Alarm Delay Setpoint		3 Sec	600		0 W025: P021 Levitronix Low RPM Alarm Delay Setpoint	
225	W026: Tank 1 Pot Life Exceeded Alarm Delay Setpoint		3 Sec	600		0 W026: Tank 1 Pot Life Exceeded Alarm Delay Setpoint	
226	W027: P021 Levitronix Warning Alarm Delay Setpoint		3 Sec	600		0 W027: P021 Levitronix Warning Alarm Delay Setpoint	
227	W028: P021 Levitronix Fault Alarm Delay Setpoint		3 Sec	600		0 W028: P021 Levitronix Fault Alarm Delay Setpoint	
228	W029: P021 Maintenance Hours Exceeded Alarm Delay Setpoint		3 Sec	600		0 W029: P021 Maintenance Hours Exceeded Alarm Delay Setpoint	
229	W030: M001A Conductivity Transmitter Not Enabled Alarm Delay Setpoint		1 Sec	600		0 W030: M001A Conductivity Transmitter Not Enabled Alarm Delay Setpoint	
230	W031: M001B Conductivity Transmitter Not Enabled Alarm Delay Setpoint		1 Sec	600		0 W031: M001B Conductivity Transmitter Not Enabled Alarm Delay Setpoint	
231	W032: M001A Temperature Transmitter Not Enabled Alarm Delay Setpoint		1 Sec	600		0 W032: M001A Temperature Transmitter Not Enabled Alarm Delay Setpoint	

4-18 System Operation and Process Procedures

HMI PARAMETERS / SETTINGS									
The values below are for information / training only. The equipment should ONLY be setup according to customer FINGERPRINTED settings. Those settings are not captured in this document.									
Param No	Description	Default	Units / Option 3 Text	Can Change Level	Can See Level	Help	NOTES		
232	W033: M001B Temperature Transmitter Not Enabled Alarm Delay Setpoint		1 Sec	600	0	W033: M001B Temperature Transmitter Not Enabled Alarm Delay Setpoint			
233	W034: Time Since Batch Started Alarm Alarm Delay Setpoint		3 Sec	600	0	W034: Time Since Batch Started Alarm Alarm Delay Setpoint			
234	W035: Time Since Batch Finished Alarm Alarm Delay Setpoint		3 Sec	600	0	W035: Time Since Batch Finished Alarm Alarm Delay Setpoint			
235	W036: Machine Override Switch Active Alarm Delay Setpoint		3 Sec	600	0	W036: Machine Override Switch Active Alarm Delay Setpoint			
236	W037: P001 Levitronix High RPM Alarm Delay Setpoint		5 Sec	600	0	W037: P001 Levitronix High RPM Alarm Delay Setpoint			
237	W038: P001 Levitronix Low RPM Alarm Delay Setpoint		5 Sec	600	0	W038: P001 Levitronix Low RPM Alarm Delay Setpoint			
238	W039: P001 Levitronix Warning Alarm Delay Setpoint		5 Sec	600	0	W039: P001 Levitronix Warning Alarm Delay Setpoint			
239	W040: P001 Levitronix Fault Alarm Delay Setpoint		5 Sec	600	0	W040: P001 Levitronix Fault Alarm Delay Setpoint			
240	W041: P001 Maintenance Hours Exceeded Alarm Delay Setpoint		3 Sec	600	0	W041: P001 Maintenance Hours Exceeded Alarm Delay Setpoint			
241	W042: M021 Conductivity Out Of Spec Warning Delay Setpoint		3 Sec	600	0	W042: M021 Conductivity Out Of Spec Warning Delay Setpoint			
250	M021 Quality Temperature Target Variance		20 %	600	0	M021 Quality Temperature Target Variance	Setting for alarming median value. If values are out of range of variance and setpoint, alarming will occur.		
251	M021 Quality Temperature In Spec. Time		5 sec	600	0	M021 Quality Temperature In Spec. Time	Setting for alarming median value. If values are out of range of variance and setpoint, alarming will occur.		
252	M021 Quality Temperature Out of Spec. Time		15 sec	600	0	M021 Quality Temperature Out of Spec. Time	Setting for alarming median value. If values are out of range of variance and setpoint, alarming will occur.		
253	M021 Quality Conductivity Critical Target Variance		0.5 %	600	0	M021 Quality Conductivity Critical Alarm Target Variance (v/- %)	Setting for alarming median value. If values are out of range of variance and setpoint, alarming will occur. AND THE TANK OUTLET VALVE WILL CLOSE.		
254	M021 Quality Conductivity In Spec. Time		5 Sec	600	0	M021 Quality Conductivity In Spec. Time	Setting for alarming median value. If values are out of range of variance and setpoint, alarming will occur.		
255	M021 Quality Conductivity Out of Spec. Time		15 Sec	600	0	M021 Quality Conductivity Out of Spec. Time	Setting for alarming median value. If values are out of range of variance and setpoint, alarming will occur.		
256	M021 Quality Temperature Setpoint	25 nominal	deg C	600	0	M021 Quality Temperature Setpoint	Setting for alarming median value. If values are out of range of variance and setpoint, alarming will occur.		
257	M021 Quality Conductivity Setpoint	53 n ominal	ms/cm	600	0	M021 Quality Conductivity Critical Alarm Setpoint	Setting for alarming median value. If values are out of range of variance and setpoint, alarming will occur.		
258	PT M21 910 High Pressure Setpoint		30 PSI	600	0	PT M21 910 High Pressure Setpoint	analog alarming criteria		
259	PT M21 910 Low Pressure Setpoint		5 PSI	600	0	PT M21 910 Low Pressure Setpoint	analog alarming criteria		
260	M021 Quality Conductivity Warning Target Variance		0.2 %	600	0	M021 Quality Conductivity Warning Alarm Target Variance (v/- %)	Setting for alarming median value. If values are out of range of variance and setpoint, alarming will occur.		
If this setting is changed, the analog signal scaling must be reviewed for all P001 input / output signals.									
Speed mode: Pump will operate to a commanded speed setpoint (setting 271)									
Process mode: The pump will operate toward a commanded discharge pressure according to setting 269.									
268	P001 Mode	Process	mode	600	0	Pump Mode (Speed / Process PSI)	applies when the setting 268 is process mode and all analog signals are scaled accordingly.		
269	P001 Pressure Setpoint		30 PSI	600	0	P001 Pump Pressure Setpoint	Pump setpoint speed for services		
270	P001 Pump To Drain Speed		3000 RPM	600	0	P001 Pump To Drain Speed			
271	P001 Pump To Fab Speed (Dispense)		3000 RPM	600	0	P001 Pump To Fab Speed (Dispense)	applies when the setting 268 is speed mode and all analog signals are scaled accordingly.		
AFTER THIS PERIOD OF TIME OF P1 OPERATION, A WARNING WILL POST FOR PUMP MAINTENANCE REMINDER									
272	P001 Pump Hours Maintenance Setpoint		8766 Hour	300	0	P001 Pump Hours Maintenance Setpoint	analog alarming criteria. Must be below 8000		
273	P001 High RPM Setpoint		7900 RPM	300	0	P001 High RPM Setpoint	analog alarming criteria. Must be above 400		
274	P001 Low RPM Setpoint		500 RPM	300	0	P001 Low RPM Setpoint			

Recipe Screens

The *Recipe Screen* is used to select either “Blend Recipes” or “Service Recipes”. After Blend or Service is selected by the Category Filter, the Object filter will allow the user to further refine their options based on sub-system. After the Category and Object are selected the user will see the available recipes that are in the database.

Figure 4-9. Recipe Screen

NOTE: REPRESENTATIVE SCREEN SHOWN.

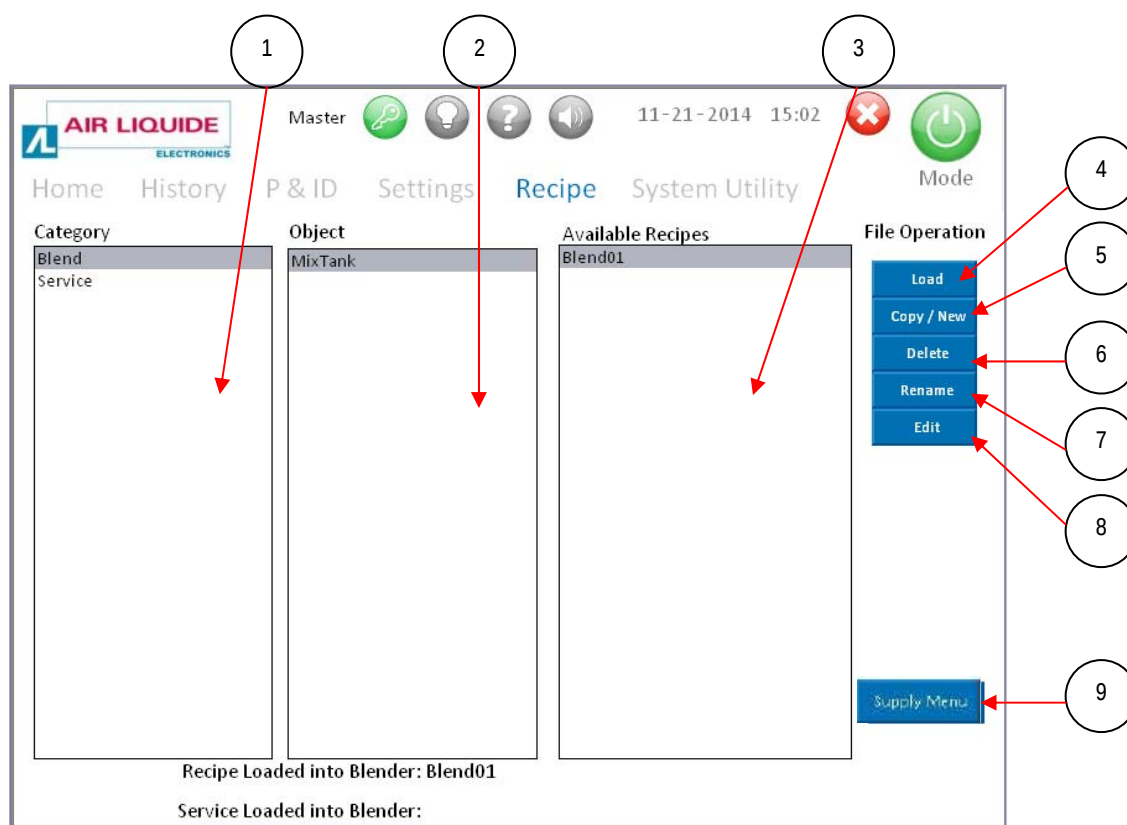


Table 4-9. Recipe screen

Callout	Description
1	Indicates the function to filter (Blend or Service).
2	Indicates the Object that is affected by the Category
3	Displays a list of available, existing recipes currently in the database. The recipes can be used as-is, or copied to a new, modified recipe.
4	Press the Load touch key to load the selected recipe.
5	Press the Copy / New touch key to copy an existing recipe or create a new recipe. In order to create a new recipe, first copy an existing recipe. Then, edit the new copy and save it as a new recipe.

Callout	Description
6	Press the De le te touch key to delete a recipe from the Avai la ble Re ci pes database.
7	Press the Re na me touch key to rename the recipe that has been selected.
8	Press the Ed it touch key to open the recipe editor for the recipe that is currently selected.
9	Press the Sup pl y Me nu touch key to open the Supplies Database editor.

Figure 4-10. Recipe Screen – Selecting the Category, Object, and Available Recipe

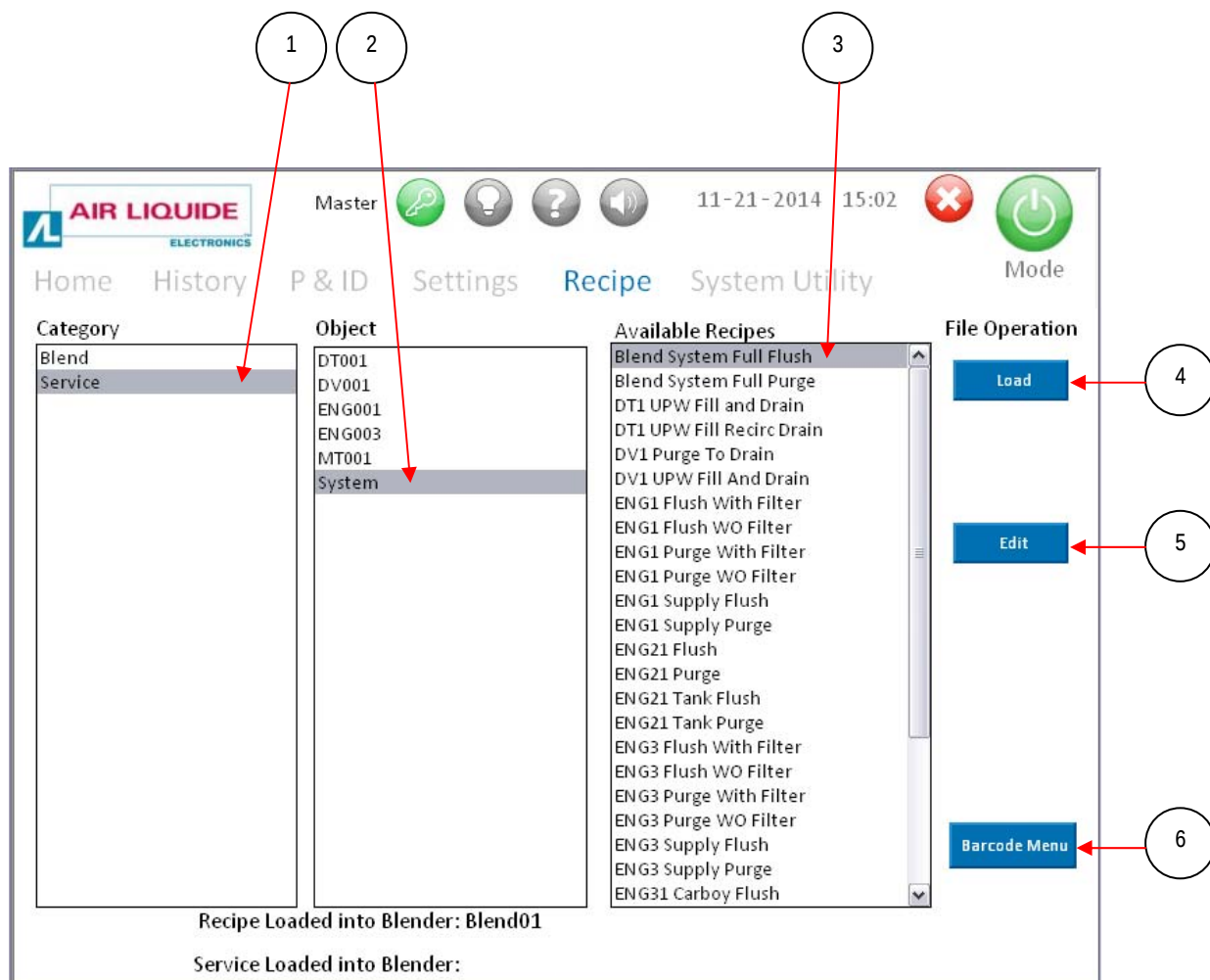
NOTE: REPRESENTATIVE SCREEN SHOWN.

Table 4-10. Recipe Screen – Selecting the Category, Object, and Available Recipe

Callout	Description
1	Indicates the type of recipe Category to perform [a Blend or a Service recipe].
2	Indicates the Object that is affected by the Category that was selected [In this example, a System Object was selected and its long list of Available Recipes is displayed in the Available Recipe column].
3	Displays a list of Available Recipes based on the Category and Object selected in the applicable columns of this screen. Select/highlight a recipe to use it.
4	Press the Load touch key to load the selected recipe. Loading a recipe is not available if the system is currently blending.
5	Press the Edit touch key to open the editor for the recipe that is currently selected.
6	Press the Supply Menu touch key to open the Supplies Database editor.

Figure 4-11. Recipe Screen – Loading Recipe

NOTE: REPRESENTATIVE SCREEN SHOWN.

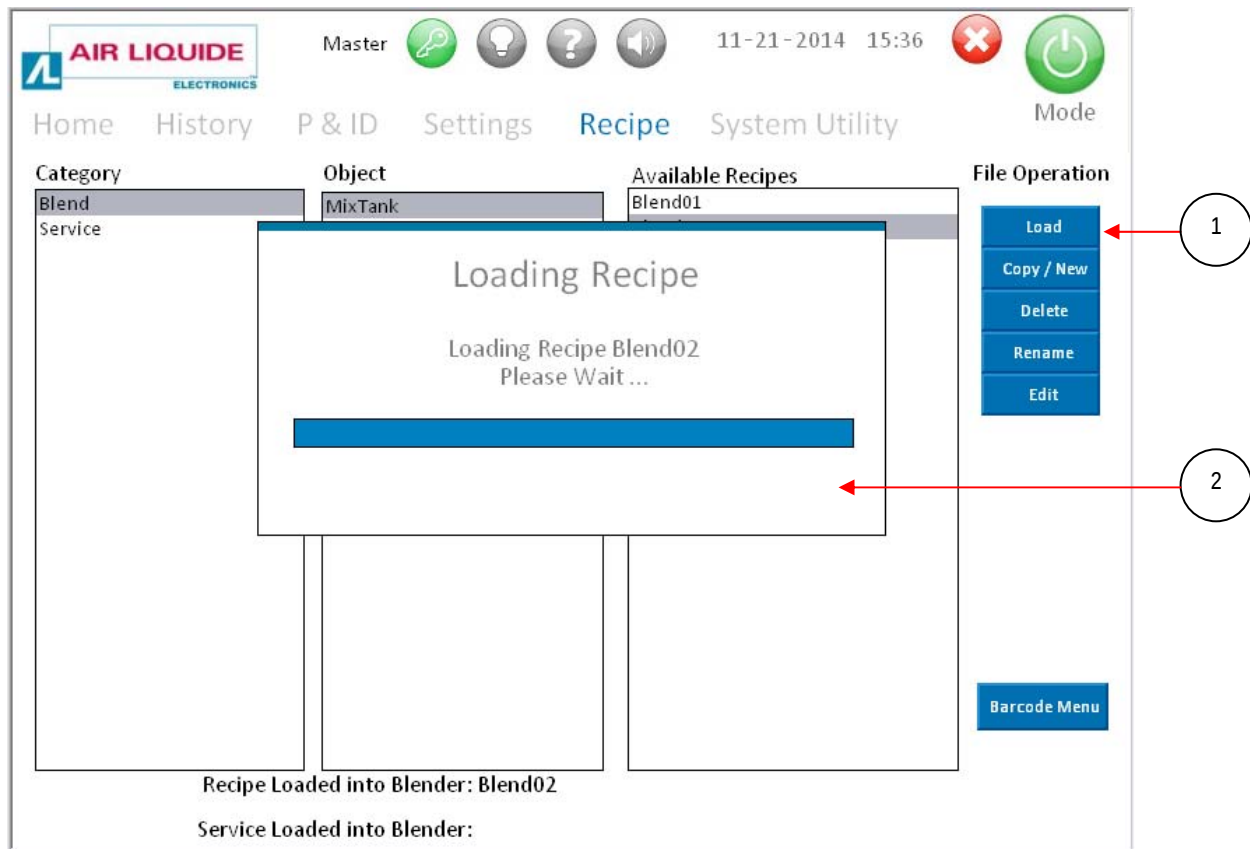


Table 4-11. Recipe screen – Loading the Recipe Step

Callout	Description
1	Press the Load touch key to load a newly created or pre-existing recipe into the system for blending.
2	Indicates that a selected recipe is being loaded into the system. This popup is displayed after a recipe line is selected and the Load Button is pressed.

Figure 4-12. Recipe Screen – Service for an Object of a Custom Recipe

NOTE: REPRESENTATIVE SCREEN SHOWN.

Service Recipes allow the user to load a set of existing service routines into the system or build a custom service recipe that can be loaded when a special service is required.

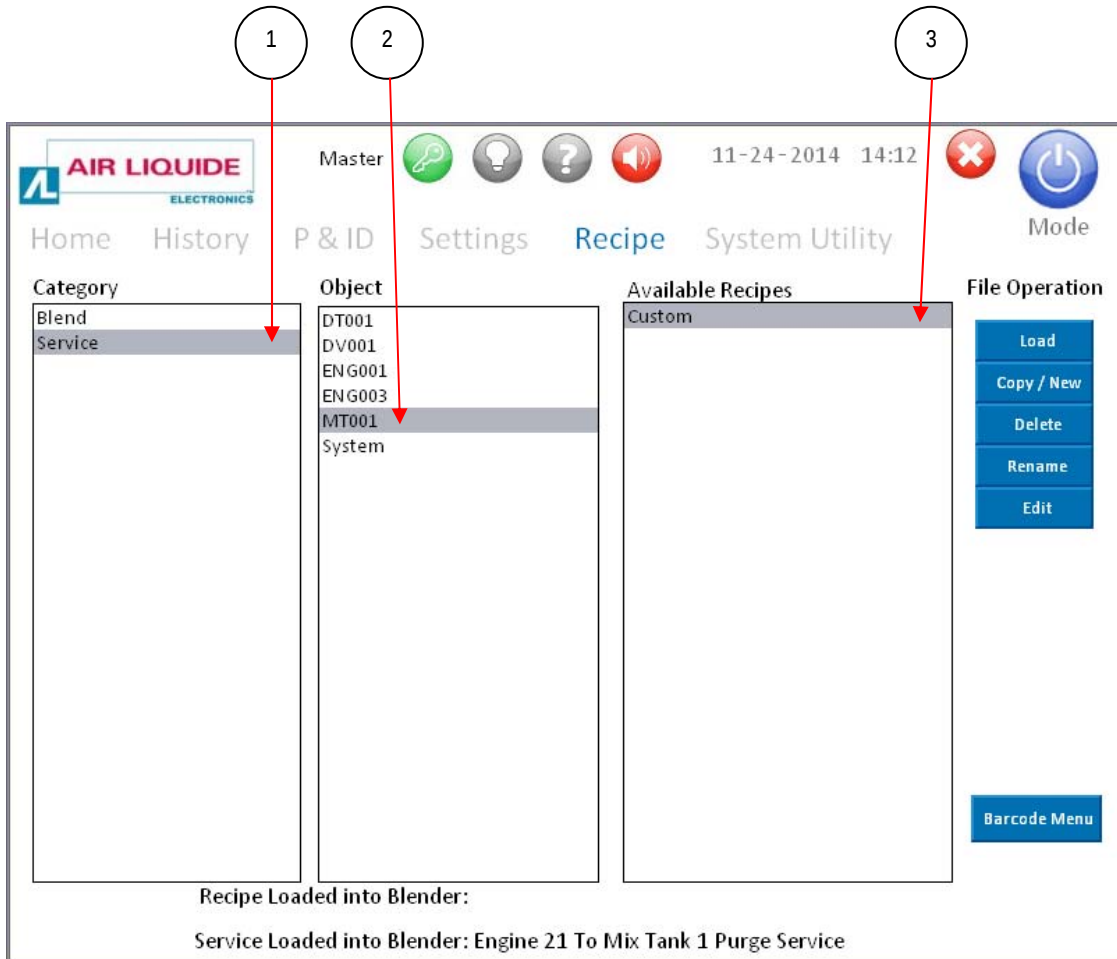


Table 4-12. Recipe screen – Service Custom Recipe Editor Selection Step

Callout	Description
1	In this example, indicates that a Service will be performed for the selected Object for a Custom recipe.
2	In this example, indicates that a Service will be performed for the tank and to its pump, P21.
3	In this example, indicates that a Custom recipe has been loaded in to the System for this Service.

Figure 4-13. Recipe Step/Action Screen – Service Custom Recipe Editor Step

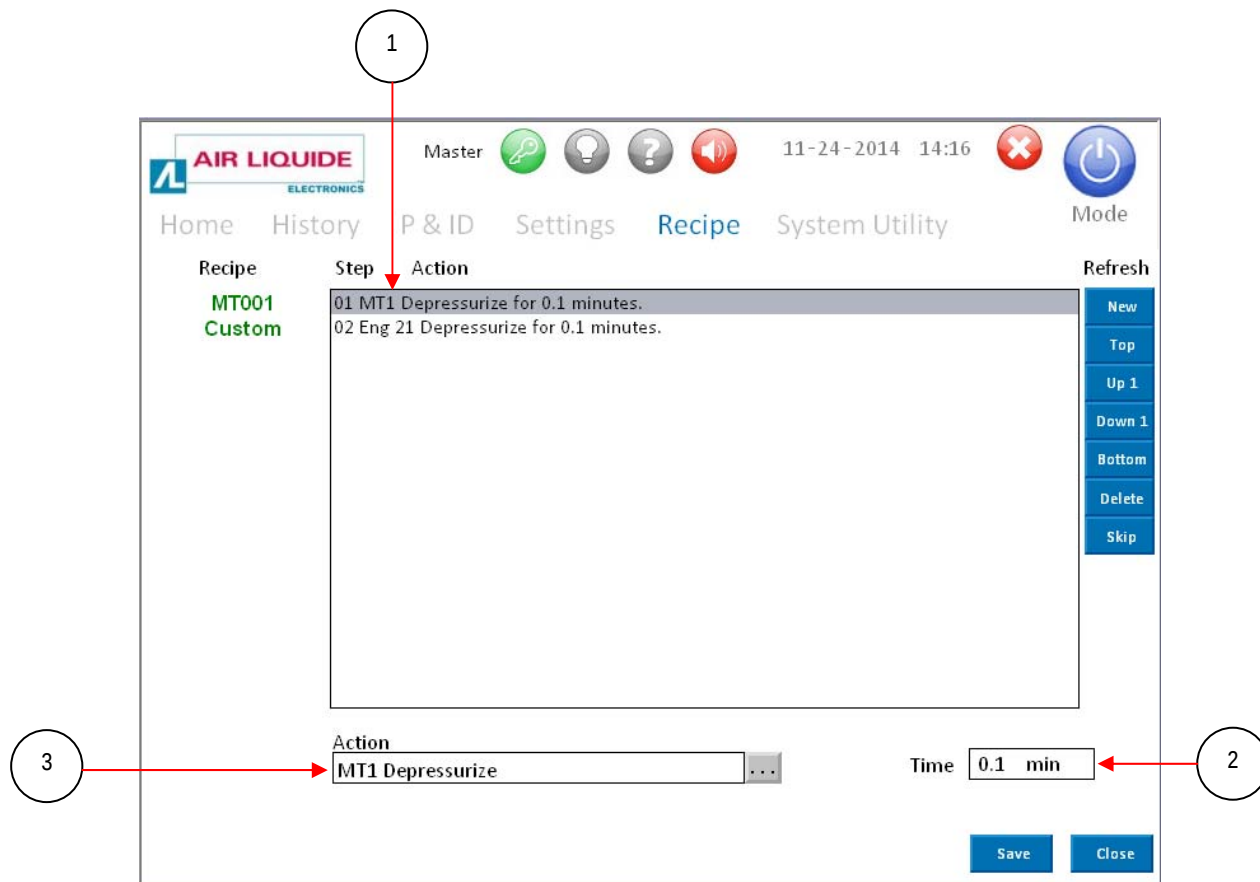
NOTE: REPRESENTATIVE EVENTS SHOWN.

Table 4-13. Recipe Step/Action Screen – Service Custom Recipe Editor Step

Callout	Description
1	This row indicates the Step and the Action that will be performed, and its length of time.
2	The selected row will be displayed on the bottom and may be edited. In this example, the length of time for step 1 may be edited. (0.1 min shown, 6 sec)
3	The selected row will be displayed on the bottom and may be edited. In this example, the step 1 action may be changed to a different action by pressing the ... key

Figure 4-14. Recipe Step/Action – Select Action and Time Required Step

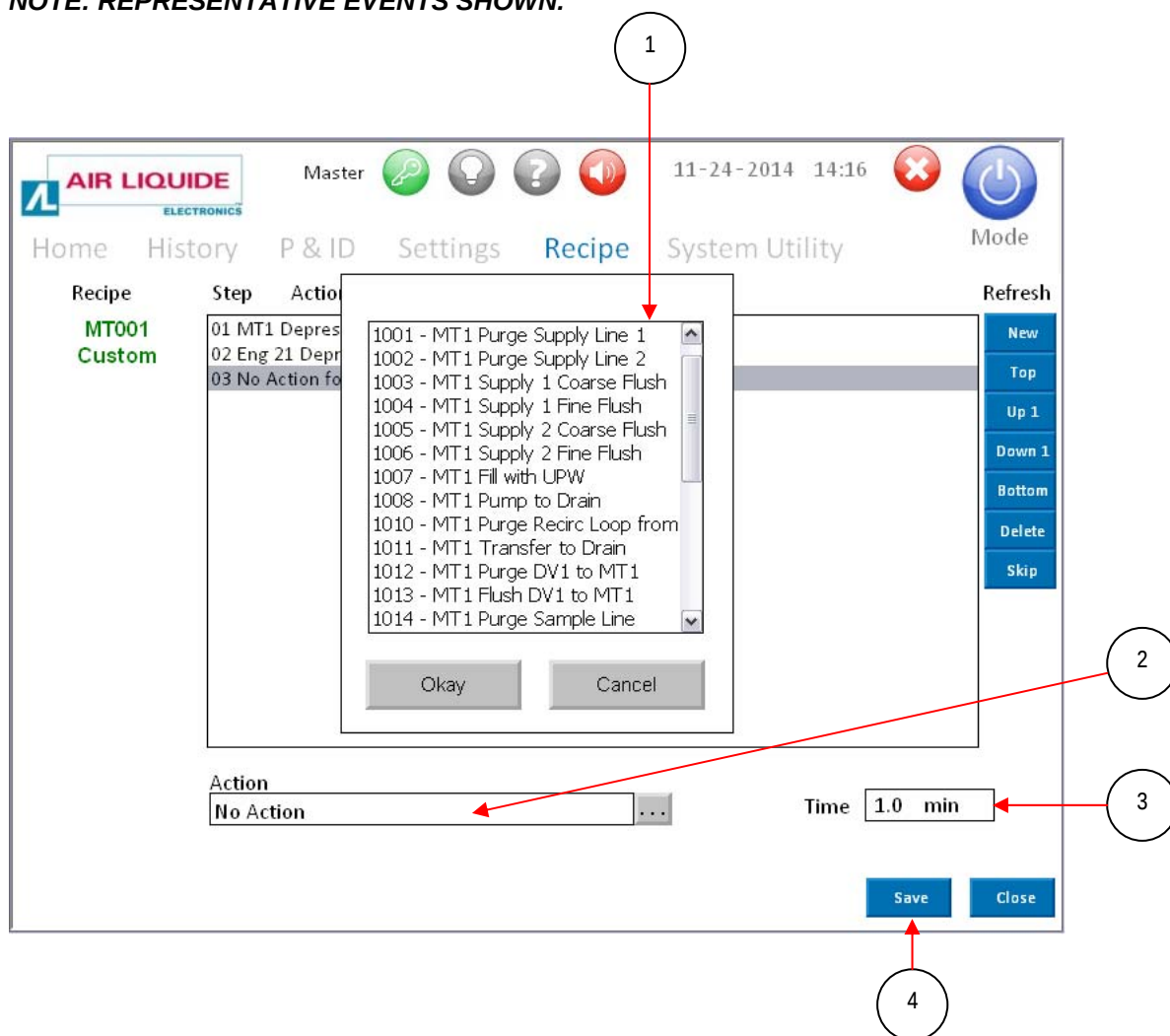
NOTE: REPRESENTATIVE EVENTS SHOWN.

Table 4-14. Recipe Step/Action – Select Action and Time Required Step

Callout	Description
1	The pull-down menu displays a list of Actions that are available after Step 03 has been selected / highlighted and after the "..." touch key is pressed. Selecting an Action allows the user to select the next service step. All of the available sub-system functions are listed in the pull down list.
2	Displays the Action to be performed. Select from available options in the drop down menu.
3	Displays the length of time for the Action [in this example, one minute].
4	Press the Save touch key after all selections have been made and are considered OK.

Figure 4-15. Recipe Step/Action – Save the Custom Recipe Step

NOTE: REPRESENTATIVE RECIPE SHOWN.

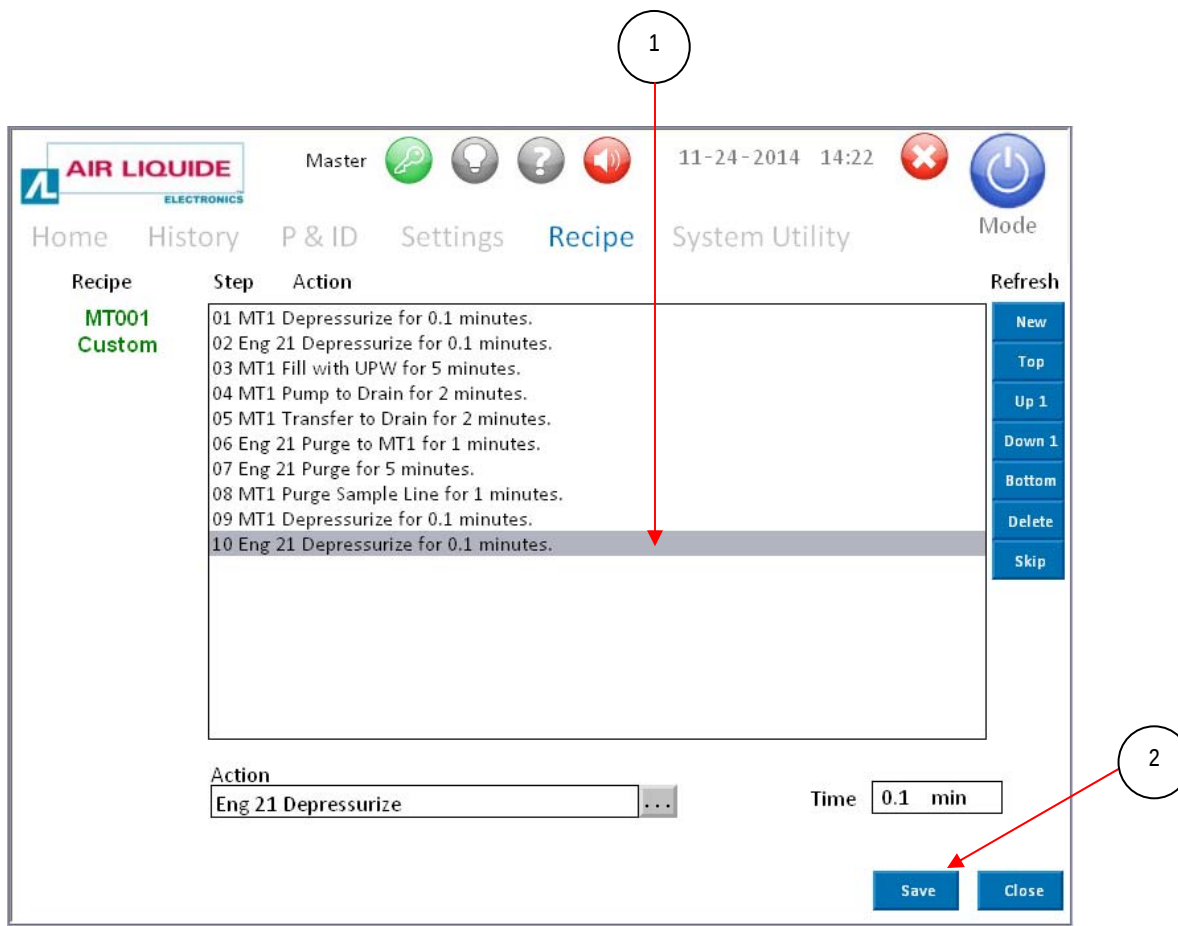


Table 4-15. Recipe Step/Action – Save the Custom Recipe Step

Callout	Description
1	The recipe actions and steps are listed here.
2	Press the Save touch key to save <u>all</u> of the Steps and Actions that are on this screen.

Figure 4-16. Recipe Category/Object/Available Recipes Screen

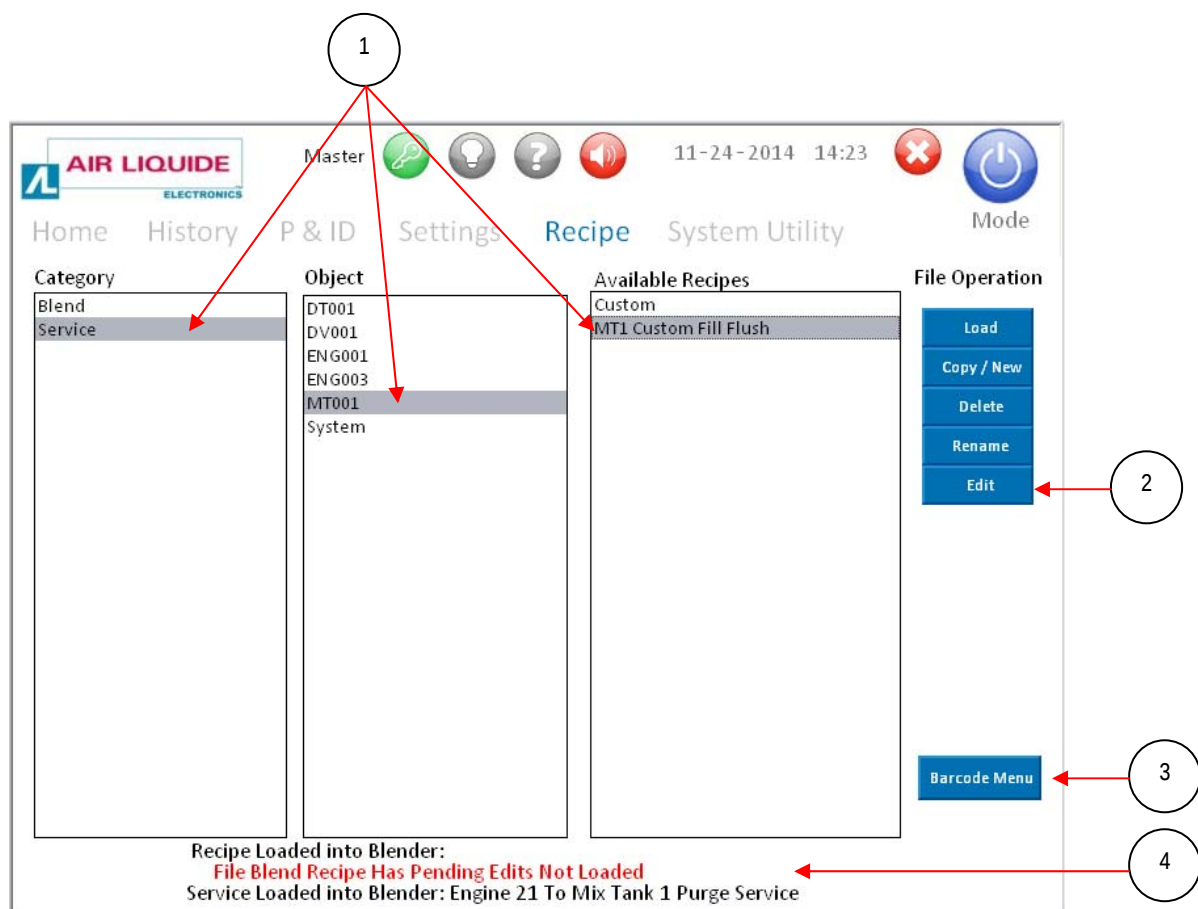
NOTE: REPRESENTATIVE EVENTS SHOWN.

Table 4-16. Recipe Category/Object/Available Recipes Screen

Callout	Description
1	In this example, indicates that a flush Service for the Object MT001 will use Available Recipe MT1 Custom Fill Flush, but that the Service recipe on file has pending edits to it that have not been loaded.
2	Press the Edit touch key to edit this recipe.
3	Press the SUPPLY MENU or Barcode Menu touch key to access the SUPPLY SCREEN or Barcode Manager – New Product/New Scan screen.
4	This area shows the current Blend Recipe and Service Recipes that are loaded into the system and waiting for operation. If a loaded recipe is edited and saved but has not been reloaded, the system will notify the user that the online and offline recipes are not equal and the file edits need to be loaded into the system before they will take effect.

Figure 4-17. Recipe – SUPPLY / Barcode – New Product Screen

NOTE: REPRESENTATIVE EVENTS SHOWN. NOT ALL UNITS WILL HAVE BARCODE OPTIONS.

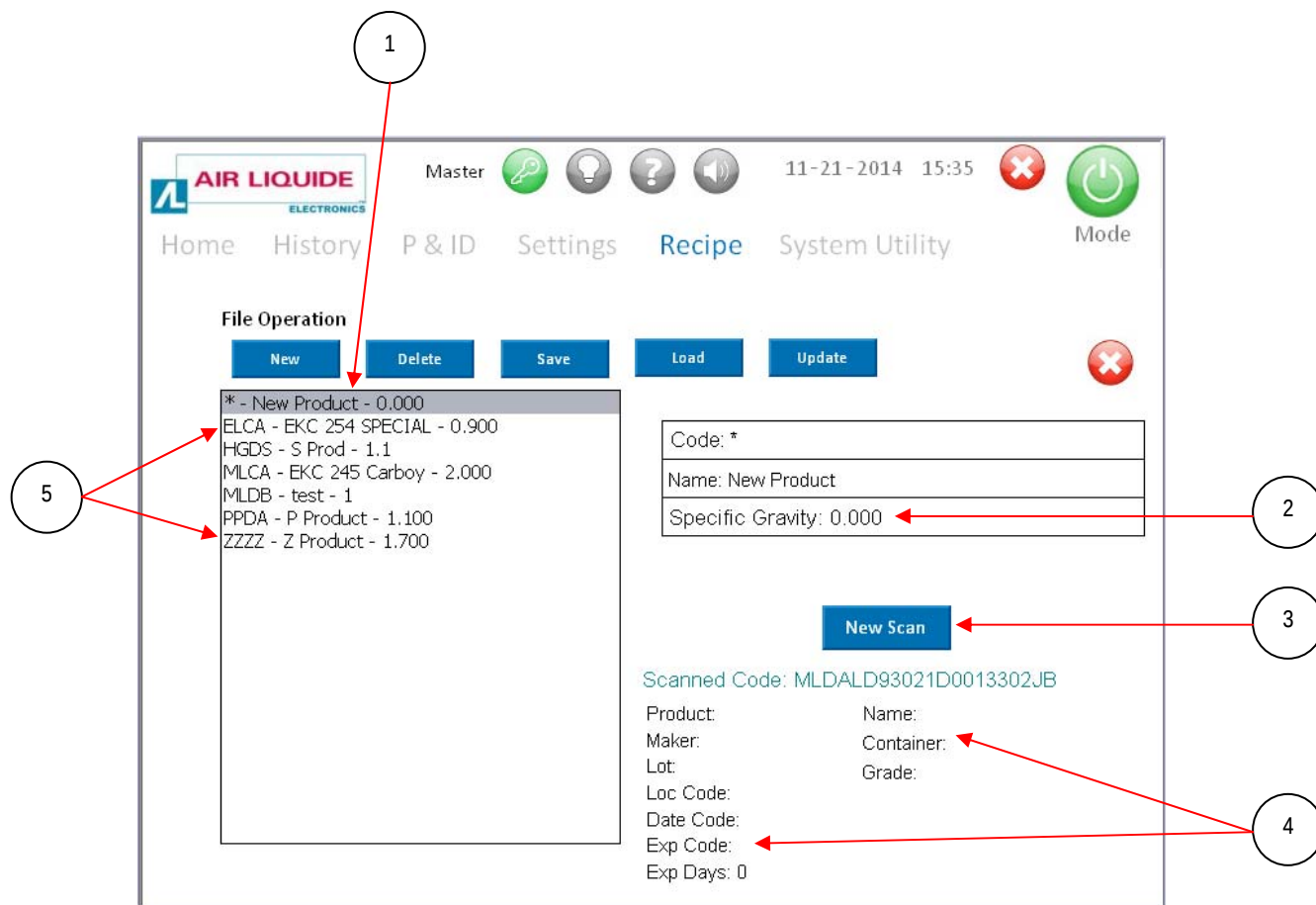


Table 4-17. Recipe – SUPPLY - Barcode – New Product/New Scan Screen

Callout	Description
1	Indicates that constituents that are in the existing database.
2	The specific gravity and properties of the new product/chemical must be manually added to the Specific Gravity data field of the screen.
3	Press the New Scan touch key and use the barcode reader to rescan the barcode label on the supply container. This is how a user will “teach” the system a new constituent.
4	BARCODE OPTION: After the barcode label on the supply container is scanned, and the system does not find any errors, the data fields on this screen are populated with data from the barcode label.
5	Indicates a list of product already in the system’s database. When a supply container of product is scanned, and the product exists in the system’s database, the system allows that product to be used in a blend recipe. New product must be added to the database, in order to be used in a blend recipe.

System Utility Screens (I/O)

Figure 4-18. System Utility Screen – Utility Pull-down Selections

NOTE: REPRESENTATIVE SCREEN SHOWN. REFER TO SPECIFIC PRINTS FOR DETAILS OF EQUIPMENT.

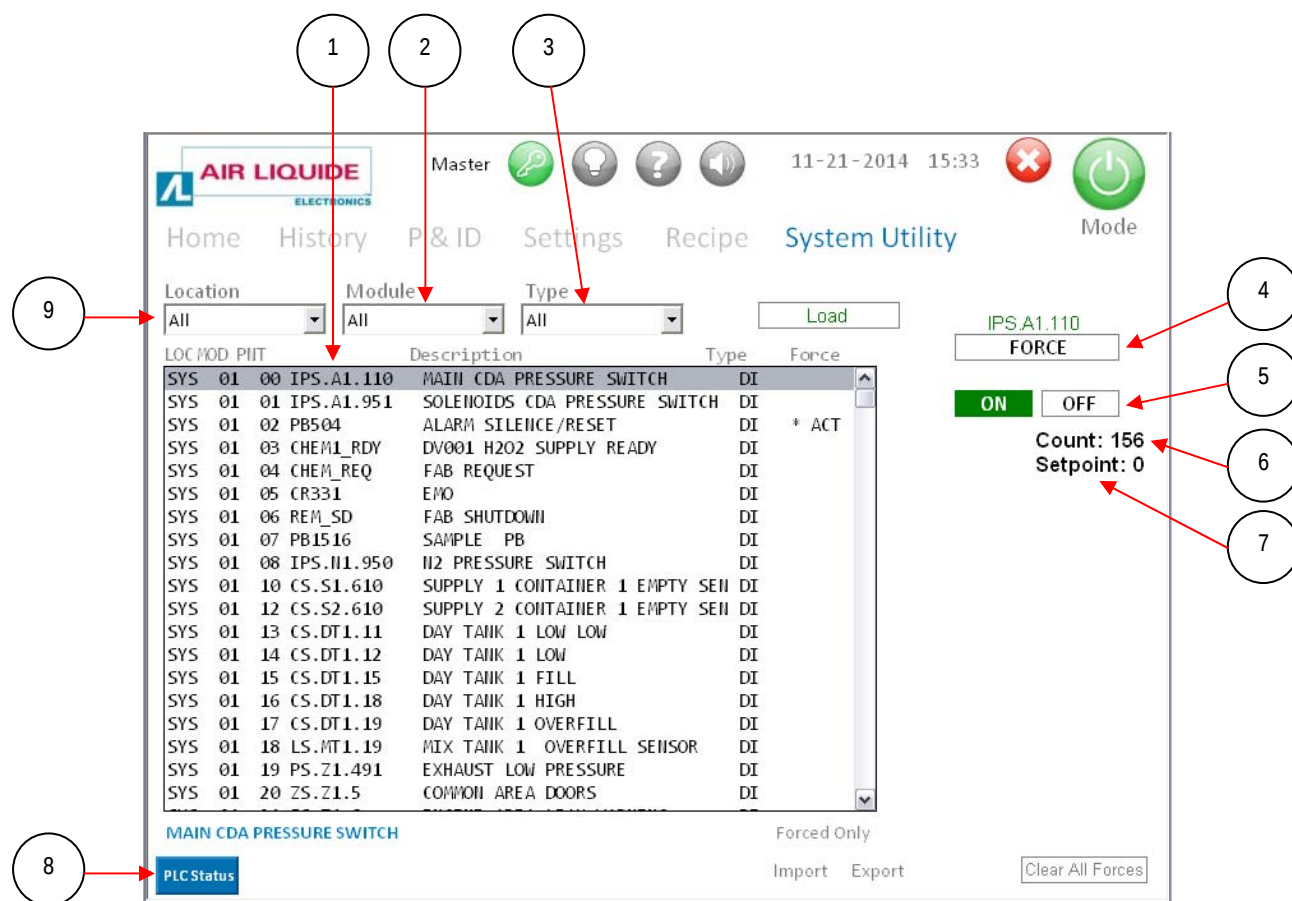
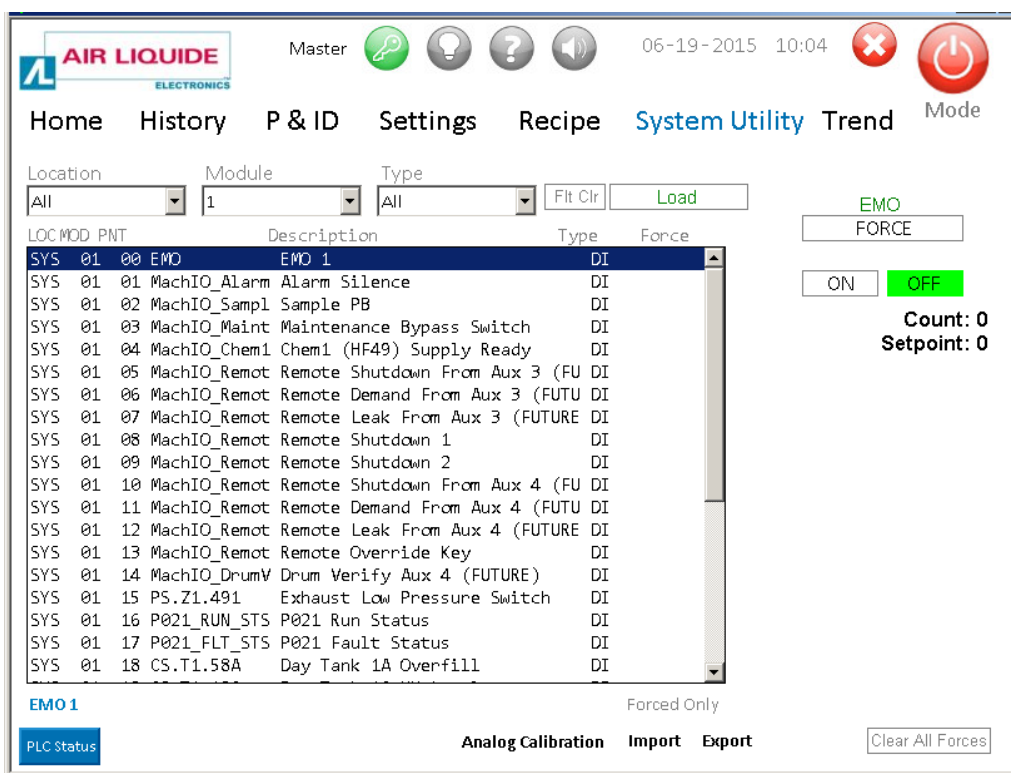


Table 4-18. System Utility Screen – Utility Pull-down Selections

Callout	Description
1	Displays a listing of utility events and the components that were affected. - SYS = System PLC rack. - MOD = the PLC module whose I/O point is indicated in the list. - PNT = the I/O point/tag at the specified PLC module. - Description = a brief function description of the I/O point on the specified PLC module. - Type = whether the I/O is digital input, digital output, analog input, or analog output. - Force = indicates if the user is manually forcing the I/O point of the PLC module.
2	The Module pull-down menu allows the user to select a module for the system utility, or to filter/narrow the choices.

Callout	Description
3	The Type pull-down menu allows the user to select a type for the system utility or to filter/narrow the choices.
4	Displays the FORCE status of the IO point. Press this button to ENABLE forcing of this point. (login must be high enough and system must be in appropriate state: ex. Not in auto)
5	Once FORCE has been pressed, ON or OFF may be pressed to FORCE the state for the selected IO point.
6	Indicates the number of times the point has cycled On and Off.
7	Indicates the parameter setpoint (if any) for the specified component that is being forced.
8	Press to display the <i>System Utility</i> screen, which indicates the status of the PLC modules.
9	The pull-down menu allows the user to select a location filter. This filter allows the user to sort by sub-system

Figure 4-19. System Utility Screen – ANALOG SCALING ACCESS



1. To access the *ANALOG SCALING POPUP*, press the text “ANALOG” on the bottom of the *SYSTEM UTILITY SCREEN*.
 2. Select the desired analog signal from the drop down menu in the upper right of the popup.
 3. The min / max raw and engineering units will be displayed on the x and y axis.
- NOTE: Changes CANNOT be made when the blender is in AUTO mode.**

Figure 4-20. System Utility Screen – ANALOG SCALING POPUP

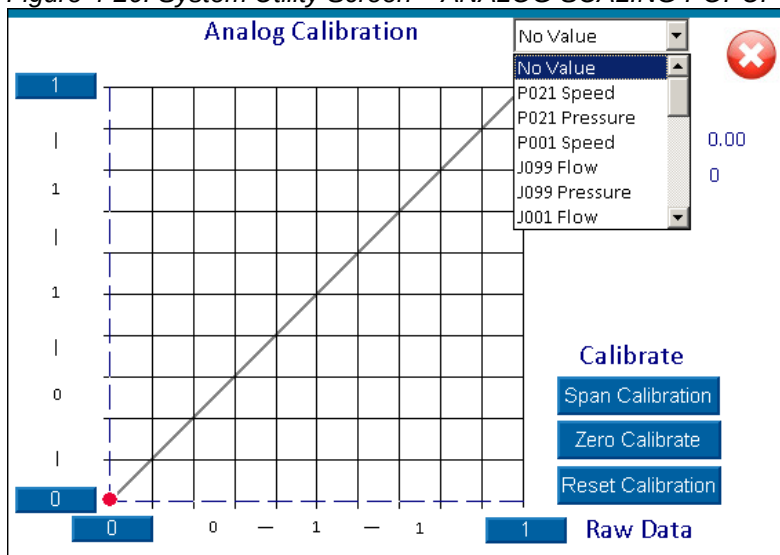


Figure 4-21. System Utility Screen – PLC Controller Forcing and Status

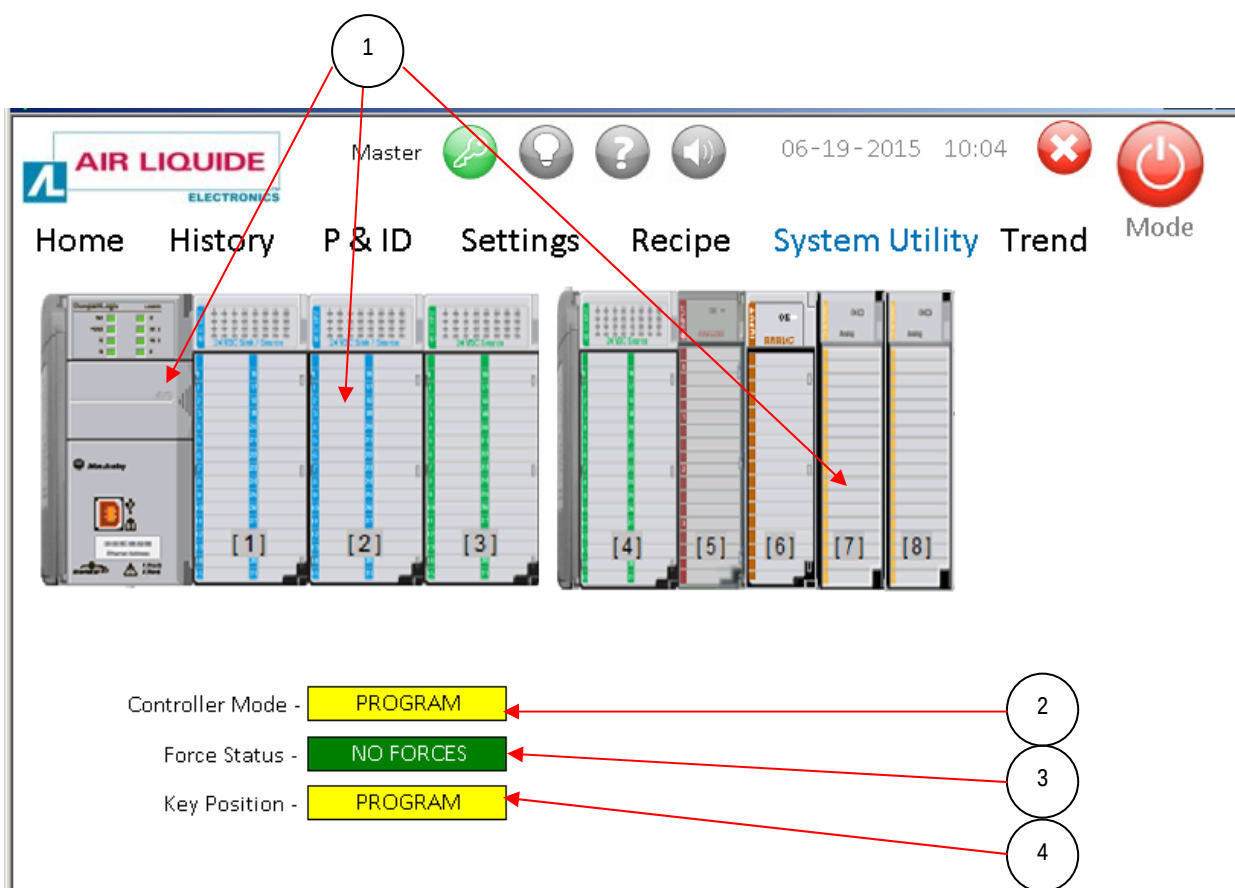


Table 4-19. System Utility Screen – PLC Controller Forcing and Status

Callout	Description
1	Provides a visual display of the PLC Controller and the I/O modules used in the System. Each module IO list may be accessed by pressing a module on this screen. Navigation is also available from the <i>System Utility Screen</i> .
2	Indicates the plc mode: (ex. RUN , PROGRAM).
3	Indicates whether a PLC I/O is being manually forced or if no inputs or outputs are being forced.
4	Indicates the controller key position status (Run, Remote, Program).

User Access Level Screen

Figure 4-22. User Access Level Screen

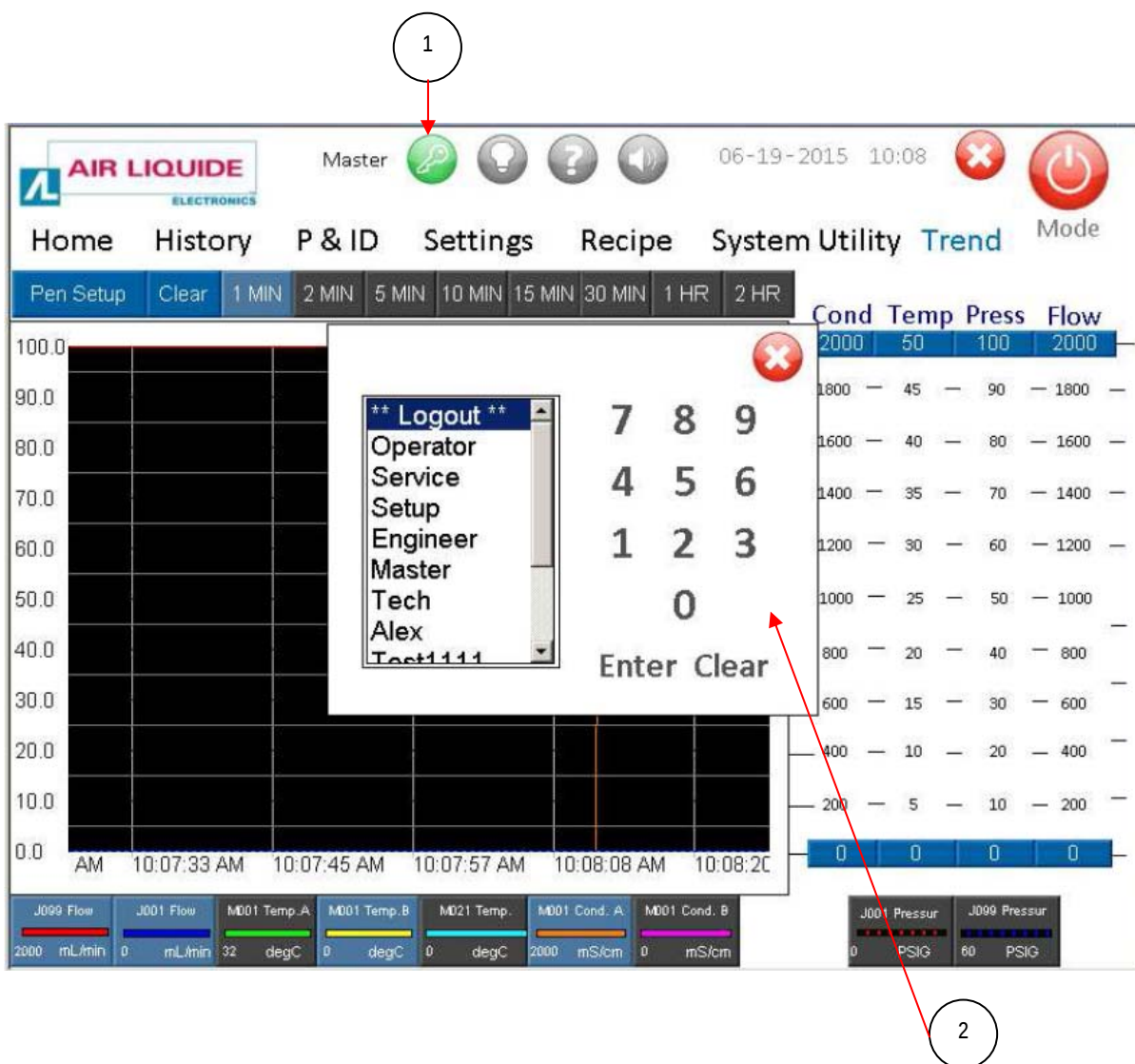


Table 4-20. User Access Level Screen

Callout	Description
1	Press the green key button to access the <i>User Access Level</i> pop-up screen. If the logged in user has the rights of an Engineer or Master they may click on the login name (here the name is "Master" to the left of the key button). This will bring up a pop-up screen to allow the user to add new users and change passwords.
2	Displays the <i>User Access Level</i> pop-up screen for selecting an Access Level.

About Screen

Figure 4-23. About Screen

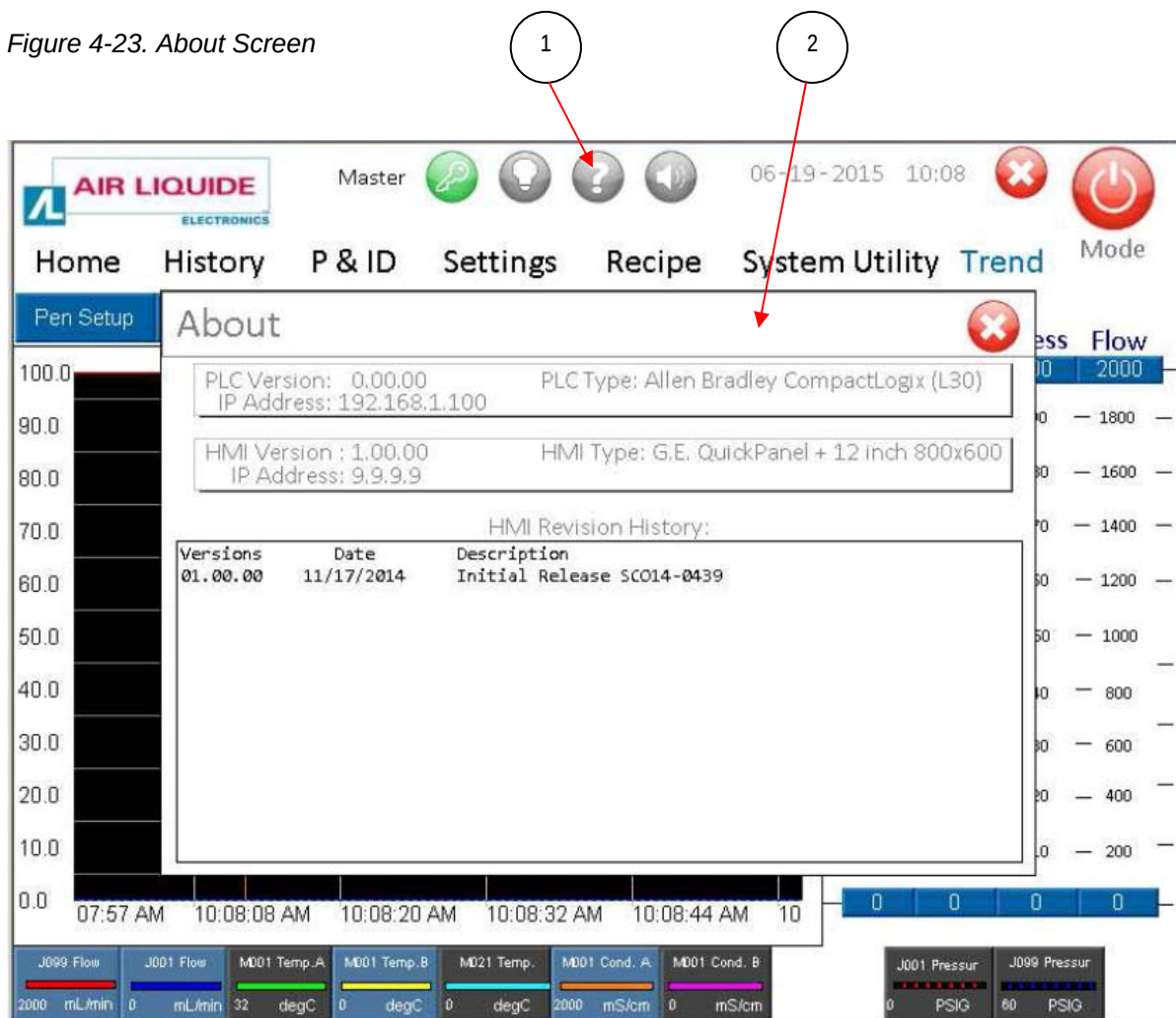


Table 4-21. About Screen

Callout	Description
1	Press this key to access the <i>About</i> popup screen.
2	Displays information including revision notes.

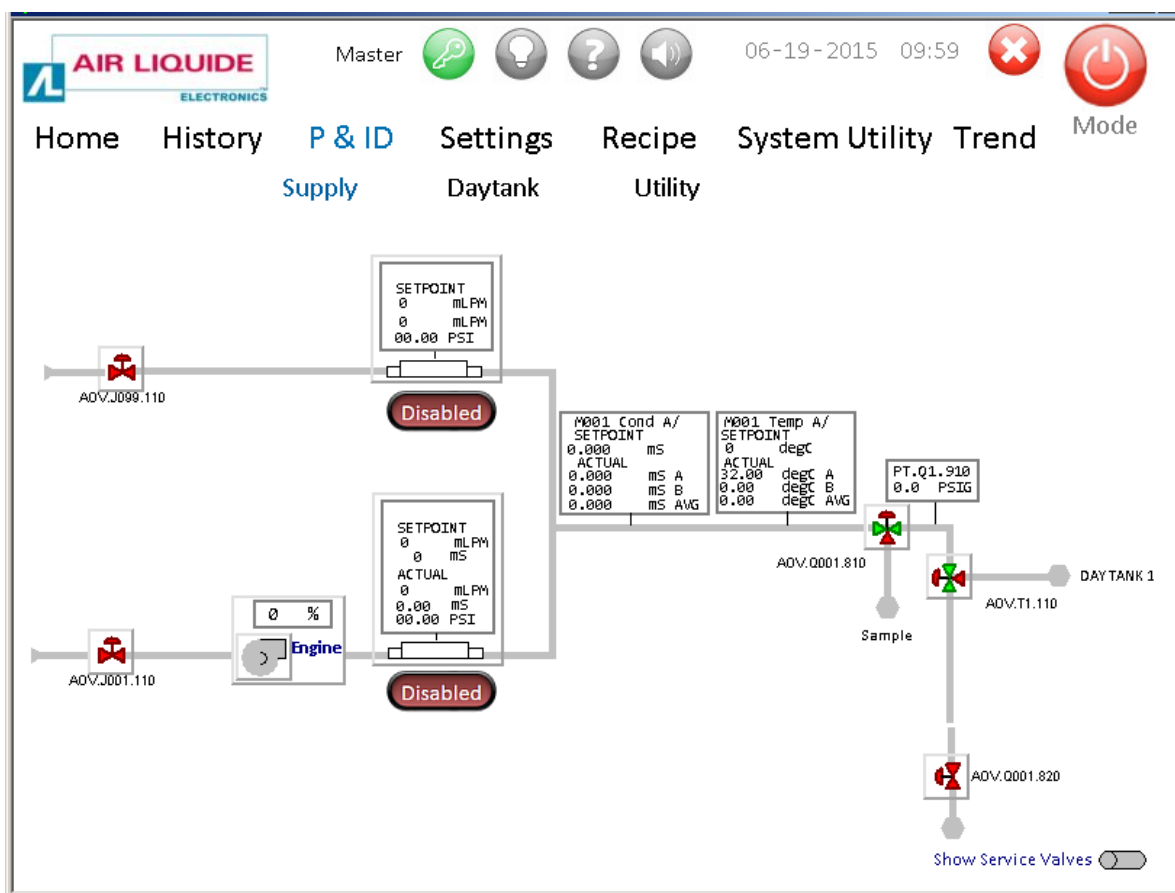
P&ID Screens

The color of the plumbing in the P&ID screen relates to the following liquids or function:

- Bright green = chemical
- Blue = water
- Black = drain
- Gray = idle [no flow occurring]

Note: In the below screenshot, the service valves are not shown. The visibility of these valves may be toggled on / off by sliding / pressing the button in the lower right corner.

Figure 4-24. P&ID Screen: SUPPLY BLEND



If there is an active warning / alarm that is associated with an analog signal or the component, then the box perimeter of that object / signal will also be a red color.

Figure 4-25. P&ID Screen – SUPPLY BLEND (with service valves shown)

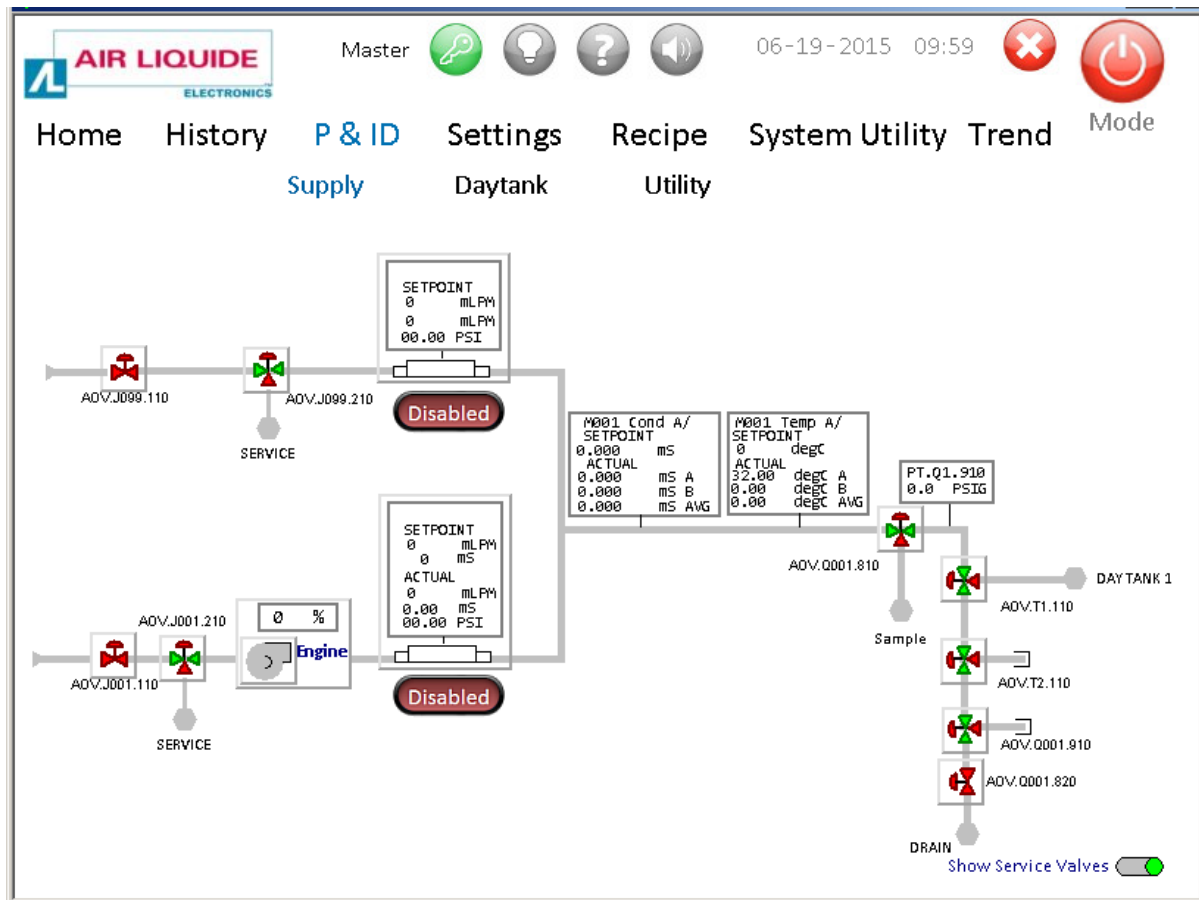


Figure 4-26. P&ID Screen – DAYTANK

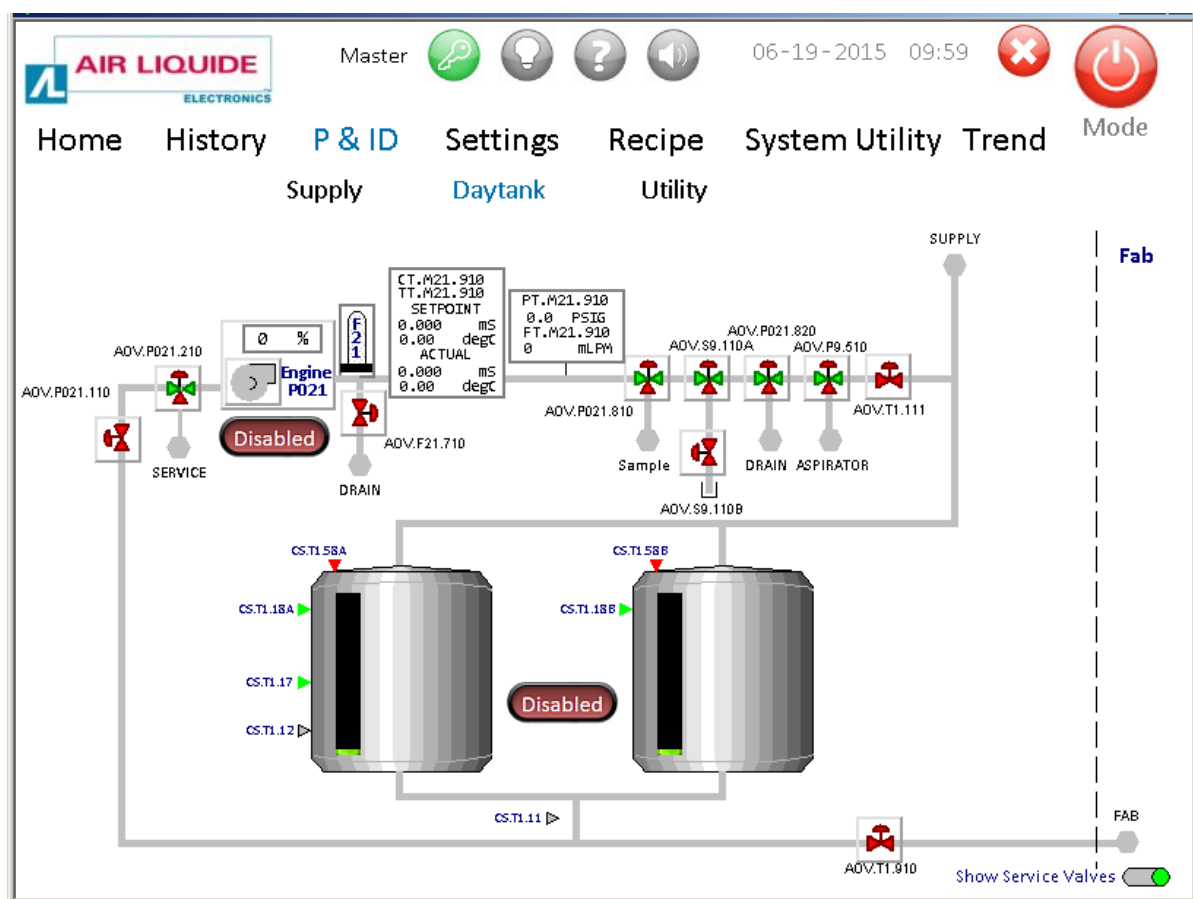


Figure 4-27. P&ID Screen – Utility

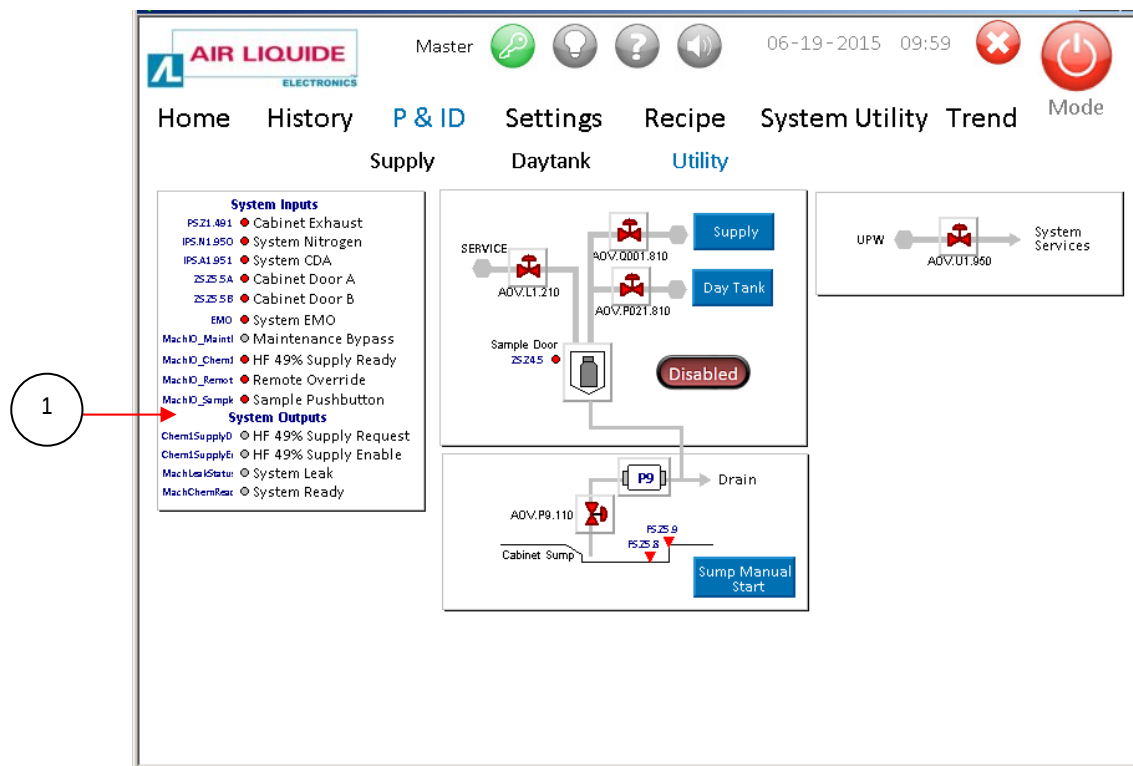


Table 4-22. P&ID Screen – Utilities

Callout	Description
1	The System Inputs/System Outputs section of the screen indicates which inputs and outputs are on (green), off (red), or offline (gray).
Sample Box icon	Press to View sample box options and to enable a sample or change sample location.
Sump Manual Start	Press to Enable the Sump Pump / Aspirator to operate for the configurable time period.
Valve / Pump Icons	Press to view specific component popup.

Figure 4-28. P&ID Screen – Filter Status

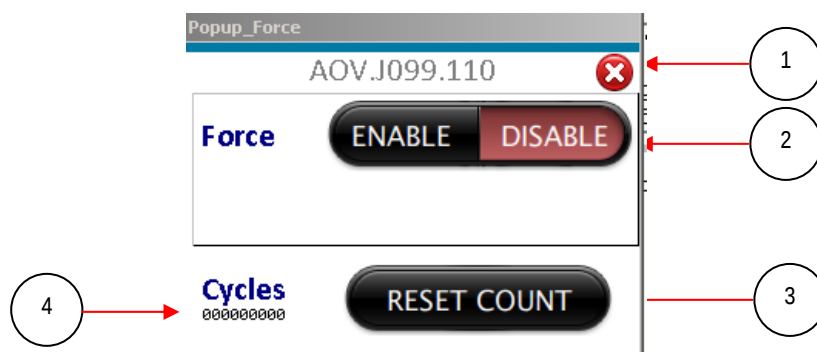
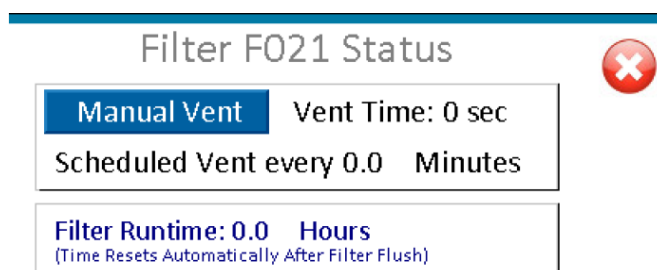


Table 4-23. P&ID Screen – Valve Forcing Pop-Up Screen

Callout	Description
1	Press the red X touch key to exit from the I/O forcing pop-up screen.
2	- Press the ENABLE side of the touch key to Enable forcing of an I/O Point. (ex. Valve). - Press the DISABLE side of the touch key to Disable forcing.
3	Press the RESET COUNT touch key to reset valve counts of the specified I/O valve.
4	Indicates the current life-cycle valve count of the I/O valve.

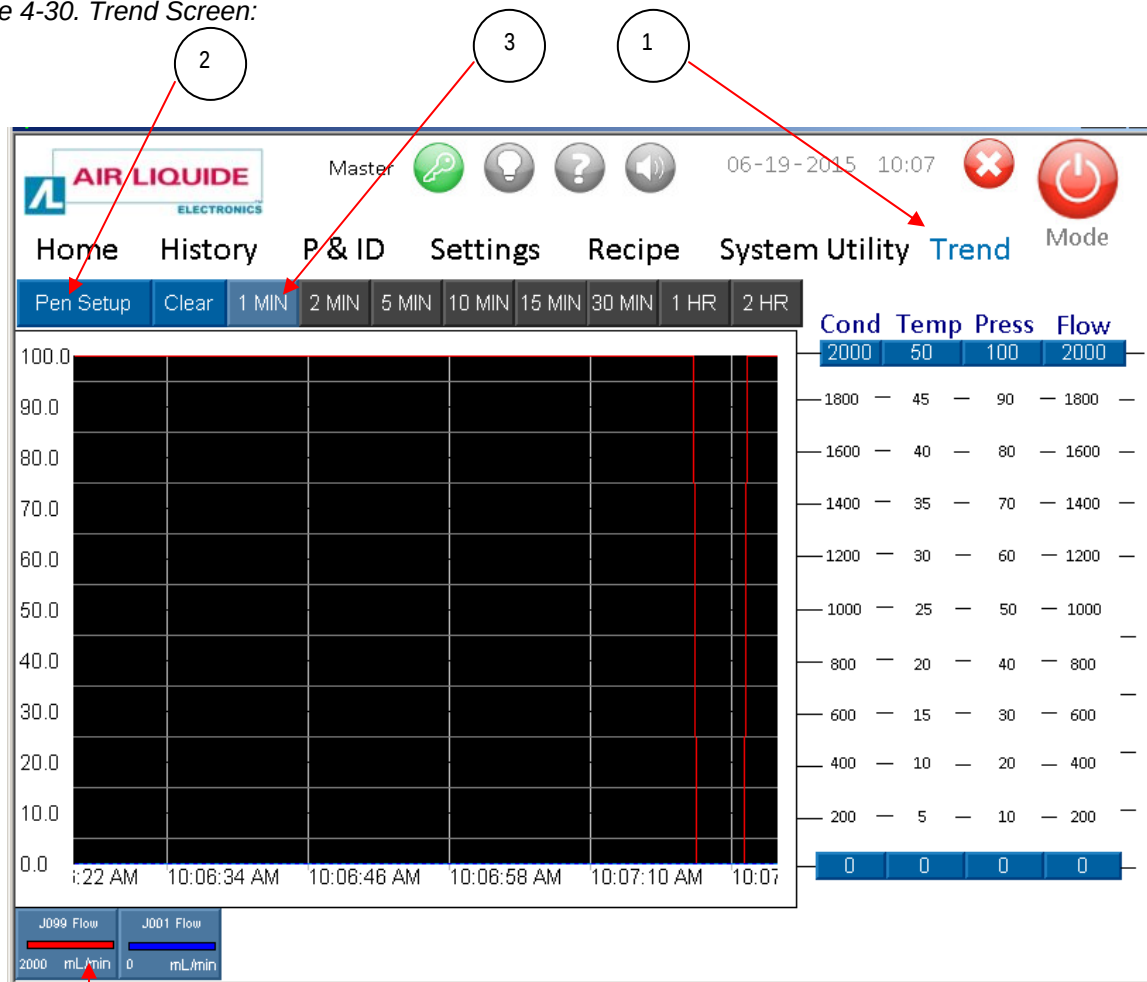
Figure 4-29. P&ID Screen – Filter Status



Callout	Description
MANUAL VENT	Press to open the filter vent valve immediately for a duration (according to the setting)

Trend Screens

Figure 4-30. Trend Screen:

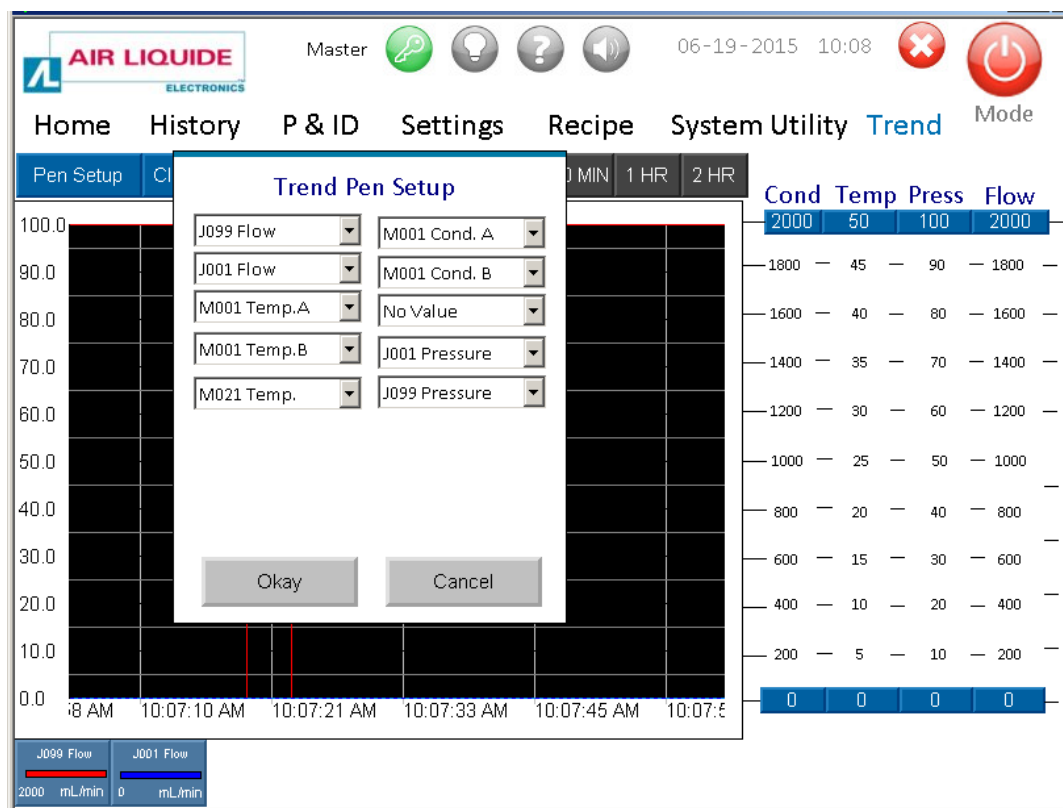


Note: The y axis scaling (engineering units) are displayed on the right side of the graph and are configurable. The left side of the graph remains 0 – 100%

Table 4-24. Trend Screen

Callout	Description
1	Press the Trend Menu Button in the upper right to access the Trend Screens.
2	Press the Pen Setup Button in the upper left in order to access the popup: and select the analog signals for trending pens.
3	Time Scales for the X axis. The graph automatically progresses, however the x axis MUST have a Time Period selected. If no selection is made, then there will be no trend pens or graphs displayed.
4	After the Pens have been selected in the Pen Setup Popup, they may be toggled temporarily on or off by pushing this button on the bottom of the graph. This enables or disables the graph plotting for that signal only.

Figure 4-31. Trend Screen – Trend Pen Setup Popup



Press the Pen Setup button in the upper left above the graph in order to access the Pen Setup popup. Each drop down menu in this popup contains the analog signals (input and output) for the blender.

System Processes

Alarms: Silencing

Purpose

To silence an active alarm.

Precautions



WARNING



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY

Personnel performing this procedure will be exposed to chemical or that has no secondary containment. Wear proper safety gear when performing this procedure.

Prerequisites

None.

Procedure

1. After an alarm has sounded and alerted the operator, press the **ALARM SILENCE HOLD TO RESET** button on the front electrical panel of the system. (This acknowledges the alarm and momentarily silences the buzzer).
2. Determine the cause of the alarm and what is required to correct the problem. Refer to [Chapter 7, Troubleshooting](#), for a complete alarm list.
3. If service is required inside the wetted area, shutdown the system and follow a suitable lockout/tagout procedure.
4. Isolate incoming chemical supplies. Follow Fab procedures regarding personal protective devices and hazard reduction.
5. After the problem is corrected, check the HMI panel for additional alarms.

Blend Creation: Recipe Setup

Purpose

To create a blend, using a sequence of recipe setup steps.

Precautions



WARNING



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY

Personnel performing this procedure will be exposed to chemical or that has no secondary containment. Wear proper safety gear when performing this procedure.

Prerequisites

Knowledge of the recipe database and how and where to access pre-existing recipes.

Discussion

The steps described in this procedure are generic. Each recipe has variables, parameters, steps, and conditions specific to that recipe and process.

Procedure

Loading a Recipe

1. In order for the system to blend a recipe, the recipe must first be loaded into the system from the recipe database. The operator must have a security access above the User Level "0" in order to access the **LOAD** option.
2. At the *Home* screen, press the **Recipe** touch key to access the main *Recipe* screen.
3. At the **Category** column, select **Blend**.
4. At the *Available Recipe* column, select the desired recipe from the list. Press the **Load** button to load the selected recipe into the system for blending.
5. A pop-up screen will appear to acknowledge the recipe is being loaded into the system. The user cannot stop a Load operation after it has started and must wait for it to complete before moving on.

Creating a New Recipe

1. In order to save a new recipe the operator must have a security access above the User Level “0”.
2. At the *Home* screen, press the **Recipe** touch key to access the main *Recipe* screen.
3. At the **Category** column, select **Blend**.
4. In order to create a new recipe an existing recipe must be selected from the *Available Recipe* column and copied to a new file by pressing the **Copy/New** pushbutton. After the file has been copied, it can then be edited, renamed and saved as a new recipe.

Editing an Existing Recipe

1. The *Recipe Manager* screen is designed to build a recipe based on sequence steps and blend criteria. The Total Flow and constituent flows must be within the hardware limitations. (ex. installed range of flow controllers).
2. In order to save a new recipe the operator must have a security access above the User Level “0”. Any user level can view existing recipes but user level “0” cannot save any changes.
3. At the *Home* screen, press the **Recipe** touch key to access the main *Recipe* screen.
4. At the **Category** column, select **Blend**.
5. At the *Available Recipe* column, select the desired recipe from the list. Press the **Edit** button to open the *Recipe Manager* screen.

6. At this point the user has the ability to edit the recipe. The recipes are arranged as a collection of information rows and action rows.
- **Add.** The Add Action is used to add a RECIPE Sequence or Action to the blend sequence. On the chemical J001 line, the user must enter the desired FINAL chemical Weight Percentage Assay. The theoretical flowrate of J001 will then be automatically calculated by the program based on this entry.
 - **NOTE:** The flow rates listed for J99 and J001 are starting flow rates and are active setpoints for the flow controllers at the initial steps / startup of a blend. The metrology closed loop will control the final output of the blender based on the metrology targets entered into the recipe (at the bottom of the screen). After the initial blend steps, the blender will control and vary the flows based on metrology target settings. At that time, the flow controller J001 is no longer operating in a fixed setpoint mode and the flowrate listed in the recipe for J001 line is a nominal value at that point. Large deviations from the listed flow will result in alarming, so it is important to align the metrology targets on the recipe screen with the expected flow rates listed in the rows of the recipe screen.
 - **Blend Pause / Wait.** The Wait Action has the effect of pausing the blend, for a user specified time, and can be added between the pre-rinse and blend steps.
 - **Pre-Rinse J99.** Regardless of the order in which this sequence is added, the result is that PRIOR to the blend, the J99 flow controller and downstream circuit will be rinsed with DIW to drain for the configurable time listed.
 - **POST RINSE J99.** Regardless of the order in which this sequence is added, the result is that AFTER the blend, the J99 flow controller and downstream circuit will be rinsed with DIW to drain for the configurable time listed. J001 will not be rinsed during this sequence.
 - **POST RINSE ALL.** Regardless of the order in which this sequence is added, the result is that AFTER the blend, the J99 flow controller, J001 flow controller, and downstream circuit will be rinsed with DIW to drain for the configurable time listed.

NOTE: J001 and J99 blend at the same steps. They are listed as different rows within the recipe for clarity, but occur at the same time.

Uploading the Current Recipe

1. If the operator is unsure what recipe is loaded into the system, the operator can upload the recipe from the system onto the screen. This can be done with any User Level, but the User Level must be above “0” in order to save the uploaded recipe.
2. At the *Home* screen, press the **Recipe** touch key to access the main *Recipe* screen.
3. At the **Category** column, select **Blend**.
4. At the *Available Recipe* column, select the desired recipe from the list. Press the **Edit** button to open the *Recipe Manager* screen.
5. From the *Recipe Manager* screen press the **Recipe Load** button and the recipe from the system will be loaded on the screen. After it is loaded the operator can perform edit, save, or cancel functions.

Blending: Recipe Verification, Changing a Recipe, Recipe Settings, and Starting a Blend

Make sure that the correct recipe is loaded into the blender by accessing the *Recipe* menu. Verify which recipe is loaded into the blender by examining the bottom of the *Blend Recipe* screen.

1. To change a recipe, choose the desired recipe from the list of **Available Recipes**.
2. Choose the **Load Recipe** box(on the left side of the HMI panel).
3. Make sure that the **Blend Command Mode** (parameter 17 in **Settings**) is set to the desired setting.

To start a blend to Drain

(During this procedure, no valves will open to the tank, and no flows will go to the tank. All liquid, either in spec or out of spec, will flow to drain.)

REMINDER: AFTER COMPLETION OF THIS PROCEDURE, PUT ALL SETTINGS BACK FOR NORMAL OPERATION.

1. Make sure that all blending components are in Auto and ready to blend.
2. Choose the Tank icon. Choose BLEND DESTINATION: BLEND TO DRAIN.
3. Choose the tank icon again. Choose the Manual Start Blend Command.
4. The blender will flow to drain until the user stops the sequence or until the blender timeout is reached. (setting: 189)
5. After the sequence is completed, change all of the settings and modes back into normal operation.

Chemical or Leak/Spill Recovery Procedure

Purpose

The purpose of this procedure is to ensure that the system is properly cleaned and prepared for reactivation after a chemical or leak or spill.

Precautions



WARNING



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Wear proper safety gear when performing this procedure.

Procedure for Chemical Leak

1. Press the **EMO** button on the front panel.
2. Call for support. Never attempt to perform a chemical or cleanup on your own.

IMPORTANT

Only properly trained maintenance personnel should perform this procedure.

3. Wear full PPE for this procedure, including a respirator and self-contained breathing apparatus (SCBA), as required.
4. Turn the **MAIN POWER** switch to the OFF position and install a LOTO device.
5. Before opening the door where the chemical leak was detected, insure no chemical “spray” is occurring.
6. Open the chemical cabinet door slowly to a 2-inch to 3-inch position and hold the door in this position for no less than 20 seconds. This allows all chemical fumes to be drawn out of the doorway area to the exhaust duct. The door can now be slowly opened the rest of way to perform any maintenance.
7. Inspect the system and identify the source of the chemical or leak/spill.
8. Close all manual isolation valves for chemical inlet and outlets, N₂ and CDA inlets. Install LOTO devices.

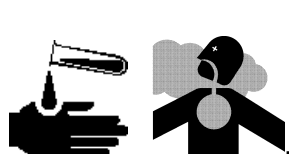
9. Take appropriate action to replace defective component or otherwise repair the cause of the chemical or leak. See the applicable procedure in [Chapter 8, Component Procedure and Adjustments](#).
10. Make sure that the cabinet drain is connected to facility drain and the facility drain valve is open.
11. **Any liquid that flows to the sump cup will not be evacuated until the blender sump pump / aspirator is on.** Use the DIW spray gun to thoroughly spray down cabinet.
12. Allow DIW to drain and dry any remaining DIW.
13. Remove the LOTO device from and turn on all manual isolation valves and close cabinet door.
14. Remove the LOTO device from the **MAIN POWER** and turn the switch to the ON position.
15. Turn and pull the **EMO** button to the ON (out) position.
16. Enable the system.
17. Place the system in the desired mode of operation.
18. Check the system for leaks.

Depressurizing the System After Pressing the EMO Switch

Purpose

The purpose of this procedure is to remove any trapped pressure from the system and to place the system in a safe, depressurized state after pressing the EMO button for any reason. This procedure is typically done in an emergency situation, when the operator wants to service the system, or for short-term/long-term shutdown or decommissioning.

Precautions



WARNING



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY

Personnel performing this procedure will be exposed to chemical or that has no secondary containment. Wear proper safety gear when performing this procedure.

Prerequisites

This procedure assumes electrical power is present or has been properly restored.

Procedure

Press the **EMO** button. This removes power to the system's outputs after a configurable time, to allow depressurization prior to establishing a safe (isolated) state.

Drain Processes: Drain the Tank

Purpose

The purpose of this operation is to manually drain tank DT1 of its liquid and send the liquid to the drain.

Precautions



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY

Personnel performing this procedure will be exposed to chemical or that has no secondary containment. Wear proper safety gear when performing this procedure.

Prerequisites

None.

Procedure

The tank can be drained by running a tank drain service recipe. Monitor the service progress at all times.

Sample – Take a Chemical Sample

Purpose

This procedure explains sampling.

Special Equipment Required

- An uncontaminated sample bottle with sealing cap that meets safe-handling requirements for chemical or sample analysis.
- Proper PPE (Personal Protection Equipment).

Prerequisites

Sample Compartment

Ensure the sample compartment is free of residual chemical or by thoroughly rinsing the compartment with the spray gun provided on the system. Take precautions to ensure nothing splashes out of the containment area.

Precautions



WARNING



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY.

Personnel performing this procedure will be exposed to chemical or that has no secondary containment. Wear appropriate personal protective equipment when performing this procedure.

Check the SDS sheet for chemical that is being sampled. Determine if a supplied-air respirator is required.

For all sampling tasks, avoid spilling or splashing chemical. If a spill occurs, clean up promptly in accordance with site regulations. Rinse gloves promptly. Ensure sampling compartment door is shut immediately after sampling operations.



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY.

For systems that process hazardous chemical, forced air-breathing apparatus should be worn by service personnel.

The forced air-breathing apparatus can consist of, but is not limited to, the following types of devices:

The customer's in-house forced air-line

An air-breathing device on a portable cart

A self-contained breathing apparatus (SCBA)

For any maintenance task that requires a respirator (cartridge-type or forced-air), maintenance personnel must be fit-tested for their own respirator. They must have a valid physical certification proving that the respirator fitting was accomplished.

Sampling chemical or is considered a two-person task, with two trained maintenance personnel performing the steps described in this procedure.

Procedure

1. At this time, make sure that a sample bottle is not placed in the Sample Compartment.
2. Touch the P&ID at the *Header* menu. Then, touch *Utility* at the sub-menu.
3. Make sure that the *Sample* function is in *Auto*.
4. Choose the sampling destination by touching the *Sampling Box* button and choosing the *destination*. (Only one selection or the other can be made).
5. Press and hold the **Sample Enable** button to start the bottle sampling function.

NOTE

Chemical will flow through the sample plumbing to the sample box drain, wetting the plumbing and preparing it for the actual bottle sample.

6. Release the *Sample Enable* button to stop the bottle sampling function.
7. Change the system mode from *Sample* to *Service* and initiate a sample box *RINSE* function to wash down the box.
8. After the rinse is completed, install an empty sample bottle into the compartment and align the tubing into the container.

9. Change the sample mode to Sampling.
10. Press the green **Start Service** button.
11. Push and hold the **Sample Enable** button for sampling.
12. Release the **Sample Enable** button to stop the bottle sampling function.
13. While wearing full PPE, carefully remove the sample bottle and cap.
14. Initiate a sample box rinse service to flush the sample box compartment to drain.

Sump Pump / Aspirator Operation

Purpose

This procedure explains how to manually activate sump pump / aspirator P9 and aspirate the cabinet sump of leaked liquid.

Precautions



WARNING



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY

Personnel performing this procedure will be exposed to chemical or that has no secondary containment. Wear proper safety gear when performing this procedure.

Prerequisites

None.

Procedure

1. At the *P&ID* screen, press the **System Utility** touch key to access the main *Utility Screen*.
2. In the **Sump Area** select the P9 icon.
3. When the *Pump Forcing* pop-up opens, select **ENABLE Forcing**.
4. After Forcing is **ENABLED** the pop-up might be opened again and the pump can be **Forced On**.
5. When the function is complete, press **DISABLE Forcing** to put the pump back into normal operation.

System Maintenance: Service Setup

Purpose

The blender is designed to allow each sub-system to operate independently of each other allowing maintenance to be performed separately on every sub-system using a sequence of service steps.

Precautions



WARNING



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY

Personnel performing this procedure will be exposed to chemical or that has no secondary containment. Wear proper safety gear when performing this procedure.

Prerequisites

In order to perform a system service the sub-systems involved in the service must be in Service mode and a blend cannot be actively running on the system being serviced.

IMPORTANT

ANY CHEMICAL THAT IS IN THE BLEND TANK WILL BE IMMEDIATELY PUSHED TO DRAIN WHEN THE TANK IS TAKEN OUT OF THE SERVICE OR OFFLINE MODES.

Procedure

Running an Existing Service Recipe from the Home or P&ID Screen

1. In order for the system to run a service it must first be loaded into the system. Only one service can be loaded at a time. The Operator must have a security access above the User Level "0" in order to access the **LOAD** option.
2. A service can be loaded from either the *Home* screen or through the *P&ID*. From the *Home* screen or the *P&ID*, press the mode icon of the sub-system that you want to service.
3. A pop-up menu will display the current sub-system mode. Click on **Change to Service Mode** if the sub-system is not already in service mode. Click the green check mark icon to accept the change.

4. The sub-system will then depressurize and change to the service mode. When the sub-system changes to the service mode click the mode icon again to access the available service functions. Scroll down the list until the required service is found. Select the service and click the green check mark icon to accept the change.
5. The loaded service recipe will appear in the **Service Description** box on the *Home* screen and the *P&ID*. From the *P&ID* screen click the **Service Start** pushbutton to start the service.
6. When the service is running **Service Description** box will show the service description (if the service is a custom service the description might not appear) and the current service step and the time remaining until the next step. Stop the service at any time by pressing the **Service Stop** pushbutton.
7. When the service is complete, press the mode icon of the sub-system that the operator wants to put back into Auto.
8. A pop-up menu will display the current sub-system mode. Click on **Change to Auto Mode**. Click the green check mark icon to accept the change.

Creating a new Custom Service Recipe

1. Custom services may be created from the *Service Recipe* screen.
2. At the *Home* screen, press the Recipe touch key to access the main *Recipe* screen.
3. At the **Category** column, select **Service**.
4. At the **Object** column, select the required sub-system.
5. At the **Available Recipe** column, select **Custom**.
6. In order to create a new service recipe an existing custom recipe must be selected from the *Available Recipe* column and copied to a new file by pressing the **Copy/New** push button. After the file has been copied, it can then be edited, renamed, and saved as a new recipe.

Editing Custom Service Recipe

1. At the *Home* screen, press the **Recipe** touch key to access the main *Recipe* screen.
2. At the **Category** column, select **Service**.
3. At the **Object** column, select the required sub-system. Only custom recipes can be edited. Any recipe from the **System** filter have been preloaded from the factory and will only allow the user to change to service step times.
4. At the **Available Recipe** column, select the **Custom** recipe to be edited. Press the **Edit** pushbutton to open the **Service Recipe Manager**.
5. The **Service Recipe Manager** is organized by Service Steps. Each step is in order of operation with a time for every step. The minimum time a service step can run is 0.1 minutes (6 seconds) in order for the system to process each step correctly. The maximum amount of time for each step is 120 minutes.

The operator can select the needed system function from *Action* pop-up screen. The *Action* pop-up screen lists all of the available actions that the selected sub-system is able to complete. By selecting the required Action, and pressing **Okay**, the action will appear in the **Step** list.

Running a Custom Service Recipe

1. From the *Home* screen or the P&ID, press the mode icon of the sub-system that you want to service.
2. A pop-up menu will display the current sub-system mode. Click on the **Change to Service Mode** if the sub-system is not already in service mode. Click the green check mark icon to accept the change.
.(The sub-system will then depressurize and change to the service mode.)
3. Press the **Recipe** touch key to access the main *Recipe* screen.
4. At the **Category** column, select **Service**.
5. At the **Object** column, select the required sub-system. Only custom recipes can be edited. Any recipe from the **System** filter have been preloaded from the factory and will only allow the user to change service step timers.
6. At the **Available Recipe** column, select the **Custom** recipe to be loaded. Press the **Load** pushbutton to load the custom service into the system.

7. The display at the bottom of the screen will show the recipe is loaded into the system.
8. The loaded service recipe will appear in the **Service Description** box on the *Home* screen and the *P&ID*. If the service is a custom service the description may not appear. From the *P&ID* screen click the **Service Start** pushbutton to start the service.
9. When the service is running **Service Description** box will show the service description (If the service is a custom service the description may not appear), the current service step and the time remaining until the next step. Stop the service at any time by pressing the **Service Stop** pushbutton.
10. When the service is complete, press the mode icon of the sub-system that you want to put back into auto.
11. A pop-up menu will display the current sub-system mode. Click on **Change to Auto Mode**. Click the green check mark icon to accept the change.

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5 Startup and Shutdown



WARNING



BEFORE PROCEEDING, READ AND UNDERSTAND THE SAFETY INFORMATION CONTAINED WITHIN THE [SAFETY](#) CHAPTER OF THIS MANUAL.

Startup Procedures	5-2
Shutdown Procedures	5-4
Short-Term Shutdown Procedure	5-4
Long-Term Shutdown Procedure	5-5
Inline Blender, with Integrated Daytanks, Disposal Procedure	5-6

This chapter contains startup procedures that are intended for initial startup after installation, and also for when a Inline Blender, with Integrated Daytanks, is brought back into service after a long-term shutdown. This chapter should be considered an overview and not an exhaustive list of the procedures. Some of the procedures require a high level of training and experience to properly execute.

This chapter also contains shutdown procedures that are intended for when a Inline Blender, with Integrated Daytanks, is going to be shut down for a few days (short-term shutdown) or when a Inline Blender, with Integrated Daytanks, is going to be shut down for more than a few days (long-term shutdown). Site-specific circumstances and requirements can determine whether a shutdown is short-term or long-term.

Startup Procedures

**WARNING**

CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Wear proper safety gear when performing this procedure.

**WARNING**

USE APPROPRIATE HEARING PROTECTION WHEN ACCESSING THE ELECTRICAL AND PNEUMATIC AREAS IF THE INLINE BLENDER, WITH INTEGRATED DAYTANKS IS OPERATING.

IMPORTANT

The startup procedures listed in this chapter will not apply to all installations. Site-specific circumstances and requirements may require deviation from these procedures.

The startup procedures include many procedures that are in other chapters in this manual. Chapter references are given for procedures that are located elsewhere.

[Table 5-1](#) lists the steps and procedures to be performed and includes a column to check when completed. The user should make a copy of this table and use it as a checklist.

NOTE

To silence any alarms during the startup procedures, momentarily press the ALARM SILENCER button. The alarm can be reset on the operator touch screen.

Table 5-1. Inline Blender, with Integrated Daytanks Startup Procedures Checklist

Step	Procedure	Chapter	✓ Check
1	Ensure that all of the installation procedures in Chapter 2 have been performed.	Chapter 2	
2	Turn all pressure regulators fully counterclockwise.	Chapter 3	
3	Verify that all of the facility connections are correct (and open) per the Facility & Installation drawing.	Prints section	
4	Power up the Inline Blender, with Integrated Daytanks.	Chapter 8	
5	Verify that the PLC and operator interface panel software are installed.	—	
6	Set pressure settings.	Appendix D	
7	Close all manual valves.	Appendix D	
8	Adjust pressure switches.	Appendix D	
9	Adjust pressure indicators.	Appendix D	

10	Adjust capacitive sensors and float switches.	Chapter 8
11	Test sensors and switches.	Chapter 8
12	Test all pneumatic and solenoid valves.	Chapter 8
13	Calibrate conductivity sensors in air.	Chapter 8
14	Flush and fill the daytanks with UPW.	Chapter 8
15	Place the Inline Blender, with Integrated Daytanks, in Manual mode. Recirculate dispense loops.	Chapter 4
16	Perform Blender Rinse (Maintenance mode).	Chapter 4
17	Force flow rate for UPW Flow Control Valve and set pressure regulator. Remove force for UPW flow control valve.	Chapter 4
18	Force flow for Chem Flow control valve. Adjust Chem1 pump air pressure. Remove force on flow control valves.	Chapter 4 , Chapter 8 and Appendix D
19	Install appropriate filters in blend and tanks piping.	Chapter 8
20	Measure backpressure during blending to tank and drain. Set backpressure regulator.	Chapter 8 and Appendix D
21	Calibrate conductivity and temperature probes.	Chapter 8
22	Place unit in Auto mode. Blend to drain.	Chapter 4
23	Flush day tanks.	Chapter 4
24	Fill the daytanks with blended chemical.	Chapter 4
25	Perform the Qualification procedure.	—

Shutdown Procedures

IMPORTANT

The shutdown procedures listed in this chapter will not apply to all installations. Site-specific circumstances and requirements may require deviation from these procedures.

Perform shutdown procedure when the system needs to be shut down.

- If the system is to be shut down for only a few days, perform the [Short-Term Shutdown Procedure](#) below.
- If the system is to be shut down for longer than a few days, perform the [Long-Term Shutdown Procedure](#) on page 5-5.
- To dispose of the system, perform the [Inline Blender, with Integrated Daytanks, Disposal Procedure](#) on page 5-6.

Short-Term Shutdown Procedure

[Table 5-2](#) lists the steps and procedures to be performed and includes a column to check when completed. The user should make a copy of this table and use it as a checklist.

Table 5-2. Short-Term Shutdown Procedures Checklist

Step	Procedure	Chapter	✓ Check
1	Leave the Inline Blender, with Integrated Daytanks, in Auto mode.	Chapter 4	
2	On a monthly basis, blend with Chem1 + UPW to drain.	—	
3	Perform the monthly PM procedures.	Chapter 8	

Long-Term Shutdown Procedure



WARNING



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Wear proper safety gear when performing this procedure.

Table 5-3 lists the steps and procedures to be performed and includes a column to check when completed. The user should make a copy of this table and use it as a checklist.

Table 5-3. Long-Term Shutdown Procedures Checklist

Step	Procedure	Chapter	✓ Check
1	Place the Inline Blender, with Integrated Daytanks, in Manual mode.	Chapter 4	
2	Flush the distribution tubing with UPW.	—	
3	a. Flush daytanks 1A and 1B. Connect a ½-inch UPW line to the chemical inlet lines and flush and fill the daytanks. b. Recirculate the UPW through each daytank's recirc loop to eliminate trapped chemical.	—	
4	a. Drain daytanks 1A and 1B of excess liquid. Remove the caps at the liquid outlet ports of MV.T1.991 and MV.T1.992. b. Connect an appropriate length and diameter of plumbing to the liquid outlet ports of MV.T1.991 and MV.T1.992. c. Place the free-end of each plumbing line into the Fab's waste-liquid drain. d. Open MV.T1.991 and MV.T1.992 and allow daytanks 1A and 1B to gravity-drain. e. After all liquid has drained from the daytanks, remove the plumbing lines from the outlet ports of MV.T1.991 and MV.T1.992. f. After all liquid has drained from the daytanks, remove the free end of the lengths of the plumbing from the Fab's waste-liquid drain.	—	
5	Remove the conductivity probes and blow down the plumbing lines of the Inline Blender, with Integrated Daytanks, with N ₂ .	—	
6	Rinse the cabinet with UPW.	—	
7	Drain the cabinet.	—	
8	Press the EMO button.	—	
9	Turn the MAIN POWER DISCONNECT SWITCH to OFF.	—	
10	Close the facility valves for N ₂ and UPW.	—	

Step	Procedure	Chapter	✓ Check
11	Close all manual valves.	Appendix D	
12	Disconnect the chemical supply connections and cap the openings.	—	

Inline Blender, with Integrated Daytanks, Disposal Procedure

To dispose of the Inline Blender, with Integrated Daytanks, follow the procedure below.

Table 5-4. Disposal Procedures Checklist

Step	Procedure	Chapter	✓ Check
1	Complete the Long-term Shutdown procedures.	Chapter 5	
2	Disconnect the Inline Blender, with Integrated Daytanks from all facility connections.	—	
3	Package the Inline Blender, with Integrated Daytanks, daytanks appropriately for shipment.	—	
4	Dispose of the Inline Blender, with Integrated Daytanks, and daytanks 1A and 1B per site requirements and local regulations.	—	

6 Preventive Maintenance



BEFORE PROCEEDING, READ AND UNDERSTAND THE SAFETY INFORMATION CONTAINED WITHIN THE [SAFETY](#) CHAPTER OF THIS MANUAL.

Preventive Maintenance Schedule	6-4
Daily PM Procedures/Log.....	6-5
Weekly PM Procedures/Log.....	6-6
Quarterly PM Procedures/Log	6-7
Semiannual PM Procedures/Log.....	6-8
Annual PM Procedures/Log	6-9
Two-Year PM Procedures/Log	6-10
Five-Year PM Procedures/Log	6-11



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY.

Personnel servicing this Inline Blender, with Integrated Daytanks, will be exposed to chemical that has no secondary containment. Wear proper safety gear when servicing the Inline Blender, with Integrated Daytanks.



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY.

Rotating the MAIN POWER DISCONNECT SWITCH to OFF may allow trapped pressure and chemical to remain in the plumbing. Depressurize the chemical lines before attempting any servicing or troubleshooting by pressing the EMO button.



OPENING A SAFETY INTERLOCK DOOR OR PANEL DURING SYSTEM OPERATION COULD EXPOSE PERSONNEL TO HAZARDOUS CHEMICALS.

Open a safety lock door or panel only after the system has been depressurized and after the chemical plumbing and valves have been flushed of chemical.



COMPONENTS THAT ARE REMOVED DURING MAINTENANCE MIGHT CONTAIN HAZARDOUS CHEMICALS OR CHEMICAL RESIDUE.

Before removing wetted components, or components that might have come in contact with chemical, flush the affected engine subsystem or the Inline Blender, with Integrated Daytanks.



WHEN PERFORMING ANY WORK IN THE ELECTRICAL AREA, ENSURE POWER IS ISOLATED PER LOCAL/SITE RULES AND OR PROCEDURES.

Care should be taken when handling of the electrical panel during installation, removal and troubleshooting:

- The touch screen panel during normal operation is seated into a recess in the cabinet and secured with screws.
- When the screws are removed or not installed, there is a potential for the panel to fall from the cabinet and cause potential injury to an operator.
- Use the two-person “buddy” system to handle the panel.



TO PREVENT BACK OR OTHER INJURIES, DO NOT ATTEMPT TO SINGLE-HANDEDLY LIFT A COMPONENT THAT WEIGHS MORE THAN 50 POUNDS.

A component that weighs more than 50 pounds requires two people to lift it. Read and understand the entire procedure before starting.



IMPORTANT

Improper handling of electronic components can result in electrostatic discharge, which can cause component failure.

To avoid damage from electrostatic discharge, follow these basic rules:

1. Handle all static-sensitive components at a designated static-safeguarded area or on a portable ESD mat.
2. Transport all static-sensitive components (for example, circuit boards) in static-shielded containers or packages.
3. Monitor and test static-protection products and materials to ensure that they are working properly.

Preventive Maintenance Schedule

IMPORTANT

The preventive maintenance schedules must be performed completely and on schedule to ensure proper system performance and to maintain warranty coverage.

Preventive maintenance (PM) is scheduled on a daily, weekly, quarterly, semiannually, annually, two-year and five-year basis.

NOTE

The recommended usage-scheduled preventive maintenance can vary, depending on the chemical used in the Inline Blender, with Integrated Daytanks

When planning your PM program, consider these suggestions to divide up the required downtime:

- Perform all inspections at one time that require the system to be down, and then bring the system back up to complete the remaining inspections.
- For quarterly, semiannual, and annual inspections, distribute the procedures over several months. For example, if the quarterly PM for your system involves checking 30 system components, split them into three identified groups of 10, and schedule each group for a different month.

It is strongly recommended that when the Inline Blender, with Integrated Daytanks, is checked for leaks that the distribution piping be checked also. System integrity is not limited to the Inline Blender, with Integrated Daytanks, only.

Scheduled PM procedures are presented in a table format with areas to record performance date and initials of person performing the procedures. There is also room to record any corrective actions and adjustments taken.

A copy of these procedures should be made each time they are to be performed, and that you record any actions taken. Completed logs should be retained for maintenance history.

IMPORTANT

It is recommended that conductivity and temperature sensors NOT be calibrated or replaced on a periodic basis.

Sensors should only be replaced based on guidance provided by the Troubleshooting Guide in [Chapter 7, Troubleshooting](#).

Daily PM Procedures/Log



WARNING



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Wear proper safety gear when performing this procedure.

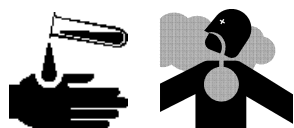
Perform the procedures listed in the table below as follows:

1. Make a copy of this page.
2. Perform the procedures listed.
3. Note the following information on your copy: your initials, date, and any corrective actions taken or adjustments made.
4. Retain the completed log for maintenance history.

Procedure	Component Tag	Mode	Chapter	Date	Initials
Check that all pressure gauge readings are within specified ranges and read zero when no pressure is present.	Various (see Chapter 3, Component Description and Locations)	Auto or Manual	Appendix D		
Verify that the Levitronix BPS-600 centrifugal pump operates properly.	P21	Auto	Appendix D		
Check for leaks and damaged components.	All components and cabinet	Auto	—		
Check the alarms log to determine the types of faults, if any, the Inline Blender, with Integrated Daytanks is experiencing.	HMI panel, plus the SCADA	Auto	Chapter 4		

Corrective Actions and Adjustments:

Weekly PM Procedures/Log



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY

Personnel performing this procedure will be exposed to slurry that has no secondary containment. Wear proper safety gear when performing this procedure.

Perform the procedures listed in the table below as follows:

1. Make a copy of this page.
2. Perform the procedures listed.
3. Note the following information on your copy: your initials, date, and any corrective actions taken or adjustments made.
4. Retain the completed log for maintenance history.

Procedure	Component Tag	Mode	Chapter	Date	Initials
Inspect cabinet air intake filters and clean or replace them if dirty.	n/a	Auto	Chapter 8		
Use a UPW spray gun to rinse cabinet interior with UPW if there is any sign of chemical deposits.	SG.U1.950, SG.U.951	Auto	—		

Corrective Actions and Adjustments:

Quarterly PM Procedures/Log


WARNING


CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Wear proper safety gear when performing this procedure.

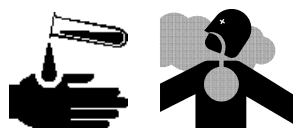
Perform the procedures listed in the table below as follows:

1. Make a copy of this page.
2. Perform the procedures listed.
3. Note the following information on your copy: your initials, date, and any corrective actions taken or adjustments made.
4. Retain the completed log for maintenance history.

Procedure	Component	Mode	Chapter	Date	Initials
Check manual valves and their positions.	Listed in Appendix D	Manual	Appendix D		
Check operation of sump-area switches.	FS.Z5.8, FS.Z5.9	Manual	Chapter 8		
Check door interlock switch operation.	ZS.Z4.5, ZS.Z5.5A, ZS.Z5.5B	Manual	Chapter 8		
Check valve weep-holes for leakage (if applicable).	All AOVs and MVs (see Chapter 3 for a listing).	Manual	Chapter 8		
Check operation of capacitive sensors.	See Chapter 3 for a listing.	Manual	Chapter 8		
Verify that all pressure indicators read "zero" with no pressure applied.	PI.J1.301, PI.J99.301, PI.Z1.491	Manual	—		
Verify operation of air-operated valves.	All AOVs (see Chapter 3 for a listing).	Manual	—		

Corrective Actions and Adjustments:

Semiannual PM Procedures/Log



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Wear proper safety gear when performing this procedure.

Perform the procedures listed in the table below as follows:

1. Make a copy of this page.
2. Perform the procedures listed.
3. Note the following information on your copy: your initials, date, and any corrective actions taken or adjustments made.
4. Retain the completed log for maintenance history.

Procedure	Component	Mode	Chapter	Date	Initials
Replace cabinet air intake filters.	n/a	Auto	Chapter 8		

Corrective Actions and Adjustments:

Annual PM Procedures/Log



WARNING



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY.

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Wear proper safety gear when performing this procedure.

IMPORTANT

The lifetime of wetted components is dependent on the chemistries and operating conditions of the system. During routine inspection of this equipment, all valves and piping must be inspected for wear and corrosion and replaced as needed. If you have questions about material compatibility for your chemistries, please contact Air Liquide Electronics.

Perform the procedures listed in the table below as follows:

1. Make a copy of this page.
2. Perform the procedures listed.
3. Note the following information on your copy: your initials, date, and any corrective actions taken or adjustments made.
4. Retain the completed log for maintenance history.

Procedure	Component	Mode	Chapter	Date	Initials
Verify EMO button operation, as well as all pushbuttons and lamps.	PB1603	Manual	Chapter 8		
Examine all valves. Look for chemical leaks. Shine a bright light through the valve bodies to determine if there is any discoloration of the internal components.	All pneumatic and manual valves (see Chapter 3 for a listing).	Manual	—		
Replace the nitrogen filter	F.N1.950	Manual	—		
Actuate/force all air-operated valves and look for movement of the pop-up indicator (if applicable) on the valve.	All AOVs (see Chapter 3 for a listing).	Manual	—		
Verify pressure switch operation.	PS.Z1.491	Manual	Chapter 8		

Corrective Actions and Adjustments:

Two-Year PM Procedures/Log

**WARNING**

CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY.

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Wear proper safety gear when performing this procedure.

Perform the procedures listed in the table below as follows:

1. Make a copy of this page.
2. Perform the procedures listed.
3. Note the following information on your copy: your initials, date, and any corrective actions taken or adjustments made.
4. Retain the completed log for maintenance history.

Procedure	Component	Mode	Chapter	Date	Initials
Replace the PLC power supplies.	Allen-Bradley MicroLogix 1769-series, PLC428, PLC432	Manual	—		
Replace the 24 VDC, 10W, and 48 VDC, 600W power supplies.	PWS215, PWS207	Manual	—		
Rebuild or replace the Levitronix BPS-600 centrifugal pump.	P21	Manual	Chapter 8		

Corrective Actions and Adjustments:

Five-Year PM Procedures/Log



WARNING



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY.

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Wear proper safety gear when performing this procedure.

IMPORTANT

The lifetime of wetted components is dependent on the chemistries and operating conditions of the system. During routine inspection of this equipment, all valves and piping must be inspected for wear and corrosion and replaced as needed. If you have questions about material compatibility for your chemistries, please contact Air Liquide.

Perform the procedures listed in the table below as follows:

1. Make a copy of this page.
2. Perform the procedures listed.
3. Note the following information on your copy: your initials, date, and any corrective actions taken or adjustments made.
4. Retain the completed log for maintenance history.

Procedure	Component	Mode	Chapter	Date	Initials
Replace all manual and pneumatic valves, indicating pressure switches, pressure indicators, and pressure regulating valves.	All those in direct chemical contact (see Chapter 3 for a listing).	Shutdown	Chapter 8		

Corrective Actions and Adjustments:

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7 Troubleshooting



BEFORE PROCEEDING, READ AND UNDERSTAND THE SAFETY INFORMATION CONTAINED WITHIN THE [SAFETY](#) CHAPTER OF THIS MANUAL.



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY.

Personnel servicing this unit will be exposed to chemical that has no secondary containment. Wear proper safety gear when servicing this unit.



OPENING A DOOR OR PANEL DURING UNIT OPERATION COULD EXPOSE PERSONNEL TO HAZARDOUS CHEMICAL.

Open a door or panel only after the system has been depressurized and after the chemical plumbing and valves have been flushed and purged of chemical.



COMPONENTS THAT ARE REMOVED DURING MAINTENANCE MIGHT CONTAIN HAZARDOUS CHEMICALS OR CHEMICAL RESIDUE

Before removing wetted components or components that might have come in contact with chemical, flush and purge the affected engine subsystem and plumbing components.

IMPORTANT

ONLY PERSONNEL WHO ARE TRAINED AND CERTIFIED BY AIR LIQUIDE SHALL SERVICE AND PERFORM MAINTENANCE ON THE INLINE BLENDER, WITH INTEGRATED DAYTANKS.

This chapter contains a list of alarms, causes, and corrective actions.

Introduction

Record Keeping

Record keeping is the most important part of the troubleshooting procedure. The following types of information should be recorded:

- **Operator's or supervisor's notes on system problems.** Include time, date, problem description, and corrective steps.

Problem Identification

Steps to determine problem:

- Isolate the problem in increasingly smaller circles of probability.
- Determine what sequence does not operate properly.
- Determine whether the problem is electrical or mechanical.

Additional Help

Contact your Air Liquide representative for more information. Contact information for Air Liquide Advanced Equipment Systems is located in the [*Technical Support Contact Information*](#) section at the beginning of this manual.

Explanation of Alarms

There are three levels of alarms:

Shutdown Alarm: Unit has been shutdown. Shutdown alarms cause the PLC to immediately shut down all activity in the system that is not safety related. When active, non-safety machine outputs are placed in their normal (safe) state. Auto mode operation is not allowed until all shutdown alarms have been corrected.

Critical Alarm: Suspends operation of sub unit and alerts the operator. The unit may be approaching a condition that, unless the operator intervenes, could result in a Shutdown alarm that will shut down the unit.

Warning Alarm: Used to notify that operator attention is required. If no operator attention is provided, a warning could escalate to a Critical or Shutdown alarm.

Audible Alarm

The operator can silence any active audible alarm by momentarily pressing the **ALARM SILENCE HOLD TO RESET** button on the front of the system. The horn will be silenced for the amount of time set in the user-configurable **ALARM SILENCER** timer.

Lamp/Horn Definitions

Table 7-1. Lamp / Horn Definitions

Lamp / Horn Status	Indication
Steady red light and horn.	Shutdown alarm
Flashing red light and horn.	Critical alarm
Steady yellow light and horn.	Warning alarm
Steady green light.	System is healthy

Troubleshooting Tables

Active Alarm messages are viewed from the *Active Alarm Summary Screen*. Press the Alarm List button on the top of any screen to go to the *Active Alarm Summary Screen*.

Figure 7-1. Alarm Summary

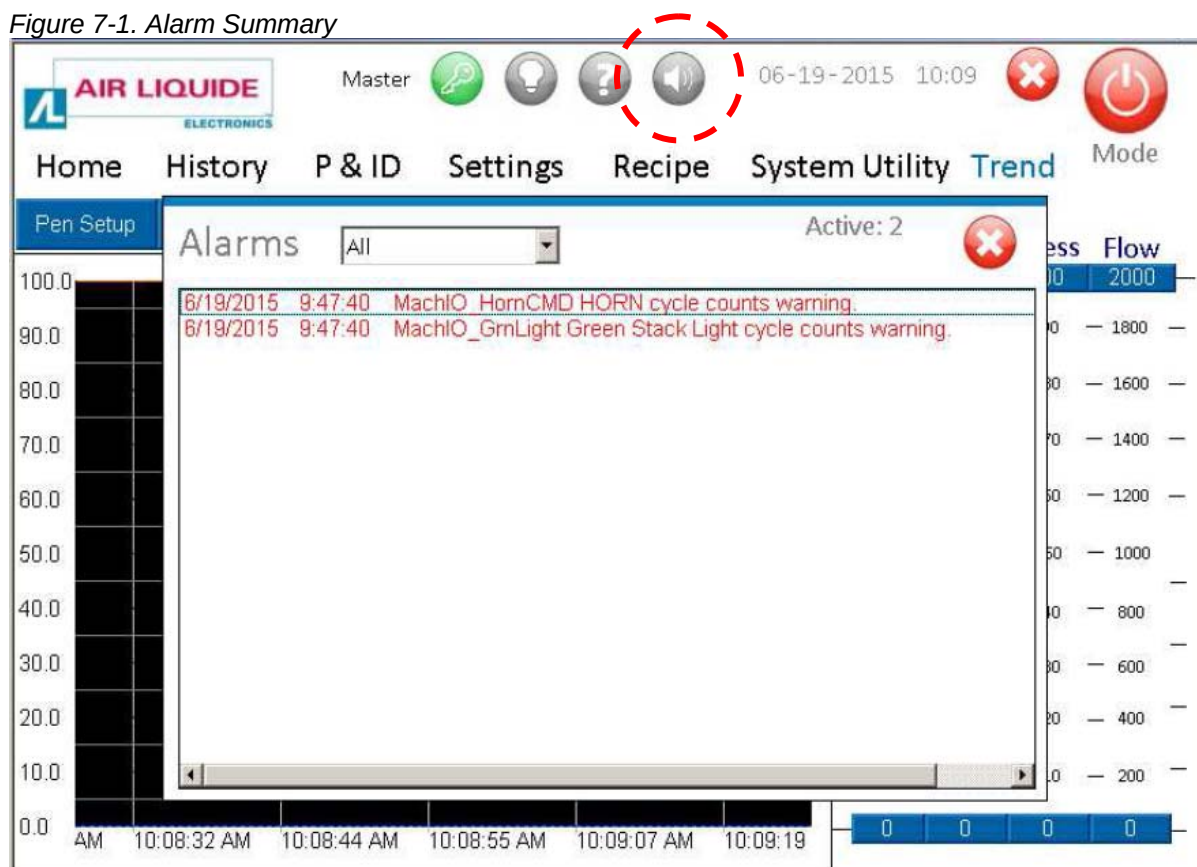


Table 7-2. Alarm Summary Screen

Field or Softkey	Purpose
DATE	The date at which the alarm was triggered.
TIME	The time at which the alarm was triggered.
DESCRIPTION	Description of alarm triggered.
<i>Softkeys</i>	
ALARM SILENCE/RESET	Silences horn, and resets alarms if possible.

History Screen

The historical ALARMS and EVENTS might be viewed by pressing the HISTORY button MENU. Lines with Red text are active.

Figure 7-2. History Screen.



Shutdown Alarms

A shutdown alarm shuts down the system.



WARNING



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY.

An uncontrolled loss of electrical power to the system allows trapped pressure to remain in the chemical plumbing. Depressurize the chemical lines before attempting any servicing or troubleshooting.

Table 7-3. Shutdown Alarm Troubleshooting Chart

ALARM and INDICATION	POSSIBLE CAUSE	CORRECTIVE ACTION
S001: EMO	1. An Emergency Stop button has been pressed. 2. Faulty contact.	1. Check that EMO button is in the out position. 2. Replace.
S002: CDA Pressure IPS.A1.951 out of range	1. Facility CDA pressure out of range. 2. CDA Pressure Switch set incorrectly. 3. CDA Pressure Switch faulty.	1. Check and correct. 2. Check the set points of the pressure switch and adjust. 3. Replace.
S003: N2 Facility IPS.N1.950 out of range	1. Facility CDA pressure out of range. 2. CDA Pressure Switch set incorrectly. 3. CDA Pressure Switch faulty.	1. Check and correct. 2. Check the set points of the pressure switch and adjust. 3. Replace.
S004: Door (ZS.Z5.5)	1. Door open. 2. Door switch faulty.	1. Close the door. 2. Replace.
S005: Drain Overfill (CS.Z1.58)	1. Drain overfilled. 2. Faulty sensor.	1. Obstruction in drain, unexpected backup of liquid from facility. 2. Replace the faulty sensor.
S006: Cabinet Exhaust (PS.Z1.491)	1. Cabinet Exhaust Pressure Switch set incorrectly. 2. Cabinet Exhaust Pressure Switch faulty. 3. Fluctuation in facility exhaust draw.	1. Check the set points of the pressure switch and adjust. 2. Replace. 3. Verify that other units using the same exhaust piping are not disrupting the exhaust setting of this unit. Adjust pressure switch if necessary.
S007: Cabinet High Level Leak (FS.Z5.9)	1. Component leak. 2. Faulty sensor. 3. AOV faulty.	1. Determine source of leak and replace. 2. If the area is dry, the sensor might have failed or is stuck. Verify adjustment of the sensor. If OK, replace the sensor. 3. Replace.

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Table 7-3. Shutdown Alarm Troubleshooting Chart

ALARM and INDICATION	POSSIBLE CAUSE	CORRECTIVE ACTION
S008: Cabinet Low Level Leak (FS.Z5.8)	1. Component leak. 2. Sump pump plugged or faulty. 3. Faulty sensor. 4. AOV faulty. 5. Leak not reaching High Level leak in set point time with low level leak active.	1. Sump pump will run automatically. Determine source of leak and replace. 2. Check and unplug or replace, if faulty. If sump pump is not operating, increase the CDA pressure. If sump pump is operating, it might be clogged. 3. If the area is dry, the sensor might have failed or is stuck. Verify adjustment of the sensor. If OK, replace the sensor. 4. Replace.
S015: Tank 1A Overfill Level	1. Tank overfilled. 2. Faulty sensor.	1. Unexpected Tank overfill. Review Level and Sequences, hardware adjustments. 2. Replace the faulty sensor.
S016: Tank 1B Overfill Level	1. Tank overfilled 2. Faulty sensor.	1. Unexpected Tank overfill. Review Level and Sequences, hardware adjustments. 2. Replace the faulty sensor.

Critical Alarms

A critical alarm indicates the unit may be approaching a condition that, unless the operator intervenes, could result in a Shutdown alarm that will shut down the system.



WARNING



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY.

An uncontrolled loss of electrical power to the system allows trapped pressure to remain in the chemical plumbing. Depressurize the chemical lines before attempting any servicing or troubleshooting.

Table 7-4. Critical Alarm Troubleshooting Chart

ALARM and INDICATION	POSSIBLE CAUSE	CORRECTIVE ACTION
C001: Supply J001 Not Ready	1. J1 object on HMI is not in AUTO mode 2. Facility pressurized supply is not ready (CHEM 1 SUPPLY READY).	1. Review the object's mode and settings on HMI. 2. Review Factory signal.
C002: Supply J099 Not Ready	J99 object on HMI is not in AUTO mode.	Review the object's mode and settings on HMI.
C003: P021 not ready	Pump HMI setup incorrectly.	Review Pump MODE and Settings on HMI.
C004: M1 Conductivity A Out of Spec. (CT.M1.910A)	1. Recipe Setup incorrectly. 2. Device setup incorrectly. 3. Device faulty.	1. Modify Settings on Recipe Setup screen so that recipe results in proper reading. 2. Check the setup. 3. Replace
C005: M1 Conductivity B Out of Spec. (CT.M1.910B)	1. Recipe Setup incorrectly. 2. Device setup incorrectly. 3. Device faulty.	1. Modify Settings on Recipe Setup screen so that recipe results in proper reading. 2. Check the setup. 3. Replace.
C006: M1 Temperature A Out of Spec. (TT.M1.910A)	1. Recipe Setup incorrectly. 2. Device setup incorrectly. 3. Device faulty.	1. Modify Settings on Recipe Setup screen so that recipe results in proper reading. 2. Check the setup. 3. Replace.
C007: M1 Temperature B Out of Spec. (TT.M1.910B)	1. Recipe Setup incorrectly. 2. Device setup incorrectly. 3. Device faulty.	1. Modify Settings on Recipe Setup screen so that recipe results in proper reading. 2. Check the setup. 3. Replace.
C013: M2 Temp Out of Spec. (TT.M2.910A)	1. Source chemical out of spec. 2. Device setup incorrectly. 3. Device faulty.	1. Use source chemical that has had time to come into equilibrium with room temperature. 2. Check the setup. 3. Replace.
C008: J001 Flow Out of Spec.	1. Device requires re-zeroing. 2. Device faulty.	1. Remove demand signal, re-zeroing is performed at completion of a blend. 2. Replace.
C009: J099 Flow Out of Spec.	1. Device requires re-zeroing. 2. Device faulty.	1. Remove demand signal, re-zeroing is performed at completion of a blend. 2. Replace.

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Table 7-4. Critical Alarm Troubleshooting Chart

ALARM and INDICATION	POSSIBLE CAUSE	CORRECTIVE ACTION
C010: J001 Pressure Out of Spec.	1. J001 Pressure is out of spec during Blend Step 4 or 5 for the configurable amount of time. 2. Settings are incorrectly matched to the Process. 3. Device faulty.	1. Review the settings. 2. Troubleshoot / Replace.
C011: J099 Pressure Out of Spec.	1. J099 Pressure is out of spec during Blend Step 4 or 5 for the configurable amount of time. 2. Settings are incorrectly matched to the Process. 3. Device faulty.	1. Review the settings. 2. Troubleshoot / Replace.
C012: J001 Pressure High.	Instrumentation pressure is higher than value in settings for specified time.	1. Check for failed components. 2. Check the settings for J001.
C013: J001 Pressure Low.	Instrumentation pressure is lower than value in settings for specified time.	1. Check for failed components. 2. Check the settings for J001.
C014: J099 Pressure High.	Instrumentation pressure is higher than value in settings for specified time.	1. Check for failed components. 2. Check the settings for J099.
C015: J099 Pressure Low.	Instrumentation pressure is lower than value in settings for specified time.	1. Check for failed components. 2. Check the settings for J099.
C016: M1 High Conductivity	1. During Blend Step 4 or 5, M1 Conductivity is higher than Value in Settings for the specified Time. 2. Flow Controller Faulty. 3. Settings are incorrectly matched to the process. 4. Conductivity device is faulty.	1. Review the settings. 2. Review blend to drain sequence and observe flows and conductivities. 3. Troubleshoot / Replace 4. Review the settings 5. Troubleshoot / Disable probes as needed to identify root cause. The blender may operate on any combination of metrology probes.
C017: M1 Low Conductivity	1. During Blend Step 4 or 5, M1 Conductivity is lower than Value in Settings for the specified Time. 2. Flow Controller Faulty. 3. Settings are incorrectly matched to the process. 4. Conductivity device is faulty.	1. Review the settings. 2. Review blend to drain sequence and observe flows and conductivities. 3. Troubleshoot / Replace. 4. Review the settings. 5. Troubleshoot / Disable probes as needed to identify root cause. The blender may operate on any combination of metrology probes.
C018: M1 High Temperature	1. During Blend Step 4 or 5, M1 Temperature is higher than Value in Settings for the specified Time. 2. Settings are incorrectly matched to the process. 3. Temperature RTD probe or transmitter device is faulty.	1. Review the settings. 2. Review the blend to drain sequence and observe flows and temperatures. 3. Review the settings 4. Troubleshoot / Disable probes as needed to identify root cause. The blender might operate on any combination of metrology probes.
C019: M1 Low Temperature	1. During Blend Step 4 or 5, M1 Temperature is lower than Value in Settings for the specified Time. 2. Settings are incorrectly matched to the process. 3. Temperature RTD probe or transmitter device is faulty.	1. Review the settings. 2. Review the blend to drain sequence and observe flows and temperatures. 3. Review the settings. 4. Troubleshoot / Disable probes as needed to identify root cause. The blender might operate on any combination of metrology probes.

Table 7-4. Critical Alarm Troubleshooting Chart

ALARM and INDICATION	POSSIBLE CAUSE	CORRECTIVE ACTION
C020: Tank T001A High-High Level	1. Tank liquid level is above CS.T1.18A. 2. Faulty sensor.	1. Review level and sequences, hardware adjustments. 2. Replace the faulty sensor.
C021: Tank T001 Low-Low Level	1. Tank liquid level is below CS.T1.11. 2. Faulty sensor.	1. Review level and sequences, hardware adjustments. 2. Replace the faulty sensor.
C022: Tank T001B High-High Level	1. Tank liquid level is above CS.T1.18B. 2. Faulty sensor.	1. Review level and sequences, hardware adjustments. 2. Replace the faulty sensor.
C023: Tank T001 Fill Timeout	Fill Time has exceeded Settings value.	1. Review the settings 2. Review the blender state and any other active alarms.
C024: Tank T001 End of Life Exceeded	Tank life has exceeded Settings value.	1. Review the settings. 2. Review the blender state and factory consumption of chemical. 3. Manage Tank contents accordingly. (ex. Drain tank and re-blend if necessary)
C025: N2 Facility IPS.N1.950	1. Facility N ₂ pressure out of range. 2. N ₂ Pressure Switch set incorrectly. 3. N ₂ Pressure Switch faulty.	1. Check and correct. 2. Check the set points of the pressure switch and adjust. 3. Replace.
C026: Remote Shutdown from Fab Use 1	Remote Shutdown 1 signal.	Review system integration with factory and required factory signals or jumpers.
C027: Remote Shutdown from Fab Use 2	Remote Shutdown 2 signal.	Review system integration with factory and required factory signals or jumpers.
C028: Remote Shutdown from Fab Use 3	Remote Shutdown 3 signal.	Review system integration with factory and required factory signals or jumpers.
C029: Remote Shutdown from Fab Use 4	Remote Shutdown 4 signal.	Review system integration with factory and required factory signals or jumpers.
C030: No Conductivity Transmitters ENABLED	No Conductivity Transmitters are ENABLED.	Review the settings.
C031: No Temperature Transmitters ENABLED	No Temperature Transmitters are ENABLED.	Review the settings.
C032: M021 Conductivity Out of Spec, Tank 1 Out of Spec (CT.M21.910)	1. Recipe Setup incorrectly. 2. Device setup incorrectly. 3. Device faulty.	1. Modify the settings on <i>Recipe Setup</i> screen or <i>Settings</i> screen. 2. Check the setup. 3. Replace.
C033: M021 Temperature Out of Spec, Tank 1 Out of Spec (TT.M21.910)	1. Temperature is out of range. 2. Device setup incorrectly. 3. Device faulty.	1. Modify the settings on <i>Settings</i> screen. 2. Check the setup. 3. Replace.
C034: M021 High Pressure , Tank 1 Recirc (PT.M21.910)	Pressure measurement is higher than Settings Value: 1. Settings Setup incorrectly. 2. Device setup incorrectly. 3. Device fault.	1. Modify the settings on the <i>Settings</i> screen. 2. Check the setup, including IO scaling. 3. Replace.
C035: M021 Low Pressure , Tank 1 Recirc (PT.M21.910)	Pressure measurement is Lower than Settings Value. 1. Settings Setup incorrectly. 2. Device setup incorrectly. 3. Device faulty.	1. Modify the settings on the <i>Settings</i> screen. 2. Check the setup 3. Replace.
C036: M021 Conductivity High, Tank 1 (CT.M21.910)	1. Recipe Setup incorrectly. 2. Device setup incorrectly. 3. Device faulty.	1. Modify the settings on the <i>Recipe Setup</i> screen or <i>Settings</i> screen 2. Check the setup. 3. Replace.

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Table 7-4. Critical Alarm Troubleshooting Chart

ALARM and INDICATION	POSSIBLE CAUSE	CORRECTIVE ACTION
C037: M021 Conductivity Low, Tank 1 (CT.M21.910)	1. Recipe Setup incorrectly. 2. Device setup incorrectly. 3. Device faulty.	1. Modify settings on the <i>Recipe Setup</i> screen or <i>Settings</i> screen 2. Check the setup. 3. Replace.
C038: M021 Temperature High, Tank 1 (TT.M21.910)	1. Recipe Setup incorrectly. 2. Device setup incorrectly. 3. Device faulty.	1. Modify settings on the <i>Settings</i> screen. 2. Check the setup. 3. Replace.
C039: M021 Temperature Low, Tank 1 (TT.M21.910)	1. Recipe Setup incorrectly. 2. Device setup incorrectly. 3. Device faulty.	1. Modify settings on the <i>Settings</i> screen. 2. Check the setup. 3. Replace.
C040: Blend To Drain Override Timeout	Blend to Drain Time has exceeded Value in Settings.	Review sequences and settings.
C041: Blend to Drain Too Long (NOT IN SPEC)	During an attempt to blend to tank, the blend to drain time exceeded the value in the SETTINGS.	Review sequences and settings. Review any other active alarm.
C042: All Tank Outlet Valves CLOSED	For standalone blender operation, the tank outlet valve is closed. For Dual Blender Usage, All tank outlet valves must be closed for this alarm to post. (AOV.T1.910 is closed for the local blender, AND the remote tank status is NOT ONLINE).	Review sequences, settings, and Fab consumption.
C043: Blend retries exceeded.	1. Specification settings are too tight.. 2. Faulty device.	1. Enlarge tolerance range on Recipe Tolerance Setpoints screen. 2. Disable use of faulty device on the <i>Settings</i> screen.
C044: Liquid in Drain (CS.Z1.17)	1. Liquid in drain. 2. Faulty sensor.	1. Obstruction in drain, unexpected backup of liquid from facility. 2. Replace.
C045: J001 Pressure Not Reached in Expected Time	During Blend Step 3, the pressure did not reach spec.	Review sequences and settings.
C046: J099 Pressure Not Reached in Expected Time	During Blend Step 3, the pressure did not reach spec.	Review sequences and settings.
C047: J001 Pressure Low	The pressure is lower than the value in the settings.	Review sequences and settings.
C048: J099 Pressure Low	The pressure is lower than the value in the settings.	Review sequences and settings.

Warning Alarms

Used to notify that operator attention is required. If no operator attention is provided, a warning could escalate to a Critical or Shutdown alarm.



WARNING

CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY.

An uncontrolled loss of electrical power to the System allows trapped pressure to remain in the chemical plumbing. Depressurize the chemical lines before attempting any servicing or troubleshooting.

Table 7-5. Warning Alarm Troubleshooting Chart

ALARM and INDICATION	POSSIBLE CAUSE	CORRECTIVE ACTION
W001: Cabinet Exhaust (PS.Z1.491)	1. Cabinet Exhaust Pressure Switch set incorrectly. 2. Cabinet Exhaust Pressure Switch faulty. 3. Fluctuation in facility exhaust draw.	1. Check the set points of the pressure switch and adjust. 2. Replace. 3. Verify that other units using the same exhaust piping are not disrupting the exhaust setting of this unit. Adjust pressure switch if necessary.
W002: Sample Door (ZS.Z4.5)	1. Sample door open. 2. Door switch faulty.	1. Close the door. 2. Replace.
W003: Liquid in Drain (CS.Z1.17)	1. Liquid in drain. 2. Faulty sensor.	1. Obstruction in drain, unexpected backup of liquid from facility. 2. Replace.
W004: Temp Sensors Exceeded Max Diff. (TT.M1.910A/B)	1. Sensor faulty. 2. Max difference setting too small.	1. Replace. 2. Modify max difference setting.
W005: Cond Sensors Exceeded Max Diff. (CT.M1.910A/B)	1. Sensor faulty. 2. Max difference setting too small.	1. Replace. 2. Modify max difference setting.
W006: Tank 1 Sensor Error (CS.T1.11/12/17/18)	1. Sensor(s) out of adjustment. 2. Faulty Sensor(s).	1. Adjust level sensors so that they work in sequence. (Low-Low must be covered before Low. Low must be covered before High. High must be covered before High-High). 2. Replace.
W007: Tank T001A High-High Level	1. Tank liquid level is above CS.T1.18A 2. Faulty sensor.	1. Review level and sequences, hardware adjustments. 2. Replace the faulty sensor.
W008: Tank T001B High-High Level	1. Tank liquid level is above CS.T1.18B 2. Faulty sensor.	1. Review level and sequences, hardware adjustments. 2. Replace the faulty sensor.
W009: Tank T001 Low-Low Level	1. Tank liquid level is below CS.T1.11 2. Faulty sensor.	1. Review level and sequences, hardware adjustments. 2. Replace the faulty sensor.

7-14 Troubleshooting

Table 7-5. Warning Alarm Troubleshooting Chart

W010: Cabinet Low Level Leak (FS.Z5.8)	1. Component leak. 2. Faulty sensor.	1. Determine source of leak and replace. 2. If the area is dry, the sensor might have failed or is stuck. Verify adjustment of the sensor. If OK, replace the sensor.
W011: Cabinet High Level Leak (FS.Z5.9)	1. Component leak. 2. Faulty sensor.	1. Determine source of leak and replace. 2. If the area is dry, the sensor might have failed or is stuck. Verify adjustment of the sensor. If OK, replace the sensor.
W012: HMI Forcing Enabled	HMI Forcing is enabled.	Press Forcing Disabled key to clear.
W013: PLC Battery	Low battery charge.	Replace the PLC battery while electrical power is on.
W014: J001 NOT in Auto mode	Object is not in AUTO on the HMI.	Review the settings on the HMI.
W015: J099 NOT in Auto mode	Object is not in AUTO on the HMI.	Review the settings on the HMI.
W016: P021 NOT in Auto mode	Object is not in AUTO on the HMI.	Review the settings on the HMI.
W017: Tank 1 NOT in AUTO mode	Object is not in AUTO on the HMI.	Review the settings on the HMI.
W018: Sample Area NOT in AUTO mode	Object is not in AUTO on the HMI.	Review the settings on the HMI.
W019: MACHINE is NOT in AUTO mode	Object is not in AUTO on the HMI.	Review the settings on the HMI.
W020: P021 High Pressure , Tank 1 Recirc (PT.M21.910)	Pressure measurement is higher than Settings Value. 1. Settings Setup incorrectly. 2. Device setup incorrectly. 3. Device faulty.	1. Modify Settings on Settings screen. 2. Check the setup, including IO scaling. 3. Replace.
W021: P021 Low Pressure , Tank 1 Recirc (PT.M21.910)	Pressure measurement is Lower than Settings Value: 1. Settings Setup incorrectly. 2. Device setup incorrectly. 3. Device faulty.	1. Modify Settings on Settings screen. 2. Check the setup. 3. Replace.
W022: P021 Low Flow , Tank 1 Recirc (FT.M21.910)	Measurement is Lower than Settings Value: 1. Settings Setup incorrectly. 2. Device setup incorrectly. 3. Device faulty.	1. Modify Settings on Settings screen. 2. Check the setup, including I/O scaling. 3. Replace.
W023: P021 High Flow , Tank 1 Recirc (FT.M21.910)	Measurement is Higher than Settings Value: 1. Settings Setup incorrectly. 2. Device setup incorrectly. 3. Device faulty.	1. Modify Settings on Settings screen. 2. Check the setup. 3. Replace.
W024: P021 Levitronix HIGH RPM	Pump is operating at a higher RPM value than the value in SETTINGS.	Review sequences and settings.
W025: P021 Levitronix LOW RPM	Pump is operating at a lower RPM value than the value in SETTINGS.	Review sequences and settings.
W026: Tank 1 Pot Life Exceeded	Tank Life Exceeds the value in SETTINGS.	Review settings and Fab consumption.

Table 7-5. Warning Alarm Troubleshooting Chart

W027: P021 Levitronix Warning	Digital INPUT WARNING SIGNAL from Pump Controller.	1. Review Pump Controller Status. 2. Clear other alarms, after they are addressed. 3. Re-enable the Pump with the push-button on the HMI door.
W028: P021 Levitronix FAULT	Digital INPUT FAULT SIGNAL from Pump Controller.	1. Review Pump Controller Status. 2. Clear other alarms, after they are addressed. 3. Re-enable the Pump with the push-button on the HMI door. Further Troubleshooting / Replacement might be required.
W029: P021 Maintenance Hours Exceeded	The Time has exceeded the value in SETTINGS for MAINTENANCE TIMER of pump P21.	
W030: M001A Conductivity Transmitter NOT ENABLED	This signal / device is disabled in SETTINGS on HMI.	
W031: M001B Conductivity Transmitter NOT ENABLED	This signal / device is disabled in SETTINGS on HMI.	
W032: M001A Temperature Transmitter NOT ENABLED	This signal / device is disabled in SETTINGS on HMI.	
W033: M001B Temperature Transmitter NOT ENABLED	This signal / device is disabled in SETTINGS on HMI.	
W034: Time Since Batch Started	The time has exceeded the value in SETTINGS, since blend has started.	1. Review the blend rate and settings. 2. Review other active alarms.
W035: Time Since Batch Finished	The time has exceeded the value in SETTINGS, since blend has Finished.	1. Review the blend rate, Fab consumption, and settings. 2. Review other active alarms.
W036: Machine Override Switch Active ALARM	The pneumatic override keyswitch within the electrical compartment is turned ON. AOV.T1.910 for this local blender is ON.	Review application and conditions.
W037: P001 Booster Pump Levitronix HIGH RPM	The reported speed is higher than the value in settings. 1. Low incoming facility chemical pressure, resulting in booster pump high RPM operation to achieve pressure setpoint, when P1 is in PROCESS mode. 2. Low chemical pressure regulator adjustment of PRV.J1.110, (PRV.J1.301 on pneumatic panel) 3. Faulty devices: Broken airline, closed chemical valve upstream, faulty regulator, higher than normal blend flowrate. (Review settings).	Review sequences and settings.

Table 7-5. Warning Alarm Troubleshooting Chart

W038: P001 Booster Pump Levitronix Low RPM	The reported speed is lower than the value in settings. 1. High incoming facility chemical pressure, resulting in booster pump low RPM operation to achieve pressure setpoint, when P1 is in PROCESS mode. 2. High chemical pressure regulator adjustment of PRV.J1.110, (PRV.J1.301 on pneumatic panel). 3. Faulty devices: Broken airline, closed chemical valve downstream, faulty regulator, lower than normal blend flowrate. (Review settings).	Review sequences and settings.
W039: Maintenance Bypass Key Switch ALARM	The Maintenance Bypass key in the electrical enclosure on the left wall is ON.	
W040: P001 Levitronix FAULT	Digital INPUT FAULT SIGNAL from Pump Controller.	1. Review Pump Controller Status. 2. Clear other alarms after they are addressed. Note: There is no ENABLE pushbutton for this lower voltage pump. Operation is AUTOMATIC from the HMI / PLC. 3. Further Troubleshooting / Replacement might be required.
W041: P001 Maintenance Hours Exceeded	The Time has exceeded the value in SETTINGS for MAINTENANCE TIMER of pump P001.	
W042: M021 Conductivity Out of Spec	The measurement is out of spec according to the values in SETTINGS.	Review Settings, Blend Recipe, Sequences

8 Component Procedures and Adjustments



WARNING



BEFORE PROCEEDING, READ AND UNDERSTAND THE SAFETY INFORMATION CONTAINED WITHIN THE [SAFETY](#) CHAPTER OF THIS MANUAL.

This chapter covers component maintenance procedures for the Inline Blender, with Integrated Daytanks.

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WARNING



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY.

Personnel servicing this unit will be exposed to chemical that has no secondary containment. Wear proper safety gear when servicing this unit.



WARNING



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY.

Rotating the MAIN POWER DISCONNECT SWITCH to OFF might allow trapped pressure and chemical to remain in the plumbing. Depressurize the chemical lines before attempting any servicing or troubleshooting.



WARNING



OPENING A SAFETY INTERLOCK DOOR OR PANEL DURING SYSTEM OPERATION COULD EXPOSE PERSONNEL TO HAZARDOUS CHEMICAL.

Open a safety lock door or panel only after the Inline Blender, with Integrated Daytanks, has been depressurized and after the chemical plumbing and valves have been flushed of chemical. Depressurize the chemical lines before attempting any servicing or troubleshooting.



WARNING



COMPONENTS THAT ARE REMOVED DURING MAINTENANCE MIGHT CONTAIN HAZARDOUS CHEMICALS OR CHEMICAL RESIDUE.

Before removing wetted components, or components that might have come in contact with chemical, flush the affected components and tubing.



WARNING



WHEN PERFORMING ANY WORK IN THE ELECTRICAL AREA, ENSURE POWER IS ISOLATED PER LOCAL/SITE RULES AND PROCEDURES.

Care should be taken when handling of the electrical panel during installation, removal and troubleshooting:

- The touch screen panel during normal operation is seated into a recess in the cabinet and secured with screws.
- When the screws are removed or not installed, there is a potential for the panel to fall from the cabinet and cause potential injury to an operator.
- Use the two-person “buddy” system to handle the panel.



WARNING



THE IMPELLER OF A LEVITRONIX CENTRIFUGAL PUMP CREATES A MAGNETIC FIELD THAT CAN BE HARMFUL TO PACEMAKER WEARERS.

Pacemaker wearers must stay back at least 12 inches (30 centimeters) from the Levitronix centrifugal pump used inside the Inline Blender, with Integrated Daytanks



For chemicals—and potential chemical combinations—identified by site procedures, forced air-breathing apparatus must be worn by maintenance personnel.

The forced air-breathing apparatus can consist of, but is not limited to, the following types of devices:

- Site's in-house forced-air line
- Air-breathing device on a portable cart
- Self-contained breathing apparatus (SCBA)

For any maintenance task that requires a respirator (cartridge-type or forced-air), maintenance personnel must be fit-tested for their own respirator. They must have a valid physical certification proving that the respirator fitting was accomplished.



To prevent back or other injuries, do not attempt to lift, by yourself, a component that weighs more than 50 pounds

A component that weighs more than 50 pounds requires two people to lift it. Read and understand an entire procedure before starting it.



IMPORTANT

Improper handling of electronic components can result in electrostatic discharge, which can cause component failure.

To avoid damage from electrostatic discharge, follow these basic rules:

1. Handle all static-sensitive components at a designated static-safeguarded area or on a portable ESD mat.
2. Transport all static-sensitive components (for example, circuit boards) in static-shielded containers or packages.
3. Monitor and test static-protection products and materials to ensure that they are working properly.

Using the Maintenance Bypass Switch

**WARNING**

TURNING THE MAINTENANCE BYPASS SWITCH ON/OFF TO “ON” PREVENTS AN ALARM FROM ACTIVATING AND OVERRIDES THE NORMAL DOOR INTERLOCK, EXHAUST LOW-PRESSURE, AND UNIT/COMPARTMENT DEPRESSURIZATION SAFETY FUNCTIONS.

BEFORE USING THIS FEATURE MAKE SURE TO THOROUGHLY UNDERSTAND THE SAFETY AND FUNCTIONAL IMPLICATIONS AND POTENTIAL DANGERS TO EQUIPMENT AND PERSONNEL.

**WARNING**

OPENING A SAFETY INTERLOCK-PROTECTED ACCESS DOOR OR PANEL DURING UNIT OPERATION COULD EXPOSE PERSONNEL TO HAZARDOUS CHEMICAL. MAKE SURE TO WEAR ALL REQUIRED PPE BEFORE OPENING A SAFETY INTERLOCK-PROTECTED ACCESS DOOR OR PANEL.

Opening a safety interlock-protected access door or panel, after turning the MAINTENANCE BYPASS SWITCH to ON, potentially exposes the maintainer to chemicals and pressures that are not secondarily contained. Make sure to wear all required PPE (including, but not limited to, full-face protection, a respirator, and hand and body protection) before using the MAINTENANCE BYPASS SWITCH and opening an access door.

The **MAINTENANCE BYPASS SWITCH**, located inside the Controls Compartment, was designed to be an infrequently used option for the unit maintainer. It is meant to be used when the maintainer wants to keep a pump or other sub-system running while maintenance is being performed.

Figure 8-1. Location of the Maintenance Bypass Switch inside the Controls Compartment (Typical)



Table 8-1. Location of the Maintenance Bypass Switch inside the Controls Compartment

Call Out	Component Description
1	Maintenance Bypass Switch

The **MAINTENANCE BYPASS SWITCH** overrides door interlocks and the exhaust low-pressure switch, and prevents an alarm from activating and the alarm horn from sounding. However, a leak-activated fault alarm cannot be bypassed, and will initiate an alarm regardless of the **MAINTENANCE BYPASS SWITCH** being turned to ON.

The time during which an alarm is bypassed by the **MAINTENANCE BYPASS SWITCH** is set, in minutes, at a system configuration screen.

It is strongly recommended that the **MAINTENANCE BYPASS SWITCH** not be used when removing and replacing chemical filter F21.

After maintenance is finished and the alarm bypass function is no longer needed, turn the **MAINTENANCE BYPASS SWITCH** to OFF.

Accumulator Replacement

The purpose of this procedure is to remove and replace the GC.Q1.910 accumulator in the chemical line in the Wetted Compartment.

Precautions



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY.

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Wear appropriate personal protective equipment when performing this procedure.

Procedure

Removal

1. Place the Inline Blender, with Integrated Daytanks, in Manual mode.
2. At the *Main Menu* screen, press the MAINTENANCE softkey.
3. Perform the [UPW Flush Pumps/Flow Controllers to Drain](#) procedure, to remove chemical from the accumulator. See the procedure in [Chapter 4, System Operation and Process Procedures](#).
4. After the service routine stops, push in the EMO switch.
5. Turn the **MAIN POWER DISCONNECT SWITCH** to OFF.
6. Disconnect the plumbing line between GC.Q1.910 and static mixer SM.Q1.910, and the plumbing line to the metrology suite (which consists of TT.M1.910A, CT.M1.910A, CT.M1.910B, and TT.M1.910B).
7. Remove the accumulator from its location, by unclamping it from its restraining clamps.

Installation

1. Install the new accumulator.
2. Connect the tubing and electrical connections to the new accumulator.
3. Close the Wetted Compartment door.
4. Pull the **EMO** button to the “out” position.
5. Turn the **MAIN POWER DISCONNECT SWITCH** to ON.
6. Press the **EMO RESET** button.
7. Place the Inline Blender, with Integrated Daytanks, in the desired mode of operation.

Air Filter Replacement

Purpose

The purpose of this procedure is to remove and replace the air-intake filters on the front doors of the Wetted Compartment.

Precautions



WARNING



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Wear proper safety gear when performing this procedure.

Procedure

NOTE

To remove the cover, apply pressure at the top and bottom at about midpoint, and pull the cover off.

1. Remove the filter cover and filter.
2. Remove the existing filter from the cover.
3. Insert the new filter in the cover with the white side against the louvers.
4. Position the cover and filter and push to them snap in place.

Capacitive Sensor Replacement & Adjustment

Purpose

The purpose of this procedure is to remove, replace, and adjust liquid-level capacitive sensors, which are located on the chemical and metrology vessels, main drain, and daytanks and their secondary containment equipment.

Prerequisites

Proper PPE.

Precautions



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Wear proper safety gear when performing this procedure.

Procedure: Removal and Replacement

1. Place the Inline Blender, with Integrated Daytanks, into Shutdown mode.
2. Press the **EMO** button.
3. Turn the **MAIN POWER DISCONNECT SWITCH** to OFF.
4. Disconnect the capacitive sensor cable wires from the terminal.
5. Remove the sensor from its mounting bracket.
6. Secure the replacement sensor in the mounting bracket with the front of the sensor touching pipe, tubing, or daytank wall.
7. Reconnect the sensor wires on the terminal.
8. Turn the **MAIN POWER DISCONNECT SWITCH** to ON.

Procedure: Adjustment

NOTE

To prevent maladjustment of sensor, use a screwdriver with a plastic handle. If the screwdriver has a metal blade, make sure you do not touch the blade when making an adjustment.

NOTE

Capacitive sensor can be adjusted with an empty tube or with the tube completely filled. To be considered a normal condition where no liquid is present, there should be no liquid in the tube any closer than 2 inches (5 cm) of the sensor.

NOTE

Replace the tubing if there is buildup of chemical in the line.

NOTE

Replace the sensor if adjustment does not provide reliable chemical level detection.

If the LED on the sensor is “on” when the procedure starts, refer to [Table 8-1](#) on page [8-13](#).

If the LED on the sensor is “off” when the procedure starts, refer to [Table 8-2](#) on page [8-13](#).

Each table lists the specific of the procedure for four cases of configurations of the capacitive sensors:

- **NC-LP:** Normally closed, adjusted originally with liquid present
- **NC-LNP:** Normally closed, adjusted originally with liquid not present
- **NO-LP:** Normally open, adjusted originally with liquid present
- **NO-LNP:** Normally open, adjusted originally with liquid not present

To adjust a sensor to the lower end of its sensitivity (Low and Low-Low sensors on vessels and tanks):

1. Place on a tube with chemical and adjust the sensor to the lowest sensitivity (counterclockwise [CCW]) rotation on the adjustment screw.
2. Slowly increase sensitivity (turn clockwise [CW]) until it sees chemical.
3. Turn CW another quarter turn beyond that.
4. Remove and install the sensor on the tube several times to ensure reliable operation.
5. Adjust CW by quarter-turn increments, as needed, and retest.

To adjust a sensor to the maximum reliable sensitivity (High and High-High sensors on vessels and tanks):

1. Place on an empty tube and adjust the sensor to the highest sensitivity (CW rotation).
2. Slowly reduce sensitivity (turn CCW) until the sensor stops seeing the tube.
3. Turn CCW another quarter turn beyond that.
4. Remove and reinstall the sensor on the tube several times to ensure reliable operation.
5. Adjust by quarter-turn increments, as needed, and retest.

Procedure for Adjusting Vessel Capacitive Sensors

Follow the procedure given in [Table 8-2](#) or [Table 8-3](#) on page [8-13](#) so that:

- Low and Low-Low sensors are set to the *lower* end of their sensitivity (refer to the notes on the previous page). This provides the most rapid response to falling liquid levels.
- High and High-High sensors are set at maximum reliable sensitivity (refer to the notes on the previous page).

Table 8-2. CS Adjustment Procedure—Sensor LED is *ON* when you start

Sensor LED is <i>**ON**</i> When You Start the Adjustment Procedure							
Adjustment Procedure Step ↓	Sensor Configuration →	NC-LP 18mm	NC-LNP 18mm	NO-LP 18mm	NO-LNP 18mm	LP 30mm	LNP 30mm
1. Using a small-blade, plastic-handle screwdriver, insert the blade into the small hole in the end of the sensor opposite to the sensor face.							
2. Slowly turn the adjustment screw [CW/CCW]...		CW	CW	CCW	CCW	CCW	CCW
3. ...until the LED sensor is [On/Off].		OFF	OFF	OFF	OFF	OFF	OFF
4. Slowly turn the adjustment screw [CW/CCW] ...		—	CCW	CW	—	CW	—
5. ... until the LED sensor is [On/Off].		—	ON	ON	—	ON	—
6. Continue turning the screw [CW/CCW] $\frac{3}{4}$ to 1 full turn.		CW	CCW	CW	CCW	CW	CCW
7. LED will remain [On/Off].		OFF	ON	ON	OFF	ON	OFF
8. Remove the liquid in front of the sensor, or remove the sensor from the tube. The LED will be [On/Off].		—	ON	—	OFF	—	OFF
9. Place liquid in front of the sensor in the tube. The LED will be [On/Off].		OFF	—	ON	—	ON	—

Table 8-3. CS Adjustment Procedure—Sensor LED is *OFF* when you start

Adjustment Procedure Step ↓	Sensor Configuration →	NC-LP 18mm	NC-LNP 18mm	NO-LP 18mm	NO-LNP 18mm	LP 30mm	LNP 30mm
1. Using a small-blade, plastic-handle screwdriver, insert the blade into the small hole in the end of the sensor opposite to the sensor face.							
2. Slowly turn the adjustment screw [CW/CCW]...		CCW	CCW	CW	CW	CW	CW
3. ...until the LED sensor is [On/Off].		ON	ON	ON	ON	ON	ON
4. Slowly turn the adjustment screw [CW/CCW] ...		CW	—	—	CCW	—	CCW
5. ... until the LED sensor is [On/Off].		OFF	—	—	OFF	—	OFF
6. Continue turning the screw [CW/CCW] $\frac{3}{4}$ to 1 full turn.		CW	CCW	CW	CCW	CW	CCW
7. LED will remain [On/Off].		OFF	ON	ON	OFF	ON	OFF
8. Remove the liquid in front of the sensor, or remove the sensor from the tube. The LED will be [On/Off].		—	ON	—	OFF	—	OFF
9. Place liquid in front of the sensor in the tube. The LED will be [On/Off].		OFF	—	ON	—	ON	—

Blender Power-Up

Purpose

The purpose of this procedure is to start an already-commissioned Inline Blender, with Integrated Daytanks.

Prerequisites

Ensure that power is available to the Inline Blender, with Integrated Daytanks.

Precautions

IMPORTANT

This procedure is only part of the procedure for starting the unit for the first time. See [Chapter 5, *Startup and Shutdown*](#), for the full startup procedure.

IMPORTANT

Only personnel who are trained and certified by Air Liquide shall service and perform maintenance on the unit.

Procedure

1. Open the Controls Compartment door and ensure that Main Power circuit breaker CB203 is in the ON (red light illuminated) position.
2. Turn the **MAIN POWER DISCONNECT SWITCH** to ON.
3. Ensure the EMO button on the front of the unit is pulled to the “out” position.
4. Press the **EMO RESET** button.
5. Press the **ALARM SILENCER** button on the front of the cabinet to clear any alarms
6. Check the alarms listed, if any, and take appropriate corrective action.

Conductivity Sensor: Remove, Replace, and Calibrate

Application

This procedure applies to a newly installed or already installed Hach probe using a Hach sc200 conductivity analyzer.

The Hach sc200 has two channels and accommodates two inductive conductivity probes. This procedure is used to calibrate either one.

Purpose

The purpose of this procedure is to calibrate a Hach 3700sc inductive conductivity sensor (probe) with its associated Hach sc200 controller.

Precautions



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY.

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Wear appropriate personal protective equipment when performing this procedure.

Special Tools

- 10.0 Ω (0.05% tolerance) precision resistor loop assembly (Air Liquide P/N 927073-001), which applies to the specific blend for which the Inline Blender, with Integrated Daytanks, is designed.
- 100.0 Ω reference resistor with 0.05% tolerance and 47.0 Ω conductivity.

IMPORTANT

If calibrating more than one sensor, use the same resistors for each sensor calibration.

Procedure

Remove Existing Conductivity Sensor

If a new conductivity sensor (probe) is to be calibrated, skip this procedure and perform the [Zero Point Calibration](#) procedure that immediately follows.

If an existing sensor is to be recalibrated, perform this procedure first.

1. Place the affected module (Blender, daytanks 1A or 1B) in Manual mode:
 - *Blender* module, if calibrating sensors CT.M1.910A/CT.M1.910B.
 - *Tank 1* module, if calibrating sensor CT.M21.910.
2. Inside the controls compartment, turn the **MAINTENANCE KEY SWITCH** to ON to open the compartment door without triggering an alarm.
3. Perform one of the following, applicable sub-procedures:
 - If calibrating sensors CT.M1.910A/CT.M1.910B (Wetted Compartment), perform the [UPW Flow thru Jx to Drain](#) procedure in [Chapter 4, System Operation and Process Procedures](#).
 - If calibrating sensor CT.M21.910, perform steps a to f:
 - a. Open the door of the Wetted Compartment.
 - b. Reconnect UPW line to the Tank 1 loop service point connections.
 - c. Open drain valve on filter housing.
 - d. Open UPW flush valve on dispense loop, and flush UPW for 15 minutes.
 - e. Close UPW flush valve and leave drain valve open to remove pressure and drain water.
 - f. Open the cell housing of the sensor (CT.M21.910) and let the remaining UPW drain to the cabinet floor.

NOTE

To gain access the CT.M21.910 cell housing, it might be necessary to remove the filter housings and assemblies.

4. Carefully remove the sensor from the cell housing by loosening the plug fitting.
5. Using a clean dry chemical cloth, wipe down the probe. Ensure that the probe is thoroughly dry and free from contamination.

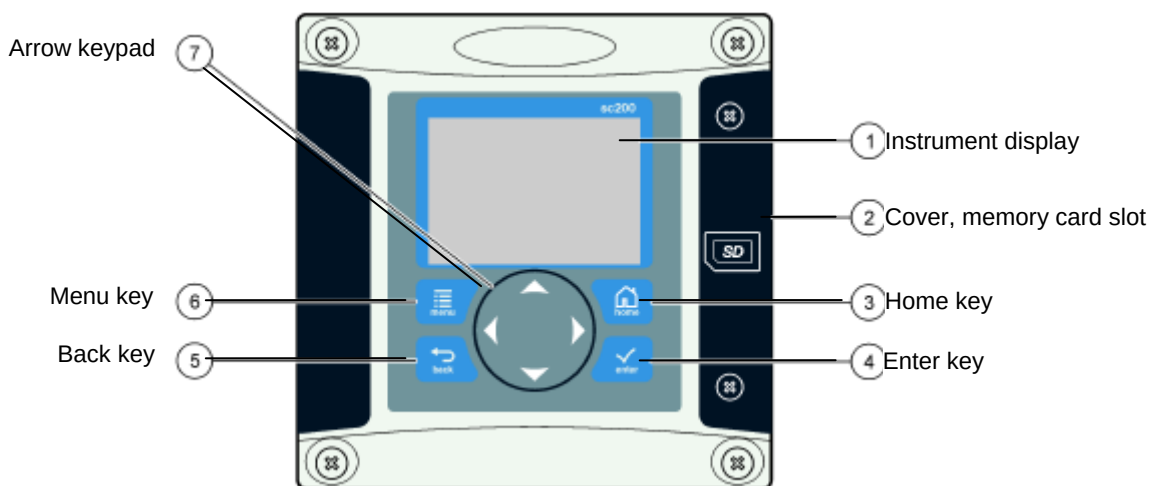
Zero Point Calibration of Conductivity Sensor

IMPORTANT

Zero Point calibration must be performed before the Span Point calibration procedure.

1. Ensure that the probe is properly connected to the Hach sc200 controller in the Controls Compartment of the Inline Blender, with Integrated Daytanks.
2. Place the clean, dry sensor in an area free from anything that would interfere with calibration, and ensure that the sensor is open in air.
3. Press the Menu key on the Hach sc200 controller.
4. Using the arrow keypad, scroll down to highlight **SENSOR SETUP** and push the Enter key.
5. Using the arrow keypad, scroll down to highlight the sensor you are calibrating and push Enter.

Hach sc200 Controller



6. Highlight **CALIBRATE** on the *COND SETUP* screen and press Enter.
7. Highlight **ZERO CAL** and press Enter.
8. Highlight **ACTIVE** and press Enter.
9. With **MOVE DRY SENSOR TO DRY AIR. PUSH ENTER TO CONTINUE** displayed, again make sure the sensor is dry, in air, and located in an area without interference. Press Enter. The display shows **PLEASE WAIT**. When done, the calibration results are displayed.
10. Review the calibration results.

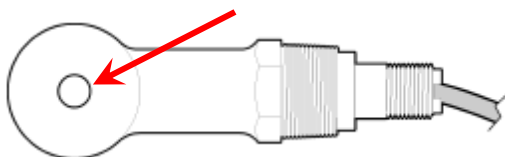
11. If the calibration passed, skip this step. If the calibration failed, return to step 9.
12. When this calibration passes, **NEW SENSOR?** is displayed. Highlight **YES** or **NO** and press **Enter**.
 - Select **YES** to reset the days of operation and previous calibration history.
 - Select **NO** to *not* reset the days of operation and previous calibration history.
13. Press **Enter** when this is displayed: **RETURN SENSOR TO PROCESS. PUSH ENTER TO CONTINUE.** (Do *not* return the sensor to chemical.)
14. Press the **Home** key. The value on the Instrument Display (#1 above) should read **0.000 mS/cm**.
15. Continue calibration by performing the following Span Point Calibration procedure.

Span Point Calibration of Conductivity Sensor

IMPORTANT

The preceding Zero Point Calibration procedure must be performed before this Span Point Calibration procedure is performed.

1. Thoroughly rinse the sensor with UPW. Dry the sensor completely using a clean, lint-free cloth.
2. Support the sensor such that it does not touch anything and there is room to insert the resistor loop and connect the electrical pins without any impediment.
3. Insert the 10.0 Ω reference resistor loop (listed in [Special Tools](#) section above) through the center of the sensor toroid (donut ring).



4. Press the **Menu** key on the Hach sc200 controller.
5. Using the arrow keypad, scroll down to highlight **SENSOR SETUP** and press the **Enter** key.
6. Using the arrow keypad, scroll down to highlight the sensor you are calibrating and press **Enter**.
7. Highlight **CALIBRATE** on the **COND SETUP** screen and press **Enter**.
8. Highlight **COND CAL** and press **Enter**.
9. Highlight **ACTIVE** and press **Enter**.
10. With **MOVE DRY SENSOR TO SAMPLE. PUSH ENTER TO CONTINUE** displayed, press **Enter**. (Do not move the sensor.)

11. During the span point calibration operation, this is displayed: **PUSH ENTER WHEN STABLE . VALUE : [current conductivity reading] TEMP : [current temperature reading]**. Wait for the controller to determine that the values are stable. When this occurs, the *ENTER SAMPLE* screen is displayed with this is displayed: **PUSH [arrow indicators] TO CHANGE VALUE [+470.000 mS/cm, for example]**.
 12. Using the arrow keypad, enter +470.000 mS/cm and press **Enter**.
 13. The calibration result is displayed on the *COND CAL* screen:
 - If this is displayed, calibration succeeded. Press **Enter** and skip to step 14: **CALIBRATION: PASS CELL VAL : [4.636, for example] PUSH ENTER TO CONTINUE .**
- NOTE**
- The CELL VAL is the cell constant. Even when displayed later with only two decimal places, the Hach sc200 controller does not round up. The full value of the cell constant is used.**
- If this is displayed, calibration failed: **CALIBRATION FAILED. CAL FACTOR OUT OF RANGE . PUSH ENTER TO REPEAT.** Press **Enter**. Ensure that the sensor is dry and that the correct resistor is being used. Then, return to step 3.
 14. When this calibration passes, **NEW SENSOR?** is displayed. Highlight **Yes** or **No** and press **Enter**:
 - Select **Yes** to reset the days of operation and previous calibration history.
 - Select **NO** to *not* reset the days of operation and previous calibration history.
 15. Press **Enter** when this is displayed: **RETURN SENSOR TO PROCESS . PUSH ENTER TO CONTINUE .** (Do *not* return the sensor to chemical.)
 16. Press the **Home** key. The value on the Instrument Display (#1 above) should read close to **470 .000 mS/cm**.
 17. Remove the 10.0 Ω reference resistor loop from the sensor toroid. The value on the display should read **0 .000 mS/cm**.

Verify Calibration and Repeat for Second Sensor (If Present)

1. To verify the calibration, insert a 100 Ω (0.05% tolerance) reference resistor into the sensor toroid. Record the value, which should be very close to 47.0 Ω .
2. If there is a second conductivity sensor used with the same Hach sc200 controller, repeat the *Zero Calibration* and *Span Calibration* procedures on that sensor (probe) using the same 10 Ω and 100 Ω reference resistors.
3. Remove the resistor.

Return Probe to Chemical

1. Carefully remove the sensor from the cell housing by loosening the plug fitting.
2. Using a clean dry chemical cloth, wipe down the probe. Ensure that the probe is thoroughly dry and free from contamination.
3. Carefully return the probe(s) to the cell housing(s). Align the probe in the housing where the slots are horizontal. Then, tighten the housing.
4. Reinstall the cell housings.

NOTE

If filter housings and assemblies were removed to gain access to the CT.M21.910 cell housing, check the filters. If the filters have become dry, rewet or replace the filter (depending on the style of the filter). Perform the required function and reassemble the filter process area.

NOTE

Before beginning the next step, validate the positions of the drain and sample valves to ensure proper flow path. IS THIS CONTROLLED BY MAINTENANCE MODE OR DONE MANUALLY? IF MANUALLY, WHICH VALVES AT WHAT POSITIONS?

5. Turn UPW flush valve on, rinse section for 30 minutes, and monitor for leaks. Tighten fittings if necessary.
6. Perform one of the following, applicable sub-procedures:
 - If sensor CT.M1.910A/CT.M1.910B (Wetted Compartment) was calibrated, perform the *UPW Flow thru Jx to Drain* procedure in [Chapter 4, System Operation and Process Procedures](#). Monitor for leaks. Tighten fittings, if necessary.
 - If sensor CT.M21.910 was calibrated:
 - a. Turn the UPW flush valve off, disconnect UPW line, cap and plug respective fittings, and leave drain valve open.
 - b. On the *Tank 1* [Tank – Recirculation/Metrology] screen force-on AOV.P21.810 to create the flow path for the incoming chemical to flow to the sample port, filter assembly and drain. Remove air by letting the section fill with chemical and flow to drain for additional 10 minutes.

- c. Close the drain valves and remove the forces: *Tank 1* screen force-off AOV.P21.810.
7. Put the Inline Blender, with Integrated Daytanks, and daytanks 1A and 1B in Auto mode.
8. Initiate a blend to the applicable tank.
9. After the qualified blend is flowing toward daytanks 1A and 1B, go to the tank and verify there are no leaks.
10. Allow the blend to fill daytanks 1A and 1B, until the tanks are topped off.
11. After the daytanks are full, recirculate the blend in the daytanks for one hour.
12. Before resuming normal operations, take a sample of the blend to confirm that the results are acceptable.
13. Return all modules to normal operating conditions.
14. On the HMI screen, return all modules of the Inline Blender, with Integrated Daytanks, to normal operating conditions.
15. Turn the **MAINTENANCE KEY SWITCH** to OFF.

View and Record Full Cell Constant Value (as required)

1. To observe the cell constant at the highest resolution (5 decimal places), press the **Menu** key at the Hach sc200 controller.
2. Using the arrow keypad, scroll down to highlight **SENSOR SETUP**. Press the **Enter** key.
3. Using the arrow keypad, scroll down to highlight the sensor being calibrated and press the **Enter** key.
4. Using the arrow keypad, scroll down to highlight **DIAG/TEST** on the *COND SETUP* screen. Press the **Enter** key.
5. Using the arrow keypad, scroll down to highlight **CAL HISTORY**. Press the **Enter** key.
6. On the *SELECT CAL RECORD* screen, the latest calibration record should be highlighted. You can select any calibration record by scrolling with the arrow key. When the intended calibration record is highlighted, press the **Enter** key.
7. Three lines down on the *CAL HISTORY* screen, the complete cell constant is displayed: **CELL CONST:** [4.71979, for example]. Record this value.

CPU Replacement

Purpose

The purpose of this procedure is to remove and replace the CPU.

Procedure

Procedure

Removal

1. Place the Inline Blender, with Integrated Daytanks, into Manual mode.
2. Turn the **MAIN POWER DISCONNECT SWITCH** to OFF.
3. Press the **EMO** button.
4. Open the Controls Compartment door.
5. Detach the Ethernet cable from the PLC CPU module.
6. Disconnect the cable from the adjacent module next to the CPU module.
7. Pull down on the retaining clip on the bottom backside of the module until the module is released from the DIN rail.
8. Tip the front of the CPU module up to lift from the DIN rail.

Installation

1. Hook the top of the CPU module on the DIN rail.
2. Push up on the retaining clip on the bottom backside of the module.
3. Connect the cable from the adjacent module.
4. Connect the Ethernet cable to the CPU module.
5. Visually inspect module to verify that it is properly seated.
6. Pull the EMO button to the “out” position.
7. Turn the **MAIN POWER DISCONNECT SWITCH** to ON.
8. Place the Inline Blender, with Integrated Daytanks, in the desired mode of operation.

Door Switch Replacement

Purpose

This procedure is used to replace a magnetic door interlock switch.

Precautions



WARNING



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY.

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Wear appropriate personal protective equipment when performing this procedure.

Procedure

1. Place the Inline Blender, with Integrated Daytanks, into Manual mode.
2. Open the affected compartment door (Wetted Compartment or the Sample Box).
3. Press the **ALARM SILENCER** button to clear any alarms.
4. Disconnect the door switch cable at its quick disconnect.
5. Remove the bracket and or fasteners holding the switch.
6. Install the replacement sensor in its bracket.
7. Re-connect the switch cable to its quick disconnect.
8. Close the compartment door.
9. Test the operation of the new interlock switch to verify correct alignment and operation.
10. Place the Inline Blender, with Integrated Daytanks, in the desired mode of operation.

Electrical Switch Replacement

Purpose

The purpose of this procedure is to remove and replace one of the following electrical switches:

- **MAIN POWER DISCONNECT SWITCH**
- **EMO** pushbutton/switches
- **EMO RESET** button
- **ALARM SILENCER** button
- **SAMPLE ENABLE PUSH AND HOLD** button
- **DIW BOOSTER PUMP ENABLE PUSHBUTTON**
- **DIW BOOSTER PUMP DISCONNECT SWITCH**

Precautions



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY

Personnel performing this procedure on a rear panel switch will be exposed to chemical that has no secondary containment. Wear proper safety gear when performing this procedure.

Procedure

Removal

1. Place the Inline Blender, with Integrated Daytanks, in Stop mode.
2. Turn the **MAIN POWER DISCONNECT SWITCH** to OFF.
3. Press the **EMO** switch.
4. Place Main Power circuit breaker CB203 in the “off” (down) position.

IMPORTANT

If replacing the DIW BOOSTER PUMP DISCONNECT SWITCH, before proceeding also:

- Place circuit breaker CB203 in the “off” (down) position.
 - Place the DIW BOOSTER PUMP ENABLE PUSHBUTTON in the OFF position.
5. Remove all power coming to the Inline Blender, with Integrated Daytanks, from the power distribution panel.
 6. Note the orientation of wiring, and any washers or gaskets, to the defective button or switch.
 7. Disconnect wiring from the defective button/switch.
 8. **If the defective switch has a plastic knob**, remove (unscrew counterclockwise) and discard it.
 9. Remove (unscrew counterclockwise) and discard the knurled metal knob or ring nut that attaches the button/switch to the panel. Remove and discard the button/switch.

Installation

1. Remove the plastic knob (if applicable), the knurled metal knob or ring nut, and the washer from the replacement button/switch.

NOTE

If replacing the MAIN POWER DISCONNECT SWITCH, it must be installed in the OFF position, with the ON position accessible.

2. Place the replacement button/switch and any gaskets or washers in position on the panel. Attach with the ring nut and metal washer and tighten.
3. Screw on (clockwise) the plastic handle/knob (if separate).
4. Attach wires to button/switch as noted in removal procedure.
5. Restore all power coming to the blender from the power distribution panel.
6. Place the main circuit breaker in the ON position.
7. Turn the **MAIN POWER DISCONNECT SWITCH** to ON.
8. Verify proper button/switch operation.
9. Place the Inline Blender, with Integrated Daytanks, in the desired mode of operation.

Ethernet Switch and Router Replacement

Purpose

The purpose of this procedure is to remove and replace either or both of the following components:

- Ethernet switch ES436
- RS4000 router MGRD445

Precautions



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY.

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Wear appropriate personal protective equipment when performing this procedure.

Procedure

Removal

1. Place the Inline Blender, with Integrated Daytanks, into Manual mode.
2. Press the **EMO** switch.
3. Turn the **MAIN POWER DISCONNECT SWITCH** to OFF.
4. Open the Controls Compartment door.
5. Using a small ladder, access the Ethernet switch or the adjacent MGRD445.
6. Label the cables and wires, as necessary, prior to removing them. Use the electrical schematic for guidance.
7. Disconnect all wires and cables from ES436 or MGRD445.
8. Rotate ES436 or MGRD445 upward and away from the DIN rail on which it is mounted.
9. Remove ES436 or MGRD445 from the Controls Compartment.

Installation

1. Position the top hook of the replacement ES436 or MGRD445 on the DIN rail.
2. Rotate the module downward until it snaps into position on the DIN rail.
3. Reconnect all wiring, using the electrical schematic for guidance.
4. After the installation is completed, close the Controls Compartment door.
5. Pull the **EMO** button to its “out” position.
6. Turn the **MAIN POWER DISCONNECT SWITCH** to ON.
7. Press the **EMO RESET** button.
8. Place the Inline Blender, with Integrated Daytanks, in the desired mode of operation.

Exhaust Pressure Indicator Replacement

Purpose

The purpose of this procedure is to remove, replace and adjust the Magnehelic exhaust pressure indicator (PI.Z1.491), located on top of the cabinet.

Precautions



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY.

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Wear appropriate personal protective equipment when performing this procedure.

Procedure

Removal and Replacement

1. Press the Inline Blender, with Integrated Daytanks' **EMO** switch.
2. Turn the **MAIN POWER DISCONNECT SWITCH** to OFF.
3. Use a ladder to reach the pressure indicator at the top front of the cabinet.
4. Open the access door nearest the switch. Separate the wires attached to the pressure indicator.
5. Remove the mounting screws that attach the pressure indicator and indicator to its mounting bracket.
6. Install a new pressure indicator into its mounting bracket. Tighten the mounting screws.
7. Connect the wires to the correct positions on the pressure indicator.
8. Turn the **MAIN POWER DISCONNECT SWITCH** to ON.
9. Pull the **EMO** switch to the "out" position.
10. Press the **EMO RESET** button.
11. Perform the following indicator adjustment procedure.

Adjustment

1. Place a hot wire anemometer or other flow-measuring device in the exhaust pipe.

NOTE

In the following step, the initial response time of the exhaust pressure indicator after adjustment is 1 to 10 seconds. Adjust the knob in small increments and wait for the response.

2. Adjust the facility exhaust supply pressure to the **minimum** flow and pressure required per the *Facilities* drawing provided in the *Prints* section.
3. When the flow indicated on the device in step 1 stabilizes at the minimum exhaust flow, use a small flat-head screwdriver to turn the adjustment knob (clockwise) on the exhaust pressure indicator until it indicates the minimum pressure per the *Facilities* drawing.
4. Open the cabinet exhaust damper to increase exhaust flow to the **operational** flow/pressure point.
5. Use the small flat-head screwdriver to turn the indicator's adjustment knob clockwise to agree with the detected **operational** flow/pressure.
6. Place the Inline Blender, with Integrated Daytanks, in the desired mode of operation.

Exhaust Pressure Switch Replacement

Purpose

The purpose of this procedure is to remove, replace, adjust and test the exhaust low pressure switch, PS.Z1.491, in the Controls Compartment.

Procedure

Removal

1. Place the blender in Stop mode.
2. Turn the **MAIN POWER DISCONNECT SWITCH** to OFF.
3. Press the **EMO** button.
4. Note the orientation of the exhaust lines to switch PS.Z1.491.
5. Note the orientation of the wiring to the switch.
6. Disconnect the wiring from the switch.
7. Remove the switch and attaching hardware.

Installation

1. Attach the replacement vacuum switch to the panel with mounting hardware, in the same orientation noted in the [Removal](#) procedure above.
2. Attach the wiring to the switch as noted in the [Removal](#) procedure above.
3. Pull the EMO button to the “out” position.
4. Turn the **MAIN POWER DISCONNECT SWITCH** to ON.
5. Press the **EMO RESET** button.
6. Place the Inline Blender, with Integrated Daytanks, in the desired mode of operation.

Adjustment/Test

1. Note the value of the exhaust flow prior to adjustments.

NOTE

The value of the exhaust flow prior to adjustments

2. Place a hot wire anemometer or other flow-measuring device in the exhaust pipe.
3. Adjust the facility exhaust damper to at least the minimum flow and pressure required per the applicable *Facilities* drawing listed in the *Prints* section of this manual.
4. Use a small flat-head screwdriver to turn the adjustment knob (clockwise) on the exhaust pressure indicator, until the LED on the indicator is lit.

NOTE

The initial response time of the exhaust pressure indicator after adjustment is 1-10 seconds. Adjust the knob in small increments and wait for the response.

5. Open the cabinet exhaust damper to increase exhaust flow to the point where the typical facilities exhaust fluctuation will not cause the LED on the exhaust pressure indicator to change state.
6. Turn the adjustment knob counter-clockwise until the LED turns off at a reliable setting.
7. Verify that the calibration is correct by lowering the exhaust flow rate below the set point and check to see that the LED lights up. Then increase the flow rate above the set point and verify that the LED turns off.
8. Place the Inline Blender, with Integrated Daytanks, in the desired mode of operation.

Flare Fitting Tightening Requirements: SMC LQ3

Purpose

The purpose of this procedure is to properly tighten the SMC Series LQ3 Hyperflare fittings after they are removed and replaced.

Precautions



WARNING



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY.

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Wear proper safety gear when performing this procedure.

Prerequisites

Ensure all plumbing to be accessed has its contents completely depressurized. Ensure proper PPE is worn.

Procedure

IMPORTANT

Whenever a chemical leak is noticed at a fitting, tighten the fitting only once. If the leak is not corrected, remove and replace the tubing and fitting.

1. Hand-tighten the LQ3 nut until it touches the baseplate of the body.
2. There should be no gap between the bottom edge of the nut and the fitting body. A wrench can be used gently on the wrench flats to achieve a snug fit.

Flare Fitting Torque Requirements: Flaretek

Purpose

The purpose of this procedure is to properly apply torque to the Entegris Flaretek fittings after they are removed and replaced.

Precautions

**WARNING**

CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY.

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Wear proper safety gear when performing this procedure.

Prerequisites

- Ensure all plumbing to be accessed has its contents completely depressurized.
- Ensure proper PPE is worn.
- Flaretek Torque Verification Device (see Appendix A, [Maintenance Supplies](#)).

Procedures

IMPORTANT

Whenever a chemical leak is noticed at a fitting, tighten the fitting only once. If the leak is not corrected, remove and replace the tubing and fitting.

Use a Flaretek torque verification device or a torque wrench to achieve the correct torque values, listed in [Table 8-4](#) for the torque values for the Flaretek fittings. Refer to the table for the minimum and maximum torque values for the Flaretek fittings.

Table 8-4. Torque Values for Flaretek Fittings

PFA Tubing Size	Minimum Torque Value	Maximum Torque Value
1/4 inch	4 inch-lbs	5 inch-lbs
3/8 inch	8 inch-lbs	12 inch-lbs
1/2 inch	11 inch-lbs	16 inch-lbs
3/4 inch	14 inch-lbs	28 inch-lbs
1 inch	30 inch-lbs	45 inch-lbs
1.25 inch	80 inch-lbs	85 inch-lbs

Filter Replacement, Chemical

Purpose

This procedure covers the replacement of the filters in the right and left compartments of the blender.

Precautions



WARNING



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY.

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Wear appropriate personal protective equipment when performing this procedure.

Special Tools

- Filter wrench (Filter wrench (refer to [Appendix A, Maintenance Supplies](#), for part numbers) for filter housing, which has a PTFE collar nut.
- Kalrez o-ring.

Prerequisites

Proper PPE.

Procedure

1. Place the Inline Blender, with Integrated Daytanks, in Manual mode.
2. Perform [UPW Flush \(Pump\) to Drain](#) to flush filter F21 with UPW for 15 minutes. See the procedure in [Chapter 4, System Operation and Process Procedures](#).
3. Remove the filter bowl bracket and remove the filter bowl.
4. Remove the filter and install a new filter.
5. Inspect the bowl o-ring and replace it, if necessary.
6. Replace the filter bowl and support bracket.
7. Start pump P21 for filter F21 and flow chemical through the filter. Allow chemical to run through it for 10 minutes.
8. Place the Inline Blender, with Integrated Daytanks, in Auto mode.

Filter Replacement: Nitrogen

Purpose

The purpose of this procedure is to remove and replace N₂ filter (F.N1.950), which is located in the Controls Compartment on the N₂ supply line.

Procedure

Removal

1. Turn the **MAIN POWER DISCONNECT SWITCH** to OFF.
2. Close the upstream facility N₂ supply valves and install a LOTO device on them.
3. Ensure that the N₂ pressure switch indicates 0 psig.
4. Note the direction of the N₂ flow on the N₂ filter before removing it.
5. Disconnect the inlet and outlet lines from the N₂ filter.
6. Remove the nitrogen filter from its attaching clamp and discard the N₂ filter.

Installation

NOTE

Ensure that filter is installed with correct flow direction.

1. Attach the N₂ filter to the N₂ filter clamp.
2. Connect the inlet and outlet lines to the N₂ filter.
3. Remove the LOTO device and slowly open the facility N₂ line. Check for leaks.
4. Turn the **MAIN POWER DISCONNECT SWITCH** to ON.
5. Turn the regulator knob to the operational pressure reading provided in [Appendix D, System Settings](#).
6. Place the Inline Blender, with Integrated Daytanks, in the desired mode of operation.

Float Switch Replacement

Purpose

The purpose of this procedure is to replace a defective float switch. See [Chapter 3, Component Description and Location](#), for the listing of float switches.

Precautions

**WARNING**

CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Wear proper safety gear when performing this procedure.

Procedure

Removal

1. Place the Inline Blender, with Integrated Daytanks, in Manual mode.
2. Press the **EMO** button at the Inline Blender, with Integrated Daytanks.
3. Turn the **MAIN POWER DISCONNECT SWITCH** to OFF.
4. Open the front door of the Wetted Compartment and locate the float switch to be replaced.
5. Disconnect the switch's cable connector.
6. Remove the screws securing the float switch and bracket assembly to the cabinet.
7. Remove the float switch from its mounting bracket.

Installation

1. Install the new float switch in its mounting bracket.
2. Attach the switch and bracket assembly to the cabinet with screws.
3. Reconnect the switch's cable connector.
4. Pull the **EMO** button to the "out" position.
5. Turn the **MAIN POWER DISCONNECT SWITCH** to ON.
6. Press the **EMO RESET** button.

Flow Controller Replacement

Purpose

The purpose of this procedure is to replace flow controller J1 or J99.

Precautions

Ensure that the system is off line and depressurized.

Procedure

Removal

1. Place the Inline Blender, with Integrated Daytanks, in Manual mode.
2. At the *Main Menu* screen, press the MAINTENANCE softkey.
3. Perform the [UPW Flush Flow Controllers to Drain](#) procedure in [Chapter 4, System Operation and Process Procedures](#).
4. After the service routine stops, push in the **EMO** switch
5. Turn **MAIN POWER DISCONNECT SWITCH** to OFF.
6. Disconnect the communications cable on the flow controller from the Controls Compartment (pigtail style). Note the color and location of each termination point. Disconnect the cable connector near the base of the flow controller (connector style).
7. Disconnect the fluid inlet and outlet tube to the flow controller.
8. Remove the flow controller from the mounting panel within the blender.

Installation

1. Mount the new flow controller to the mounting panel in the blender.
2. Route the communications cable into the electrical compartment and terminate all wires (pigtail-style). Reconnect the cable connector near the base of the flow controller (connector-style).
3. Connect the fluid inlet and outlet tubes.
4. Turn the **MAIN POWER DISCONNECT SWITCH** to ON.
5. Pull the **EMO** switch to the “out” position.
6. Place the Inline Blender, with Integrated Daytanks, in the desired mode of operation.

HMI Panel: Replacement

Purpose

The purpose of this procedure is to remove and replace the General Electric 12-inch QuickPanel+ HMI panel.

NOTE

Two people are required to remove or replace the panel. One person is needed to hold the panel in place while the other person removes or installs attaching hardware and cabling.

Procedure

Removal

1. Place the Inline Blender, with Integrated Daytanks, in Manual mode.
2. Turn the **MAIN POWER DISCONNECT SWITCH** to OFF.
3. Press the **EMO** button.
4. Open the Controls Compartment door.
5. Remove the HMI-backup CompactFlash card from the back of the HMI panel.
6. Disconnect the interface cable from the back of the HMI panel.
7. Disconnect the power harness wiring from the rear of the HMI panel.
8. Remove the wire ties holding wiring to the back of the HMI panel.
9. With a second person holding it in place, remove the mounting hardware that attaches the HMI panel to the door and remove the HMI panel.

Replacement

1. Insert the HMI-backup CompactFlash card from the back of the HMI panel.
2. Attach the panel, with a second person holding it in place, as follows:
 - a. Insert the HMI panel into the opening.
 - b. Attach the HMI panel with the mounting hardware.
 - c. Tighten the mounting hardware to secure the HMI panel in place.

IMPORTANT

In the following step, ensure that the cable is connected to the same connector. If it is connected to the wrong connector, it might corrupt the HMI software.

3. Connect the interface cable to the connector on the rear of the HMI panel.
4. Connect the power harness wiring to the rear of the HMI panel.
5. Use wire ties to attach connecting wires to the wire supports on the back of the HMI panel. Add wire ties as desired for optimal use and maintenance.
6. Pull the **EMO** button to the “out” position.
7. Turn the **MAIN POWER DISCONNECT SWITCH** to ON.
8. Press the **EMO RESET** button.
9. Reset the date and time, user list display, and other startup parameters according to site procedures.
10. Place the Inline Blender, with Integrated Daytanks, in the desired mode of operation.

HMI Panel: Loading SDMemory Files

Purpose

The purpose of this procedure is to locate and load the SDMemory files to a General Electric 12-inch QuickPanel+ HMI panel.

The SDMemory files must be loaded every time that a GE QuickPanel HMI panel is replaced.

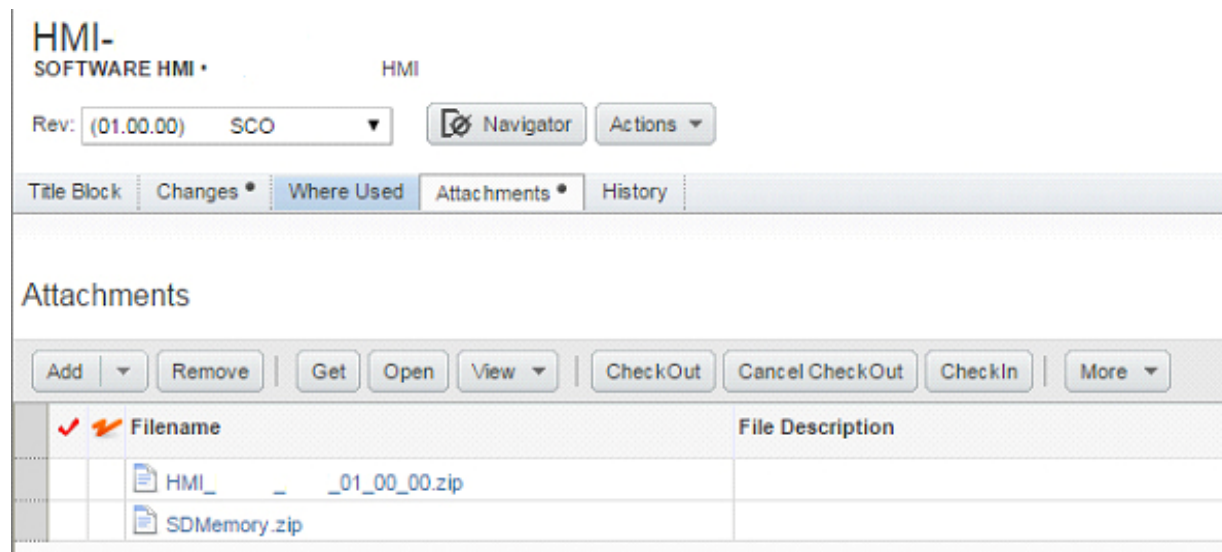
Prerequisites

Air Liquide Agile password and knowledge of how to navigate through Agile.

Procedure: Accessing and Loading the SDMemory Files

1. Log onto the Air Liquide Agile database.
2. Locate the SDMemory file, as an attachment, at the “HMI” item within the system’s Software package. See [Figure 8-2](#) below.

Figure 8-2. Locating the SDMemory.zip file in Agile (typical screen, without customer-specific data)



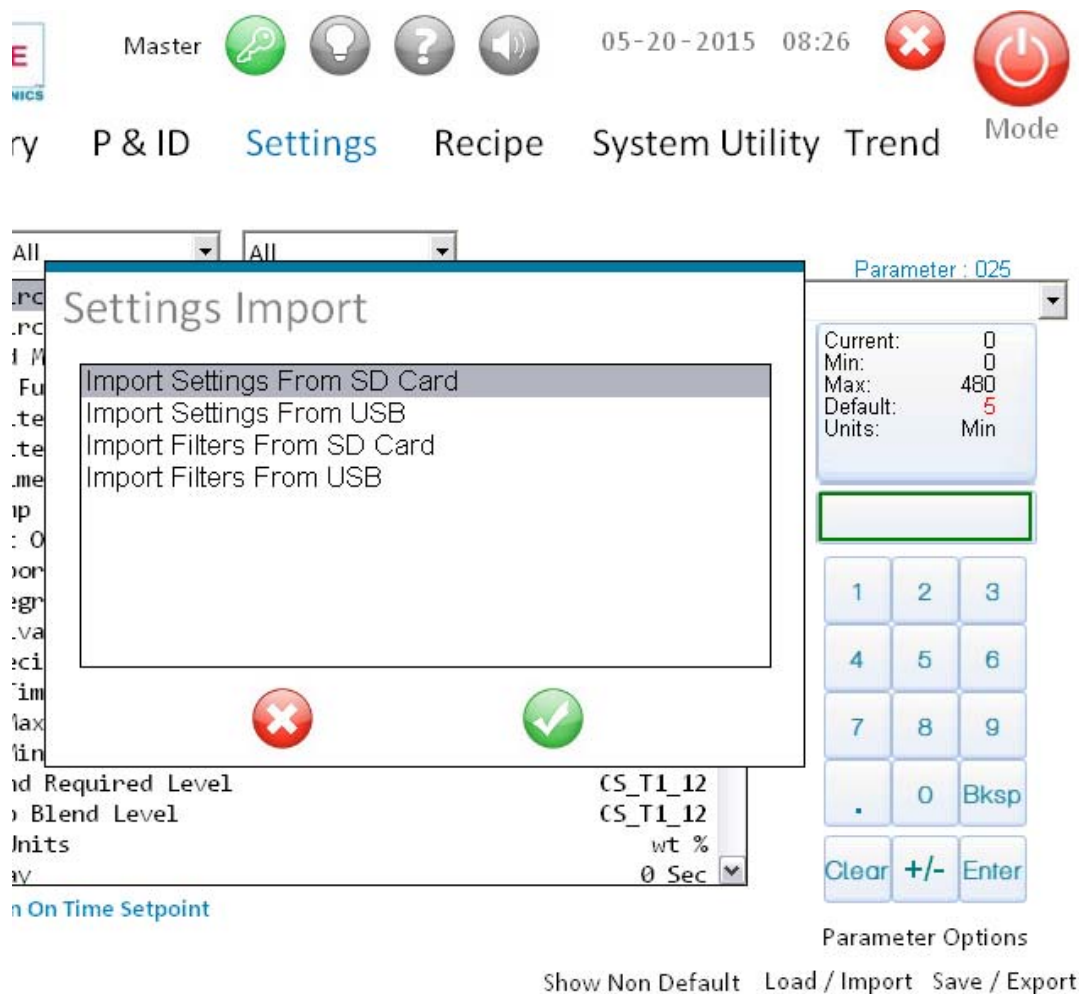
3. Make sure that there are no archived versions of the SDMemory file within the same folder as the new SDMemory file to be loaded.
4. After putting the .zip file in the folder on your computer, right-click and choose >7-ZIP > Extract here.
5. On the HMI, close the Runtime by hitting the red “X” beside the Mode button in the upper right corner (If the red “X” button is not visible, log-in to the system at a higher user-level, if permissible).

Figure 8-3. Location of the red "X" button



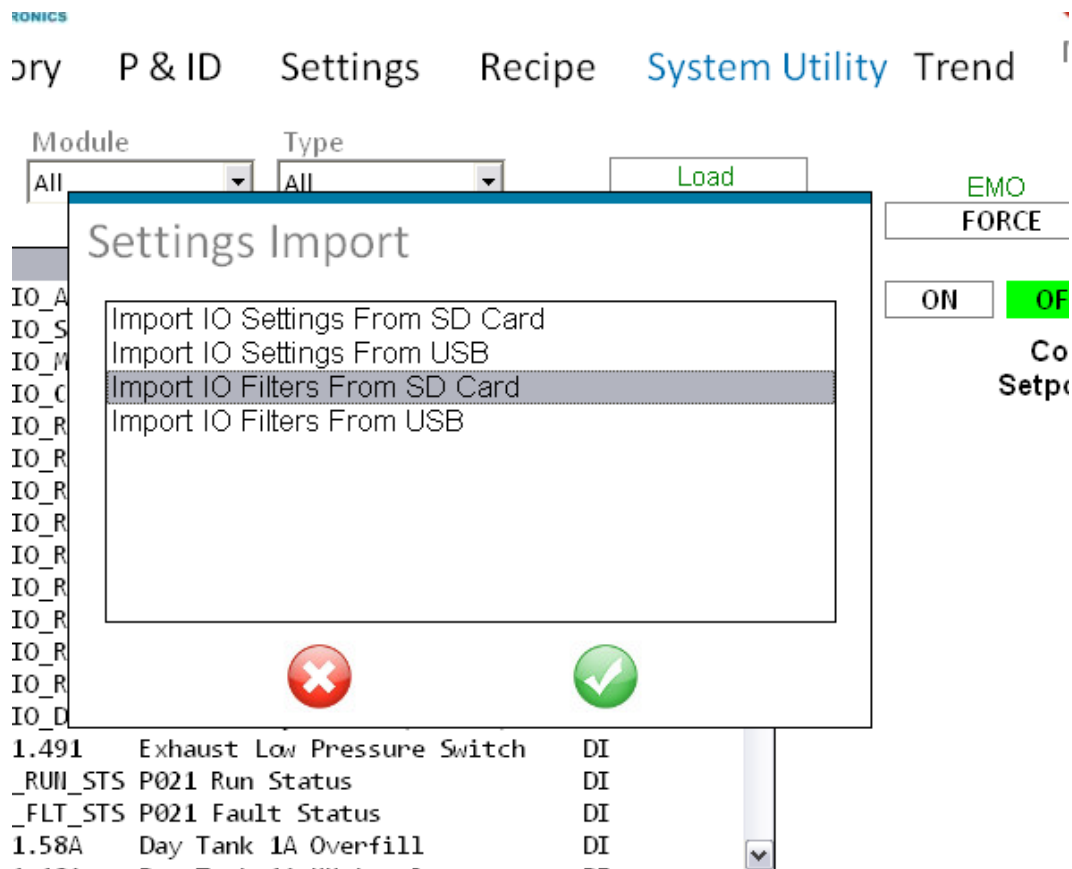
6. When the runtime is finished, remove the SD Card from the HMI and plug it into your laptop.
7. Using either your computer's SD port or a separate SD card reader, transfer the **SDMemory** file directly onto the card.
8. Re-install the SD card into the HMI panel.
9. On the HMI, log-in at the appropriate level to see the **Settings Options** in the bottom right corner of the screen. Choose > **Settings**.
10. Allow the settings to load. Then, choose > **Load/Import > Import Settings From SD Card**.
11. After the settings are loaded perform the same task for > **Import Filters From SD Card**. See [Figure 8-4](#) on page 8-41.

Figure 8-4. Import Settings From SD Card selection



12. After the function is finished, choose System Utility, Allow I/O to load. Then choose Import > Import IO Settings From SD Card > Yes at prompt box.
13. After the settings are loaded perform the same task for > Import IO Filters From SD Card. See [Figure 8-5](#) on page 8-43.

Figure 8-5. Import IO Filters From SD Card selection



PLC Input or Output Module Replacement

Purpose

The purpose of this procedure is to remove and replace a PLC input or output module.

Procedure

Removal

1. Place the Inline Blender, with Integrated Daytanks, in Manual mode.
2. Turn the **MAIN POWER DISCONNECT SWITCH** to OFF.
3. Press the **EMO** button.
4. Open the Controls Compartment door.
5. Identify the module to be replaced.
6. Remove the ribbon cable jumpers on either side of the PLC module.
7. Remove all wires from the PLC module. Make sure all wires are labeled before removal, and make note of wire terminations.
8. Pull down on the grey retaining clip on the bottom backside of the module, until the module is released from the DIN rail.
9. Tip the front of the PLC module up to lift from the DIN rail.

Replacement

IMPORTANT

Verify that the replacement module is the same as the module it is replacing.

1. Place the replacement PLC module on the DIN rail.
2. Push up on the retaining clip on the bottom backside of the module.
3. Connect the ribbon cables on both sides of the module.
4. Connect all wires to the PLC module, in accordance with the electrical schematic listed in the [Prints](#) section.
5. Visually inspect module to verify that it is properly seated.
6. Pull the **EMO** button to the “out” position.
7. Turn the **MAIN POWER DISCONNECT SWITCH** to ON.
8. Place the Inline Blender, with Integrated Daytanks, in the desired mode of operation.

Power Supply Replacement

Purpose

The purpose of this procedure is to remove and replace the 24 Vdc, 10A power supply PS215.

Procedure

Removal

1. Place the Inline Blender, with Integrated Daytanks, into Manual mode.
2. Turn the **MAIN POWER DISCONNECT SWITCH** to OFF.
3. Press the **EMO** button.
4. Open the Controls Compartment door.
5. Locate the power supply at the top left of the compartment.
6. Label, if necessary, the wires from the power supply, prior to removing them from the terminal connectors. Use the electrical schematic, listed in the [Prints](#) section, for guidance.
7. Remove all wires from the power supply.
8. Remove the gray end-stop at the end of the DIN rail.

NOTE

If PS215 is being removed, first remove circuit breaker 215A.

9. Lift and rotate the power supply up and off the DIN rail.
10. Carefully lower and remove the power supply from the controls compartment.

Installation

1. Hook the back of the replacement power supply onto the DIN rail.
2. Rotate the power supply downward, until it snaps into position on the DIN rail.
3. Use the gray end-stop to secure the power supply on the DIN rail.
4. Reconnect all wiring, using the electrical schematic for guidance.
5. After the installation is completed, close the Controls Compartment door.
6. Pull the **EMO** button to its “out” position.
7. Turn the **MAIN POWER DISCONNECT SWITCH** to ON.
8. Press the **EMO RESET** button.
9. Place the Inline Blender, with Integrated Daytanks, in the desired mode of operation.

Pressure Regulators: Replace & Adjust

Application

This procedure applies to the CDA and N₂ pressure regulating valves and backpressure regulators.

The purpose of this procedure is to remove and replace pressure regulating valves and backpressure regulators that are used to regulate CDA and N₂ pressure.

See [Chapter 3, *Component Description and Location*](#), for a listing of pressure regulating valves.

Purpose

The purpose of this procedure is to remove, replace and adjust the applicable component listed above.

Procedure

Removal

1. Place the Inline Blender, with Integrated Daytanks, in Manual mode.
2. Turn the **MAIN POWER DISCONNECT SWITCH** to OFF.
3. Press the **EMO** button.
4. Open the door to the applicable compartment.
5. Identify the component to be removed. (See [Chapter 3, *Component Description and Location*](#), for guidance).
6. Close the facility supply line to the defective regulator.
7. Remove pressure from line to be serviced by adjusting the controlling pressure regulating valve to obtain a reading of 0 psi on the pressure indicator or indicating pressure switch.

NOTE

Pressure indicators or indicating pressure switches only indicate pressure downstream from a regulator. Pressure can remain upstream.

8. Verify that pressure does not exist in the line before loosening any connections.
9. Remove the UN₂ or CDA pilot lines to the component.
10. Remove the defective component.

IMPORTANT

Do not separate the indicator/regulator from the isolator (if so equipped). Existing connectors and fittings can be cleaned and reused. Remove any thread sealer from fitting threads. Do not remove or tighten isolators installed on replacement indicator/ regulator.

Installation

1. Wrap threads on the replacement component with Teflon tape.
2. Install the replacement component.
3. Open the facility supply line (if it is closed).
4. Check connections on replacement component for leaks.
5. Adjust the controlling pressure regulator to obtain a pressure indicator or indicating pressure switch reading noted in the [Removal](#) procedure.
6. Turn the **MAIN POWER DISCONNECT SWITCH** to ON.
7. Clear alarms using the HMI panel.
8. Turn the **EMO** switch to the “out” position.
9. Place the Inline Blender, with Integrated Daytanks, in the desired mode of operation.

Pressure Indicator or Indicating Pressure Switch Replacement & Adjustment

Purpose

The purpose of this procedure is to remove, replace and adjust a pressure indicator (PI) or indicating pressure switch (IPS). See [Chapter 3, Component Description and Location](#), for a listing).

Procedure

Removal

1. Place the Inline Blender, with Integrated Daytanks in Manual mode.
2. Turn the **MAIN POWER DISCONNECT SWITCH** to OFF.
3. Press the **EMO** button.
4. Open the door to the applicable compartment.
5. Identify the pressure switch to be removed. (See [Chapter 3, Component Description and Location](#), for guidance).
6. Turn the corresponding regulator until the pressure indicator or indicating pressure switch reads 0.
7. Disconnect the corresponding service line from the pressure indicator or indicating pressure switch.

NOTE

In the following step, cut cable ties as necessary to free the switch wiring.

8. Unplug the wired connection on the pressure indicator or indicating pressure switch.
9. Loosen the switch mounting bracket and remove the pressure indicator or indicating pressure switch.

Replacement

1. Position the pressure indicator or indicating pressure switch in the mounting bracket and tighten the mounting screws.

NOTE

In the following step, add cable ties to the wiring, as necessary, to secure wires properly.

2. Plug-in the switch wiring.
3. Connect the corresponding service line to the pressure indicator or indicating pressure switch.

Program a Digital Indicating Pressure Switch

Figure 8-6. Digital Pressure Switch

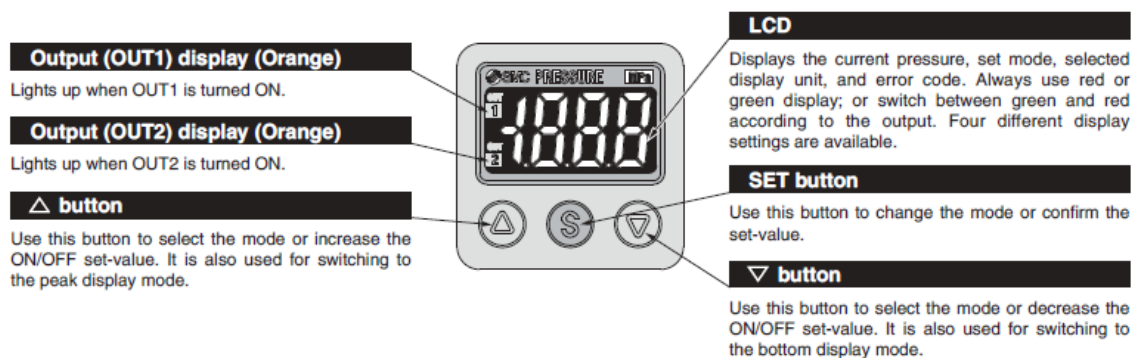


Table 8-5. Pressure Switch Error Codes

Error Code	Description	Notes and Corrective Action
Er1	(OUT1) Over 80 mA load current	Turn off power, remove source of over current, and re-input power.
Er2	(OUT2) Over 80 mA load current	Turn off power, remove source of over current, and re-input power.
Er3	Residual pressure error. When performing a zero reset, pressure differences have occurred.	Re-perform a zero reset after reinstating the unit to atmospheric pressure.
Er0	Internal data error	Turn off power, and re-input power.
Er4	Internal data error	Turn off power, and re-input power.
Er6	Internal data error	Turn off power, and re-input power.
Er7	Internal data error	Turn off power, and re-input power.
Er8	Internal data error	Turn off power, and re-input power.
Er9	Internal data error	Turn off power, and re-input power.
HHH	Supply pressure exceeds the maximum set pressure	Reset applied pressure to a level within the set pressure range.
LLL	Supply pressure is below the minimum set pressure	Reset applied pressure to a level within the set pressure range.

Initial Setting Procedure

Before using the SMC indicating pressure switch, review the following components of the switch:

- The LCD display shows the current pressure condition, setting mode, display units, and error codes.
 - The ▲(up) button is used to change mode or increase the ON/OFF set value.
 - The ▼ (down) button is used to change mode or decrease the ON/OFF set value.
 - The **SET** button is used to switch modes and input the set value.
1. At the pressure switch, push and hold the **SET** button for at least two seconds. This action places the switch in Initial Set mode.
 2. Measurement Unit Selection: Repeatedly press the **UP** or **DOWN** button until **PSi** is displayed and press the **SET** button.
 3. OUT1 Output Type: Press the **UP** or **DOWN** button until **1no** (Normally Open) is displayed and press the **SET** button.
 4. Response Time Set: Press the **UP** or **DOWN** button until **2.5** (2.5ms) is displayed and press the **SET** button.
 5. Pressure Set Mode Select: Press the **UP** or **DOWN** button until **mAn** (Manual set mode) is displayed and press the **SET** button. This returns the switch to Measurement Mode.
 6. Press the **SET** button for 1 second to enter the Pressure Set Mode.

7. P-1 Low Pressure Setpoint: Press the **UP** or **Down** button to enter the digits for the low-pressure limit point to the value in [Table 8-6](#) below. Press the **SET** button once and release to display set point 2.

8. P-2 High Pressure Setpoint: Press the **UP** or **DOWN** button to enter the digits for the high-pressure limit to the value in [Table 8-6](#) below.

9. Repeatedly press the **SET** button until the switch returns to Measurement Mode.

Table 8-6. Pressure Settings, Low and High

Component	P_1 [Low Pressure Setpoint]	P_2 [High Pressure Setpoint]
PRV.A1.951 N ₂ pilot pressure switch to solenoids	65 psi	85 psi
PRV.N1.950 N ₂ supply pressure switch	20 psi	30 psi

Pressure Transmitter Replacement

Purpose

The purpose of this procedure is to remove and replace an NT4100 pressure transmitter: PT.M21.910 or PT.Q1.910.

Precautions



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY

Personnel performing this procedure will be exposed to chemical that has no secondary containment. The transducer must be treated as if chemical is present, as it may have either contained chemical or been in contact with chemical. Wear proper PPE when performing this procedure.

Prerequisites

1. Purge the chemical line to the pressure transducer and remove all chemical residues from the fittings.
2. Ensure that the chemical plumbing to the pressure transducer is completely isolated and depressurized.
3. Ensure that the pump associated with this transducer is offline.
4. Access the *Parameters Selection* or *Settings* screen of the HMI panel and confirm that the **Maint. Bypass Alarm** softkey is set for a sufficient time to complete the maintenance work.

Procedure

Removal

1. Open the Controls Compartment door. Turn the **MAINTENANCE BYPASS KEY SWITCH ON/OFF** to the ON position.
2. Open the door to the compartment that contains the pressure transducer. Disconnect the transducer cable at its quick disconnect.



Carefully and slowly perform the next step, as trapped pressure might remain in the connecting lines.

3. Slowly disconnect the Flaretek fittings from the pressure transducer.
4. Disconnect the mounting bracket from the body of the transducer, if applicable.
5. Remove the defective pressure transducer as follows:
 - a. Wrap absorbent wipes completely around the component connection to the chemical tubing.
 - b. Holding the tube against the connection, reach under the wipe and slowly unthread the flare nut until it is disconnected. Continue to hold the tube against the transducer.
 - c. Ensuring that absorbent wipes surround the connection, slowly pull the transducer away from the tube.
 - d. Clean up any spilled chemical, and, in a manner approved by the facility, dispose of any wipes that are saturated with chemical.

Replacement

IMPORTANT

When tightening Flaretek tube fittings, refer to the [Flare Fitting Torque Requirements: Flaretek](#) procedure on page 8-33.

1. Insert the pressure transducer into the flared fitting piping connection until both ends of the tubing are snug. Tighten the nut over the flare fitting, using the applicable torque pressures (inch-lbs).
2. Connect the electrical cabling to the replacement pressure transducer at the quick-disconnect.
3. Attach the mounting bracket to the transducer body, if applicable.
4. Close the Controls Compartment door.
5. In the controls compartment, turn the **MAINTENANCE BYPASS KEY SWITCH ON/OFF** to the OFF position.
6. At the HMI panel, enable the mix tank and the pump associated with the compartment where the maintenance was completed.

Pump Replacement: Levitronix BPS-i100

Application

This procedure applies to the Levitronix BPS-i100 pump (P1), which is used as the Chem1 booster pump.

Purpose

This procedure applies to Levitronix BPS-i100 pump P1, which is used to boost pressurized Chem1 from the Fab to flow controller J1.

Precautions



WARNING

CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY PERSONNEL PERFORMING THIS PROCEDURE WILL BE EXPOSED TO CHEMICAL THAT HAS NO SECONDARY CONTAINMENT.

Wear appropriate personal protective equipment when performing this procedure. Be aware that chemical might remain in the pump and plumbing lines.



WARNING

HIGH MAGNETIC FIELD STRENGTH OF PUMP IMPELLER

The Levitronix BPS-i100 contains a rotor magnet with high field strength. Persons who are assisted by electronic devices such as pacemakers need to keep 12 inches (30 cm.) away as to not incur contusions.

Procedure

Pump Removal

1. Perform an N₂ purge, with the applicable procedure below from [Chapter 4, System Operation and Process Procedures](#), for 2 minutes or for a desired length of time.
2. Open the Wetted Compartment access door. Silence any alarms.
3. Ensure power has been removed from the Levitronix BPS-i100 pump controller. Disconnect the two electrical cables that feed the pump.
4. Disconnect the chemical inlet and outlet lines from the pump.
5. Remove the pump and base assembly.
6. Loosen and remove the bolts that hold the pump onto its mounting base.
7. Separate the pump from its mounting base.

Pump Installation

1. Place the new pump on its mounting base.
2. Install the pump and base together into the applicable cabinet.
3. Connect the chemical inlet and outlet lines to the pump.
4. Connect the two electrical lines to the pump.
5. Close the Wetted Compartment access door.
6. Run the pump and check for leaks and proper pump operation.
7. As applicable, see the Levitronix BPS-i100 User Manual in [Appendix E, OEM Documentation](#), for guidance.

Pump Replacement: Levitronix BPS-600

Application

This procedure applies to Levitronix recirculation pump P21.

Purpose

The purpose of this procedure is to remove and replace a Levitronix Model BPS-600 centrifugal pump.

Precautions



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY PERSONNEL PERFORMING THIS PROCEDURE WILL BE EXPOSED TO CHEMICAL THAT HAS NO SECONDARY CONTAINMENT.

Wear appropriate personal protective equipment when performing this procedure. Be aware that chemical might remain in the pump and plumbing lines.



HIGH MAGNETIC FIELD STRENGTH OF PUMP IMPELLER

The Levitronix BPS-600 contains a rotor magnet with high field strength. Persons who are assisted by electronic devices, such as pacemakers, need to keep 12 inches (30 cm.) away as to not incur contusions.

Procedure

Pump Removal

1. Perform an N₂ purge, with the applicable procedure below from [Chapter 4, System Operation and Process Procedures](#), for 2 minutes or for a desired length of time.
2. At the side of the Inline Blender, with Integrated Daytanks, turn the **DIW BOOSTER PUMP DISCONNECT SWITCH** to OFF.
3. Open the Wetted Compartment door. Silence any alarms.
4. Ensure power has been removed from the Levitronix pump controller. Disconnect the two electrical cables that feed the pump.
5. Disconnect the chemical inlet and outlet lines from the pump.
6. Remove the pump and base assembly.
7. Loosen and remove the bolts that hold the pump onto its mounting base.
8. Separate the pump from its mounting base.

Pump Installation

1. Place the new pump on its mounting base.
2. Install the pump and base together into the applicable cabinet.
3. Connect the chemical inlet and outlet lines to the pump.
4. Connect the two electrical lines to the pump.
5. Close the Wetted Compartment access door.
6. On the applicable unit, turn the **DIW BOOSTER PUMP DISCONNECT SWITCH** to ON.
7. Press the **DIW BOOSTER PUMP ENABLE PUSHBUTTON**.
8. Run the pump and check for leaks and proper pump operation.
9. As applicable, see the Levitronix BPS-600 User Manual in [Appendix E, OEM Documentation](#), for guidance.

Sensors/Switches Test

Purpose

The purpose of this procedure is to verify that sensors and switches are working properly.

Precautions



WARNING

CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Wear proper safety gear when performing this procedure.

Procedure

For any component listed below, perform the operational test shown and verify the result. These tests/results assume liquid is not currently present at sensor/switch points.

Table 8-7. Sensor/Switches Test Procedures

Component Designators	Component Description	Operational Test/OK Result
Capacitive Sensors—NC: CS.T1.58A, CS.T1.58B	Daytank 1A and 1B level sensors at Overflow.	Place a finger on the sides of the sensor bracket to simulate liquid. Verify that the digital input LED is lit when there is no fluid.
Capacitive Sensors—NC: CS.T1.17A, CS.T1.17B, CS.T1.18A, CS.T1.18B	Daytank 1A and 1B level sensors at High, High-High.	Place a finger on the sides of the sensor bracket to simulate liquid. Verify that the digital input LED is lit when there is no fluid.
Capacitive Sensors—NC: CS.Z1.17, CS.Z1.58	Liquid in drain; Liquid in drain overflow level sensor; facilities waste drain overflow sensor.	Place a finger on the sides of the sensor bracket to simulate liquid. Verify that the digital input LED is lit when there is no fluid.
Capacitive Sensors—NO: CS.T1.11, CS.T1.12A, CS.T1.12B	Level sensors at Low and Low-Low for daytank 1A and 1B.	Place a finger on the sides of the sensor bracket to simulate liquid. Verify that the digital input LED is lit when fluid is present.
Float switches—NOHC: FS.Z5.8, FS.Z5.9	Inline Blender cabinet.	1. All float switches trigger Shutdown alarms. Disable the EMO signal to the SCADA first! 2. Lift the float. Verify digital input LED is unlit at high (leak) position.
Magnetic switches—NOHC: ZS.Z4.5, ZS.Z5.5A, ZS.Z5.5B	Door interlocks on Wetted and Sample compartments.	Open the door. Verify that the digital input LED is unlit. Close the door to verify door alignment.
Pressure switches—NO: IPS.N1.950	N ₂ pressure to daytank blankets.	Use the pressure regulator to reduce CDA pressure to the pumps, below the operating range (see Appendix D, System Settings). Verify that the LED is unlit.
Pressure switch—NOHC: IPS.A1.951	N ₂ -to-solenoids pressure.	Use the CDA pressure regulator to reduce the solenoid pressure below the operating range (see Appendix D, System Settings). Verify that the LED is unlit.
Pressure switch—NOHC: PS.Z1.491	Cabinet exhaust draw-low.	Remove the tubing to the exhaust. Verify that the LED is unlit.

Static Mixer Replacement

The purpose of this procedure is to remove and replace the static mixer (SM12.910), which is located at the rear of the Wetted Compartment.

Precautions

**WARNING**

CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY.

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Wear appropriate personal protective equipment when performing this procedure.

Procedure

Removal

1. Place the Inline Blender, with Integrated Daytanks, in Manual mode.
2. Perform the [UPW Flush Pumps/Flow Controllers to Drain](#) procedure in [Chapter 4, System Operation and Process Procedures](#).
3. Turn the **MAIN POWER DISCONNECT SWITCH** to OFF.
4. Press the **EMO** switch.
5. Open the Wetted Compartment door.
6. Note the location of the helical static mixer(s) at the back wall of the compartment.
7. Carefully loosen the plumbing fittings above and below the applicable static mixer. Beware of liquid that might drain/drip from the plumbing line. Wipe up any spilled liquid.

Installation

1. Position a new static mixer into position.
2. Tighten the plumbing fittings above and below tubing that contains the static mixer. Make sure that the accumulator GC.Q1.910 is positioned correctly and secured tightly.
3. Close the Wetted Compartment door.
4. Pull the **EMO** button to its “out” position.
5. Turn the **MAIN POWER DISCONNECT SWITCH** to ON.
6. Press the **EMO RESET** button.
7. Place the Inline Blender, with Integrated Daytanks in the desired mode of operation.

Temperature Probe/Transmitter Replacement

Purpose

The purpose of this procedure is to remove and replace one of the temperature probes in the chemical lines:

- TT.M1.910A (to temperature transmitter TT1203).
- TT.M1.910B (to temperature transmitter TT1209).
- TT.M21.910 (to temperature transmitter TT1215).

Precautions



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY.

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Wear appropriate personal protective equipment when performing this procedure.

Prerequisites

Obtain the THZ DH configuration software, RTD probe calibration sheet, and serial number to the Hart communication adapter.

Procedure

1. At the *Operating Mode* screen, press the **Maintenance** softkey.
2. Turn the **MAIN POWER DISCONNECT SWITCH** to OFF.
3. Open the Controls Compartment door and locate the twelve black THZ-DH temperature transmitters at the left side. (The transmitters are mounted on a swing-out panel, which allows access to the wiring).
4. Note the location of the wires at the back of the THZ-DH temperature transmitters. Refer to the electrical schematic for guidance.
5. Disconnect the wires and remove the applicable THZ-DH module from the swing-out panel.
6. Replace the THZ-DH temperature transmitter and reconnect its wires.
7. Connect the THZ-DH temperature transmitter to the Hart serial communication adapter.
8. Download the setup file TDZ_THZ_v2_4.exe (3.8M) from the Moore Industries website: http://www.miinet.com/custservice/config_software.shtml
9. When loading the RTD sensor calibration points, match the sensor serial number tag to the calibration data sheet provided with the device.
10. Using the screen shots shown in [Figure 8-7](#) through [Figure 8-12](#) as reference only, configure the actual information for the THZ-DH transmitters.
11. Attach the MicroLink Hart Protocol Modem – USB Interface device (P/N 101-0007) clips to the transmitter +/- terminals to communicate between the transmitter and the laptop.
12. In the *Setup* screen and **Input** tab (see [Figure 8-7](#) on page 8-63):
 - a. Choose **Resistance** type sensor, which automatically displays 0-4000 ohm range.
 - b. Choose **4-wire**.
 - c. Input the **Transmitter** designator located next to the transmitter in the electrical cabinet.
 - d. Input the **Tag Number** of the transmitter from the sensor tag in the **Descriptor** line, and the **Serial Number** in the **Message** line.
 - e. At the **Input** tab, **Lower Range Value**, enter the ohm-value for 15 deg C from the sensor calibration certificate.
 - f. At the **Input** tab, **Upper Range Value** enter the ohm-value for 35 deg C from the sensor calibration certificate.

- g. Choose the correct power frequency.
 - h. Choose Enabled for Broken Wire.
13. In the Output tab (see [Figure 8-8](#) on page 8-64):
 - a. Input 4 mA for the Lower Range Output Current, and 20 mA for the Upper Range Output Current.
 - b. Input 15 seconds for the PV Damping and 15 seconds for Mix.
 - c. Choose Fail High.
 - d. Choose Primary Variable for Display Mode.

The *Output Tab Sample Screen* shown at page 8-64 is an example of the configuration setup, which updates after downloading to the transmitter.

Figure 8-7. Example of a Setup Screen and Input Tab

The screenshot displays the 'Untitled - THZ/TDZ PC Configuration Program' window. The interface is divided into several sections:

- Program Status:** Shows 'IDLE' and 'Device Id.'.
- Input Configuration:** Includes dropdowns for 'Resistance', '0 - 4000 ohms', and '4 wire'.
- Variables:** Fields for 'mA', 'Percent', and 'SV'.
- Device Status:** A section with a red indicator light and 'Address' field set to '13'.
- Communications:** Includes 'No Response' and 'Address' fields.
- Input Settings:** Contains 'Limits = 0.0 to 4000.0 Ohms', 'Min Span = 10.0 Ohms', 'Lower Range Value' (1066.65), and 'Upper Range Value' (1129.03).
- Temperature:** Radio buttons for 'Deg C' and 'Deg F'.
- Filter:** Radio buttons for '50 Hz' and '60 Hz'.
- Broken Wire:** A checkbox labeled 'Enabled' which is checked.
- Device Configuration:** A section on the left showing 'Resistance 2 wire', 'PV Upper Range : 850.00 Ohms', 'PV Lower Range : -200.00 Ohms', 'Damping : 0.0 Secs', and 'Output Scale : 4.0 to 20.0 mA'.

Figure 8-8. Example of an Output Tab Sample Screen

Untitled - THZ/TDZ PC Configuration Program.

File View Transfer Comms Monitoring Burst Mode Help

Program Status: IDLE

Device Id:

Tag: TTX941 Programmed Date: 26 Oct 2010

Descriptor: M1.910B Address: 13

Message: S/N 25680#1

Device Configuration:

Resistance: 2 wire

PV Upper Range : 850.00 Ohms

PV Lower Range : -200.00 Ohms

Damping : 0.0 Secs

Output Scale : 4.0 to 20.0 mA

Input Configuration:

Resistance

0 - 4000 ohms

4 wire

Variables:

mA Percent

SV

Device Status:

Communications:

No Response

Address: 13

Output Settings:

Lower Range Output Current

Upper Range Output Current: 20

PV Damping (0-30 secs): 30

Failure Mode:

☐ Fail Low (3.6mA)

☒ Fail High (23.6mA)

Display Mode:

☒ Primary variable

☐ Current

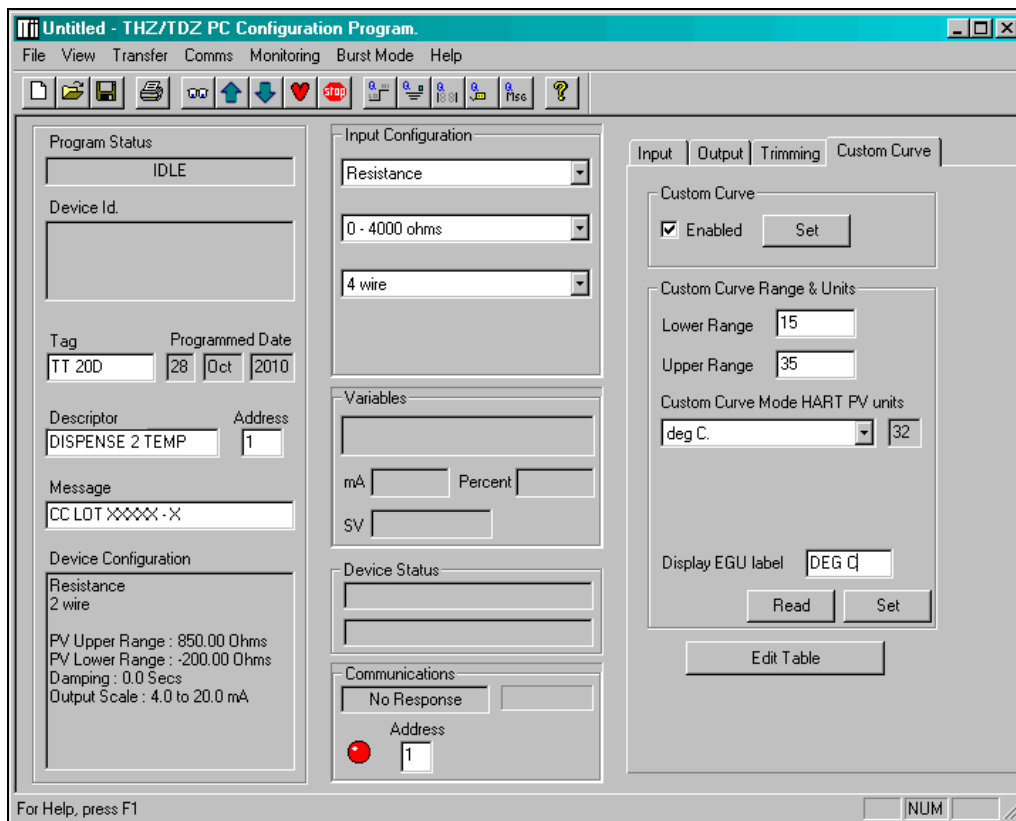
☐ Percentage

☐ Display Toggle

14. In the Custom Curve tab:

- a. Click **Enabled** to configure calibration curve. Clicking on **Set** updates the Device Configuration in the lower left of the screen, but does not download the data to the transmitter.
- b. Input **15** and **35** as the **Lower Range** and **Upper Range** of the curve.
- c. Choose **deg C** from the dropdown, and make sure the number following is “**32.**”
- d. Input “**C.**” These are the units displayed on the transmitter face.
- e. Click on **SET** below the **Display EGU** label to fix the values in the program.
- f. Click on **Edit Table** to input the calibration data in the Custom Table.

Figure 8-9. Example of a Custom Curve Screen



15. In the *Custom Table* screen:

- a. Choose 3 points for the custom curve.
- b. Input the calibration points for 15, 25, and 35 deg C from the sensor calibration certificate that comes with each calibrated sensor. Make sure the serial number matches the probe connected to the transmitter being configured.
- c. Download transfers the data to the transmitter. A message will display when the download is successfully completed.

Figure 8-10. Example of a Custom Table Screen

Untitled - THZ/TDZ PC Configuration Program.

File View

Custom Table

File

Program

Device

Tag

TT 20D

Descript

DISPEN

Message

CC LOT

Device

Resistar

2 wire

PV Upp

PV Low

Dampin

Output

	X	Y		X	Y
0	xxxx.xx	15	16	0	0
1	xxxx.xx	25	17	0	0
2	xxxx.xx	35	18	0	0
3	0	0	19	0	0
4	0	0	20	0	0
5	0	0	21	0	0
6	0	0	22	0	0
7	0	0	23	0	0
8	0	0	24	0	0
9	0	0	25	0	0
10	0	0	26	0	0
11	0	0	27	0	0
12	0	0	28	0	0
13	0	0	29	0	0
14	0	0	30	0	0
15	0	0	31	0	0

PAGE

Number of Points

3

Custom Curve

☒ Enabled Set

Upload Download

Clear Table Cancel

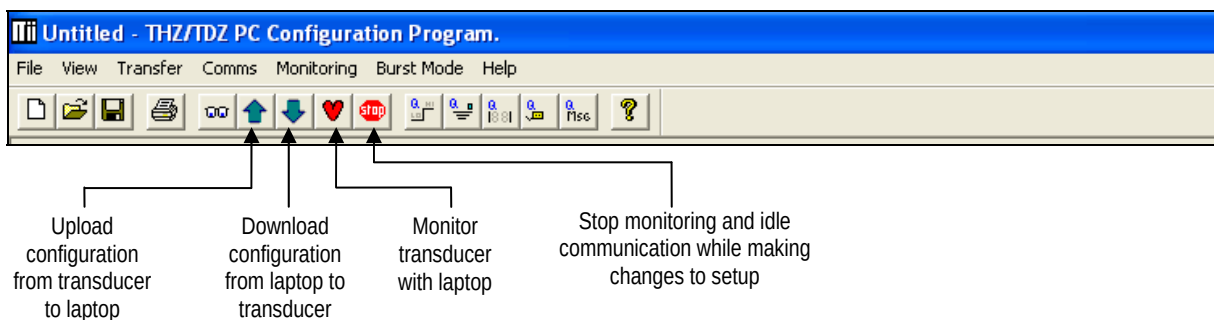
For Help, press F1

NUM

16. When the Input, Output, and Custom Curve tabs are configured and the Custom Table data has been successfully downloaded:

- a. Click on the download arrow.

Figure 8-11. Configuration Program Control Functions



- b. When the download is complete, the values displayed on the screen may appear corrupted. **IGNORE THESE CORRUPTED VALUES.** To test that the correct data has been properly downloaded, click on the **UPLOAD** arrow, and review all the input screens to make sure the correct data was downloaded.
 - c. If any screens are incorrect, repeat the setup and download steps.
 - d. When the download is completed, click on the Heart icon to check the program correctly monitors the transmitter output.
 - e. Click **Stop** to idle the program, and continue with **Trimming**.
17. On the **Trimming** tab:
 - a. Ignore **Sensor Trimming**. This is a factory setup function.
 - b. **Output Trimming** is done after the transmitter is correctly configured and monitoring correctly. Make sure the program is in **IDLE** (click the **STOP** icon) mode to make the following changes.
 - c. Go to the I/O page on the HMI, and locate the temperature transmitter displays.
 - d. Return to the laptop, select **FIX CURRENT @ 4MA**, and read the temperature value displayed on the HMI.
 - e. If the displayed value is less than 4000 counts, input a value in the **OUTPUT CURRENT** box that is slightly less 4.00, and click on **FIX CURRENT @ 4MA**.
 - f. If the temperature does not reach 4000 counts, click again until it reads 4000 or **Minimum Eng Units** on the HMI.
 - g. Click **UN-FIX CURRENT**.

- h. Click on **FIX CURRENT @ 20mA**, and repeat the procedure above until the HMI display reads 20000 counts or **Maximum Eng Units**.
- i. Click **UN-FIX CURRENT**.

Figure 8-12. Example of a Trimming Tab

The screenshot displays the 'THZ/TDZ PC Configuration Program' interface. The 'Trimming' tab is selected, showing various configuration options for sensor and output trimming. The 'Sensor Trimming' section includes 'Trim Mode' (Factory/User), 'Trim Points' (1/2), and 'Trimming Points' (Point 1 and Point 2) with 'Set Value' and 'Set' buttons. The 'Output Trimming' section includes 'Enter measured loop current', 'Trim 4mA', 'Un-Fix Current', 'Fix Current @ 20mA', 'Output Current' (3.98), and 'Reset' buttons. The 'Loop Test' section includes '4', 'Fix', and 'Unfix' buttons. The 'Input Configuration' section includes 'Resistance', '0 - 4000 ohms', and '4 wire' dropdowns. The 'Variables' section includes 'mA', 'Percent', and 'SV' fields. The 'Device Status' section includes a 'No Response' button and an 'Address' field (13). The 'Communications' section includes a 'No Response' button and an 'Address' field (13). The 'Program Status' section includes 'IDLE', 'Device Id.', 'Tag' (TTX941), 'Programmed Date' (26 Oct 2010), 'Descriptor' (M1.910B), 'Address' (13), 'Message' (S/N 25680#1), and 'Device Configuration' (Resistance 2 wire, PV Upper Range: 850.00 Ohms, PV Lower Range: -200.00 Ohms, Damping: 0.0 Secs, Output Scale: 4.0 to 20.0 mA).

Valve Replacement & Test

Purpose

The purpose of this procedure is to remove and replace the pneumatic valves, manual valves, and check valves of the Inline Blender, with Integrated Daytanks.

Precautions



WARNING



CONTACT WITH A HAZARDOUS CHEMICAL OR ITS FUMES CAN CAUSE SEVERE PERSONAL INJURY

Personnel performing this procedure will be exposed to chemical that has no secondary containment. Valves must be treated as if chemical is present, as they have either contained or been in contact with chemical. Wear proper safety gear when performing this procedure.



CAUTION



For chemicals—and potential chemical combinations—identified by site procedures, forced air-breathing apparatus must be worn by maintenance personnel.

The forced air-breathing apparatus can consist of, but is not limited to, the following types of devices:

- Site's in-house forced-air line
- Air-breathing device on a portable cart
- Self-contained breathing apparatus (SCBA)

For any maintenance task that requires a respirator (cartridge-type or forced-air), maintenance personnel must be fit-tested for their own respirator. They must have a valid physical certification proving that the respirator fitting was accomplished.

NOTE

Replacing a valve is considered a two-person ("buddy system") task, with two trained maintenance personnel performing the steps described in this procedure.

Procedure

Removal

1. If possible, flush the lines where the replacement component will be installed (see [Chapter 4, System Operation and Process Procedures](#)).
2. Place the module in Manual mode.
3. Turn the **MAIN POWER DISCONNECT SWITCH** to OFF.
4. Label all connections to defective valve to ensure proper connections to the replacement valve.

IMPORTANT

In the following step, be careful not to damage the pilot tube barbed fitting. If it is damaged, it must be replaced.

5. If the defective valve is a pneumatic valve, disconnect the 5/32-in. O.D. pilot tube by turning quick-disconnect fitting 3/4-turn counterclockwise.
6. Remove one line from the valve as follows:
 - a. Wrap absorbent wipes completely around the valve/tube connection.
 - b. Hold the tube against the valve and reach under the wipe and slowly unthread the flare nut until it is disconnected. Continue to hold the tube against the valve.
 - c. Ensuring that absorbent wipes surround the connection, slowly pull the line away from the valve.
 - d. Clean up any spilled chemical, and, in a manner approved by the facility, dispose of any wipes that are saturated with chemical.
7. Repeat step 6 on each tube connected to the valve.
8. Remove the valve and place it where it can be safely cleaned before disposal.
9. Clean up any spilled chemical.
10. If the removed valve is a pneumatic valve:
 - a. Remove the pilot tube fitting.
 - b. Clean any thread sealer from the fitting threads.
 - c. Wrap new thread tape over the fitting threads.
 - d. Install fitting on the new valve.

Replacement

IMPORTANT

Keep a log of the valves that were replaced, including their P&ID tag number, cycles, part number, and the date.

IMPORTANT

Ensure that the replacement valve has not been exposed to contamination. Use the old mounting hardware only if it is not corroded.

1. Install the new valve in same position and orientation as the original valve.
2. Reattach the tubing to the new valve as follows:
 - a. Inspect the flaring on the tubing for cuts, compression marks, or other damage. If necessary, replace with new tubing.
 - b. Insert the tubing in valve ports.
 - c. Verify that tubing is installed at the correct valve ports.
 - d. Fasten the tubing to the valve ports with flare nuts.

IMPORTANT

When tightening LQ3 tube fittings, refer to the [Flare Fitting Tightening Requirements—SMC LQ3](#) procedure on page 8-32.

When tightening Flaretek tube fittings, refer to the [Flare Fitting Torque Requirements—Flaretek](#) procedure on page 8-33.

3. If the valve is air-operated, connect the 5/32-in. O.D. pilot tube and turn quick-disconnect 3/4 turn clockwise.
4. Turn the **MAIN POWER DISCONNECT SWITCH** to ON.
5. Restore the Inline Blender, with Integrated Daytanks to an operational mode that will allow manual checking of replaced valve to verify that it opens and closes properly (as described in [Appendix C, Input and Output Lists](#), in this manual), and also that valve does not leak. If the replaced valve is a pneumatic valve, continue with the following [Test Air-Operated Valve](#) procedure.

Test Air-Operated Valve

Pneumatic valves are activated by solenoid valves. Refer to [Appendix C, Input and Output Lists](#), or the electrical schematic to identify the solenoid that corresponds to each pneumatic valve. This procedure checks out the valves by forcing them to open or close.

1. Ensure that CDA supply is available to the unit.
2. On the *In/Out Page Selection* screen, press the *Discrete Outputs* screen that contains the valve. Then, locate the valve to be tested.
3. Press the **FORCE** softkey to toggle the valve to from **OFF** to **ON** or from **ON** to **OFF**.
4. Verify the applicable digital output LED lights.
5. Where possible, verify the light on the solenoid valve comes on. If no light is present, verify that the pneumatic valve is operational as follows:
 - On pneumatic valve with a red poppet, verify that poppet actuates.
 - On pneumatic without a red poppet or a visual indicator, touch the valve to feel it actuate, or use a flashlight to view diaphragm movement when valve is actuated.
6. When done, clear the force on the *Discrete Outputs* screen (or equivalent).
7. Return the Inline Blender, with Integrated Daytanks to operational mode.

A Maintenance Supplies



WARNING



BEFORE PROCEEDING, READ AND UNDERSTAND THE SAFETY INFORMATION CONTAINED WITHIN THE [SAFETY](#) CHAPTER OF THIS MANUAL.

Special Tools and Test Equipment

Some maintenance procedures described in this manual require the use of special tools and test equipment. It is recommended that the items listed in [Table A-1](#) be available for repair and maintenance work.

Table A-1. Special Tools and Test Equipment

Qty	Part Number	Description
1	301681-005	Flare Tool, 1-inch
1	302519-001	Flare Tool, Combination (two recommended)
1	302474-001	Crimper, Amp Connector
2	304087-001	Magnet (for door interlock)
1 of each size	306069-XXX	Flaretek Sockets
1	316423-001	Filter Wrench for filter housing with PTFE collar nut
1	1921	Tube Cutter
2	1922	Blade, Tube Cutter
1	—	Multimeter
1 of each size	314013-XXX	Tool, Flaretek Torque Verification Device
1	927073-001	10.0 Ω Calibration reference resistor loop assembly

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B System Specifications



BEFORE PROCEEDING, READ AND UNDERSTAND THE SAFETY INFORMATION CONTAINED WITHIN THE [SAFETY](#) CHAPTER OF THIS MANUAL.

For system specifications, refer to the facilities diagrams provided in the [Prints](#) section.

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C Input and Output Lists



BEFORE PROCEEDING, READ AND UNDERSTAND THE SAFETY INFORMATION CONTAINED WITHIN THE [SAFETY](#) CHAPTER OF THIS MANUAL.

This appendix contains information on the Programmable Logic Controller (PLC) inputs and outputs.

Refer the electrical schematic listed in the [Prints](#) section of this manual for detailed information about the system inputs and outputs.

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D System Settings



BEFORE PROCEEDING, READ AND UNDERSTAND THE SAFETY INFORMATION CONTAINED WITHIN THE [SAFETY](#) CHAPTER OF THIS MANUAL.

This appendix contains suggested equipment settings for the Inline Blender, with Integrated Daytanks. These settings can be changed on the touch screen (HMI) with the proper access level and password.

Appendix Contents

Pressure Settings	D-2
Manual Valves Settings	D-2
Valve Lifetime Cycle Counts and Maximum Usage	D-3

Pressure Settings

The following table lists suggested settings for pressure controls.

NOTE

All pressures listed below are suggested values. Actual pressures will vary depending on the UPW pressure and flow delivered to the blender.

Table D-1. Pressure Regulating Valve/Indicating Pressure Switch/Pressure Indicator Settings

Pressure Controlled	Controlling Regulator	Pressure Indicator	Initial/Startup Pressure	Maximum Pressure	Minimum Pressure
N ₂ supply	PRV.N1.950	IPS.N1.950	30 psi	30 psi	20 psi
N ₂ to J1	PRV.J1.301	PI.J1.301	50 psi	65 psi	30 psi
N ₂ to J99	PRV.J99.301	PI.J99.301	50 psi	65 psi	30 psi
N ₂ to solenoid valves	PRV.A1.951	IPS.A1.951	75 psi	85 psi	65 psi

Manual Valves Settings

This table lists the operating positions for the manual valves.

Table D-2. Manual Valve Settings

Valve #	Operating Position	Description
MV.J99.110	Open	Manual isolation valve for pressurized chemical from the Fab.
MV.J99.801	Open	Post-flow controller J99 to drain manual isolation valve.
MV.P21.801	Open	Manual isolation inlet to pump P21 or to the pre-pump drain.
MV.Q1.910	Closed	Manual isolation valve to future option AUX OUT 3 [capped].
MV.S9.110	Closed	Manual isolation valve to future option AUX FILL 4 [capped].
MV.T1.110	Open	Blended chemical to daytanks 1A or 1B.
MV.T1.111A	Open	Manual isolation inlet valve for daytank 1A.
MV.T1.111B	Open	Manual isolation inlet valve for daytank 1B.
MV.T1.910	Open	Daytanks 1A or 1B drain manual isolation valve.
MV.T1.991	Closed	Daytanks 1A or 1B manual isolation valve to T1 Chem Out 1 [capped].
MV.T1.992	Closed	Daytanks 1A or 1B manual isolation valve to T1 Chem Out 2 [capped].
MV.T2.110	Closed	Manual isolation valve to future BLEND OUT 2 outlet [capped].

Valve Lifetime Cycle Counts and Maximum Usage

The recommended maximum lifecycle counts for the valves used with the system are given below. Air Liquide Advanced Equipment Systems Division recommends replacing valves when they reach their maximum lifetime or cycle count.

IMPORTANT—MAXIMUM LIFETIMES OF VALVES

The lifetime of wetted components is dependent on the chemistries and operating conditions of the system. During routine inspection of this equipment, all manual valves, pneumatic valves, and needle valves and piping must be inspected for wear and corrosion and replaced as needed. Regardless of the cycle counts listed in [Table D-9](#), DO NOT EXCEED these recommended maximum lifetimes:

Chemical	MAXIMUM Lifetime of Valves
• Hydrochloric acid • Hydrofluoric acid • Sulfuric acid • Nitric acid	3 years max.
Any chemical OTHER THAN concentrated hydrochloric acid, hydrofluoric acid, or nitric acid	5 years max.

If you have questions about material compatibility for your chemistries, please contact Air Liquide Electronics (ALE).

Table D-3. Recommended Maximum Valve Cycle Counts and Lifetimes

Valve Manufacturer	Maximum Cycle Count	Maximum Lifetime (per Table above)
Entegris	2,000,000	3 or 5 yr depending on chemical, per the table above
St. Gobain/Furon	1,000,000	3 or 5 yr depending on chemical, per the table above
GEMÜ AOVs	1,500,000	3 or 5 yr depending on chemical, per the table above
GEMÜ MV01	10,000	3 or 5 yr depending on chemical, per the table above
Plastomatic	500,000	2 yr, per the table above
SMC	3,000,000	3 or 5 yr depending on chemical, per the table above

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E OEM Documentation



BEFORE PROCEEDING, READ AND UNDERSTAND THE SAFETY INFORMATION CONTAINED WITHIN THE [SAFETY](#) CHAPTER OF THIS MANUAL.

OEM documents ([Table E-1](#)) are included as part of the documentation package. Air Liquide Electronics makes no warranty regarding the content of these documents. The documents included are the latest revision available at the time of manufacture of the original equipment.

For further information on specific components, please contact the OEM manufacturer directly.

Table E-1. OEM Documents

Component	OEM Product	Document Type
Booster Pump	Levitronix BPS-i100 Centrifugal Pump	User Manual
Conductivity controller	Hach sc200	User Manual
Pressure transducer	Entegris NT Integrated Flow Controller Model 6501	Technical Brochure
Recirculating Pump	Levitronix BPS-600 Centrifugal Pump	User Manual

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Booster Pump

Component

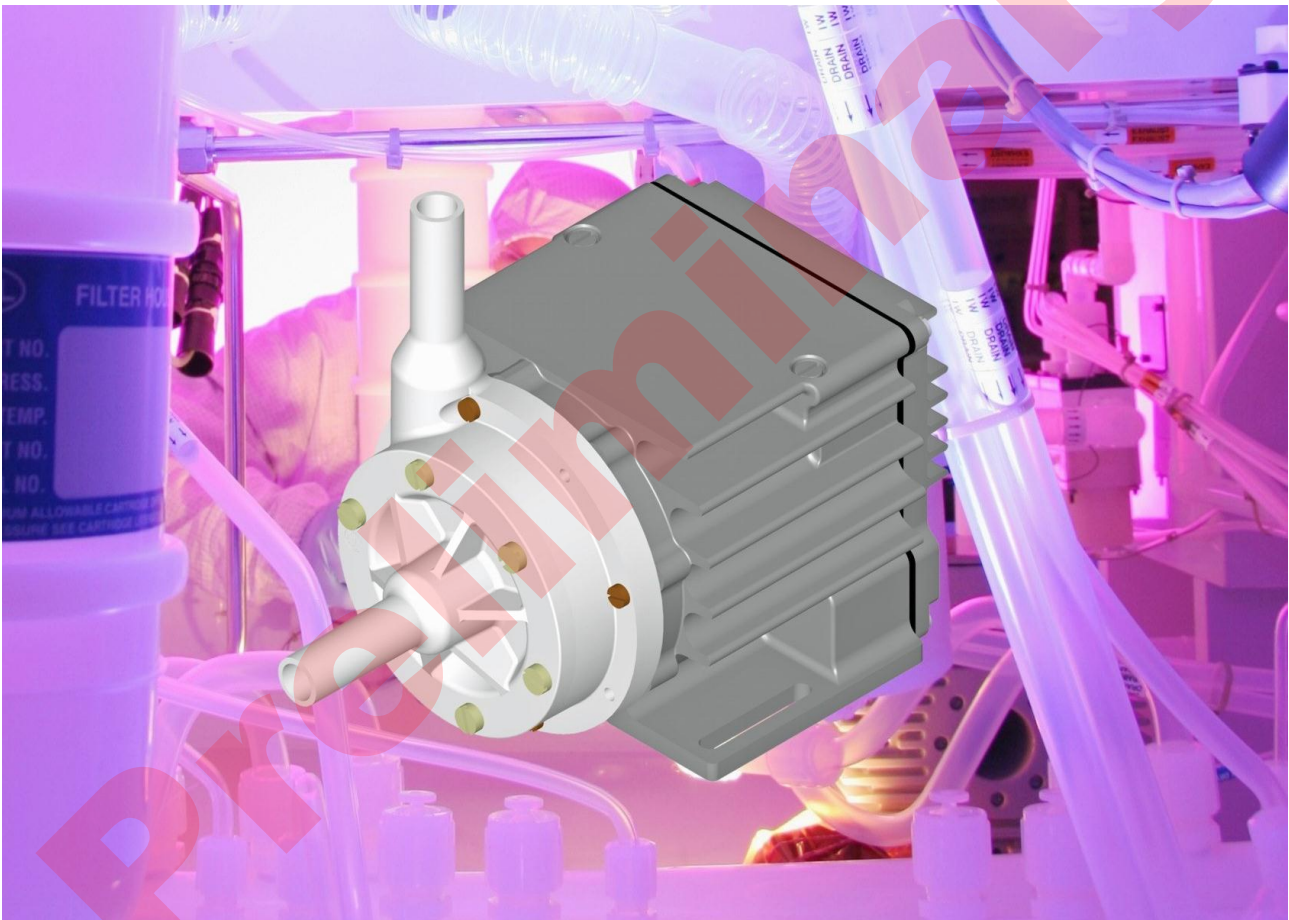
Levitronix BPS-i100 User Manual

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BPS-i100

1.7 bar (25 psi)
20 liters/min (5.3 gallons/min)

USER MANUAL



This manual contains information necessary for the safe and proper use of the *BPS-i100*. Included are specifications for the standard configurations of the pump system and instructions regarding its use, installation, operation, adjustment, inspection and maintenance. For special configurations of the pump system refer to accompanying information. Please familiarize yourself with the contents of the manual to ensure the safe and effective use of this product. After reading this manual, please store the manual where the personnel responsible for operating the pump system can readily refer to it at any time.

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1 Safety Precautions

CAUTION
Do not under any circumstances open the driver. Levitronix® does not assume responsibility for any damage, which occurs under such circumstances.

	CAUTION
	High magnetic field strength of pump impeller. The pump system contains a rotor magnet with high field strength. This may alter or damage the calibration of sensitive electronic devices and measuring instruments in the immediate surroundings. Keep at a safe distance from computers, monitors and all magnetic data storage media (e.g. disks, credit cards, audio and video tapes etc.)

	! WARNING
	Hazardous voltage may be present. In case of the usage of an inadequate AC/DC power supply, mains voltages may be present (even if the system is designed for up to 24 VDC). The usage of a galvanic separated power supply, which is certified by a 3rd party (UL or CE), is highly recommended. Do not under any circumstances open the powered driver.

	! WARNING	
	High magnetic field strength of pump impeller. The pump system contains a rotor magnet with high field strength. Pacemakers may be influenced and magnetic forces may lead to contusions. Keep distance to pacemakers and handle impeller with care.	

	! WARNING	
	TOXIC CHEMICALS may be present. When using the system to pump chemicals skin contact and toxic gases may be hazardous to your health. Wear safety gloves and other appropriate safety equipment.	

2 Specifications

2.1 Specification of Components

Figure 1 shows the main system components (driver and pump head) and Figure 2 illustrates the accessories for the pump system.

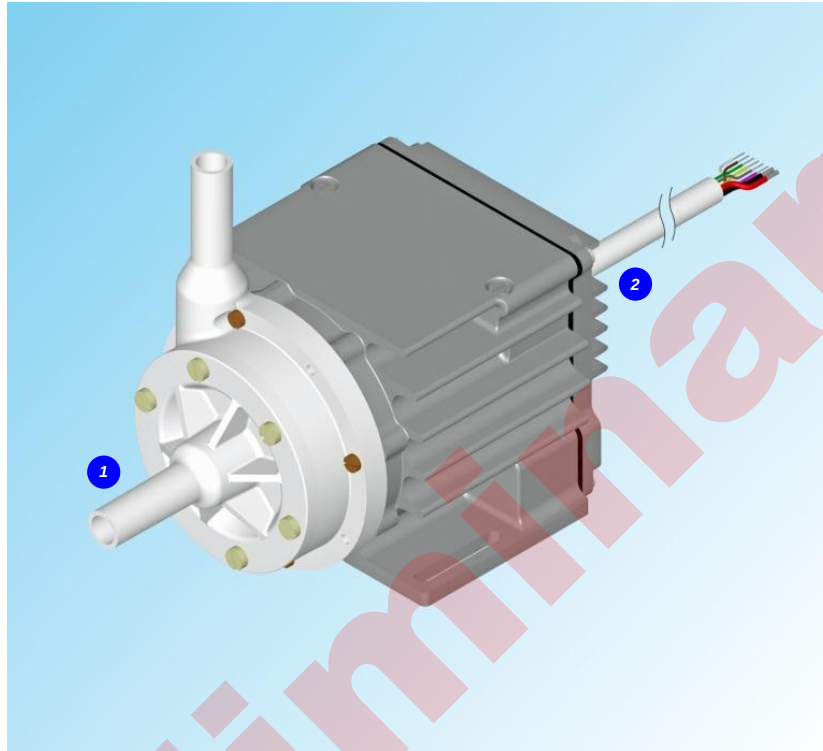


Figure 1: Pump system with main standard components

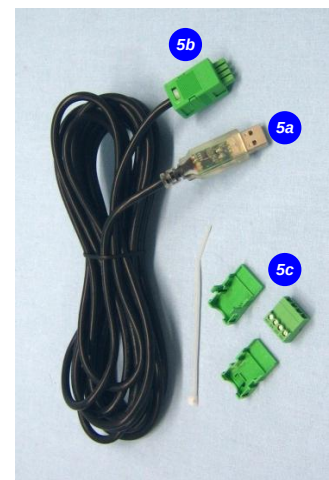


Figure 2: Accessories

System Name	Article #	Pumphead	Driver	Note
BPS-i100.2	TBD	LPP-200.7	IPD-100.2-50-01	PFA Pumphead, ETFE coated motor, 5m FEP cable with open wires

Table 1: Standard system configurations

Pos.	Component	Article Name	Article #	Characteristics	Value / Feature
1	Pumphead	LPP-200.7	100-90708	Impeller / Pump Housing Sealing Ring Fittings	High Purity PFA Kalrez® perfluoroelastomer ¹ ½" Tube fitting
				Max. Flow Max. Diff.-Pressure Max. Viscosity	20 liters/min / 5.3 gallons/min 1.7 bar / 25 psi 20 cP
				Max. Liquid Temp.	70°C / 158°F
2	Integrated Pump Driver	IPD-100.2-50-01	TBD	Voltage / Power	24 VDC ±10%, 120W
				Housing	- ETFE (grey, highly chemical resistant) coated Aluminum - Waterproofed (IP65)
				Cable	FEP jacket, open wires, cable length is 5m
				Interfaces	1x analog input 4 - 20 mA 1x analog input 0 - 10 V PLC with 1x analog output 0 - 10 V 2x digital input 0 - 24 V (optocoupler) 2x digital output 24V/100mA (open drain)
					RS485 interface (for service and extended operation) Modbus protocol
				Standard Firmware	G1.48

Table 2: Specification of standard components

1: Kalrez® is a registered trademark of DuPont Dow Elastomers

Pos.	Component	Article Name	Article #	Characteristics	Value / Feature
3 (a,b,c, d,e)	Impeller Exchange Kits	IEK-200.3	TBD	Impeller LPI-200.5 (a) Sealing O-Ring (b) Pump Casing Screws (c) Pump Motor Screws (d) Exchange Tool IET-1.1 (e)	PFA O-Ring, Kalrez, 50.52 x 1.78 6 pieces, stainless steel PTFE coated, M4x16 4 pieces, stainless steel PTFE coated, M3x14 POM-C
4	AC/DC Power Supply	TSL-120-124 (Traco)	100-40001	Voltage / Power Output Voltage Input Basic Dimensions Mounting	24 VDC / 120 W 93-132 / 187-264 VAC (autorange) 125 x 125 x 61 mm Mountable on DIN rail 35mm
				Certification	UL / CE
5	USB-RS485 Adaptor TR	USB-RS485-WE-5000-BT-TR	100-30336	Structure Purpose	USB connector (5a) with termination resistor connector (5b) and connector for external RS485 wire connections (5c) Communication over fieldbus of driver with PC

Table 3: Specification of accessories

2.2 System Overview and General Specification

2.2.1 Standard System Configuration for Speed Control

When using the pump system in speed control mode, the speed can precisely be set via an analog input. Various digital inputs and outputs allow controlling and monitoring of the system. A RS485 enables communication with a PC with the Levitronix® Service Software. Hence parameterization, firmware updates and failure analysis are possible. The RS485 can also be used as a fieldbus to implement more intelligent concepts of pump control.

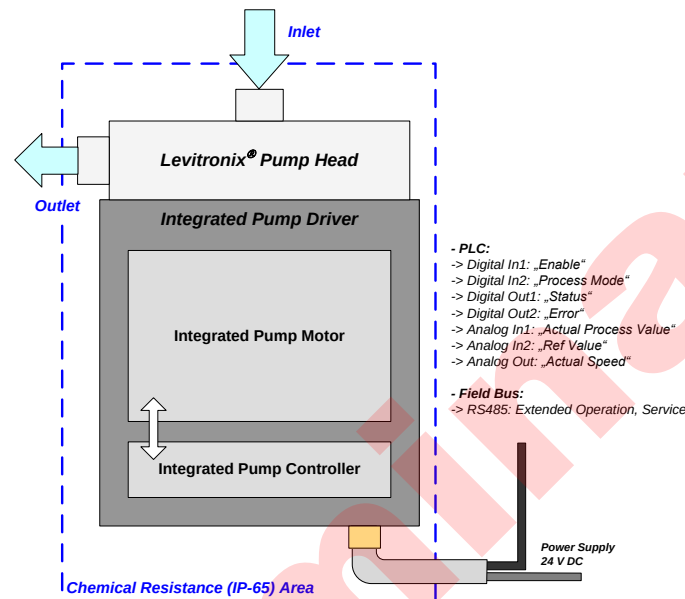


Figure 3: Standard system configuration for speed control (details in Section 4.1.1)

2.2.2 Standard Configuration for Process Control

The PLC interface of the BPS-i100 pump system enables the implementation of precise closed loop flow or pressure control in connection with either a flow or pressure sensor (see Figure 4). Precise ultrapure flow control systems can be realized with the BPS-i100 pump system in combination with LEVIFLOW® flowmeters.

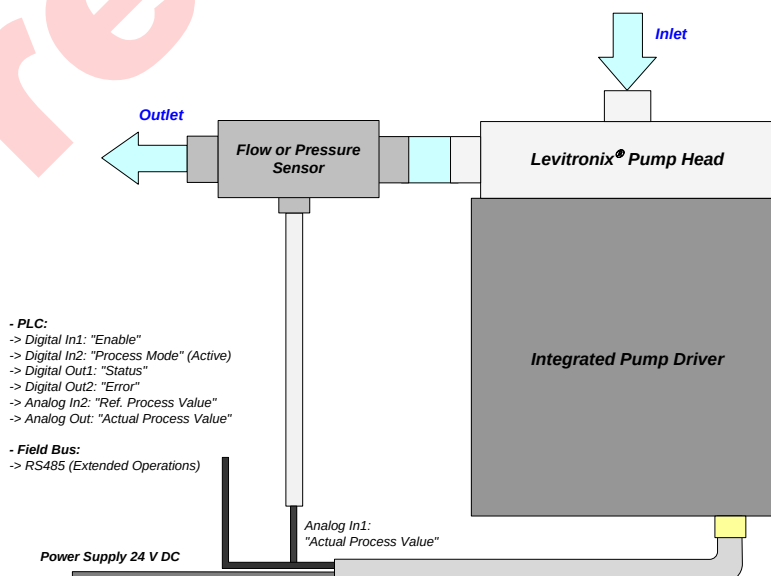


Figure 4: Standard system configuration for process control
(Pressure or flow control with external sensor, details in Section 4.1.1)

2.3 General Environmental Conditions

Usage	Indoor and Outdoor
Altitude	Up to 2000 <i>m</i>
Operating ambient temperature	0 to 40 °C
Storage ambient temperature	-20 to 65 °C
Operating humidity range (relative humidity)	15 – 95% (non condensing)
Storage humidity range (relative humidity)	15 – 95% (non condensing)
DC supply fluctuations	± 10% of nominal voltage
Transient over-voltages typically present on the mains supply	Surge immunity according to EN 61000-4-5 (tested together with certified AC/DC power supply)
Pollution degree	2

Table 4: Environmental conditions for pump system

2.4 Pressure-Flow Curves

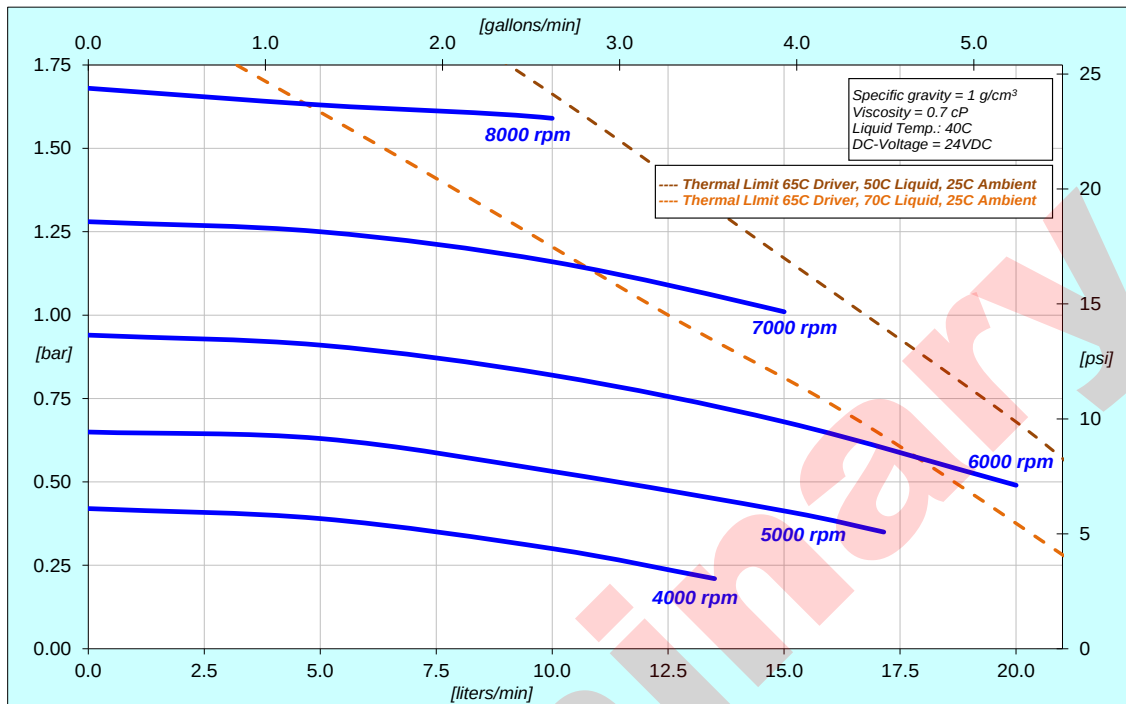


Figure 5: Pressure/flow rate curves
(Measured with pump head LPP-200.7)

2.5 Maximum Static Pressure for Pumpheads

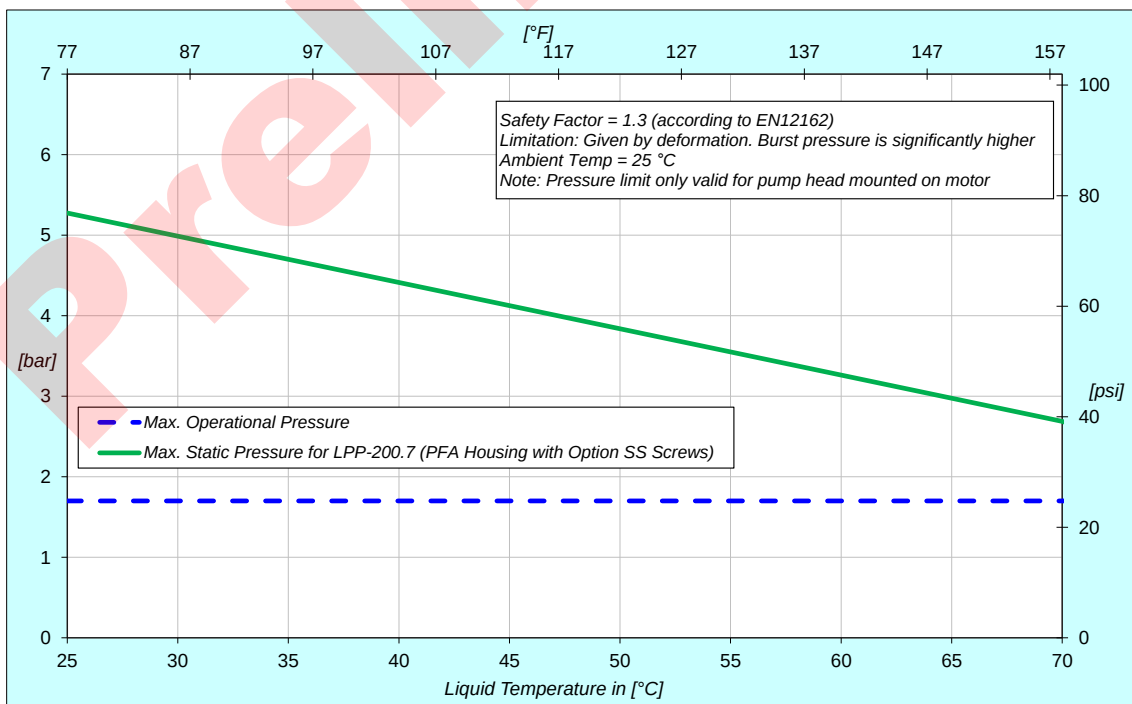


Figure 6: Specification for maximum static pressure of pump head LPP-200.7
- Pressure limits only valid for pump head mounted on motor

2.6 Basic Dimensions of Main Components

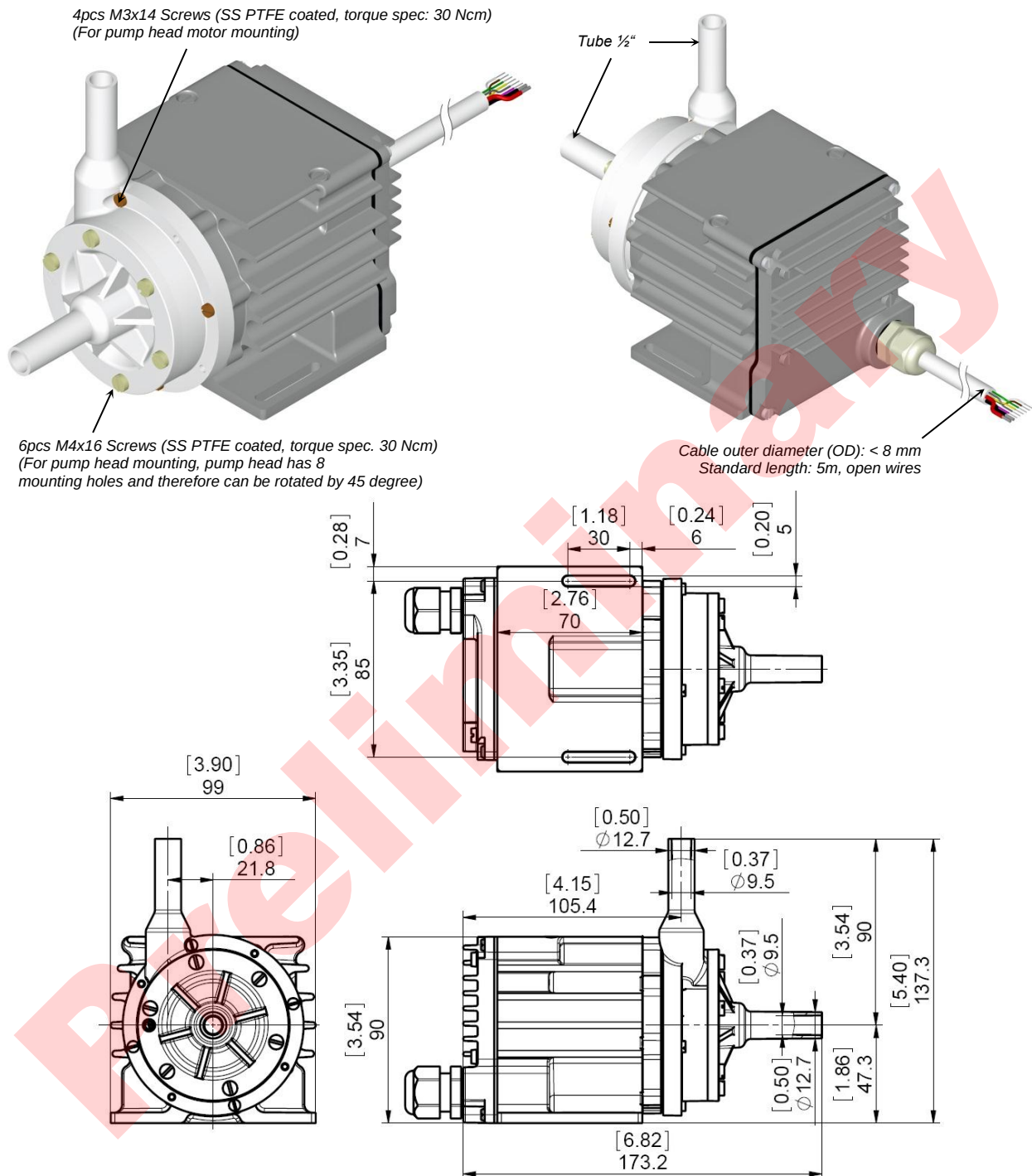


Figure 7: Basic dimensions (in [mm] and [inch]) of driver IPD-100.2-50 with pump head LPP-200.7
(For other configurations refer to according drawings)

Cable Jacket	Cable OD Driver Cables	Minimum Bending Radius Permanent Installation	Minimum Bending Radius Sometimes Moved
FEP	< 8 mm	7x Cable OD	15 x Cable OD

Table 5: Specifications for min. bending radius of driver cable

Note: If not mentioned explicitly all the cables are not suited for constant dynamic bending and movement!

3 Engineering Information

3.1 Sealing and Material Concept

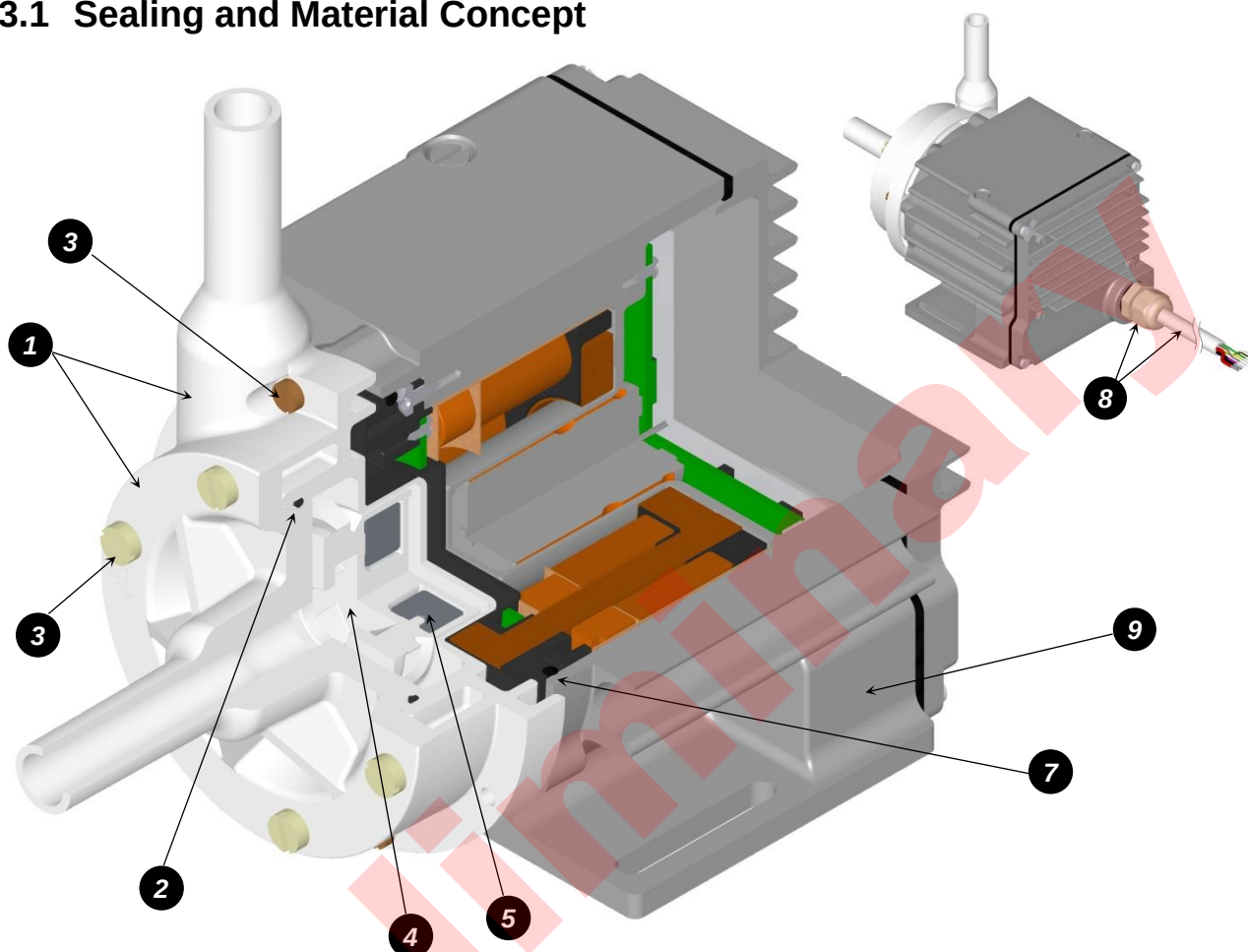


Figure 8: Sealing and material concept
(Driver IPD-100.2-50 with pump head LPP-200.7)

System Component	Item		Materials
	No	Description	
Pump head LPP-200.7	1	Pump housing lid and bottom	PFA
	2	Static sealing O-ring of pump casing	Kalrez®
	3	6pcs M4x16mm screws for pump casing	SS (stainless steel) PTFE coated screws
	4	Impeller LPI-200.5	PFA
	5	Rotor magnet	Rare-earth material
	6	4pcs M3x14mm for pump/motor mounting	SS (stainless steel) PTFE coated screws
Driver IPD-100.2-50	7	Sealing O-ring for motor housing	FPM (= FKM)
	8	Cable bushing Cable material	PVDF housing with FPM sealing FEP cable jacket
	9	Driver housing	ETFE coating, waterproof (IP-65) Coils and electromagnetic circuit potted with an epoxy compound (UL94 V0).

Table 6: Materials used in the IPD-100.2-50 driver and LPP-200.7 pump head
(For other configurations refer to according specifications and drawings)

3.2 Power Consumption

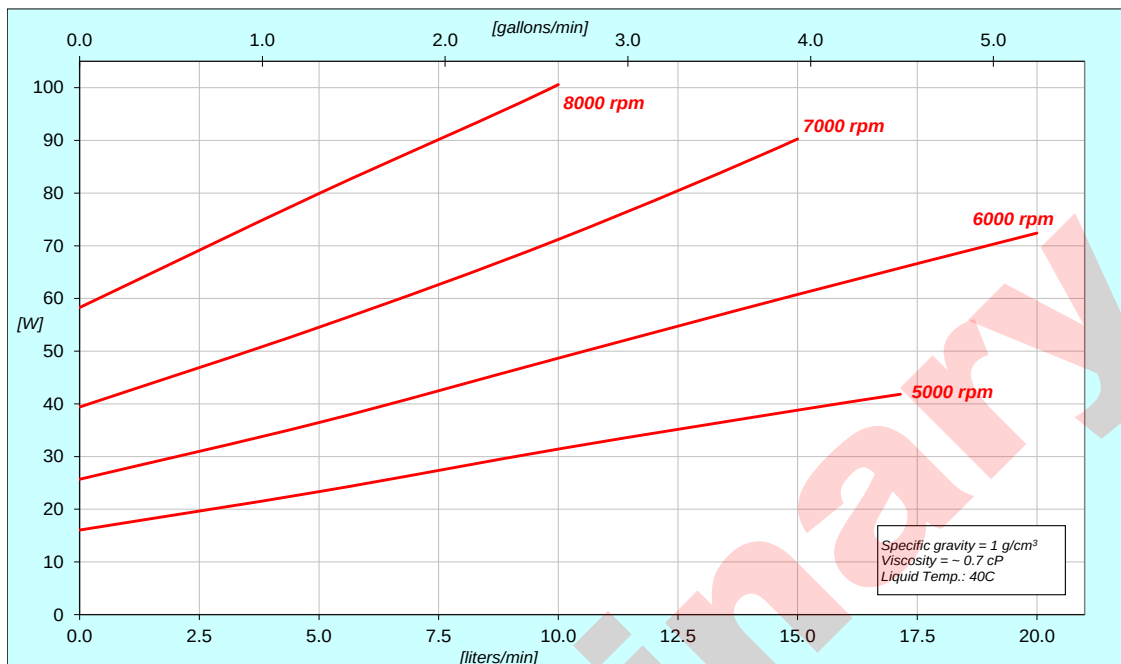


Figure 9: Electrical power consumption
(Pump head LPP-200.7 with driver IPD-100.2-50)

If other another power supply than the one recommended by Levitronix is used it is recommended to test it under power-on and dynamic conditions before using it on the final tool.

3.3 Temperature Monitoring

To avoid overheating of the system, the controller and motor temperatures within the driver are monitored. If one of both temperatures exceeds 80°C (158°F) for a duration of more than 10 minutes, the system goes into an error state and the pump stops. At 90°C (176°F), the system stops immediately.

If 65°C is exceeded a warning is given within the driver announcing to the user that the driver is running near the thermal limit (see explanation in Section 3.4). The warning signal can be monitored with the Levitronix® Service Software or configured on one of the digital outputs.

3.4 Thermal Management

The driver temperature depends on the ambient and liquid temperature, as well as on the hydraulic operation point. Figure 10, Figure 11 and Figure 12 illustrate the temperature characteristics of the motor depending on these parameters.

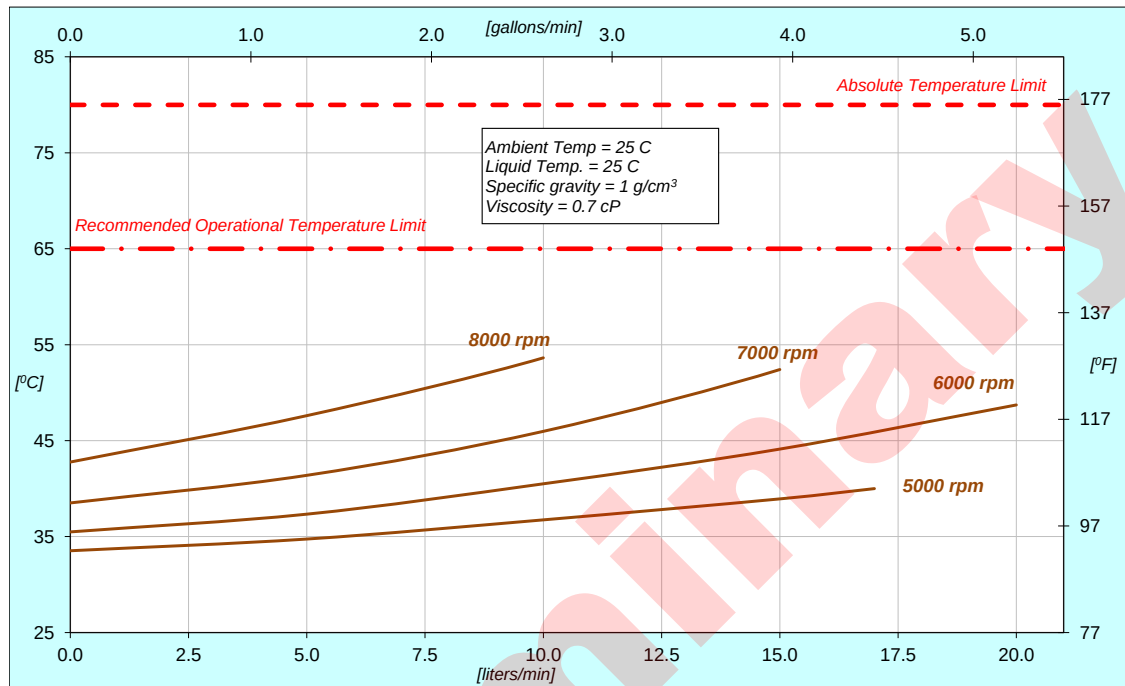


Figure 10: Temperature curves for the IPD-100.2-50 driver @ 25 C liquid temperature
(Pumping with pump head LPP-200.7, temperature is measured inside of the integrated motor and controller, contact temperature of surface is below this temperature)

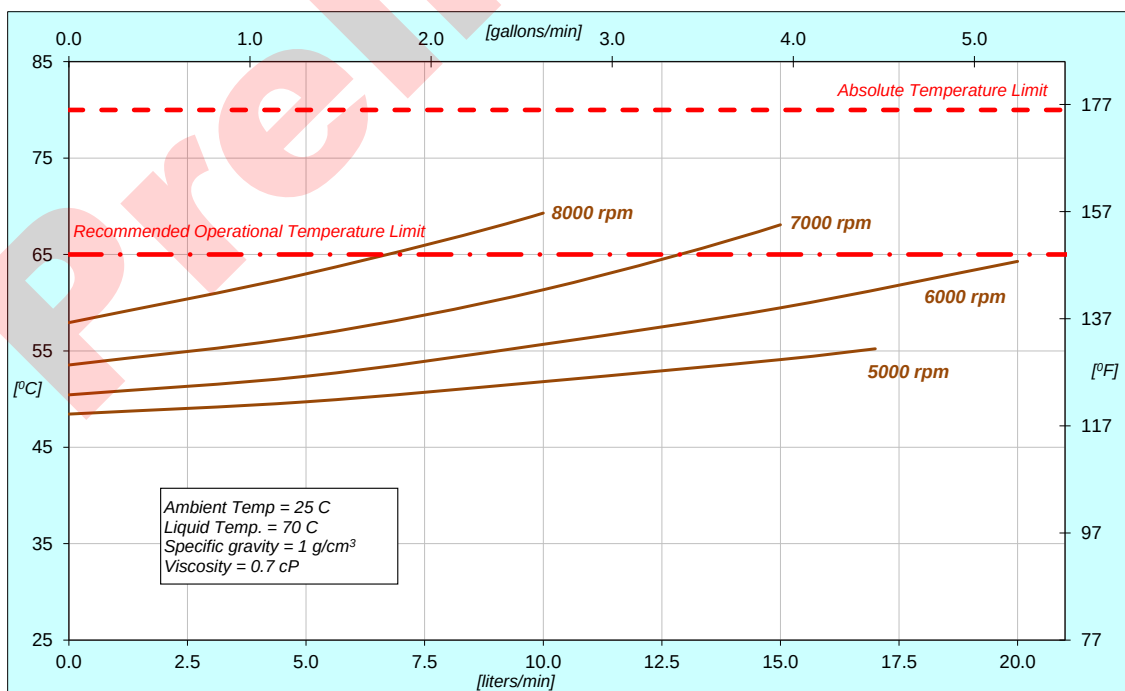


Figure 11: Temperature curves for the IPD-100.2-50 driver @ 70 C liquid temperature
(Pumping with pump head LPP-200.7, temperature is measured inside of the integrated motor and controller, contact temperature is below this temperature)

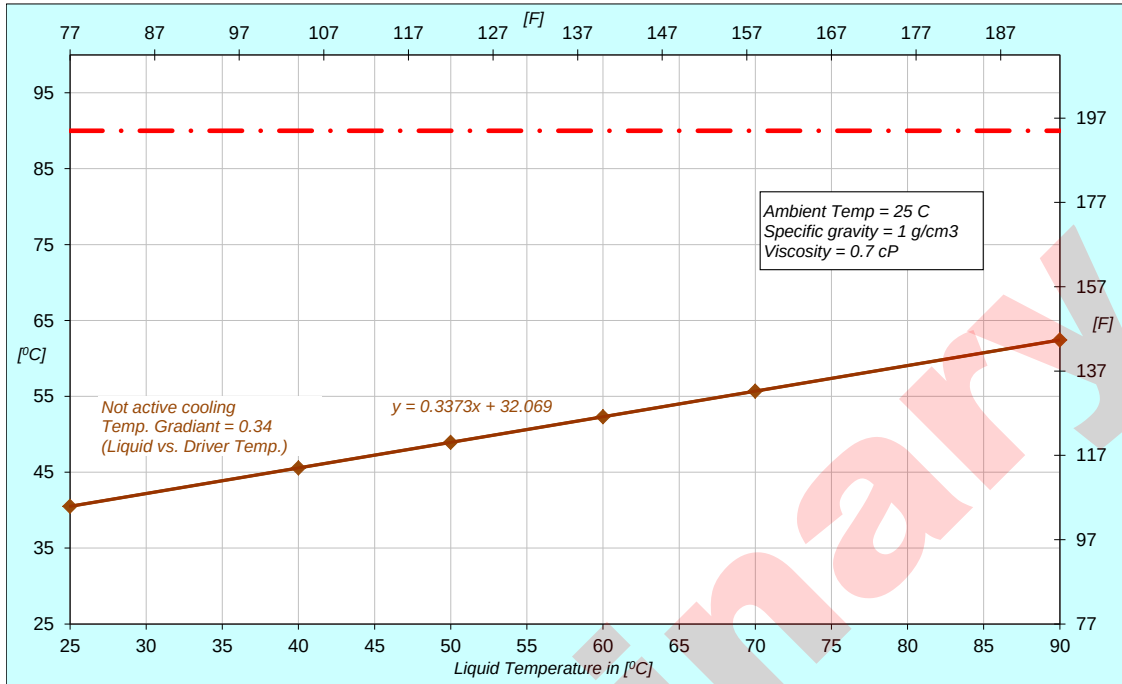


Figure 12: Influence of liquid temperature on driver temperature
(Measurement at 7000 rpm 11 lpm but gradients are representative for other operational points)

The above curves are measurements of the motor temperature at certain liquid and ambient temperatures. Equation (Eq. 1) shows how to calculate the motor temperature for other liquid and ambient temperatures based on these curves.

$$T_D(T_L, T_A) \approx \underbrace{T_D(T_L = 25^\circ\text{C}, T_A = 25^\circ\text{C})}_{\text{see Figure 12}} + (T_L - 25^\circ\text{C}) \cdot \underbrace{tg_{LM}}_{\text{see Figure 14}} + (T_A - 25^\circ\text{C})$$

(Eq. 1)

T_D = Driver temperature

T_L = Liquid temperature

T_A = Ambient temperature

tg_{LM} = Temperature gradient liquid/motor

In order to account for thermal variations (like ambient temperature, closed chemical cabinets or corners without ventilations) and to not significantly reduce the MTBF of the motor it is recommended to keep about 15 °C safety distance to the absolute thermal limit of the driver (80 °C) when designing the thermal concept of the pump system.

It is recommended to avoid thermal stagnation in the room or cabinet where the driver is placed. Any type of circulation decreases the driver temperature significantly.

4 Installation

4.1 Electrical Installation

4.1.1 Overview Wire Designation of Driver Cable

Wire Name	Wire Color	Description	Standard Designation	Typical Levels	Note
P+	red	+ 24 VDC	Supply	Voltage: 24 VDC $\pm 10\%$ Power: 120 W	P- to be connected to earth.
P-	black	Power Input Ground / Earth			
Ain1	blue	Analog Input 1 (Current Input)	Actual Process Value	4..20 mA = 0..100%	The configuration of the analog inputs can be changed with Levitronix® Service Software.
Ain2	red	Analog Input 2 (Voltage Input)	Reference Value	0..10 V = 0..10000 rpm (Speed Mode) -> Speed limit = 8000rpm -> 8V -> Cut-off (min.) speed = 300 rpm 0..10 V mA = 0..100% (Process Mode)	The configuration of the analog inputs can be changed with Levitronix® Service Software.
Ain_GND	pink	Analog Input Ground	--	--	Reference for analog inputs.
Din1	grey	Digital Input 1	Enable (Reset)	5-24 V $\Rightarrow$ active 0 V $\Rightarrow$ not active	The "Enable" signal switches the pump system on and off. Resets pump from error state with 300-700ms pulse.
Din2	yellow	Digital Input 2	Process Mode	5-24 V $\Rightarrow$ active 0 V $\Rightarrow$ not active	Switch between process mode and speed mode.
Din_COM	green	Common Digital Input	--	--	--
Aout1	black	Analog Output (Voltage Output)	Actual Speed Actual Process Value	0..10 V = 0..10000 rpm (Speed Mode) 0..10 V = 0..100% (Process Mode)	Direct connection, no protection. Galvanic isolation on user side is required. AGND is reference. The configuration of the analog inputs can be changed with Levitronix® Service Software.
Dout1	brown	Digital Output 1	Status	Closed circuit $\Rightarrow$ active, system on Open circuit $\Rightarrow$ not active, system off	This signal indicates the state of the pump system. The configuration of the analog inputs can be changed with Levitronix® Service Software.
Dout2	white	Digital Output 2	Error	Closed circuit $\Rightarrow$ not active, system on Open circuit $\Rightarrow$ active, system off	When active, the system drives the impeller to zero rpm and shuts down. With a reset pulse (see "Enable" description) the system can be re-initialized. The configuration of the analog inputs can be changed with Levitronix® Service Software.
AGND	violet	Analog Ground	--	--	Reference for Aout1, Dout1 and Dout2
RS485+	gray-pink	RS485 +	Field Bus	Modbus protocol	Termination resistors recommended
RS485-	red-blue	RS485 -			
NC	white-green	Do not connect	--	--	For future options
NC	brown-green	Do not connect	--	--	For future options
Shield	shield wire	Shielding	Shielding		To be connected to earth (see wire P-, black)

Table 7: Signals of the driver cable with designation for standard firmware G1.48

- For other configurations of PLC Inputs and Outputs refer to alternate firmware documentation
- Configurations can be done with Levitronix® Service Software (see Manual with Doc.# PL-2034-00)
- Power wires (P+, P-) have cross section 1.5 mm² and all others 0.14 mm²

4.1.2 Overview Electrical Schematics of Driver Interface

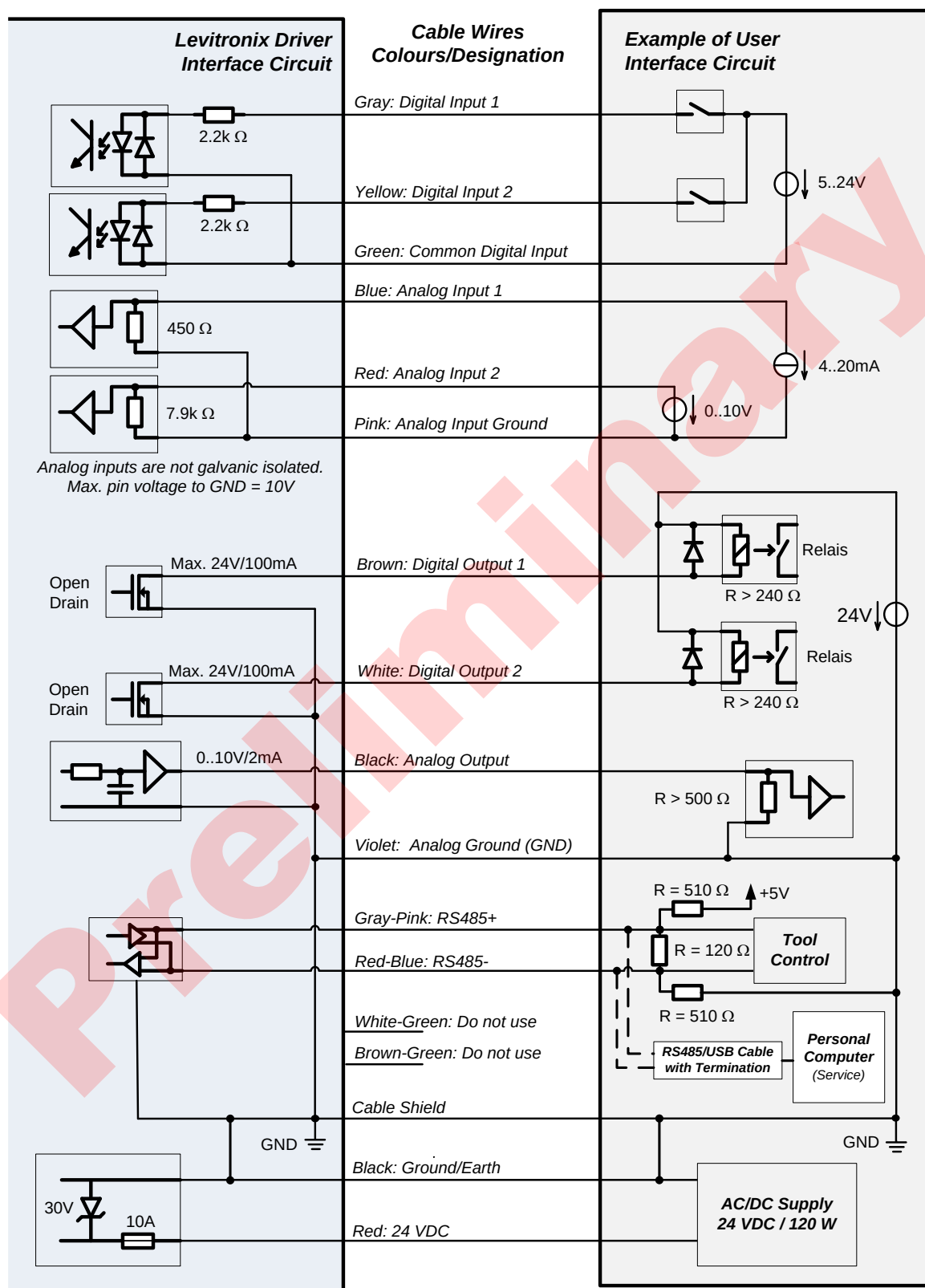


Figure 13: Electrical schematics of driver interfacing

- RS485/USB converter cable with termination resistors to be ordered according to Table 3
- Do not use multiple master devices on the RS485 at the same time

4.1.3 Installation Instructions for Power Supply

	! WARNING Hazardous voltage may be present. <i>Always isolate the electrical power supply before making or changing connections to the unit.</i> <i>In case of the usage of an inadequate AC/DC power supply, mains voltages may be present (even if the system is designed for 24 VDC). The usage of a galvanic separated power supply, which is certified by a 3rd party (UL or CE), is highly recommended.</i>	
---	--	---

1. Depending on the required hydraulic operational point (see *Figure 5*), the pump system requires 24V with a maximum power of 120 W. At a lower performance power supplies with smaller power or bigger supplies to supply several pump systems simultaneously may be used. Consult *Figure 9* to get the power consumption depending on the flow. If other power supplies than the ones recommended by Levitronix are used it is highly recommended to test these under dynamic conditions (acceleration and braking of the pump rotor speed).
2. Make sure that the polarity is correct and that AC/DC power supply is off.

4.1.4 Installation PLC Interface Signals

To operate the pump system with a PLC the analog input can be used to set either the speed or the process value (flow or pressure). The digital and analog outputs can be used to monitor the pump status and operating parameters (see *Table 7*).

CAUTION
<i>The analog inputs and outputs are not galvanic isolated from the controller electronics. To avoid ground loops and malfunctions, use floating analog signals.</i>

1. Power off the system
2. Connect the designated wires of the driver cable according to *Table 7*. Assignments and functions of the I/Os can be changed with the controller firmware version (refer to according firmware documentation).
3. Follow *Figure 13* as reference for hardware configuration of the PLC in/outputs.
4. Protect the un-used wires against short-circuit to each other

4.1.5 Installation of RS485 Interface

For usage of the RS485 as a control or service interface termination resistances shall be used at the end of the cable in order to have a stable communication and avoid disturbance effects (see *Figure 13*). For *Service and Debugging* purposes with the Levitronix[®] Service Software and PC a USB/RS485 converter cable with integrated termination resistances can be ordered according to *Table 3*. Do not use multiple master devices at the same time.

4.2 Mechanical Installation

The driver can be fixed with four screws on the motor feet (see *Figure 7*). Mounting can be done in either the horizontal or vertical position.

When mounting the pump head fittings into a circuit it has to be taken care that no mechanical stress is acting on the fittings, which can result in distorting creeping effects.

5 Operation

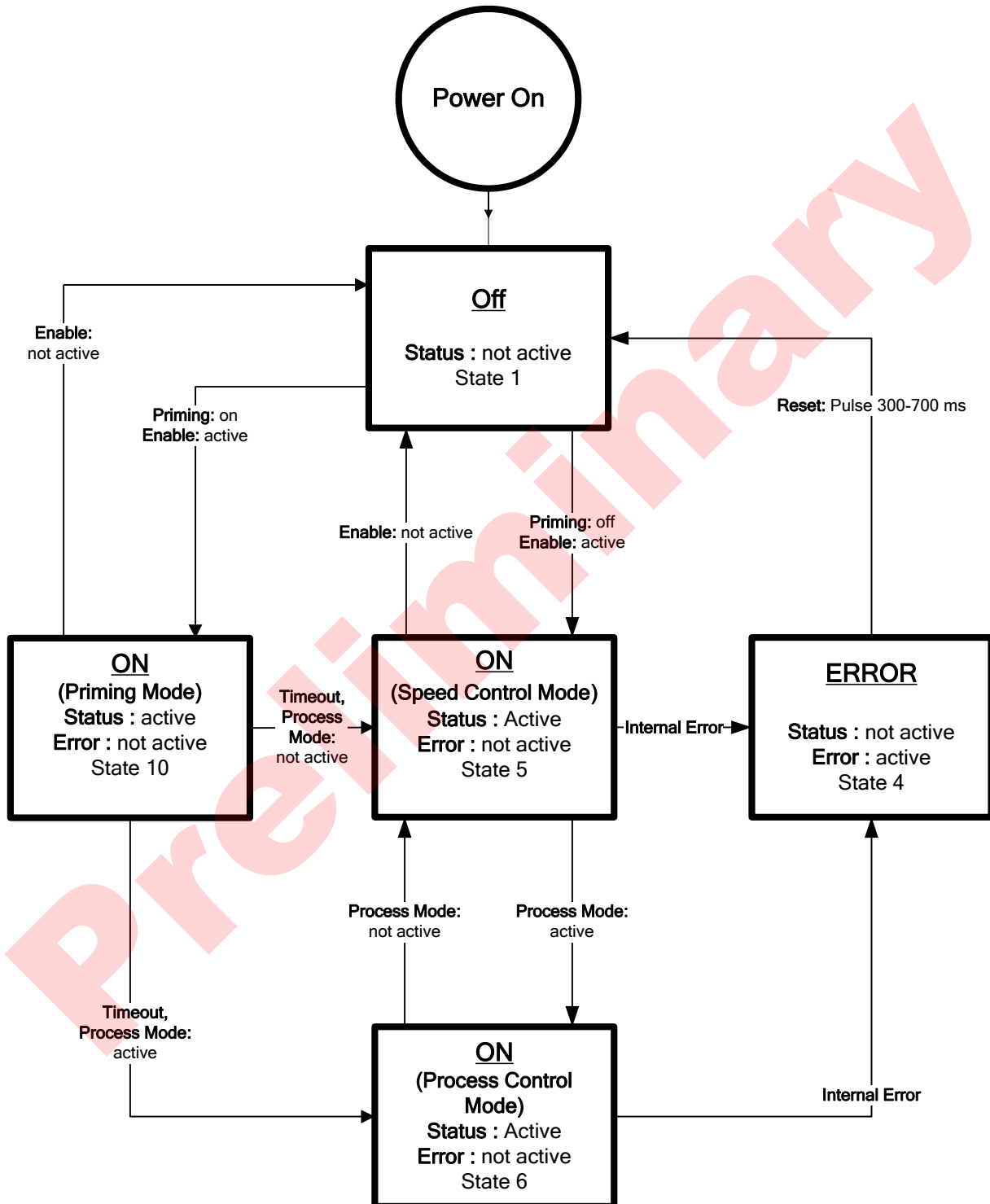


Figure 14: PLC interface state diagram for standard firmware G1.48
(For other configurations refer to alternate firmware documentation)

State “Off”:

The pump system is switched off and the motor has no power. In this state, Levitronix® Service Software has full control.

State “ON” (speed control mode):

The pump system is switched ON and the impeller is rotating with the referenced speed. The motor has electrical power when in this state.

State “ON” (process control mode):

The pump system is switched ON and the impeller is rotating with the referenced flow/pressure. The motor has electrical power when in this state.

On (Priming mode):

The pump system is switched ON and the impeller is rotating with the priming speed. The motor has electrical power when in this state. This mode can only be accessed by activating priming feature within EEPROM-editor in Levitronix® Service Software. Priming speed and Rimeout-time can also be configured within EEPROM-editor.

State “Error”:

If an error according to Table 8 occurs in the pump system, the system defaults to the Error state. The according digital output on the PLC Interface Module is activated. The pump system is switched OFF. By a pulse off 300-700 ms on the Enable/Reset input the system gets back to the Off state.

Errors	Effect on Designated Digital Output of the PLC
No rotor	Error = Open circuit
Temperature over 90°C	Error = Open circuit
Temp. was higher than 80°C for more than 10 minutes.	Error = Open circuit
Over-current	Error = Open circuit
Power channel interrupted	Error = Open circuit
Internal sensor cable interrupted	Error = Open circuit
DC link (supply voltage) out of range (< 18 or > 30 V DC) If the voltage is out of range the system starts to reduce the speed and a warning is generated. When reaching 0 rpm and the voltage is still out of range the system is disabled and an error is generated. In case the voltage is again within the range during speed reduction the system switches to normal operation and no Error is generated.	Error = Open circuit

Table 8: Errors with indication on PLC interface for standard firmware G1.48
(For other configurations refer to alternate firmware documentation)

6 Inspection and Maintenance

6.1 Replacement Interval of the Impeller

The impeller has a limited lifetime depending on the chemical type, concentration and temperature of the fluid which is pumped. Therefore a preventive periodical exchange of the impeller is recommended. Contact the Levitronix® Technical Service Department (see Section 8) for further information on replacement times.

6.2 Impeller Replacement Procedure

6.2.1 Preparation

Before starting the impeller replacement procedure the parts and tools illustrated in Figure 15 and Figure 16 should be prepared. Impeller exchange kits, which contain these parts and tools are available at Levitronix® (see Table 3). Please verify that you have the right types of impellers, O-rings and screws.

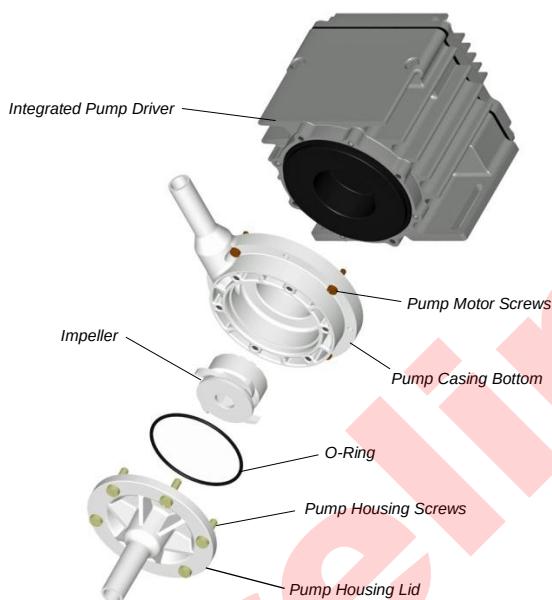


Figure 15: Explosion view of pump head with driver

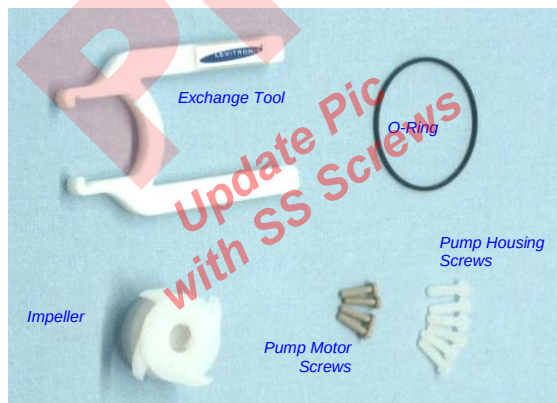


Figure 16: Components for impeller replacement

The following warnings and cautions should be read carefully before starting the replacement of the impeller.

⚠ WARNING



The impeller could splash TOXIC or CORROSIVE CHEMICALS because of the strong magnetic forces. Flush the pump housing before opening it.

⚠ WARNING



HARMFUL CHEMICALS may be present.

Skin contact and toxic gases may be hazardous to your health. Wear safety gloves and other appropriate safety equipment.

⚠ CAUTION



The rotating impeller could cause injury. Do not run the pump system when opening the pump head.

CAUTION



Pay attention to the magnetic forces when handling the impeller. The attraction of magnetic parts and particles should be avoided in order to keep the impeller and the pump head clean and free of contamination.

6.2.2 Instructions for Replacement

1. Power down the pump system and remove the AC power. If necessary, allow the housing to cool down to a workable temperature.
2. Unscrew the top of the pump head and remove it along with the sealing ring.



3. Remove the impeller with the *Impeller Exchange Tool*. Hook the claws of the *Impeller Exchange Tool* into two opposing orifices of the impeller.



4. Inspect the wet area of the pump head carefully. In case of material damage, also replace the pump casing.
5. Place the new impeller into the pump casing using the *Impeller Exchange Tool*.



6. If necessary, remove the existing *O-Ring* and gently press the new *O-Ring* into the lid of the pump casing.

CAUTION

Use the correct *O-Ring* type for your process. If necessary, consult the Levitronix® Technical Service Department.

Do NOT twist or roll the *O-Ring* as this may cause leaking to occur.

7. Press the lid with the *O-Ring* flush into the bottom of the pump casing.



8. Carefully tighten the 6 screws. The screws should not be used to press the lid with the *O-Ring* into the bottom of the pump casing. Do not apply too much torque. The torque specifications are:

Maximum torque for pump screws
SS M4: 30 Ncm SS M3: 30 Ncm

SS = Stainless Steel

9. Start up the system and check if the impeller is rotating properly and the pump head doesn't leak.
10. If the pump head leaks, check and make sure the lid and the *O-Ring* are properly pressed into the bottom of the pump casing. It may be necessary to change the *O-Ring* if it has been damaged.

7 Troubleshooting

7.1 Troubleshooting for Operation with Driver

The integrated PLC provides a *Status* and *Error* signal feedback according to *Table 8*. However, the source of error cannot be identified by these signals.

For more detailed analysis the *Levitronix® Service Software* can be used with a PC and a USB to RS485 interface cable.

7.2 Troubleshooting with Service Software

The *Levitronix® Service Software* allows communication with the pump system in connection with a PC and a USB interface. The USB to RS485 adaptor cable as specified in *Table 3* can be used to setup communication.

The software can be used for performing detailed troubleshooting. For usage of the *Service Software* refer to the *Service Software User Manual* (Document #: *PL-2034-00*), which is available in the download section on the *Levitronix®* web-page or contact the *Levitronix® Technical Service Department* (see under *Section 8*).

8 Technical Support

For troubleshooting, support and detailed technical information contact *Levitronix® Technical Service Department*:

Levitronix®
Technical Service Department
Technoparkstr. 1
CH-8005 Zurich
Switzerland

Phone for US:	888-569 07 18
Phone for outside US:	+1 888-569 07 18
E-Mail:	support@levitronix.com

9 Appendix

9.1 Regulatory Status

9.1.1 CE Marking

The *Bearingless Pump System BPS-i100*, in his various configurations, is in conformity with the essential requirements of the *EMC Directive 2004/108/EC* and the *Machinery Directive 2006/42/EC*. The following particular harmonized standards of the *EMC Directive 2004/108/EC* are tested and confirmed at a certified laboratory (Bytics AG, Test Laboratory, Switzerland, Swiss Certification No.: STS 404):

EN61000-6-4	Generic standards, Emission standard for industrial environments
EN55011	Limits and methods of measurement of electromagnetic disturbance characteristic of industrial, scientific and medical (ISM) radiofrequency equipment. Conducted and radiated emissions.
EN61000-6-2	Generic standards, Immunity for industrial environments
EN61000-4-2	ESD
EN61000-4-3	Radiated RF Immunity
EN61000-4-4	Fast transient / Burst
EN61000-4-5	Surge
EN61000-4-6	Immunity to conducted disturbances, induced by RF

The following particular standards of the *Machinery Directive* are followed for the design, validation and risk assessment:

EN809	Pumps for Fluids: basic requirements are followed.
EN12162	Procedure for hydrostatic pressure testing in fluid pumps: max. pressure testing.
EN1050	Safety for machinery – principles for risk assessments.

9.2 Symbols and Signal Words

Symbol / Signal Word	Description	Type	Source
DANGER	Indication of an imminently hazardous situation that, if not avoided, will result in death or severe injury. Limited to the most extreme situation	Signal word	SEMI S1-0701
WARNING	Indication of a potentially hazardous situation which, if not avoided, could result in death or severe injury.	Signal word	SEMI S1-0701
CAUTION	Indication of potentially hazardous situations which, if not avoided, could result in moderate or minor injury. Also alert against unsafe practice. Without safety alert indication of hazardous situation which, if not avoided, could result in property damage.	Signal word	SEMI S1-0701
	Safety alert for "Warning" and "Caution"	Safety alert	SEMI S1-0701
	Safety alert for "Danger"	Safety alert	SEMI S1-0701
	Caution (refer to accompanying documents) (is used on article labels for reference to manual)	Refer to manual	ISO 3864
	Toxic material, poison	Hazard identification	IEC 61310
	Corrosive material, corrosion	Hazard identification	IEC 61310
	Cut/sever hand, sharp object	Hazard identification	ANSI Z535.3
	Strong magnetic field	Hazard identification	SEMI S1-0701
	Danger: electricity, electrical hazard	Hazard identification	IEC 61310, ISO 3864
	Wear safety gloves	Hazard avoidance Mandatory action	IEC 61310
	Wear face shield	Hazard avoidance Mandatory action	SEMI S1-0701
	Unplug power line	Hazard avoidance Mandatory action	SEMI S1-0701
	No pacemakers	Hazard avoidance Prohibition	SEMI S1-0701
	ATEX Logo	Used for hazard identificat. in warnings	--

Table 9: Safety symbols and signal words

Conductivity Controller

Component

Hach sc200 User Manual

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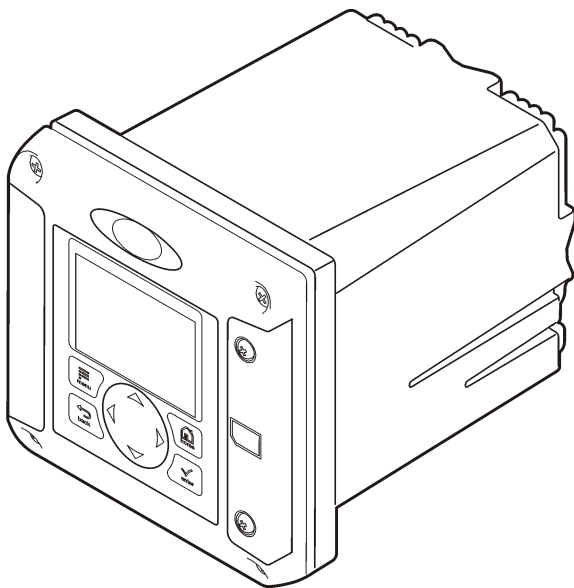


DOC023.53.80040

sc200 Controller

06/2011, Edition 3

User Manual



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Specifications

Specifications are subject to change without notice.

Specification	Details
Component description	Microprocessor-controlled and menu-driven controller that operates the sensor and displays measured values.
Operating temperature	-20 to 60 °C (-4 to 140 °F); 95% relative humidity, non-condensing with sensor load <7 W; -20 to 50 °C (-4 to 104 °F) with sensor load <28 W
Storage temperature	-20 to 70 °C (-4 to 158 °F); 95% relative humidity, non-condensing
Enclosure ¹	NEMA 4X/IP66 metal enclosure with a corrosion-resistant finish
Power requirements	AC powered controller: 100-240 VAC ±10%, 50/60 Hz; Power 50 VA with 7 W sensor/network module load, 100 VA with 28 W sensor/network module load (optional Modbus RS232/RS485 or Profibus DPV1 network connection).
	24 VDC powered controller: 24 VDC—15%, + 20%; Power 15 W with 7 W sensor/network module load, 40 W with 28 W sensor/network module load (optional Modbus RS232/RS485 or Profibus DPV1 network connection).
Altitude requirements	Standard 2000 m (6562 ft) ASL (Above Sea Level)
Pollution degree/Installation category	Polution Degree 2; Installation Category II
Outputs	Two analog (0-20 mA or 4-20 mA) outputs. Each analog output can be set to 0-20 mA or 4-20 mA, and can be assigned to represent a measured parameter such as pH, temperature, flow or calculated values. Optional 3 additional analog outputs. Secure Digital Memory card for use in data logging and software updates.

Specification	Details
Relays	Four SPDT, user-configured contacts, rated 5A 250 VAC (resistive). Contacts are rated 250 VAC, 5 Amp resistive maximum for the AC powered controller and 24 VDC, 5A resistive maximum for the DC powered controller. Relays are designed for connection to AC Mains circuits (i.e., whenever the controller is operated with 115 - 240 VAC power) or DC circuits (i.e., whenever the controller is operated with 24 VDC power).
Dimensions	½ DIN—144 x 144 x 180.9 mm (5.7 x 5.7 x 7.12 in.)
Weight	1.7 kg (3.75 lb)
Compliance information ²	CE approved (with all sensor types). Listed for use in general locations to UL and CSA safety standards by ETL (with all sensor types). Certain AC mains powered models are listed for use in general safety locations to UL and CSA safety standards by Underwriters Laboratories (with all sensor types).
Digital communication	Optional Modbus RS485/RS232 or Profibus DPV1 network connection for data transmission
Data logging	Secure Digital Card or special RS232 cable connector for data logging and performing software updates
Warranty	2 years

General information

In no event will the manufacturer be liable for direct, indirect, special, incidental or consequential damages resulting from any defect or omission in this manual. The manufacturer reserves the right to make changes in this manual and the products it describes at any time, without notice or obligation. Revised editions are found on the manufacturer's website.

¹ Units that have the Underwriters Laboratories (UL) certification are intended for indoor use only and do not have a NEMA 4X/IP66 rating.

² DC powered units are not listed by UL.

Safety information

Please read this entire manual before unpacking, setting up or operating this equipment. Pay attention to all danger, warning and caution statements. Failure to do so could result in serious injury to the operator or damage to the equipment.

To make sure that the protection provided by this equipment is not impaired, do not use or install this equipment in any manner other than that specified in this manual.

Use of hazard information

 DANGER
Indicates a potentially or imminently hazardous situation which, if not avoided, will result in death or serious injury.
 WARNING
Indicates a potentially or imminently hazardous situation which, if not avoided, could result in death or serious injury.
 CAUTION
Indicates a potentially hazardous situation that may result in minor or moderate injury.
NOTICE
Indicates a situation which, if not avoided, may cause damage to the instrument. Information that requires special emphasis.

Precautionary labels

Read all labels and tags attached to the instrument. Personal injury or damage to the instrument could occur if not observed.

	This symbol, if noted on the instrument, references the instruction manual for operation and/or safety information.
	This symbol, when noted on a product enclosure or barrier, indicates that a risk of electrical shock and/or electrocution exists.
	Delicate internal electronic components can be damaged by static electricity, resulting in degraded performance or eventual failure.
	Electrical equipment marked with this symbol may not be disposed of in European public disposal systems after 12 August of 2005. In conformity with European local and national regulations (EU Directive 2002/98/EC), European electrical equipment users must now return old or end-of-life equipment to the Producer for disposal at no charge to the user. Note: For return for recycling, please contact the equipment producer or supplier for instructions on how to return end-of-life equipment, producer-supplied electrical accessories, and all auxiliary items for proper disposal.

Certification

Canadian Radio Interference-Causing Equipment Regulation, IECS-003, Class A:

Supporting test records reside with the manufacturer.

This Class A digital apparatus meets all requirements of the Canadian Interference-Causing Equipment Regulations.

Cet appareil numérique de la classe A respecte toutes les exigences du Règlement sur le matériel brouilleur du Canada.

FCC Part 15, Class "A" Limits

Supporting test records reside with the manufacturer. The device complies with Part 15 of the FCC Rules. Operation is subject to the following conditions:

1. The equipment may not cause harmful interference.
2. The equipment must accept any interference received, including interference that may cause undesired operation.

Changes or modifications to this equipment not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment. This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to Part 15 of the FCC rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference, in which case the user will be required to correct the interference at their expense. The following techniques can be used to reduce interference problems:

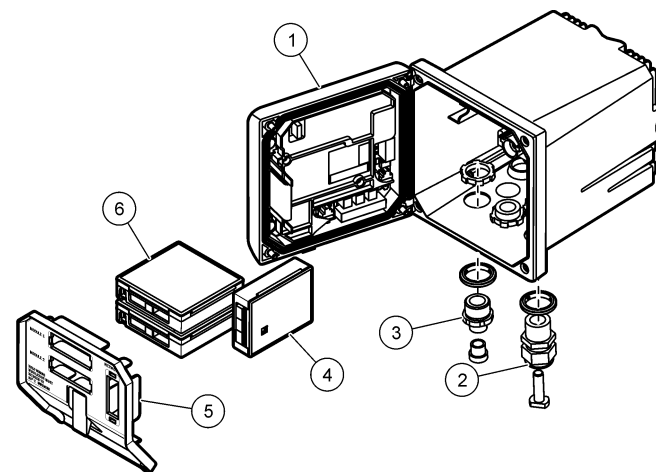
1. Disconnect the equipment from its power source to verify that it is or is not the source of the interference.
2. If the equipment is connected to the same outlet as the device experiencing interference, connect the equipment to a different outlet.
3. Move the equipment away from the device receiving the interference.
4. Reposition the receiving antenna for the device receiving the interference.
5. Try combinations of the above.

Product overview

The controller displays sensor measurements and other data, can transmit analog and digital signals, and can interact with and control other devices through outputs and relays. Outputs, relays, sensors and sensor modules are configured and calibrated through the user interface on the front of the controller.

Figure 1 shows the product components. Components may vary according to controller configuration. Contact the manufacturer if parts are damaged or missing.

Figure 1 System components



1 Controller	4 Network module (optional)
2 Strain relief assembly (optional depending on controller version)	5 High-voltage barrier
3 Digital connection fitting (optional depending on controller version)	6 Sensor modules (optional)

Sensors and sensor modules

The controller accepts up to a maximum of two sensor modules or two digital sensors (depending on the controller configuration), along with one communication module. A single digital sensor and a single sensor module can be installed in combination. A variety of sensors can be wired to the sensor modules. Sensor wiring information is given in the specific sensor manuals and in the user instructions for specific modules.

Relays, outputs and signals

The controller has four configurable relay switches and two analog outputs. An optional analog output module can increase the number of analog outputs to five.

Device scans

With two exceptions, the controller automatically scans for connected devices without user input when it is powered on. The first exception is when the controller is powered on for the first time before initial use. The second exception is after the controller configuration settings have been set to their default values and the controller is powered on. In both cases, the controller first displays the language, date and time edit screens. After the language, date and time entries are accepted, the controller performs a device scan.

Controller enclosure

The controller enclosure is NEMA 4X/IP66-rated and has a corrosion-resistant finish designed to withstand corrosive environmental constituents such as salt spray and hydrogen sulfide. Protection against environmental damage is strongly recommended for outdoor use.

Note: Units that have the Underwriters Laboratories (UL) certification are intended for indoor use only and do not have a NEMA 4X/IP66 rating.

Controller mounting options

The controller can be mounted to a panel, to a wall or to a vertical or horizontal pipe. A neoprene sealing gasket is included and can be used to reduce vibration. The gasket can be used as a template for panel mounting before the inner gasket component is separated.

Installation

Mounting components and dimensions

⚠ CAUTION

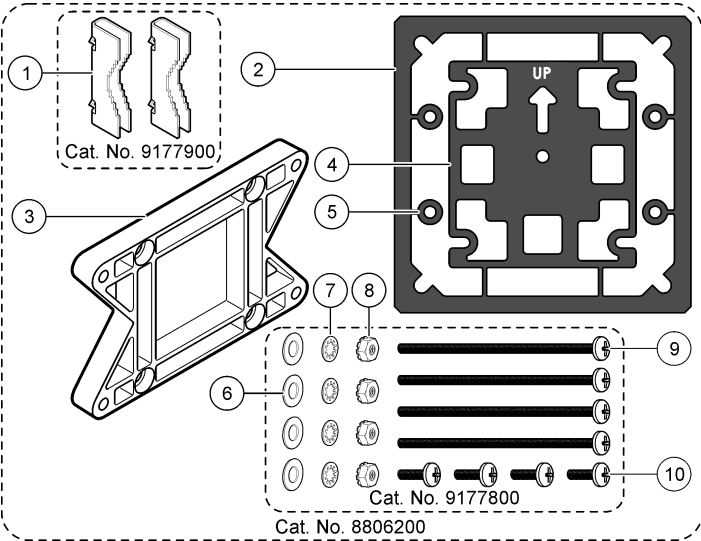
Personal injury hazard. Only qualified personnel should conduct the tasks described in this section of the manual.

The controller can be installed on a surface, panel or pipe (horizontal or vertical). For mounting options and instructions, refer to [Figure 2](#), [Figure 3](#) on page 9, [Figure 4](#) on page 9, [Figure 5](#) on page 10 and [Figure 6](#) on page 10.

For horizontal pipe mounts, the mounting feet ([Figure 2](#)) must be attached to the mounting bracket in a vertical position.

For both horizontal and vertical pipe mounts, attach the mounting bracket to the controller as shown in [Figure 5](#) on page 10.

Figure 2 Mounting components



1 Mounting foot (2x)	6 Flat washer, 1/4-inch ID (4x)
2 Sealing gasket for panel mount, Neoprene	7 Lock washer, 1/4-inch ID (4x)
3 Bracket for wall and pipe mounting	8 M5 x 0.8 Keps hexnut (4x)
4 Vibration isolation gasket for pipe mount	9 Pan head screws, M5 x 0.8 x 100mm (4x) (Used for variable diameter pipe mount installations)
5 Vibration isolation washer for pipe mount (4x)	10 Pan head screws, M5 x 0.8 x 15 mm (4x)

Controller mounting

Figure 3 Surface mounting dimensions

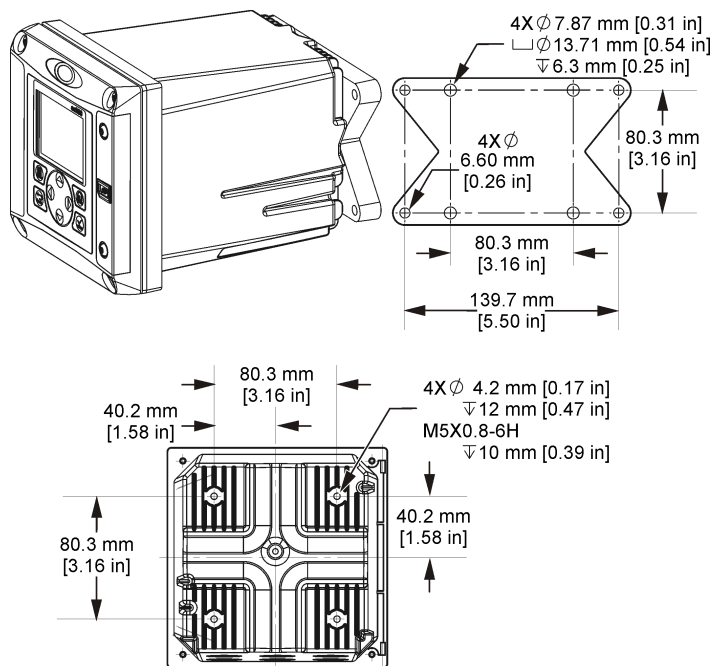


Figure 4 Panel mounting dimensions

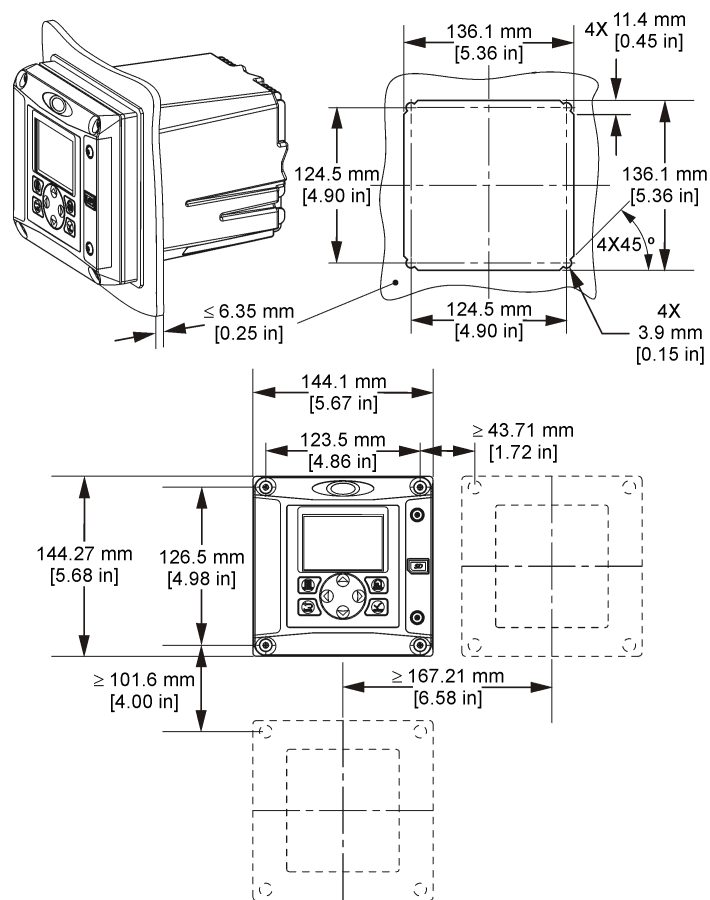


Figure 5 Pipe mounting (vertical pipe)

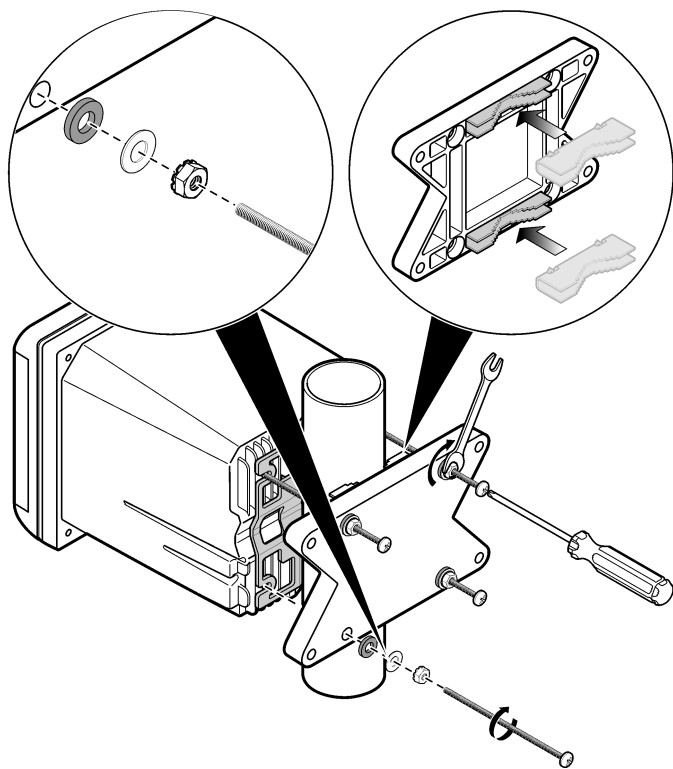
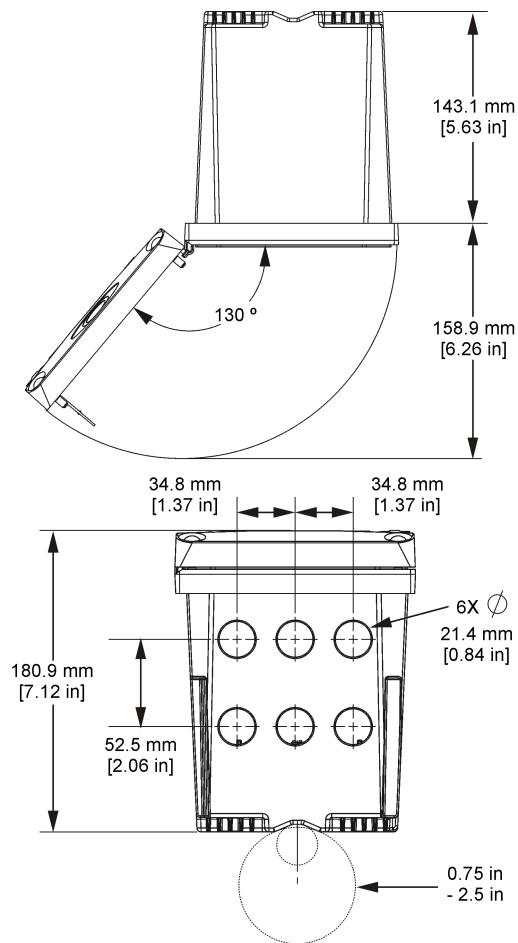


Figure 6 Top and bottom views



High-voltage barrier

High-voltage wiring for the controller is located behind the high-voltage barrier in the controller enclosure. The barrier must remain in place except when installing modules or when a qualified installation technician is wiring for power, alarms, outputs or relays. Do not remove the barrier while power is applied to the controller.

Electrostatic discharge (ESD) considerations

NOTICE



Potential Instrument Damage. Delicate internal electronic components can be damaged by static electricity, resulting in degraded performance or eventual failure.

Note: To minimize hazards and ESD risks, maintenance procedures not requiring power to the analyzer should be performed with the power removed from the instrument.

The manufacturer recommends taking the following steps to prevent ESD damage to the instrument:

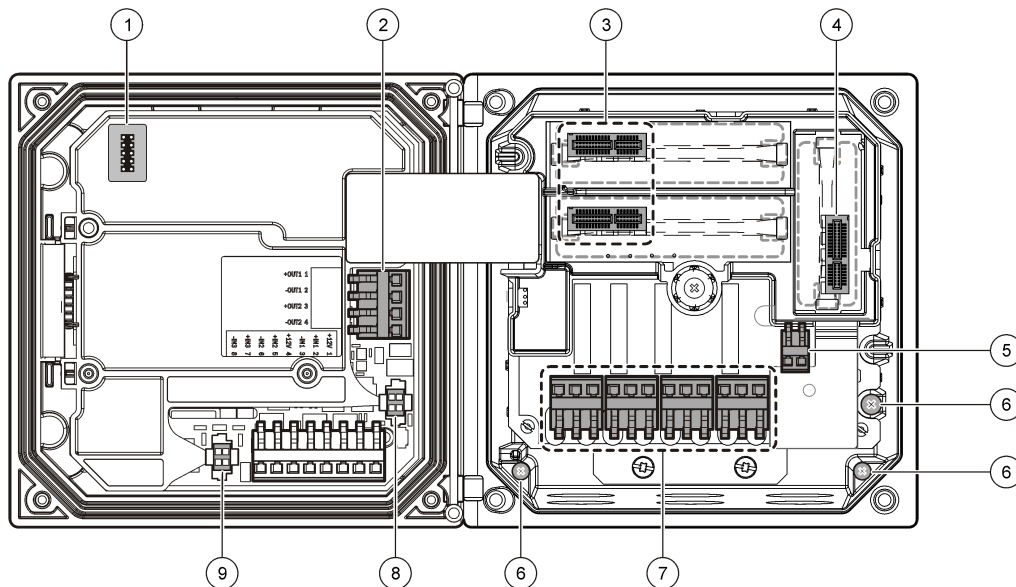
- Before touching any instrument electronic components (such as printed circuit cards and the components on them) discharge static electricity from your body. This can be accomplished by touching an earth-grounded metal surface such as the chassis of an instrument, or a metal conduit or pipe.
- To reduce static build-up, avoid excessive movement. Transport static-sensitive components in anti-static containers or packaging.
- To discharge static electricity from your body and keep it discharged, wear a wrist strap connected by a wire to earth ground.
- Handle all static-sensitive components in a static-safe area. If possible, use anti-static floor pads and work bench pads.

Wiring overview

Figure 7 shows an overview of the wiring connections inside the controller with the high voltage barrier removed. The left side of the figure shows the back side of the controller cover.

Note: Remove connector caps from the connectors before module installation.

Figure 7 Wiring connections overview



1 Service cable connection	4 Communication module connector (e.g., Modbus, Profibus, optional 4-20 mA module, etc.)	7 Relay connections ³
2 4-20 mA output ³	5 AC and DC power connector ³	8 Digital sensor connector ³
3 Sensor module connector	6 Ground terminals	9 Digital sensor connector ³

³ Terminals can be removed for improved access.

Wiring for power

⚠ WARNING	
	Potential Electrocution Hazard. Always disconnect power to the instrument when making electrical connections.

⚠ WARNING	
	Potential Electrocution Hazard. If this equipment is used outdoors or in potentially wet locations, a Ground Fault Interrupt device must be used for connecting the equipment to its mains power source.

⚠ DANGER	
	Electrocution Hazard. Do not connect AC power to a 24 VDC powered model.

⚠ WARNING	
	Potential Electrocution Hazard. A protective earth (PE) ground connection is required for both 100-240 VAC and 24 VDC wiring applications. Failure to connect a good PE ground connection can result in shock hazards and poor performance due to electromagnetic interferences. ALWAYS connect a good PE ground to the controller terminal.

NOTICE	
Install the device in a location and position that gives easy access to the disconnect device and its operation.	

The controller can be purchased as either a 100-240 VAC powered model or a 24 VDC powered model. Follow the appropriate wiring instructions for the purchased model.

The controller can be wired for line power by hard-wiring in conduit or wiring to a power cord. Regardless of the wire used, the connections are made at the same terminals. A local disconnect designed to meet local electrical code is required and must be identified for all types of installation. In hard-wired applications, the power and safety ground service drops for the instrument must be 18 to 12 AWG.

Notes:

- The voltage barrier must be removed before making any electrical connections. After making all connections, replace the voltage barrier before closing the controller cover.
- A sealing type strain relief and a power cord less than 3 meters (10 feet) in length with three 18-gauge conductors (including a safety ground wire) can be used to maintain the NEMA 4X/IP66 environmental rating.
- Controllers can be ordered with AC power cords pre-installed. Additional power cords may also be ordered.
- The DC power source that supplies power to the 24 VDC powered controller must maintain voltage regulation within the specified 24 VDC-15% +20% voltage limits. The DC power source must also provide adequate protection against surges and line transients.

Wiring procedure

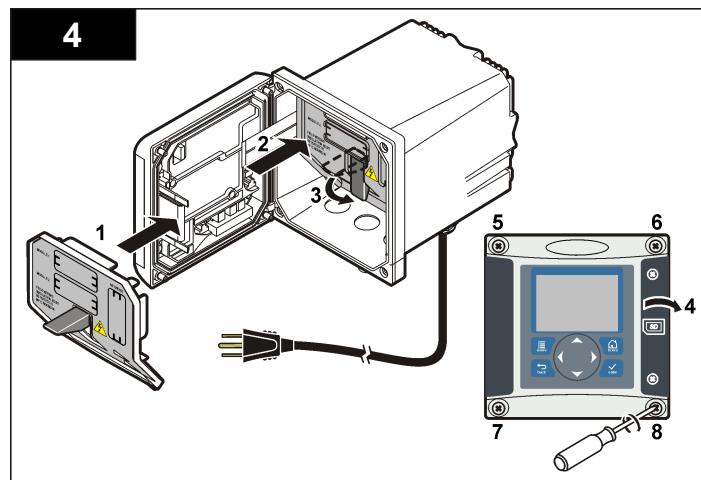
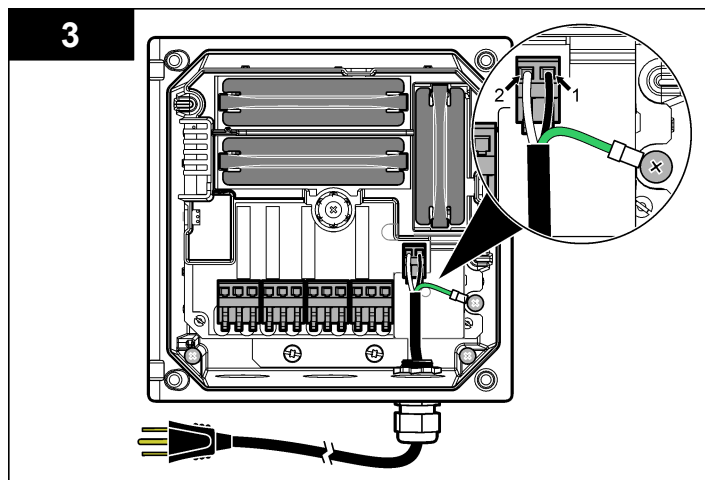
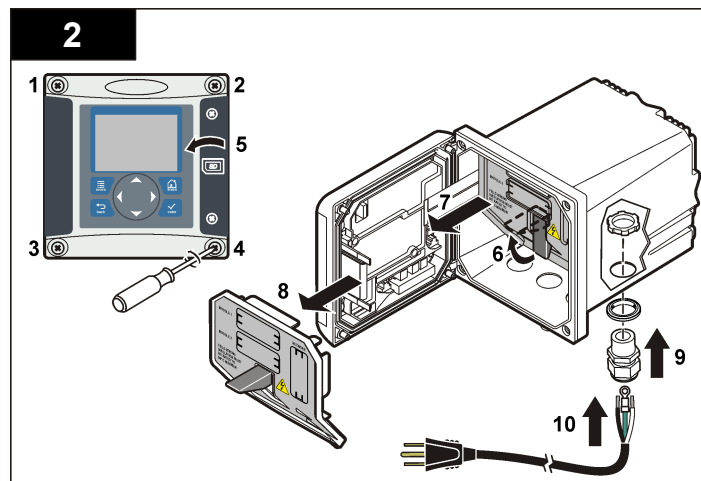
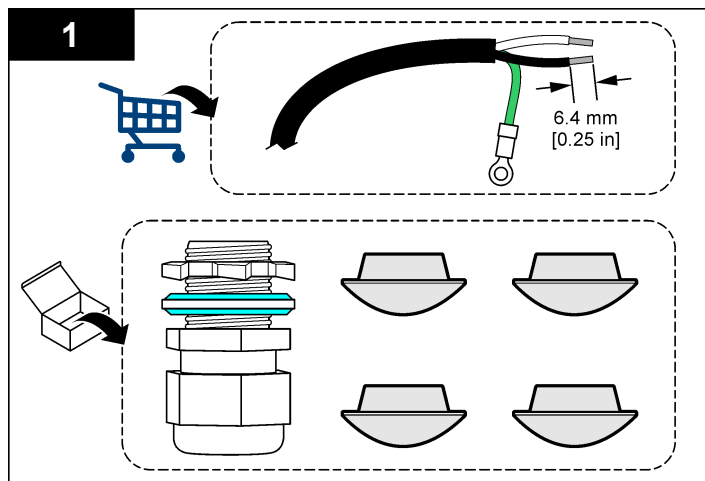
Follow the [numbered steps](#) on page 14 and [Table 1](#) or [Table 2](#) to wire the controller for power. Insert each wire into the appropriate terminal until the insulation is seated against the connector with no bare wire exposed. Tug gently after insertion to make a secure connection. Seal any unused openings in the controller box with conduit opening sealing plugs.

Table 1 AC power wiring information (AC powered models only)

Terminal	Description	Color—N. America	Color—EU
1	Hot (L1)	Black	Brown
2	Neutral (N)	White	Blue
—	Protective Earth (PE) Ground lug	Green	Green w/yellow stripe

Table 2 DC power wiring information (DC powered models only)

Terminal	Description	Color—N. America	Color—EU
1	+24 VDC	Red	Red
2	24 VDC return	Black	Black



Alarms and relays

The controller is equipped with four unpowered, single pole relays rated 100-250 VAC, 50/60 Hz, 5 amp resistive maximum. Contacts are rated 250 VAC, 5 amp resistive maximum for the AC powered controller and 24 VDC, 5A resistive maximum for the DC powered controller. The relays are not rated for inductive loads.

Wiring relays

⚠ WARNING	
	Potential Electrocution Hazard. Always disconnect power to the instrument when making electrical connections.
⚠ WARNING	
	Potential fire hazard. The relay contacts are rated 5A and are not fused. External loads connected to the relays must have current limiting devices provided to limit current to < 5 A.
⚠ WARNING	
	Potential fire hazard. Do not daisy-chain the common relay connections or jumper wire from the mains power connection inside the instrument.
⚠ WARNING	
	Potential electrocution hazard. In order to maintain the NEMA/IP environmental ratings of the enclosure, use only conduit fittings and cable glands rated for at least NEMA 4X/IP66 to route cables in to the instrument.

AC line (100—250 V) powered controllers

The wiring compartment is not designed for voltage connections in excess of 250 VAC.

24 VDC powered controllers

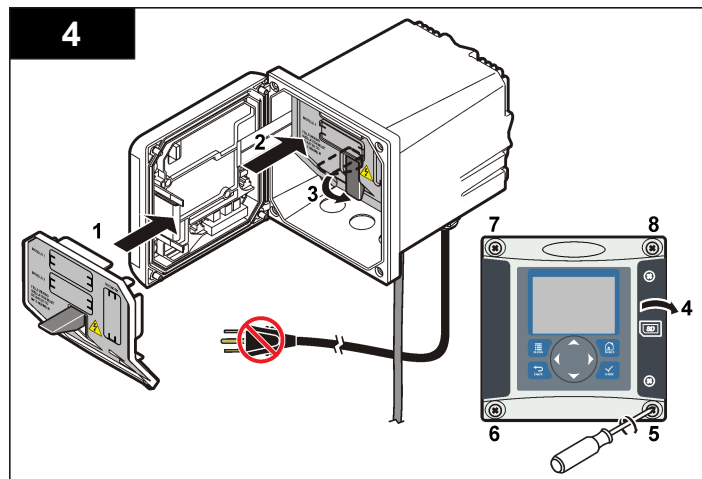
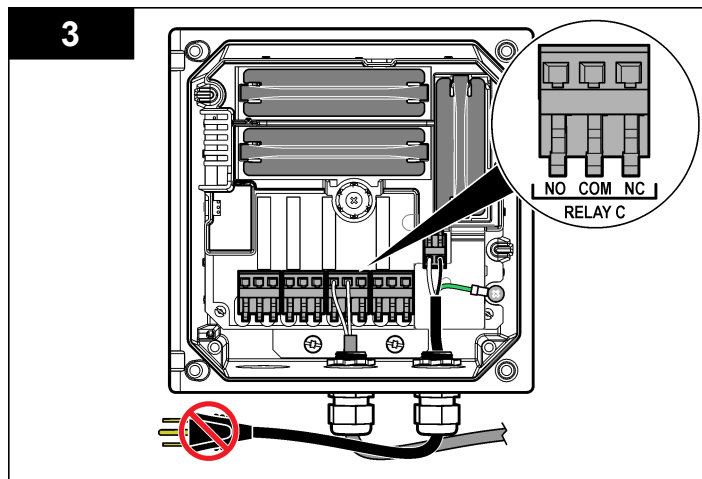
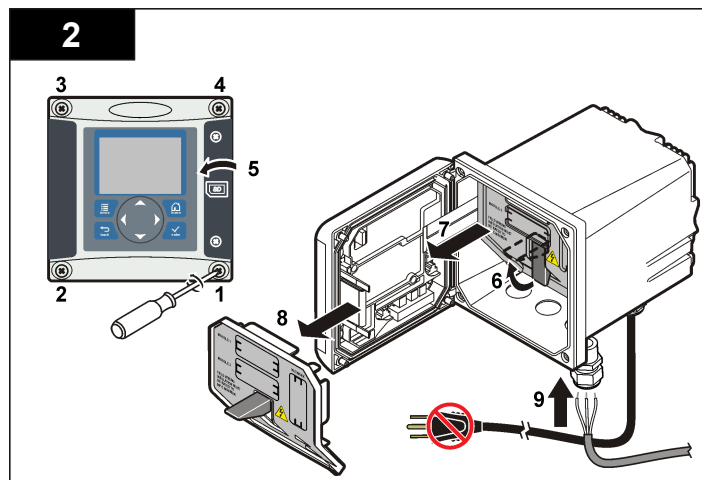
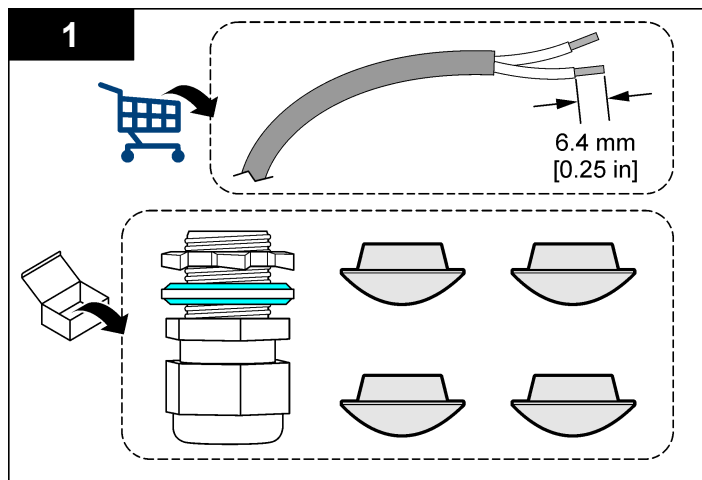
⚠ WARNING	
	Potential electrocution hazard. AC mains powered controllers (115 V–230 V) are designed for relay connections to AC mains circuits (i.e., voltages greater than 16 V-RMS, 22.6 V-PEAK or 35 VDC).
⚠ WARNING	
	Potential electrocution hazard. 24 V powered controllers are designed for relay connections to low voltage circuits (i.e., voltages less than 16 V-RMS, 22.6 V-PEAK or 35 VDC).

The 24 VDC controller relays are designed for connection to LOW voltage circuits (i.e., voltages less than 30 V-RMS, 42.2 V-PEAK or 60 VDC). The wiring compartment is not designed for voltage connections above these levels.

The relay connector accepts 18-12 AWG wire (as determined by load application). Wire gauge less than 18 AWG is not recommended.

The Normally Open (NO) and Common (COM) relay contacts will be connected when an alarm or other condition is active. The Normally Closed (NC) and Common relay contacts will be connected when an alarm or other condition is inactive (unless the Fail Safe is set to Yes) or when power is removed from the controller.

Most relay connections use either the NO and COM terminals or the NC and COM terminals. The numbered installation steps show connection to the NO and COM terminals.



Analog output connections

⚠ WARNING



Potential Electrocution Hazard. Always disconnect power to the instrument when making electrical connections.

⚠ WARNING



Potential electrocution hazard. In order to maintain the NEMA/IP environmental ratings of the enclosure, use only conduit fittings and cable glands rated for at least NEMA 4X/IP66 to route cables in to the instrument.

Two isolated analog outputs (1 and 2) are provided (Figure 8). Such outputs are commonly used for analog signaling or to control other external devices.

Make wiring connections to the controller as shown in Figure 8 and Table 3.

Note: Figure 8 shows the back of the controller cover and not the inside of the main controller compartment.

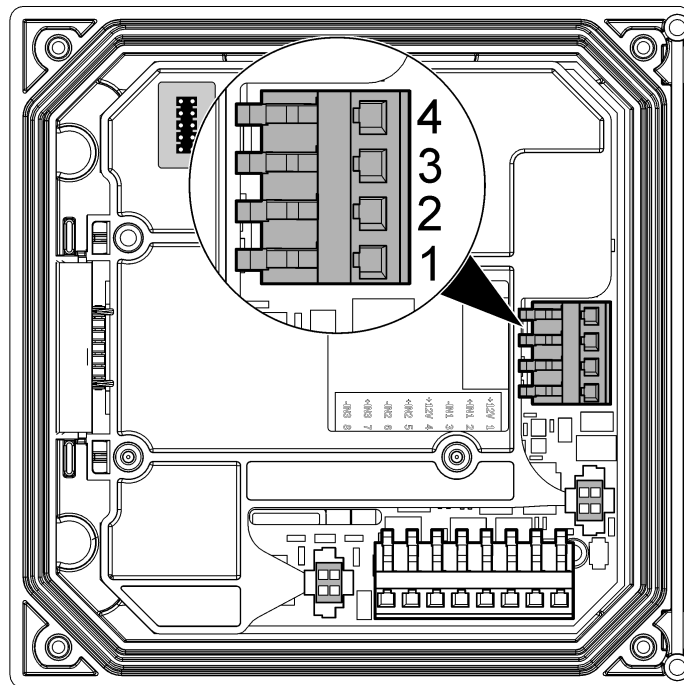
Table 3 Output connections

Recorder wires	Circuit board position
Output 1+	1
Output 1-	2
Output 2+	3
Output 2-	4

1. Open the controller cover.
2. Feed the wires through the strain relief.
3. Adjust the wire as necessary and tighten the strain relief.
4. Make connections with twisted-pair shielded wire and connect the shield at the controlled component end or at the control loop end.
 - Do not connect the shield at both ends of the cable.

- Use of non-shielded cable may result in radio frequency emission or susceptibility levels higher than allowed.
 - Maximum loop resistance is 500 ohm.
5. Close the controller cover and tighten the cover screws.
 6. Configure outputs in the controller.

Figure 8 Analog output connections



Connect a digital sc sensor

Note: To connect an analog sensor, refer to the instructions supplied in the module or sensor manual.

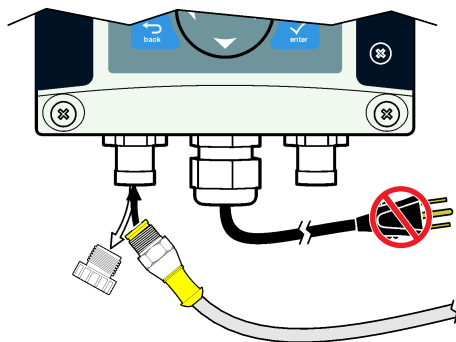
A digital sc sensor can be connected to the controller using the keyed quick-connect fitting (Figure 9). A digital sensor can be connected with the controller powered on or off.

When a sensor is connected with the controller powered on, the controller does not automatically perform a device scan. To make the controller perform a device scan, navigate to the Test/Maintenance menu and select Scan Devices. If a new device is found, the controller performs the installation process without further user action.

When a sensor is connected with the controller powered off, the controller will perform a device scan when it is powered on again. If a new device is found, the controller performs the installation process without further user action.

Retain the connector cap to seal the connector opening in case the sensor must be removed.

Figure 9 Digital sensor quick connect



Connect the optional digital communication output

The manufacturer supports Modbus RS485, Modbus RS232 and Profibus DPV1 communication protocols. The optional digital output module is installed in the location indicated by item 4 in Figure 7 on page 12. Refer to the instructions supplied with the network module for more details.

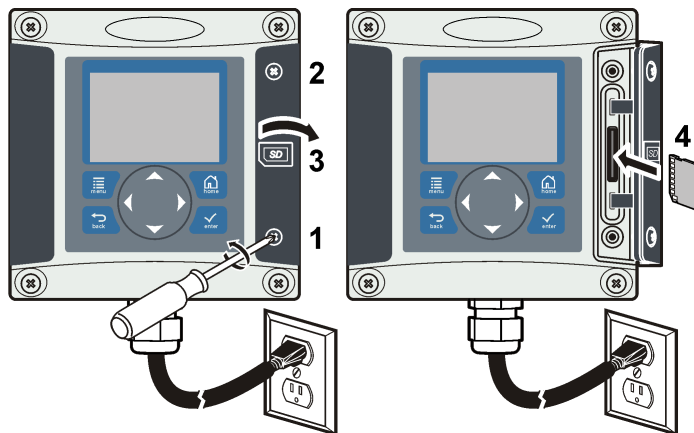
For information about Modbus registers, refer to <http://www.hach-lange.com> or <http://www.hach.com>.

Install a Secure Digital (SD) memory card

For instructions on how to install an SD card in the controller, refer to Figure 10. Information on how to use the SD memory card can be found in Using the secure digital memory (SD) card on page 34.

To remove an SD card, push down on the edge of the card and release, then pull the card up and out of the slot. After the card is removed, close the slot cover and tighten the cover screws.

Figure 10 SD card installation

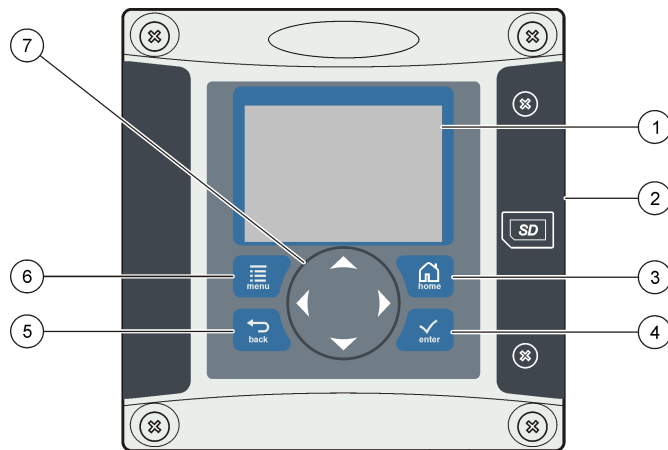


User interface and navigation

User interface

The keypad has four menu keys and four directional keys as shown in [Figure 11](#).

Figure 11 Keypad and front panel overview



1 Instrument display	5 BACK key. Moves back one level in the menu structure.
2 Cover for secure digital memory card slot	6 MENU key. Moves to the Settings Menu from other screens and submenus.
3 HOME key. Moves to the Main Measurement screen from other screens and submenus.	7 Directional keys. Used to navigate through the menus, change settings, and increment or decrement digits.
4 ENTER key. Accepts input values, updates, or displayed menu options.	

Inputs and outputs are set up and configured through the front panel using the keypad and display screen. This user interface is used to set up and configure inputs and outputs, create log information and calculated values, and calibrate sensors. The SD interface can be used to save logs and update software.

Reading and measurement screens

When two sensors are connected, push the **HOME** key and then push the Right or Left arrows to toggle through the available measurement display options. A combination screen is the default display mode when two sensors are connected.

Push the **UP** or **DOWN** arrows to toggle the status bar at the bottom of the measurement display. The footer bar displays controller, sensor, or network card errors and warnings; sensor and network card process events; secondary measurements and outputs.

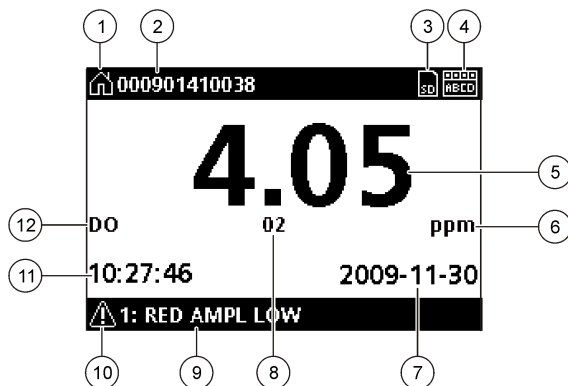
If a menu has more options than can be displayed at one time, a scroll bar will appear on the right side of the display. Push the **UP** or **DOWN** arrows to scroll through the available menu items.

Display

[Figure 12](#) shows an example of the main measurement screen with a DO sensor connected to the controller.

The front panel display screen shows sensor measurement data, calibration and configuration settings, errors, warnings and other information.

Figure 12 Example of Main Measurement screen



1 Home screen icon	7 Date
2 Sensor name	8 Measurement parameter
3 SD Memory card icon	9 Analog output values, error and warning messages and secondary measurements.
4 Relay status indicator	10 Warning icon
5 Measurement value	11 Time
6 Measurement units or warning icon	12 Measurement name

Table 4 Icon descriptions

Icon	Description
Home screen	The icon may vary depending on the screen or menu being displayed. For example, if an SD card is installed, an SD card icon appears here when the user is in the SD Card Setup menu.
SD memory card	This icon appears only if an SD card is in the reader slot. When a user is in the SD Card Setup menu, this icon appears in the upper left corner.

Table 4 Icon descriptions (continued)

Icon	Description
Warning	A warning icon consists of an exclamation point within a triangle. Warning icons appear in the display footer along with a number that indicates the associated device. • 0=controller • 1=sensor 1 • 2=sensor 2 • 3=network card
Error	An error icon consists of an exclamation point within a circle. When an error occurs, the error icon and the measurement screen each flash again and again in the main display.

System startup

When initially powered up, the Language, Date Format and Date/Time screens appear in order. After these options are set, the controller performs a device scan and displays the message **Scanning for devices. Please wait...** If a new device is found, the controller performs an installation process before displaying a main measurement screen.

If the scan finds previously installed devices without configuration changes, the main measurement screen of the device in the number one position appears immediately after the scan is complete.

If a device has been removed from the controller or is not found during the next power-cycled or menu-driven scan, the controller displays a **Device missing** message and prompts to delete the missing device.

If no sensor is connected to an installed analog module, the controller will indicate an error. If devices are connected but not found by the controller, refer to [Troubleshooting](#) on page 38.

Set the language, date and time for the first time

The controller displays the language, date and time edit screens when the controller is powered on for the first time, and when it is powered on after the configuration settings have been set to their default values.

After the language, date and time options are set for the first time, update the options as necessary through the setup menu.

1. In the Language screen, highlight a language in the options list and push the **ENTER** key. English is the default language for the controller. The selected language is saved. The Date Format screen appears.
2. In the Date Format screen, highlight a format and push the **ENTER** key. The date and time format is saved. Next, the Date /Time screen appears.
3. In the Date/Time screen, push the **RIGHT** or **LEFT** arrow keys to highlight a field, then push the **UP** and **DOWN** arrow keys to update the value in the field. Update the other fields as necessary.
4. Push the **ENTER** key. The changes are saved and the controller performs a start-up scan for devices. If connected devices are found, the controller displays the main measurement screen for the device in the number one position. If the controller fails to find connected devices, refer to [Troubleshooting](#) on page 38.

Adjust the display contrast

1. From the Settings Menu, select sc200 Setup.
2. Select Display Contrast.
3. Use the **UP** and **DOWN** arrow keys to adjust the contrast to a value between the minimum of +1 and the maximum of +9.

Controller configuration information

General information about configuration options is listed in the table. Step-by-step information for how to configure some of the options can be found on the CD.

1. To navigate to the menu options, from the Settings Menu, select sc200 Setup.

Option	Description
Security setup	Sets the passcode preferences
Output setup	Configures the controller analog outputs
Relay setup	Configures the controller relays
Display contrast	Adjusts the contrast of the controller display
Set Date/Time	Sets the controller time and date
Datalog setup	Configures data logging options. (Available if Calculation has been setup. At least one sensor must be attached to enter a calculation.) • Set Mode—Snap shot, Average, Maximum, Minimum. • Set Interval—5 sec, 30 sec, 1 min, 2 min, 5 min, 10 min, 15 min, 30 min
Error Hold Mode	Hold Outputs —Holds outputs at last known value when controller loses communication with the sensor. Transfer Outputs —Switches to transfer mode when controller loses communication with the sensor. Outputs transfer to a pre-defined value.
sc200 Information	S/W VER: —Displays the current version of controller software. Bootloader VER: —Displays the current Bootloader version. The Bootloader is a file that loads the main operating system for the controller. S/N: —Displays the controller serial number.
Edit name	Assigns a name to the controller
Language	Assigns the language used in the controller

2. Select an option and push **ENTER** to activate the menu item.

Advanced operations

Security setup

Enable or disable the passcode

By default the passcode option is disabled and all configuration settings and calibrations can be changed. When the passcode function is enabled, access to Sensor calibration and Test/Maint menus requires a passcode. To enable the passcode:

1. From the Settings Menu, select sc200 Setup and push the **ENTER** key.
2. Select Security Setup and push the **ENTER** key.
3. Select Set Passcode and push the **ENTER** key.
4. Select Disabled or Enabled and push the **ENTER** key.
The passcode is enabled.
5. Push the **BACK** key to return to the sc200 Setup Menu, or push the **MENU** key to return to the Settings Menu.

Edit the passcode

The passcode is factory set to SC200. The Edit Passcode menu option appears in the Security Setup menu only after the passcode feature is enabled and a valid passcode has been entered. A passcode consists of up to six upper or lower-case alpha, numeric and special characters. Passcodes are case-sensitive.

To edit the passcode:

1. Make sure the passcode is enabled. Refer to [Enable or disable the passcode](#) on page 22 for information on how to enable the passcode.
2. From the Settings menu, select Security Setup and push **ENTER**.
3. Use the arrow keys to enter the current valid passcode and push **ENTER**.
The Edit Passcode option appears in the Security Setup menu.
4. Select Edit Pass Code and push **ENTER**.
The Edit Pass Code screen appears.
5. Use the arrow keys to edit the passcode and push **ENTER**.
The new passcode is saved and the Security Setup menu appears.

*Note: All menus stay accessible until the **HOME** key is pushed or the controller is restarted.*

6. Push the **HOME** key or perform a controller restart.
The new passcode settings are saved, and the new passcode is required to enter the Security Setup, Datalog Setup and Test/Maint menus.

Configure a 4-20 mA input module

An analog module must be installed in the controller.

1. Determine what output the the connected device is using (0-20 mA or 4-20 mA). This information will be used to set the scale.
2. Determine what the 20 mA value is equal to (e.g, 100 psi).
3. Determine what the low end (0 or 4 mA) value is equal to (e.g., 10 psi). This information will be used to set the display range.
4. From the Settings Menu, select Sensor Setup.
5. Select Configure.
6. Update the options.
 - a. Highlight an option and push **ENTER**.
 - b. Make a selection or update the entries.
 - c. Push **ENTER** to save the changes.

Option	Description
Edit name	Edits the module name
Edit units	Edits the measurement units
Edit parameter	Edits the parameter name

Option	Description
Display range For the 0-20 mA scale: - Set the 20 mA value - Set the 0 mA value For the 4-20 mA scale: - Set the 20 mA value - Set the 4 mA value	Sets the values used for the selected scale (0-20 mA or 4-20 mA)
Signal average	Sets how often signals are averaged. Higher values produce a smoother signal but increase the time it takes for a signal to respond to a change in the process value.
Set resolution— X.XXX, XX.XX, XXX.X, XXXX	Sets the number of decimal places used in the display.
Select scale— 4-20 mA or 0-20 mA	Sets scale used for the 4-20 mA input
Data log interval—5 sec, 30 sec, 1 min, 2 min, 5 min, 10 min, 15 min, 30 min, 60 min	Sets how often data is logged to the internal controller memory.
Reset defaults—Push ENTER to reset configuration settings or push the BACK key to cancel.	Resets configuration settings to the default values.

Configure a 4-20 mA output module

The Network Setup option appears in the Settings Menu only if an analog output module or other network module such as Modbus, HART or Profibus is installed in the controller.

Outputs for analog output modules are set at 4-20 mA. Outputs can be assigned to represent a measured parameter such as pH, temperature, flow or calculated values.

1. From the Settings menu, select Network Setup.

2. Select Edit Name and enter a name for the module. Push **ENTER** to save the name.
3. Select an output (A, B, C) and push **ENTER**.

Note: Outputs A and B can be configured for HART communications.

- a. Highlight an option and push **ENTER**.
- b. Make a selection from the list or update the entries.
- c. Push **ENTER** to save the changes.

Option	Description
Select Source	Selects the output to configure—None, sensor 1 name, sensor 2 name, calculation (if set up). For sensor output, Select Parameter sets the measurement options. When the measurement is autorange, Set Range sets the range.
Set Low Value	Sets the 4 mA value (default: 0.000). (Range and units depend on sensor)
Set High Value	Sets the 20 mA value (default: 1.000). (Range and units depend on sensor)
Set Transfer	Sets the transfer value. Range 3.0 to 23.0 mA (default 4.000).
Set Filter	Sets a time-average filter value of 0 (default) to 120 seconds.
Set HART Address	Sets the address. Range 0 to 63.

Note: If a device malfunction is detected, the output goes to the user-selected transfer value.

Configure the controller analog outputs

The controller analog outputs can be assigned to represent the measured parameter or secondary measurements such as temperature and calculations. To configure the options, highlight a menu option, push **ENTER** and select an option or update the entries. Push **ENTER** after an option is selected or the entries are updated.

1. From the Settings menu, select sc200 Setup.

2. Select Output Setup.
3. Select Output 1 or Output 2.
4. Choose Select Source and select a source from the list. Typically the source is one of the sensors attached to the system. If an analog input card is installed, the analog input may be used as a source.
5. From the Output Setup menu, choose Select Parameter and choose an option from the list. Parameters will vary depending on the type of sensors installed.
6. From the Output Setup menu, select Set Function and choose a function. Further setup options will vary depending on which function is chosen.

Option	Description
Linear	Signal is linearly dependent on the process value
PID	Signal works as a PID (Proportional, Integral, Derivative) controller
Logarithmic	Signal is represented logarithmically within the process variable range
Bilinear	Signal is represented as two linear segments within the process variable range

7. From the Output Setup menu, select Activation. Use the information in the table below the chosen function to configure the options.
8. If Transfer is or will be selected as the Error Hold Mode, or if the Transfer will be used during calibration or other functions within the sensor menu, select Set Transfer from the Output Setup menu and enter the transfer value.
9. From the Output Setup menu, select Set Filter and enter the filter value.

10. From the Output Setup menu, select Scale and choose the scale (0-20 mA or 4-20 mA).

- **Linear**

Option	Description
Set low value	Sets the low endpoint of the process variable range
Set high value	Sets the high endpoint of the process variable range

- **PID**

Option	Description
Set mode (Auto or Manual)	Auto—the signal is automatically controlled by the algorithm within the analyzer using proportional, integral, and derivative inputs. Manual—the signal is controlled by the user through manual adjustment of the % change value identified in the menu as Manual Output.
Phase (Direct or Reverse)	The direction in which the signal responds to process change. Direct—signal increases as the process increases. Reverse—signal increases as process decreases.
Set setpoint	Creates a desired control point of process
Prop band	A function of the difference between the measured signal and the desired setpoint.
Integral	The period of time from the injection point of a reagent to contact with the measuring device.
Derivative	Used to compensate for the 2 nd order effects of the process. The majority of applications can be controlled without the use of the derivative setting.
Transit time	Stops all PID control for a selected period of time as the sample travels from the control pump to the measurement sensor.

- **Logarithmic**

Option	Description
Set 50% value	Sets the value corresponding to 50% of the process variable range.
Set high value	Sets the upper value of the process variable range

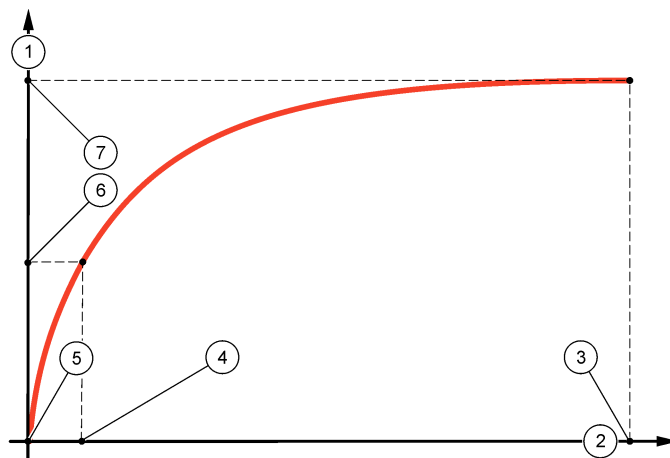
• **Bilinear**

Option	Description
Set low value	Sets the low endpoint value of the process variable range.
Set high value	Sets the high endpoint value of the process variable range.
Set knee point value	Sets the value at which the process variable range divides into another linear segment.
Set knee point current	Sets the value of the current at the knee point value.

Logarithmic output mode

Figure 13 shows in graph form the operation of the logarithmic output mode.

Figure 13 Logarithmic output

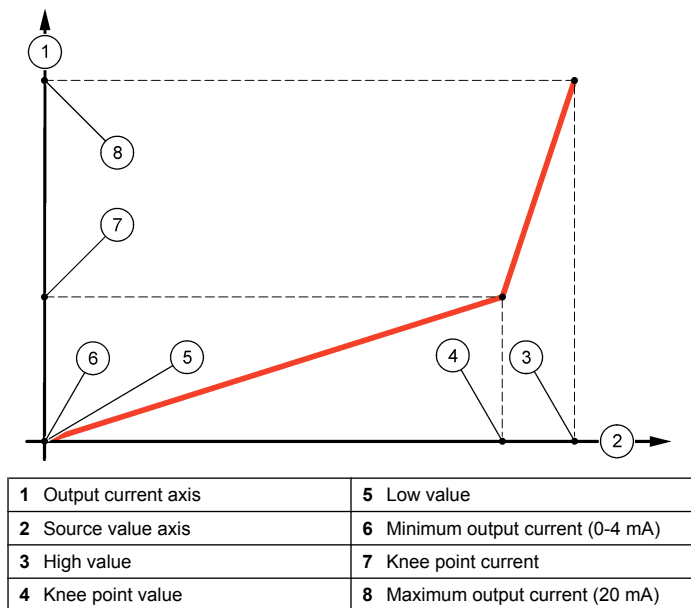


1 Output current axis	5 Minimum output current (0-4 mA)
2 Source value axis	6 50% output current
3 High value	7 Maximum output current (20 mA)
4 50% value	

Bilinear output mode

Figure 14 shows in graph form the operation of the bilinear output mode.

Figure 14 Bilinear output



Configure relays

The Normally Open (NO) and Common (COM) relay contacts will be connected when an alarm or other condition is active. The Normally Closed (NC) and Common relay contacts will be connected when an alarm or other condition is inactive (unless the Fail Safe is set to Yes), or when power is removed from the controller. To select a menu option, highlight the option and push **ENTER**.

1. From the sc200 Setup menu, select Relay Setup.
2. Select a relay from the list.

3. From the Relay Setup menu, choose Select Source and push **ENTER**. Normally, a source is one of the sensors attached to the system, but the controller can also function as a source. If an analog input module is installed, the source may be the analog input.
4. From the Relay Setup menu, select Set Parameter and choose from the list of parameters. The list of parameter options will vary with the type of attached sensor.
5. From the Relay Setup menu, select Set Function and choose from the list. Further setup will depend on the function chosen.

Option	Description
Scheduler Function (available if the controller is selected as the relay source)	Relay switches at certain times independently of any process value
Alarm Function	Relay activates when upper or lower alarm value is exceeded
Feeder Control Function	Relay indicates if a process value exceeds or falls below a setpoint
Event Control Function	Relay toggles if a process value reaches an upper or lower limit
Pulse Width Modulation (PWM) Control Function	Relay uses a Pulse Width Modulation control depending on a process value
Frequency control	Relay switches with a frequency depending on a process value
Warning	Relay indicates warning and error conditions in probes

6. From the Relay Setup menu, select Set Transfer and choose Active or Inactive.
7. From the Relay Setup menu, select Fail Safe and choose Yes or No.
8. From the Relay Setup menu, select Activation.
The activation options for the selected function appear. Use the information in the table below each function to update the options.

9. Test the relay function to make sure it is properly energizing the connected device. Relay testing is performed through the Test/Maint option of the Settings menu.

- **Scheduler Function**

Option	Description
Hold outputs	Holds outputs in the present ON or OFF state.
Run days	Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday
Start time	Sets the start time
Interval	Sets the time between activation cycles (Default value: 5 min)
Duration	Sets the period of time the relay is energized (Default value: 30 sec)
Off delay	Sets the time for additional hold/output time after the relay has been turned off

- **Alarm function**

Option	Description
Low alarm	Sets the value where the relay will turn on in response to decreasing measured value. For example, if the low alarm is set for 1.0 and the measured value drops to 0.9, the relay activates.
High alarm	Sets the value where the relay will turn on in response to increasing measured value. For example, if the high alarm is set for 1.0 and the measured value increases to 1.1, the relay activates.
Low deadband	Sets the range where the relay remains on after the measured value increases above the low alarm value. For example, if the low alarm is set for 1.0 and the low deadband is set for 0.5, the relay remains on between 1.0 and 1.5. Default is 5% of the range.

Option	Description
High deadband	Sets the range where the relay remains on after the measured value decreases below the high alarm value. For example, if the high alarm is set for 4.0 and the high deadband is set for 0.5, the relay remains on between 3.5 and 4.0. Default is 5% of the range.
Off delay	Sets a time (0-300 seconds) to delay the relay from normally turning off.
On delay	Sets a delay time for the relay to turn on (Default: 5 seconds)

- **Feeder Function**

Option	Description
Phase	Defines the relay status if the process value exceeds the setpoint. High (default)—turns the relay on when the process value exceeds the setpoint. Low—turns the relay on when the process value falls below the setpoint.
Set setpoint	Sets the process value at which the relay toggles (Default value: 10)
Deadband	Sets a hysteresis so the relay will not swing unregulated when the process value converges to the setpoint.
Overfeed timer	Sets a time period for de-activating an active relay if the process setpoint cannot be reached. Once an overfeed alarm is present, it must be manually reset.
Off delay	Sets a delay time for the relay to turn off (Default: 5 seconds)
On delay	Sets a delay time for the relay to turn on (Default: 5 seconds)

- **Event Control**

Option	Description
Set setpoint	Sets the value where the relay will turn on

Option	Description
Deadband	Sets a hysteresis so the relay will not swing unregulated when the process value converges to the setpoint
OnMax timer	Sets the maximum time the relay can remain on (Default: + 0 min)
OffMax timer	Sets the maximum time the relay can remain off (Default: + 0 min)
OnMin timer	Sets the time the relay will remain on independent from the measured value (Default: + 0 min)
OffMin timer	Sets the time the relay will remain off independent from the measured value (Default: + 0 min)

• Pulse Width Modulation Function

Option	Description
Set mode	Auto—the relay output works as a PID controller. Manual—the relay output has an On/Off ratio as set in the Manual Output menu.
Phase	Reverses the leading sign of the control deviation for the PID controller (Default: Direct)
Set setpoint	Creates a setpoint value
Dead zone	A band around the setpoint. In this band the PID controller does not change the PWM On/Off ratio output signal. The dead zone is determined as the setpoint +/- the dead zone.
Period	Sets the cycle duration of the PWM output signal (Default: 5 seconds)
Min width	Sets the minimum PWM ratio (Default: 0%)
Max width	Sets the maximum PWM ratio (Default: 100%)
Prop band	Sets the proportional part of the PID controller
Integral	Sets the integral part of the PID controller (Default: 15 minutes)

• Frequency Control Function

Option	Description
Set mode	Auto—The relay works as a PID controller. Manual—the relay output has an On/Off ratio as set in the Manual Output menu.
Phase	Reverses the leading sign of the control deviation for the PID controller (Default: Direct)
Set setpoint	Sets the process value which is controlled by the PID controller
Dead zone	Band around the setpoint. In this band, the PID controller does not change the output frequency. The dead zone is determined as the setpoint +/- the dead zone.
Pulse width	Sets the cycle duration (0-600 seconds) of the PWM output signal. (Default: 5 seconds)
Minimum pulses	Sets the minimum number of pulses per minute at which the relay can operate
Maximum pulses	Sets the maximum number of pulses per minute at which the relay can operate
Prop band	Sets the proportional part of the PID controller
Integral	Sets the derivative part of the PID controller (Default: 15 minutes)

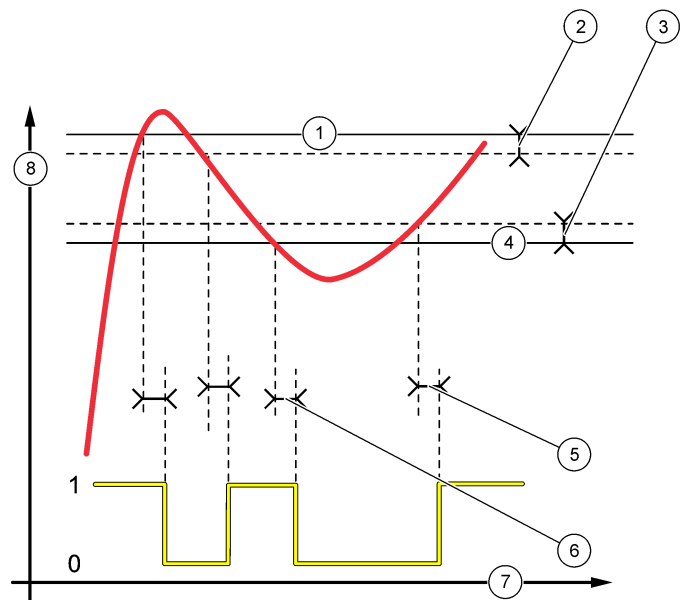
• Warning

Option	Description
Warning	Sets the level for warning activation

Relay function graphs

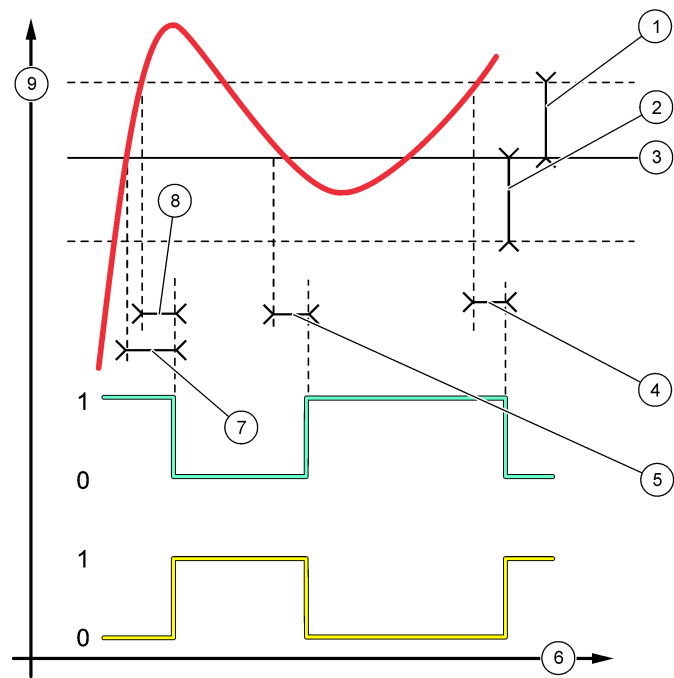
Figure 15, Figure 16, Figure 17, Figure 18, Figure 19, Figure 20, Figure 21, Figure 22 and Figure 23 show in graph form the behavior of a relay when in different function modes.

Figure 15 Alarm mode



1 High alarm	5 ON delay
2 High deadband	6 OFF delay
3 Low deadband	7 Time (x-axis)
4 Low alarm	8 Source (y-axis)

Figure 16 Feeder control mode



1 Deadband (Phase = Low)	6 Time (x-axis)
2 Deadband (Phase = High)	7 ON delay (phase set high)
3 setpoint	8 OFF delay (phase set low)
4 OFF delay (phase set high)	9 Source (y-axis)
5 ON delay (phase set low)	

Figure 17 Feeder control mode (Phase low, Overfeed timer)

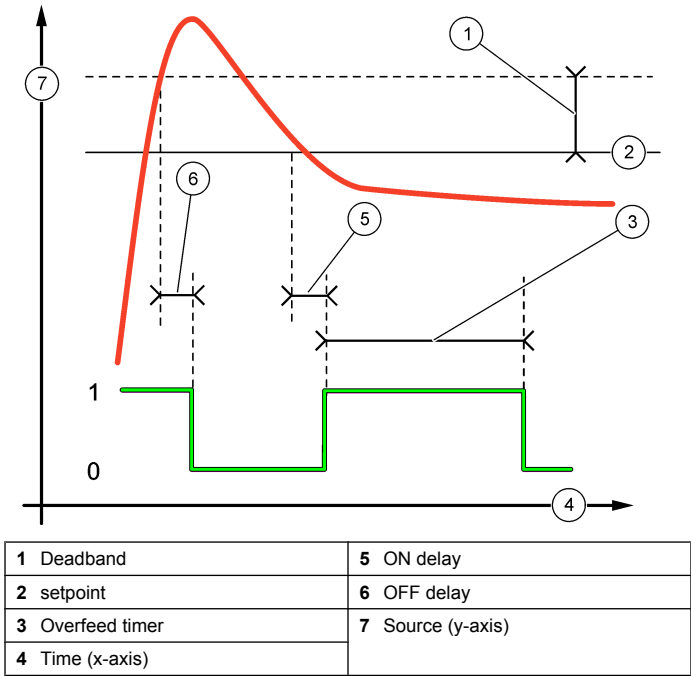


Figure 18 Event control mode (no delay)

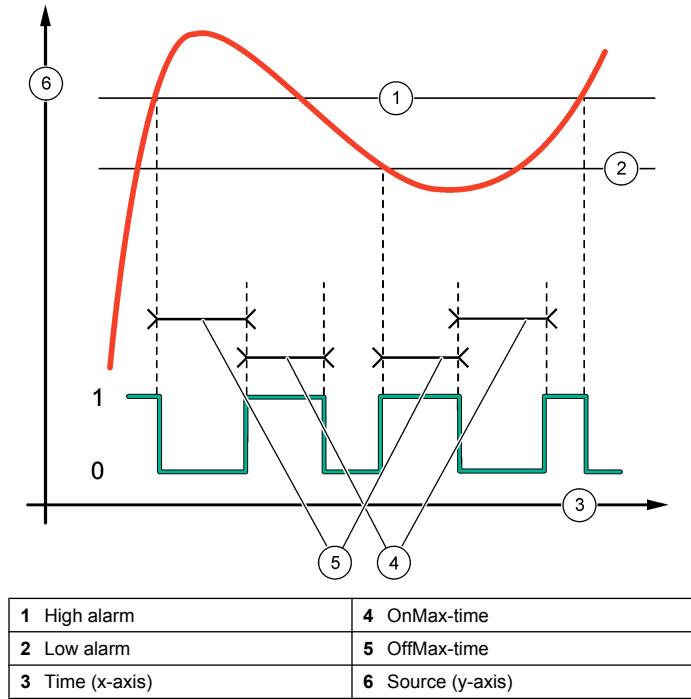
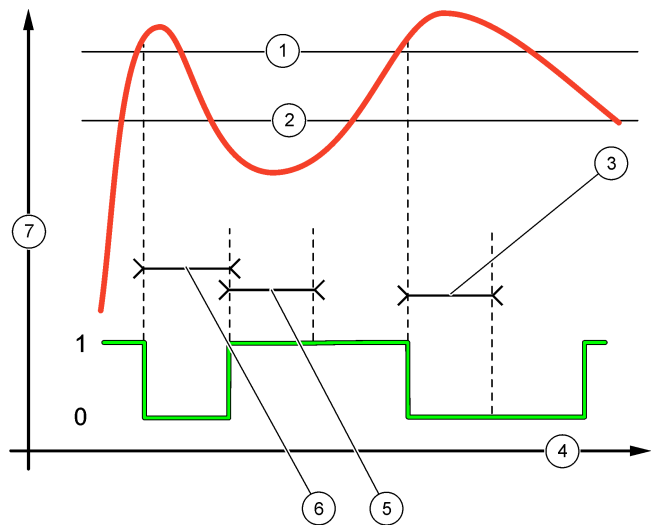
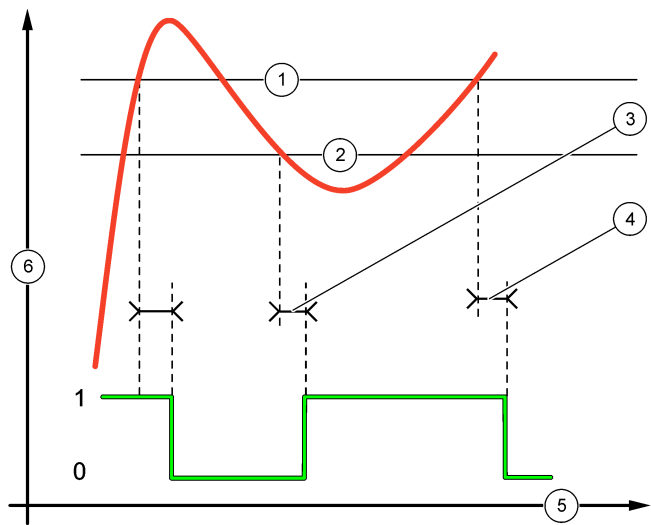


Figure 19 Event control mode (OnMin timer, OffMin timer)



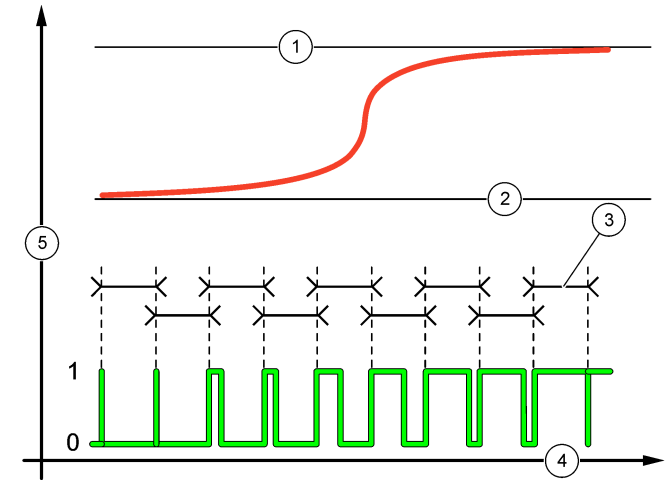
1 High alarm	5 OnMin timer
2 Low alarm	6 OffMin timer
3 OffMin timer	7 Source (y-axis)
4 Time (x-axis)	

Figure 20 Event control mode (ON/OFF delay)



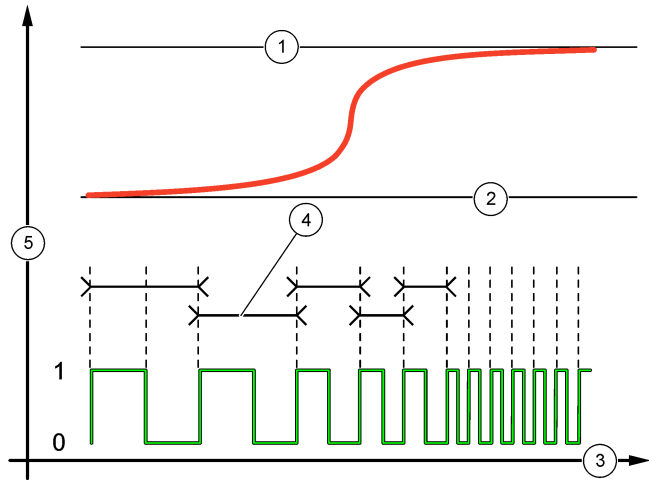
1 High alarm	4 OFF delay
2 Low alarm	5 Time (x-axis)
3 ON delay	6 Source (y-axis)

Figure 21 Pulse Width Modulation control (linear mode)



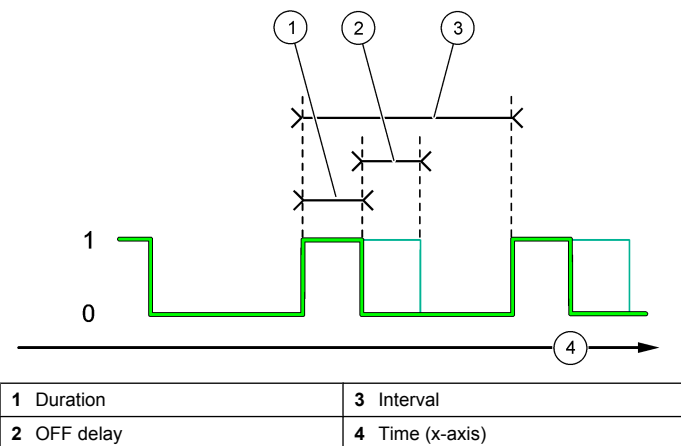
1 High alarm	4 Time (x-axis)
2 Low alarm	5 Selected source (y-axis)
3 Period	

Figure 22 Frequency control mode



1 High limit	4 Cycle duration
2 Low limit	5 Selected source (y-axis)
3 Time (x-axis)	

Figure 23 Relay timer mode



Set up the discrete inputs

Use these inputs to switch closure inputs or logic level voltage inputs.

1. Press the **MENU** key.
2. Select sc200 Setup and push **ENTER**.
3. Select Discrete Input Setup and push **ENTER**.
4. Select the desired channel (Input 1, Input 2 or Input 3) and push **ENTER**.
5. Select a control logic option and push **ENTER**.

Option	Description
Disable	This channel is disabled and not used.

6. Select a warning option and push **ENTER**.

Option	Description
Off	An active discrete input does not trigger a device warning.
On	An active discrete input triggers a device warning.

7. Select an output mode option and push **ENTER**.

Option	Description
Active	Output level continues to represent operating conditions.
Hold	Output level is held static.
Transfer	Output level moves to a pre-configured value.

8. Select the sensors that will have their outputs (analog and relay) affected when one of the discrete inputs becomes active. Push **ENTER**.

9. Use the arrows to select the On Delay value (the duration time delay between the discrete input activation and the configured response of the controller). Push **ENTER**.

10. Use the arrows to select the Off Delay value (the duration time delay between the discrete input de-activation and the configured response of the controller). Push **ENTER**.

11. Repeat steps 4–10 for each desired channel.

12. If a discrete input needs to be changed after initially set up:

- a. Repeat steps 1–4 and an Input Settings menu appears with the following options:

- Control Logic
- Set Warning

- Output Mode
 - On Delay
 - Off Delay
- b. Select the desired option and push **ENTER**.
- c. Make the desired changes and push **ENTER** to save the changes and return to the Inputs Setting menu.

Set up a calculation

Two sensors (digital sensors or sensor modules) must be installed in the controller.

1. From the sc200 Setup menu, select Calculation and push **ENTER**. The calculation menu options appear.
2. Select a menu option and choose from the displayed list or update the entry. Refer to the table below for more information about each option.
3. Push **ENTER** to save the selection or setting and return to the Calculation menu.
4. Push the **HOME** key, **MENU** key or the **BACK** key to exit to other menus.

Option	Description
Set variable X	Selects the sensor that corresponds to the variable set as "X"
Set parameter X	Selects the measurement type that corresponds to the variable set as "X"
Set variable Y	Selects the sensor that corresponds to the variable set as "Y"
Set parameter Y	Selects the measurement type that corresponds to the variable set as "Y"
Set formula	None, X-Y, X/Y, [X/Y]%, [X+Y]/2, X*Y, [X-Y]%/X
Display format	Sets the number of decimal places displayed in a calculation result

Option	Description
Set units	Sets the units of measurement
Set parameter	Assigns the name of the measurement

Set the datalog mode and interval

Datalog Setup is available if a calculation has been set up.

1. From the Settings menu, select sc200 Setup and push **ENTER**.
2. Select Datalog Setup and push **ENTER**.
3. Select Set Mode and push **ENTER**.
4. Select an option (Snap Shot, Average, Maximum, Minimum) and push **ENTER**.
5. From the Datalog Setup menu, select Set Interval and push **ENTER**.
6. Select an interval from the list and push **ENTER**.

Using the secure digital memory (SD) card

An SD card must be installed in the controller.

- The SD card can be used to update software and firmware and to download event and data logs. If the SD card is installed while the controller is in the Settings Menu, push the **HOME** key and then the **MENU** key to verify the option is visible. The SD icon will also be visible in the upper status bar of the main measurement screen when a card is installed.
- Data log files on the SD card are available in XML and binary formats.
- DataCom is used to convert files from binary to CSV format. Refer to the DataCom manual for more information on how to use the application. The DataCom manual, software updates and other downloadable resources are available at <http://www.hach.com/download-resources>.

Updating software

Notes:

- The controller does not automatically transfer information to or from an SD card.
- Each controller will have an associated folder on the SD system. To make sure software updates are in the correct folder for the controller in use, it is best to use a separate dedicated SD card for each controller.

1. From the Settings Menu, select SD Card Setup and push the **ENTER** key.
2. Select Upgrade Software and push the **ENTER** key.

Note: If the Upgrade Software option does not appear, do the steps in [Firmware updates with SD cards](#) on page 36.

3. Select a device from the list and push the **ENTER** key. The list of options includes the controller and all connected devices that have software placed in the appropriate folder on the SD card.
4. If more than one version of the upgrade software is available, select a version and push the **ENTER** key.
5. Push the **ENTER** key to begin the software transfer. The display will show "Transferring files. Please wait..." The percentage of completion appears in the bottom left corner of the display. The upgrade cannot be halted once it has begun.
 - When the transfer is successful, the display will show "Transfer complete" along with a prompt to push **ENTER** to restart the controller or to push the **BACK** key and exit to the SD Card Setup menu. Controller updates take effect when the controller is restarted. A restart is not necessary for sensor updates.
 - If the transfer is unsuccessful, the display will show "Transfer failed" and an error message. Press the **ENTER** key to acknowledge the warning and exit out of the menu.

Saving data and event logs with SD cards

Notes:

- Data and event logs can be downloaded to an SD card and viewed with any device capable of reading an SD card.
- Data logs store the measurement data at selected intervals in a packed binary format (.fig file).

- Event logs store a variety of events that occur on the devices such as configuration changes, alarms and warning conditions. Event logs are set up during the sensor or module configuration process. Event logs are stored in a CSV format.

1. From the Settings Menu, select SD Card Setup and push the **ENTER** key.
2. Select Save Logs and push the **ENTER** key.
3. If more than one device appears on the screen, all devices are selected by default. To deselect an item, highlight the selection and push the left arrow key. Select the devices from which logs will be saved and push the **ENTER** key.
4. Select the time period from which logs are to be saved.

Option	Description
Last Day	All logs from the last full 24 hours, starting from 12:00 a.m., and any additional time remaining on the current day
Last Week	All logs from the last full week (7 days) starting from 12:00 a.m., and any additional time remaining on the current day
Last Month	All logs from the last full month (30 days) starting from 12:00 a.m., and any additional time remaining on the current day
All	Save all logs in memory
New	All logs that are new since the last time logs were saved to the SD card

5. Push the **ENTER** key to confirm the choice, and push the **ENTER** key again to begin the file transfers.
6. Allow time for the files to transfer. The display will show Transferring files. Please wait... and the percentage of files transferred. If the transfer is successful, the display will show "Transfer complete." If the transfer is not successful, the display will show "Transfer failed."
7. Do one of the following:
 - a. Push the **ENTER** key to exit back to the SD Card Setup menu.
 - b. Push the **HOME** key to return to the measurement screen.
 - c. Push the **BACK** keys to return to the Settings Menu.

Access data and event log files on the SD card

A PC, a USB or other SD card reader device, Excel 2003 or higher (for XML files) or the Data Com application (for binary .flg files) are necessary to view the event and data logs stored on an SD card.

Data logs have the following structure: Device Name, Device Serial Number, Device Identification, Data Log, Time Stamp.

Event logs have the following structure: Device Name, Device Serial Number, Device Identification, Event Log, Time Stamp.

To view data or event log files stored on the SD card:

1. Attach the card reader device to the PC (if necessary) and install the SD card that contains the files in the reader device.
2. In the SD card directory, open the HACH folder.
3. Select the Logs folder.
4. Select a device folder.
The event and data log files in the folder are shown.
5. To view XML data log files:
 - a. Make sure the HachDatalog.xml style sheet exists in the device folder.
 - b. Open the Excel application.
 - c. Go to File, Open.
 - d. Select the data log file.
 - e. In the Import XML dialog box, select **Open the file with the following style sheet applied** and select **HachDatalog.xml**.
 - f. Click OK to view the data.
6. To view binary data log (.flg) files:
 - a. Make sure the device driver (.flg.drv) file exists in the device folder.
 - b. Open Data Com.
 - c. In the File Viewer section, click Open.
 - d. Select the data log file.
The data log file is shown in the box and a comma separated values (csv) file with the same file name is created. This csv file can be opened in Excel.

Firmware updates with SD cards

The latest firmware updates can be placed on an SD card. The SD card can then be used to update the controller or device firmware.

A PC and a USB card reader or other device capable of reading an SD card are necessary.

1. Find the zip file at <http://www.hach.com/scsoftware> and copy it to the PC.
2. Extract the file to the SD card.
3. Remove the SD card and follow the instructions in this manual to update the controller and device firmware.

Backup settings to an SD card

Saves the configuration of a device to the SD card.

1. Push the **MENU** key.
2. Select SD Card Setup and push **ENTER**.
3. Select Manage Configuration and push **ENTER**.
4. Select Backup Settings and push **ENTER**.
5. Select the devices to be backed up. All devices are selected by default. To deselect an item, highlight the selection and push the left arrow key. Push **ENTER** to begin the file transfers. If backup files already exist on the SD card, a confirmation window appears. Select the devices again and push **ENTER**. Wait for the "Transfer complete" message.
6. Push **ENTER** again to return to the Manage Configuration menu.

Restore settings to the controller

This menu selection only appears if a (serial number-specific) backup file for the controller or one of the sensors connected to it exists on the SD Card. This menu selection loads the configuration of a specific device from the SD card to the same device (serial number-controlled function).

1. Push the **MENU** key.
2. Select SD Card Setup and push **ENTER**.
3. Select Manage Configuration and push **ENTER**.

4. Select Restore Settings and push **ENTER**.
5. Select the device that will be restored. All devices are selected by default. To deselect an item, highlight the selection and push the left arrow key. Push **ENTER** to begin the file transfers.
6. When the transfer is complete, push **ENTER**.
7. To have the settings take effect immediately, restart the controller. Push the **BACK** key to exit the Manage Configuration menu.

Transfer settings to another device

Allows the configuration settings for a device to be transferred to an SD card and then to another device of the same type.

1. Push the **MENU** key.
2. Select SD Card Setup and push **ENTER**.
3. Select Manage Configuration and push **ENTER**.
4. Select Transfer Settings and push **ENTER**. Two options appear:
 - Retrieve Settings
 - Copy Settings
5. To retrieve settings from the controller (or a device connected to it) and put the settings on the SD card:
 - a. Select Retrieve Settings and push **ENTER**.
 - b. Select the devices that contain the information to be transferred. All devices are selected by default. To deselect an item, highlight the selection and push the left arrow key. Push **ENTER** to begin the file transfers. Wait for the "Transfer complete" message.
 - c. If files already exist on the SD card, a confirmation window appears. Select the devices again and push **ENTER**. Wait for the "Transfer complete" message.
 - d. Push **ENTER** to return to the Manage Configuration menu.
6. To copy settings from the SD card to a controller (or a device connected to it):
 - a. Select Transfer Settings and push **ENTER**.
 - b. Select Copy Settings and push **ENTER**.

- c. Select the devices on the SD card. All devices are selected by default. To deselect an item, highlight the selection and push the left arrow key. Push **ENTER** to begin the file transfers.
7. When the transfer is complete, push **ENTER** to restart the connected devices.
8. Push **ENTER** to restart the controller or push **BACK** to return to the Manage Configuration menu.

Using the service port

The service port is used to download data files from the controller. To download data and update software, use the service port in combination with DataCom and a service cable (LZX887). The service port can also be used to update controller and sensor firmware.

Using DataCom

When using the service port, it is necessary to use DataCom. DataCom is a PC Application Utility that downloads data log and event log files from the controller and installed sensors. Files are downloaded from the controller through the controller service port or they can be placed on a Secure Digital Memory (SD) card installed in the controller. In addition, DataCom is used to upload software for the controller and sensors. The DataCom application must be installed on a PC to read the files.

Refer to the DataCom manual for more information on how to use the application. The DataCom manual, software updates and other downloadable resources are available at <http://www.hach.com/download/resources>.

Update the display language

The display language can be changed through the Setup menu.

1. From the Settings Menu, select sc200 Setup and push **ENTER**.
2. Select Language and push **ENTER**.
The list of language options appears. English is the default language for the controller.
3. Highlight the language to be used for the controller and push **ENTER**.

The selected language is saved and is used for the controller display.
The display returns to the sc200 Setup menu.

Update the date and time

The date and time settings can be changed through the Setup menu.

1. From the Settings menu, select sc200 Setup and push **ENTER**.
2. Select Set Date/Time and push **ENTER**.
3. Select Date Format from the Set Date/Time screen and push **ENTER**.
4. Select a format and push **ENTER**.
5. Select Date/Time from the Set Date/Time screen and push **ENTER**.
6. Update the entries.
 - a. Use the right and left arrow keys to highlight a field.
 - b. Use the up and down arrow keys to change the values in the field and push **ENTER**.
 - c. At the end of the date field, push the right arrow to wrap down to the time fields.
 - d. Use the up and down and arrow keys to update the time fields.
7. Push **ENTER** to save the changes.
The controller returns to the Set Date/Time menu.

Maintenance

⚠ CAUTION

Personal injury hazard. Only qualified personnel should conduct the tasks described in this section of the manual.

Cleaning the controller

⚠ DANGER



Always remove power from the controller before performing maintenance activities.

Note: Never use flammable or corrosive solvents to clean any part of the controller. Use of these solvents may degrade the environmental protection of the unit and may void the warranty.

1. Make sure the controller cover is securely closed.
2. Wipe the controller exterior with a cloth dampened with water, or with a mixture of water and mild detergent.

Fuse replacement

Fuses are not user-serviceable items. The need for fuse replacement in controllers indicates severe technical failure and is therefore considered to be a service activity. If a blown fuse is suspected, contact Technical Support.

Battery replacement

The lithium ion backup battery is not user replaceable.

Troubleshooting

Problem	Resolution
No current output	Verify current output configuration.
	Test current output signal using the Test/Maintenance submenu. Input a current value and verify the output signal at the controller connections.
	Contact Technical Support.
Incorrect current output	Verify current output configuration.
	Test current output signal using the Test/Maintenance submenu. Input a current value and verify the output signal at the controller connections. If the output is incorrect, perform an output calibration.

Problem	Resolution
No relay activation	Make sure relay connections are secure.
	If using an external power source, make sure the relay wiring is correct.
	Make sure the relay configuration is correct.
	Test the relay activation through the Test/ Maintenance menu. The relay should energize and de-energize as selected.
	Make sure the controller is not in calibration mode and that the relay is not being held.
	Reset the Overfeed Timer to make sure the timer has not expired.
Secure Digital Memory (SD) card not recognized by the controller	Make sure the SD card is properly oriented. The copper traces should face toward the controller display.
	Make sure the SD card is fully seated in the slot and the spring lock is engaged.
	Make sure the SD card is properly formatted with a Fat 32 format. The MMC format is not supported. Follow the instructions of the card manufacturer to format the SD card on a PC.
	Make sure the card is not larger than 32 GB.
	Make sure an SD card is being used. Other types of cards (such as xSD, micro SD, mini SD) will not work properly.

Problem	Resolution
Information not saving, or not saving properly to the SD card.	Make sure the SD card is properly formatted with the FAT 32 format. The MMC format is not supported. Follow the instructions of the card manufacturer to format the SD card on a PC.
	If the SD card has previously been in use, format the card with the Fat 32 format, install the card in the controller, and try downloading files.
	Try a different SD card.
SD card full	Read the SD card with a PC or other card reader device. Save important files and then delete some or all of the files on the SD card.
Controller cannot find software updates on the SD card.	Make sure an appropriate folder is created by installing the SD card in the controller. An update folder will automatically be created.
	Install the SD card on a PC and make sure the software files are located in the appropriate update folder.
	If the same SD card is used with multiple controllers, each controller will have a separate folder on the system. Make sure the software updates are in the folder dedicated to the controller in use.
Display is lit but shows no characters or characters are faint or blurry.	Adjust the display contrast
	Make sure protective film has been removed from display.
	Clean the outside of the controller, including the display screen.
Controller will not power up, or powers up intermittently	Make sure the AC power connections are properly terminated in the controller.
	Make sure the power strip, line power, wall plug are all properly plugged in.
	Contact Technical Support

Problem	Resolution
Network or sensor module not recognized	Make sure the module is properly installed.
	Make sure the module selector switch is set to the proper number.
	Remove sensor module and install the module into the second analog slot. Apply power to the controller and allow the controller to perform a device scan.
	Contact Technical Support.
Sensor not recognized <i>Note: Example of possible display message: ****</i>	If the sensor is an analog sensor and a corresponding module is installed in the controller, refer to the instructions supplied with the Network or Sensor Module.
	Make sure the digital connector wiring harness is seated on the inside of the door assembly and that the wiring harness is not damaged.
	If the digital sensor is connected to the controller with a digital termination box, user supplied junction box, digital extension cables, or a user-supplied extension cable, connect the sensor directly to the controller and perform a device scan. If the controller recognizes the sensor, check that all the wiring in the junction boxes or extension cables is correct.
	Make sure that only two sensors are installed in the controller. Although two analog module ports are available, if a digital sensor and two analog modules are installed, only two of the three devices will be seen by the controller.
	Contact Technical Support
Device Missing error message appears	Perform a Device Scan from the Test/ Maintenance menu.
	Power cycle the controller

Test and Maintenance menu

- From the Settings Menu, select Test/Maint and push **ENTER**.

Option	Description
Scan devices	Performs a scan for active and missing devices.
Output cal	Calibrate 4 mA output (Min: 0 Max: 25000)
• Output 1	Calibrate 20 mA output (Min: 35000 Max: 65533)
• Output 2	
Hold outputs	Set activation—Launch or release Set Outmode—Hold Outputs (default) or Transfer Outputs Set Channels—All (default) or select from hardware list
Test output	Min: 0 mA (default +04.00) Max: 25.00 mA
• Output 1	
• Output 2	
Status	View status of all modules, sensors and relays.
Test relay—A, B, C, D	Energize or De-energize the selected relay
Overfeed reset	Resets the Overfeed Timer.
Reset default config	Resets controller configuration settings to default values
Restart sc200	Performs a controller restart

Option	Description
Simulation (only displays if sensors or modules are connected)	Source- • <Module 1> • <Module 2> (Footer displays current source selection) Parameter—Source measurement type (footer displays current source selection) Sim value—Use arrow keys to change value (footer displays current source selection)
Modbus stats	Displays Error and Good count stats for selected port. • Sensor port 1, 2, 3 or 4 • Network port • Service port • Clear stats
System data	Displays the current system current, temperature and voltage data.

Warning and error conditions

Follow the steps below to acknowledge controller warnings.

1. From the Settings menu, select Diagnostics and push **ENTER**.
2. Select the device (controller, sensor, network card) with the warning or error and push **ENTER**.
3. Select the warning, error or event list and push **ENTER**.
4. Select Yes and push **ENTER** to acknowledge the warning.
Note: Errors cannot be acknowledged.
5. For more information on a specific warning, error or event, refer to the device manual.

Device scan information

Display message	Action
Installing device...please wait	The controller has found a new device. No action is necessary. The controller automatically performs an installation process for the new device and displays the main measurement screen of the device installed in the number one position.
Device missing <device id>	A previously installed device has been removed from the controller or is not detected. • Push the Enter key to continue. • Push the left arrow key to select or de-select a missing device. • Push the Enter key to delete the missing device. The controller will display the main measurement screen for the device in the number one position. <i>Note: This message also displays when a device is missing and a new device has been installed. After the missing device is deleted, the controller automatically installs the new device and displays Installing device...please wait. The controller then displays the main measurement screen for the device installed in the number one position.</i>

Replacement parts and accessories

Description	Item number
4 GB Secure Digital Memory (SD) card	9218100
Connector kit for digital sensor	9201000
Controller installation kit	8806200

Description	Item number
DataCom cable	LZX887
Manual CD (non-EU)	DOC083.97.80040
Manual CD (EU)	DOC083.98.80040
Mounting bracket inserts	9177900
Plug, conduit opening (set of 3)	5868700
Power cord kit, with strain relief, 125 VAC, U.S.-style plug	9202900
Power cord kit, with strain relief, 230 VAC, European-style plug	9203000
Screw driver	6134300
SD card reader	9218200
SD card cover kit for sc200 controller	9200900
Screws for controller installation kit	9177800
Cord grip kit (1)	9178000
Sealing washer for cord grip assembly	1033814
UV protection screen	8809200
Weather and sun shield w/ UV protection screen	9220600
Sensor and communication modules	
Conductivity module	9013000
Flow module	9012700
4-20 mA input module	9012800
pH and DO module	9012900
4-20 mA output module	9334600
Modbus network module	9013200

Description	Item number
Profibus network module	YAB104
Profibus M12 connector kit	9178500
Profibus M12 socket (hard wire to quick connector adapter)	9178200
Profibus M12 T-splitter	9178400

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Flow Controller

Component

***Entegris NT Integrated Flow Controller Model 6501
Technical Brochure***

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NT® INTEGRATED FLOW CONTROLLER, MODEL 6501

For continuous and batch control



performs automated closed-loop flow control

Liquid Flow Controller

The NT® IFC, Model 6501 provides enhanced reliability for applications where nonstandard pressure conditions may exist. Whether it's automation, process control or safety concerns that require accurate flow control of liquid chemicals and CMP slurry, the instrumentation must be clean, accurate and reliable. Using the latest electronic technology and high purity materials, Entegris has designed a leading-edge liquid flow controller to allow for greater control of your process flow variables.

- PTFE wetted surfaces for high purity applications
- Nonmetallic components for corrosion resistance
- Integral pressure transducer for additional process information
- One percent (1%) full scale accuracy for critical dispense applications
- Compact footprint for easy field installs with limited space
- Fast response for accurate dispense rates

Constructed for Compatibility

The patented NT® Integrated Flow Controller was developed for use in ultra high purity liquid chemical instruments and slurry applications.

The instrument's valve seat and diaphragm are designed to minimize dead volume and fluid shear, reducing the possibility of process contamination. Featuring PTFE for wetted parts and inert materials for nonwetted parts, the NT® Integrated Flow Controller is resistant to harsh chemical environments and external spraydowns.

Advanced Technology

The NT® Integrated Flow Controller utilizes dual PTFE valve diaphragms for fluid containment and contamination protection. Featuring the latest motorized valve and flowmeter technology, encapsulated internal electronics control all aspects of the flow controller. The unit is activated by a setpoint signal (i.e., 4-20 mA, 0-10 VDC, 0-5 VDC) to maintain fluid flow at the desired setpoint.

Applications

Entegris is solving today's flow control challenges using the NT® Integrated Flow Controller. Combined with Entegris' differential pressure based flowmeter and leading-edge control valve technology, the closed-loop flow controller is ideal for:

- Continuous flow control for critical dispense applications
- CMP slurry dispense to replace existing peristaltic pumps
- Batch control for chemical spiking and blending
- On-demand chemical mixing applications

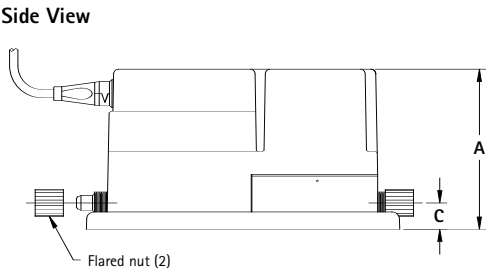
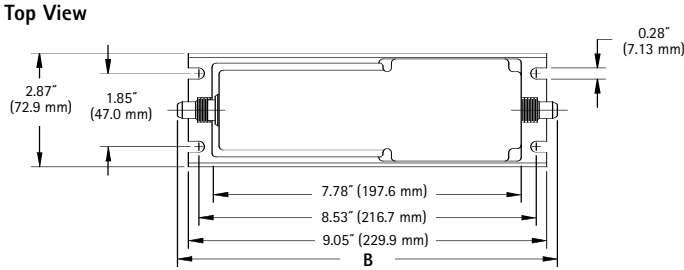
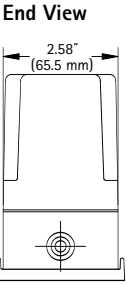
Specifications

Description		
Materials of construction	Wetted parts	Body: PTFE Diaphragms: PTFE Sensor interface: PFA or CTFE Primary o-ring: Kalrez®
	Nonwetted parts	Polypropylene, PVDF and Viton® (In addition to materials listed above.)
Flow measurement	±1% of full scale from 20 to 100% of flow range ±2.5% of full scale from 10 to 20% of flow range [Calibrated using deionized water at 73°F (23°C)]	
Repeatability	±1% of full scale	
Reliability	Wetted parts, >3 million cycles	
Pressure measurement	0-60 PSIG (0-413.7 kPa)	
Pressure accuracy	±1% of full scale	
Operating pressure	10 PSIG (68.9 kPa) to 60 PSIG (413.7 kPa)	
Output signals	Two 4-20 mA electrically isolated outputs, one for flow and one for pressure	
Response time	<3 seconds from 10 to 95% of full scale flow range	
Over-pressure limit	100 PSIG (689.5 kPa)	
Process temperature	50°F (10°C) to 149°F (65°C)	
Electrical input	24 VDC (±10%) at 1 amp	
Setpoint input signal	4-20 mA, 0-10 VDC, 0-5 VDC	
Enclosure	NEMA 5/IP54	
Approvals	CE	

Note: Specifications are subject to change without notice. Please consult the factory for the most current information or additional information regarding the NT® IFC, Model 6501.

Dimensional Information

Inlet/Outlet Port Connection	Dimensions		
	A	B	C
1/4" Flaretek®	4.63" (117.6 mm)	9.61" (244.1 mm)	0.76" (19.3 mm)
3/8" Flaretek®	4.63" (117.6 mm)	9.75" (247.7 mm)	0.76" (19.3 mm)
1/2" Flaretek®	4.72" (119.9 mm)	9.91" (251.7 mm)	0.85" (21.6 mm)
3/4" Flaretek®	5.02" (127.5 mm)	10.15" (257.8 mm)	1.01" (25.7 mm)



Ordering Information

Part Number

6501-T2-F02-D06-A-P1-U1

Primary/Secondary Seal

U1 = Kalrez® 4079/Viton® (default)
U2 = Kalrez® 1050/Viton®
U3 = Kalrez® 6375 UP/Viton®

Sensor Interface

P1 = CTFE (default)
P2 = PFA

Setpoint Input Signal, Controller Type

A = 4-20 mA, continuous
B = 0-10 VDC, continuous
C = 0-5 VDC, continuous
K = 4-20 mA, batch
L = 0-10 VDC, batch
M = 0-5 VDC, batch

Electrical Connector Type

B06 = FEP-jacketed 6' pigtail electrical cable
B12 = FEP-jacketed 12' pigtail electrical cable
B30 = FEP-jacketed 30' pigtail electrical cable
D00 = Polypropylene connector (cable not included)
D06 = Polypropylene connector and 6' PVC cable
D12 = Polypropylene connector and 12' PVC cable
D30 = Polypropylene connector and 30' PVC cable

Inlet/Outlet Port Connection

F02 = 1/4" Flaretek® tube fitting
F03 = 3/8" Flaretek® tube fitting
F04 = 1/2" Flaretek® tube fitting
F06 = 3/4" Flaretek® tube fitting

Flow Range*

T0 = 0-50 ml/min.
T1 = 0-125 ml/min.
T2 = 0-250 ml/min.
T3 = 0-500 ml/min.
T4 = 0-1250 ml/min.
T5 = 0-2.5 l/min.
T6 = 0-5 l/min.
T7 = 0-10 l/min.
T8 = 0-20 l/min.
T9 = 0-40 l/min.

* Flow ranges are scaled to zero flow, measurement is from 10 to 100% of full scale flow range.

Product specified with a flared tube connection is packaged with two PVDF nuts. For alternative nut materials, or custom configurations and specifications, please contact the factory.

(Note: Specifications are subject to change without notice. Please consult the factory for the most current information.)

For Additional Information

For more information on NT® Integrated Flow Controllers or our complete line of fluoropolymer fluid handling solutions, contact your local Entegris distributor or Entegris, Inc.

To review our complete line of sensing and control product solutions visit Entegris' Web site at www.entegrisfluidhandling.com or contact Entegris Customer Service.

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The materials integrity management company

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Recirculation Pump

Component

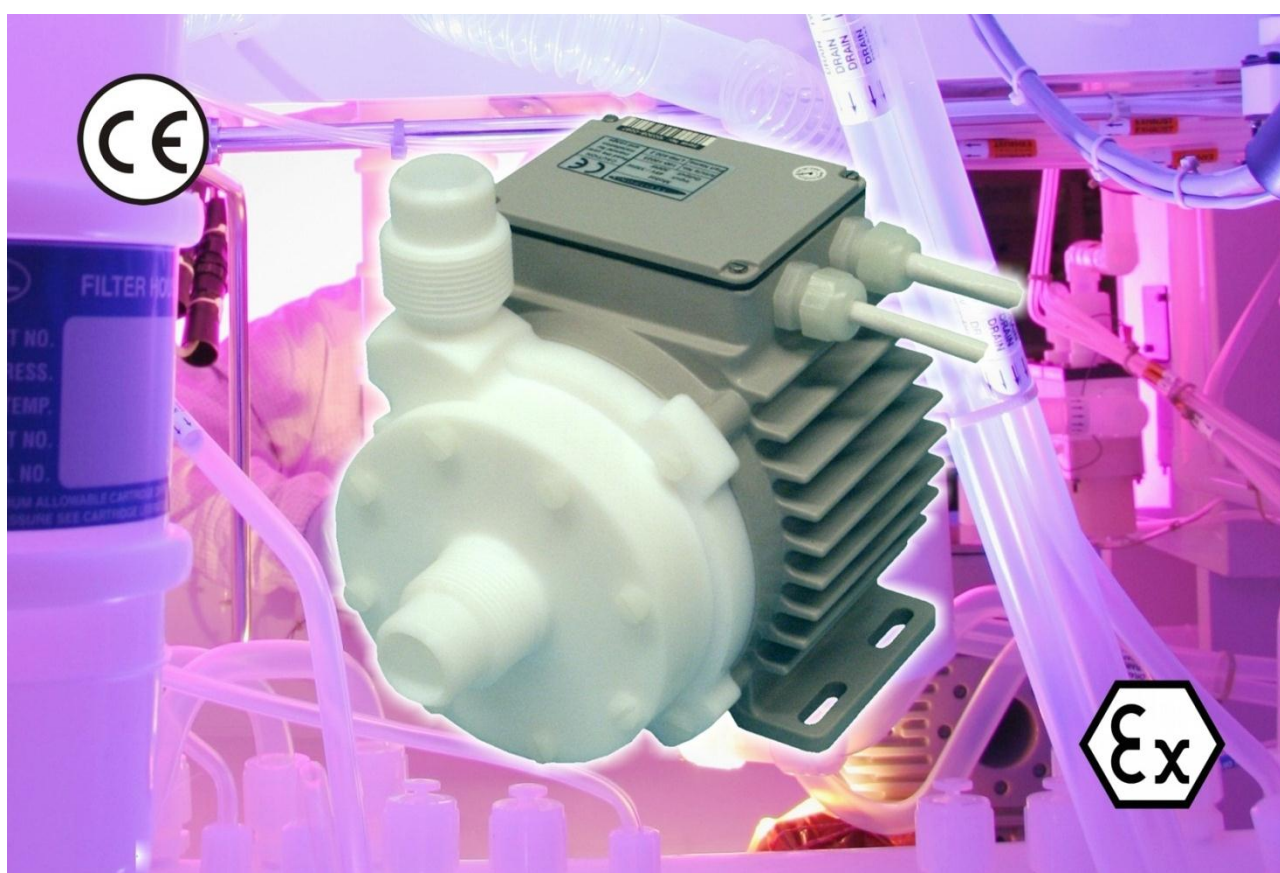
Levitronix BPS-600 Centrifugal Pump User Manual

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BPS-600

3.2 bar (46 psi)
75 liters/min (20 gallons/min)

USER MANUAL



This manual contains information necessary for the safe and proper use of the *BPS-600*. Included are specifications for the standard configurations of the pump system and instructions regarding its use, installation, operation, adjustment, inspection and maintenance. For special configurations of the pump system refer to accompanying information. Please familiarize yourself with the contents of the manual to ensure the safe and effective use of this product. After reading this manual, please store the manual where the personnel responsible for operating the pump system can readily refer to it at any time.

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1 Safety Precautions

CAUTION

Do not under any circumstances open the controller or motor. Levitronix® does not assume responsibility for any damage, which occurs under such circumstances.



CAUTION

High magnetic field strength of pump impeller.

The pump system contains a rotor magnet with high field strength. This may alter or damage the calibration of sensitive electronic devices and measuring instruments in the immediate surroundings. Keep at a safe distance from computers, monitors and all magnetic data storage media (e.g. disks, credit cards, audio and video tapes etc.)



⚠ WARNING

Hazardous voltage may be present.

In case of the usage of an inadequate AC/DC power supply, mains voltages may be present (even if the system is designed for 48VDC). The usage of a galvanic separated power supply, which is certified by a 3rd party (UL or CE), is highly recommended.

The controller must be grounded and placed in a spill protected environment. Do not under any circumstances open the powered controller. The usage of galvanic separated AC/DC supply is highly recommended.



⚠ WARNING

High magnetic field strength of pump impeller.

The pump system contains a rotor magnet with high field strength. Pacemakers may be influenced and magnetic forces may lead to contusions. Keep distance to pace makers and handle impeller with care.



⚠ WARNING

TOXIC CHEMICALS may be present.

When using the system to pump chemicals skin contact and toxic gases may be hazardous to your health. Wear safety gloves and other appropriate safety equipment.



⚠ WARNING

Motors for ATEX applications. Only specific types of motors LPM-600 are classified for the use in ATEX Zone 2 classified locations. Refer to the corresponding section in the manual.



2 Specifications

2.1 Specification of Components

Figure 1 shows the main system components (motor, controllers, pump head) and Figure 2 illustrates the according accessories.

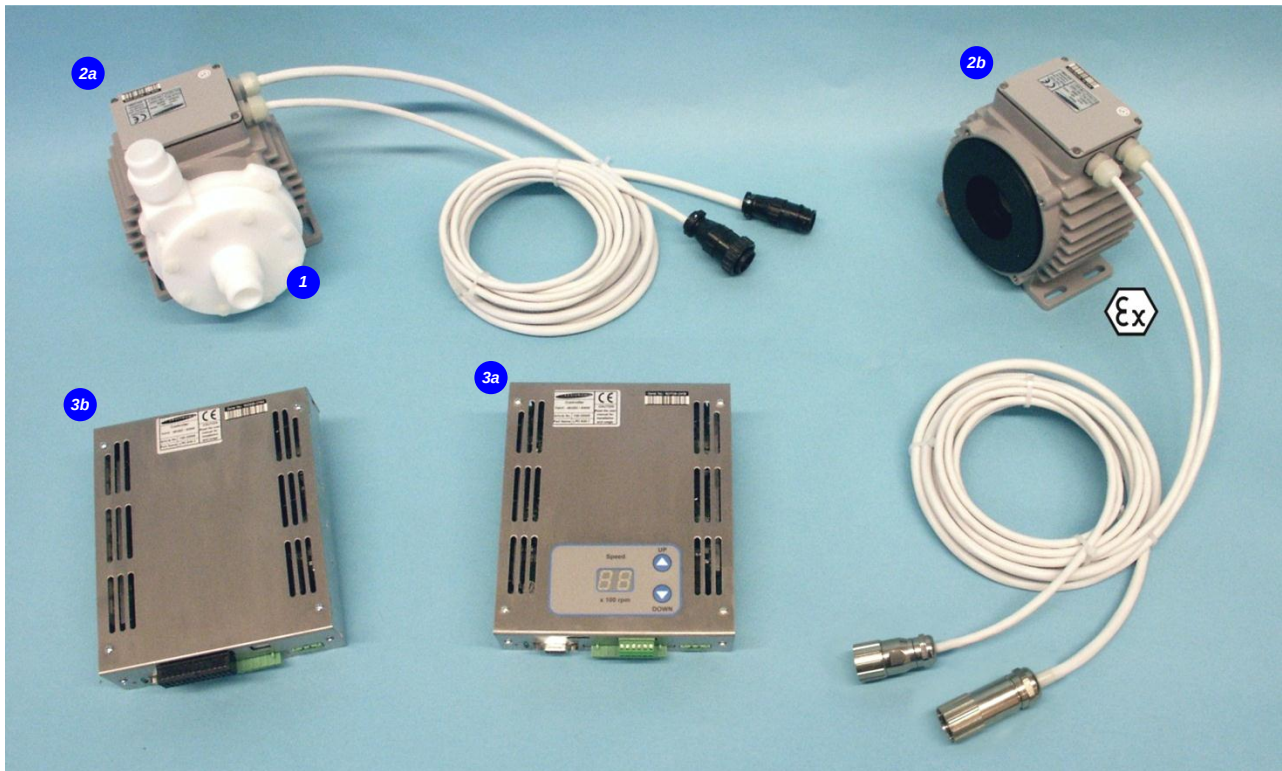


Figure 1: Standard components



Figure 2: Accessories

System Name	Article #	Pumphead	Motor	Controller	Note
BPS-600.1	100-90171	LPP-600.1	LPM-600.2	LPC-600.1	Adaptor/Extension (0.5 - 10m) cables according to Table 3 (position 4a and 4b) have to be ordered as separate article with specified length.
BPS-600.2	100-90172		LPM-600.2	LPC-600.2	
BPS-600.5 (ATEX)	100-90356		LPM-600.4 (ATEX)	LPC-600.1	Adaptor/Extension (0.5 - 10m) cables according to Table 3 (Position 5a and 5b) have to be ordered as separate article with specified length. ATEX Cable Sealing System to be ordered according to Table 4 (Pos. 8)
BPS-600.6 (ATEX)	100-90357		LPM-600.4 (ATEX)	LPC-600.2	

Table 1: Standard system configurations

Pos.	Component	Article Name	Article #	Characteristics	Value / Feature
1	Pumphead	LPP-600.1	100-90242	Impeller / Pump Housing Sealing Ring Fittings	PFA / PTFE Kalrez® perfluoroelastomer ¹ Flaretek 1"
				Max. Flow Max. Diff.-Pressure Max. Viscosity	75 liters/min / 20 gallons/min 3.2 bar / 46 psi 100 cP
				Max. Liquid Temp.	90°C / 194°F
2a	Motor	LPM-600.2	100-10025	Housing	- ETFE (chemical resistant) coated Aluminum - waterproofed (IP67 without connectors)
				Cable / Connectors	2x 3m cables with FEP jacket / 2x circular (AMP types)
2b	Motor (ATEX)	LPM-600.4	100-10038	ATEX Marking	CE II 3G Ex c nAc IIC T5 CE II 3D Ex c tc IIIC T100°C IP67
				Cable / Connectors	2x 3m cables with FEP jacket / 2x circular (M23, IP67)
3a	Standalone Controller (User Panel)	LPC-600.1	100-30005 (Controller with power supply cable and Enable connector incl. in 100-90315)	Voltage / Power	48V DC / 600 W
				Interfaces for Standalone Controller	Panel to set speed (automatic storage on internal EEPROM) PLC with 1x analog input ("Speed") 4 - 20 mA 1x digital input ("Enable") 0 - 24 V (optocoupler) 1x digital output ("Status") 0 - 24 V (relais)
				Standard Firmware	D6.25
3b	Extended Controller (PLC and USB)	LPC-600.2	100-30004 (Controller with power supply cable and PLC connector incl. in 100-90314)	Interfaces for Extended Controller	- up to 4 digital inputs 0 - 24V (optocoupler) - up to 4 digital outputs 0 - 24 V (relais) - up to 2 analog inputs 4 - 20mA - up to 2 analog outputs 0 - 10 V - up to 2 analog outputs 0 - 5 V
					USB interface (for service and system monitoring)
				Standard Firmware	D6.48

Table 2: Specification of standard components
¹: Kalrez® is a registered trademark of DuPont Dow Elastomers

Pos.	Component	Article Name		Article #		Characteristics	Value / Feature
		Sensor Cable	Power Cable	Sensor	Power		
4a	Extension Adaptor Cable for Sensor (a) and Power (b) Wires	MCAS-600.1-05 (0.5m)	MCAP-600.1-05	190-10122	190-10118	Jacket Material Connector Types Connector Material	PVC Circular AMP to D-SUB Plastics (PA)
4b		MCAS-600.1-30 (3m)	MCAP-600.1-30	190-10123	190-10119		
		MCAS-600.1-50 (5m)	MCAP-600.1-50	190-10124	190-10120		
		MCAS-600.1-70 (7m)	MCAP-600.1-70	190-10101	190-10102		
		MCAS-600.1-100 (10m)	MCAP-600.1-100	190-10125	190-10121		
5a	Extension Adaptor Cable for Sensor (a) and Power (b) Wires	MCAS-600.3-05 (0.5m)	MCAP-600.3-05	190-10158	190-10154	Jacket Material Connector Types Connector Material	PVC Circular M23 (IP-67) to D-SUB Metallic – Nickel coated
5b		MCAS-600.3-30 (3m)	MCAP-600.3-30	190-10159	190-10155		
		MCAS-600.3-50 (5m)	MCAP-600.3-50	190-10130	190-10129		
		MCAS-600.3-70 (7m)	MCAP-600.3-70	190-10160	190-10156		
		MCAS-600.3-100 (10m)	MCAP-600.3-100	190-10161	190-10157		

Table 3: Specification of adaptor/extension cables

Pos.	Component	Article Name	Article #	Characteristics	Value / Feature
6	Air Cooling Module	ACM-600.2	190-10140	Material / Connection Port	PP (+ 40% Talkum) / NPT 1/4"
				Air Pressure / Consumption	~1 - 3 bar (14 – 43 psi) / 100 Liter/min @ 1 bar (14.5 psi)
7 (a+b+c+d)	Impeller Exchange Kits	IEK-600.1	100-90515	Impeller LPI-600.2 (7a) O-Ring (7b) Pump Housing Screws (7c) Motor Mounting Screws Exchange Tool IET-3.1 (7d)	PFA O-Ring, Kalrez, 72.62x3.53 PVDF, 8pcs M6x25 PVDF, 4pcs M6x25 POM-C
8	ATEX Cable Sealing System	ACS-A.1 (Roxtec)	100-90292	Sleeve (a) and Gasket (b) Frame (c) Cable Module (d)	Stainless Steel and EPDM Roxylon (EPDM rubber) Roxylon (EPDM rubber) Note: Lubricant (e) and measurement plates (f) are included.
9	AC/DC Power Supply	TSP 600-148-M (M = Modified Levitronix design from Traco)	100-40013 (Traco ID Number: T1068-01A)	Voltage / Power Output Voltage Input	48 VDC / 600 W 85 – 265 VAC (automatic detection)
				Certification or Standards	CB, UL, CSA, Semi F47

Table 4: Specification of accessories

2.2 Standard System Configurations

2.2.1 Stand-Alone System Configuration

The stand-alone pump system configuration (*Figure 3*) consists of a controller with an integrated user panel to set the speed manually. The speed is automatically stored in the internal EEPROM of the controller. As an option, the speed can also be set with an analogue signal.

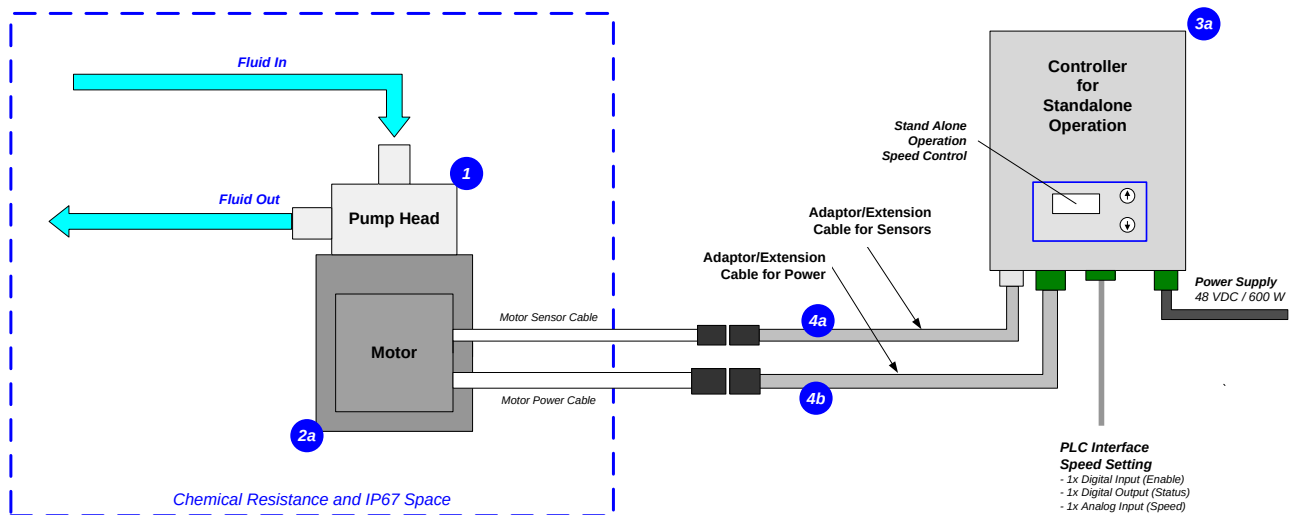


Figure 3: Standard system configuration for standalone operation
(Speed setting with integrated user panel)

2.2.2 Extended System Configuration

The extended version of the pump system (*Figure 4*) consists of a controller with an extended PLC interface. This allows setting the speed by an external signal and enables precise closed-loop flow or pressure control in connection with either a flow or a pressure sensor. A USB interface allows communication with a PC in connection with the *Levitronix® Service Software*. Hence parameterization, firmware updates and failure analysis are possible.

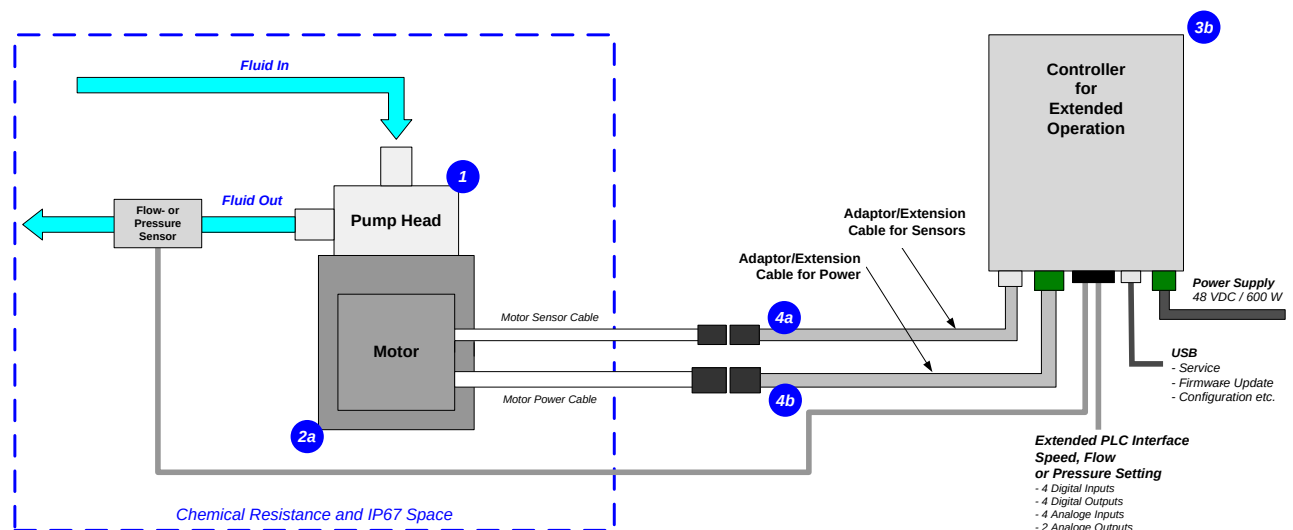


Figure 4: Extended operation (flow or pressure control) with extended controller

2.2.3 ATEX System Configuration

Together with the standard pumphead and controllers an ATEX certified motor allows installation of motor and pumphead within an ATEX Zone 2 area (see *Figure 5*). The ATEX motor comes delivered with special connectors and relevant extension cables (see *Table 3*). One option to lead the cables outside of the ATEX area is an ATEX certified cable sealing system as listed in *Table 4* (see *Pos. 8*).

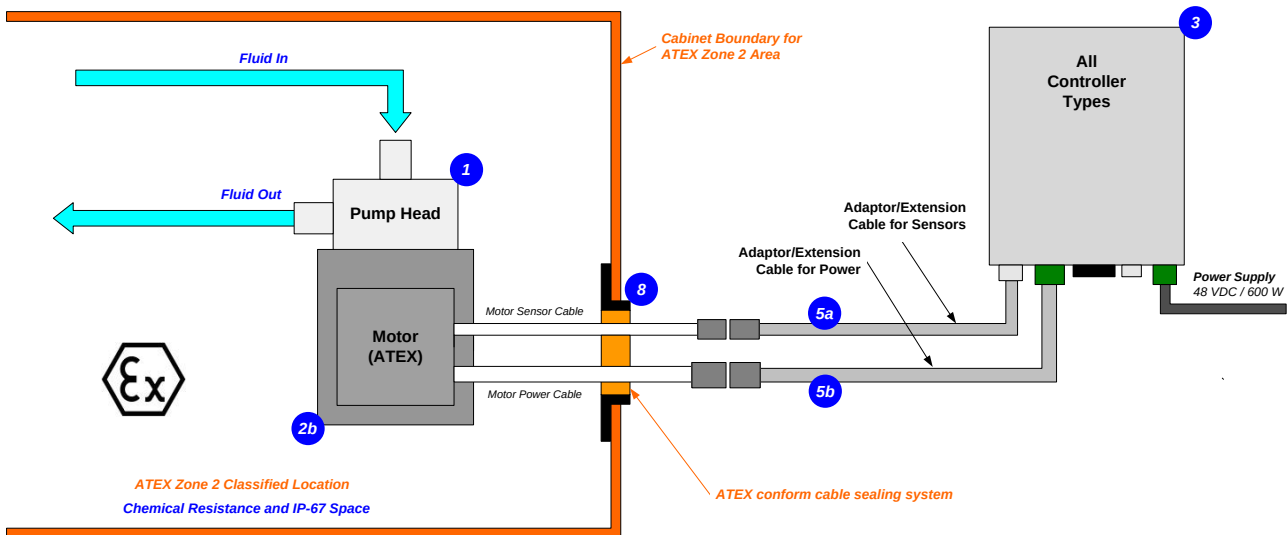


Figure 5: System configuration for ATEX applications

2.3 General Environmental Conditions

Usage	Indoor (motor with pump head can be placed outdoor)
Altitude	Up to 2000 m
Operating ambient temperature	0 to 40 °C
Storage ambient temperature	-20 to 80 °C
Operating humidity range (relative humidity)	15 – 95% (non condensing)
Storage humidity range (relative humidity)	15 – 95% (non condensing)
DC supply fluctuations	± 10% of nominal voltage
Transient over-voltages typically present on the mains supply	Surge immunity according to EN 61000-4-5 (tested together with certified AC/DC power supply)
Pollution degree	2

Table 5: Environmental conditions for pump system

2.4 Pressure-Flow Curves

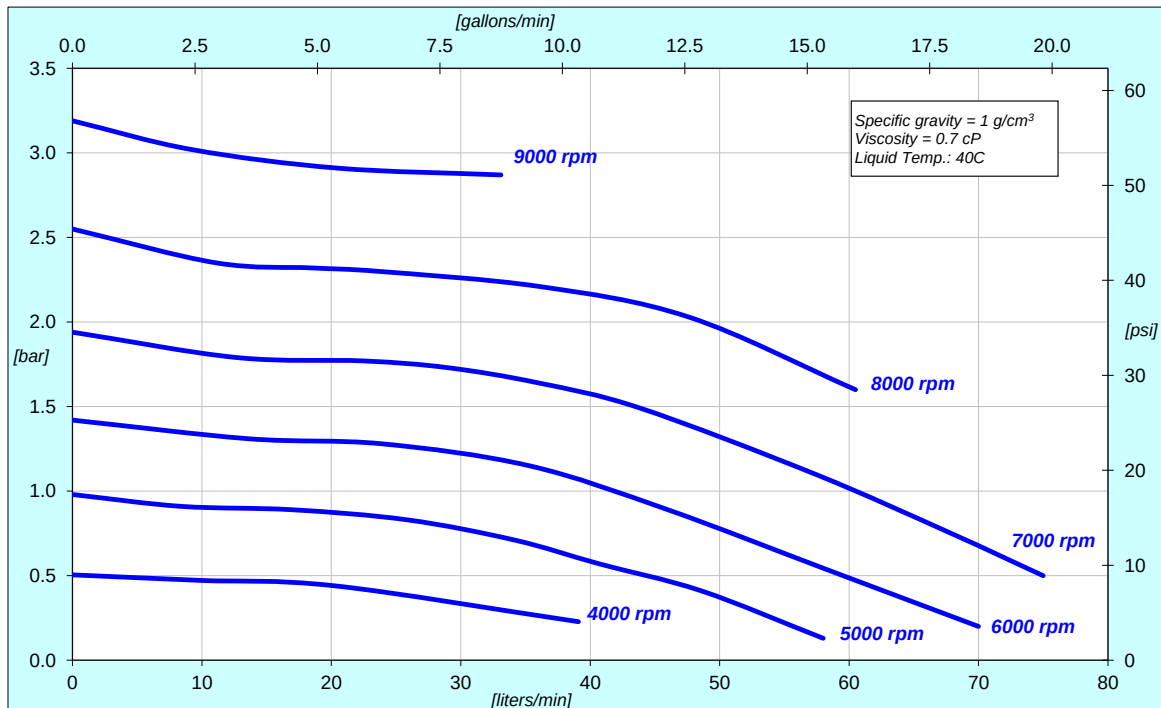


Figure 6: Pressure/flow rate charts
(Measured with Pumphead LPP-600.1)

2.5 NPSHr Curves

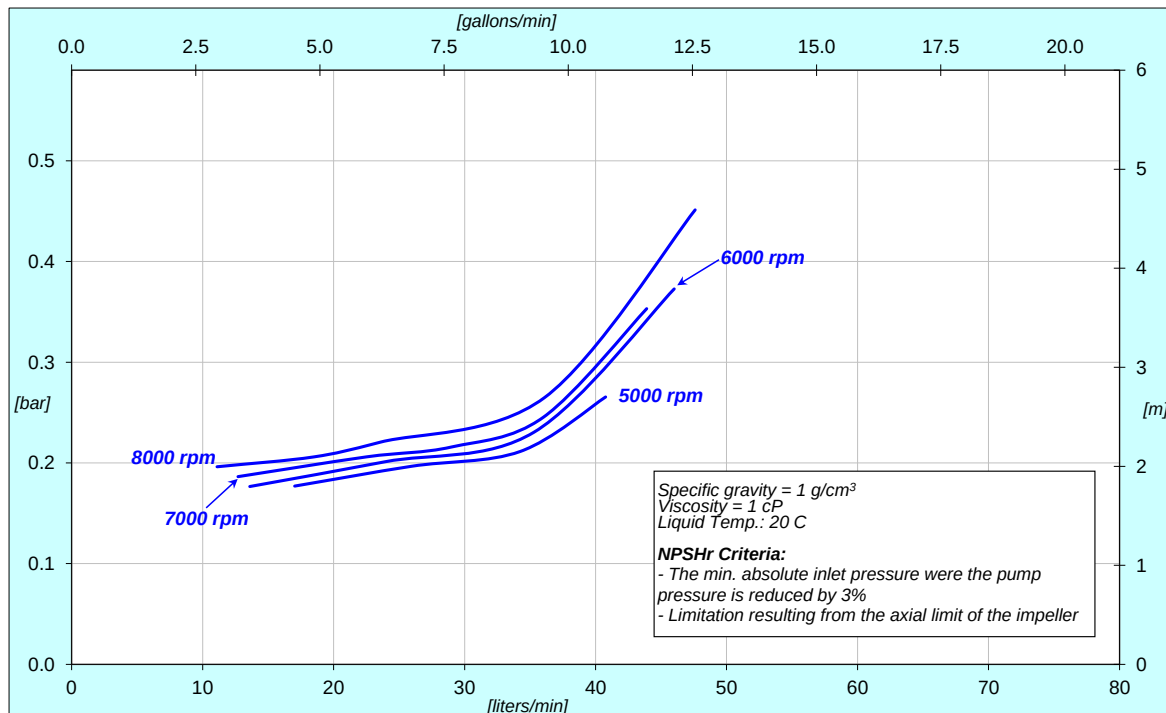


Figure 7: NPSHr curves
(Measured with Pumphead LPP-600.1)

2.6 Maximum Static Pressure for Pumphead

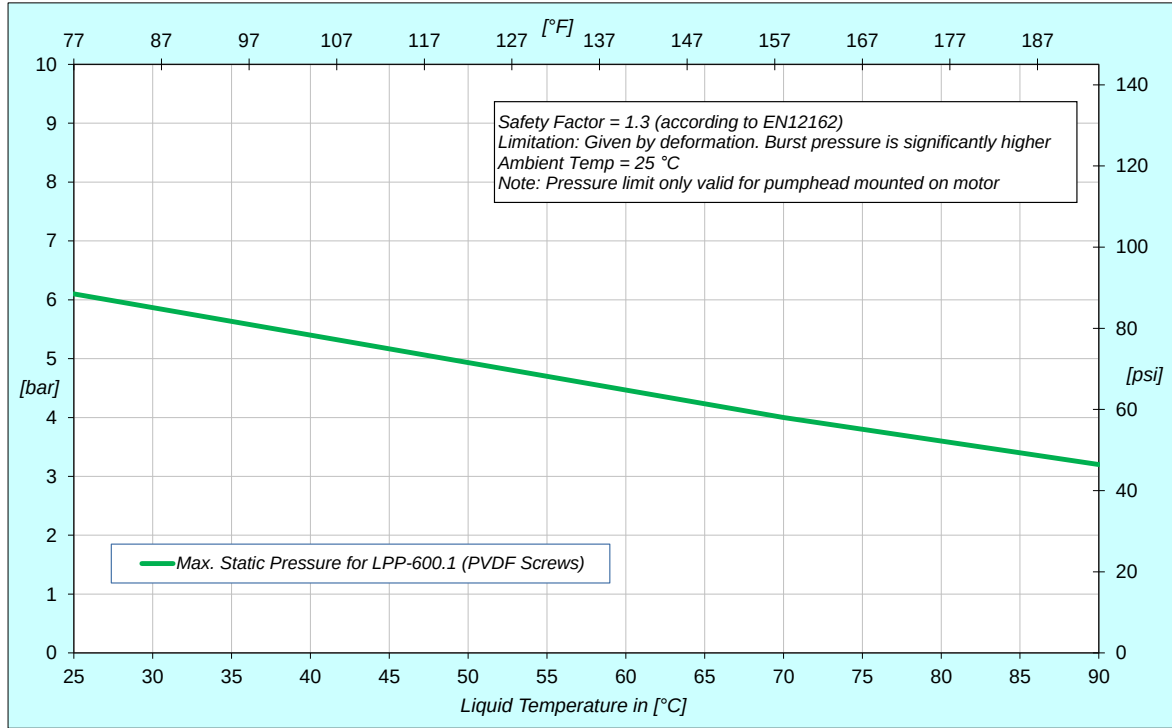


Figure 8: Specification for maximum static pressure of pumpheads LPP-600.1
 (Pressure limits only valid for pumphead mounted on motor)

2.7 Basic Dimensions of Main Components

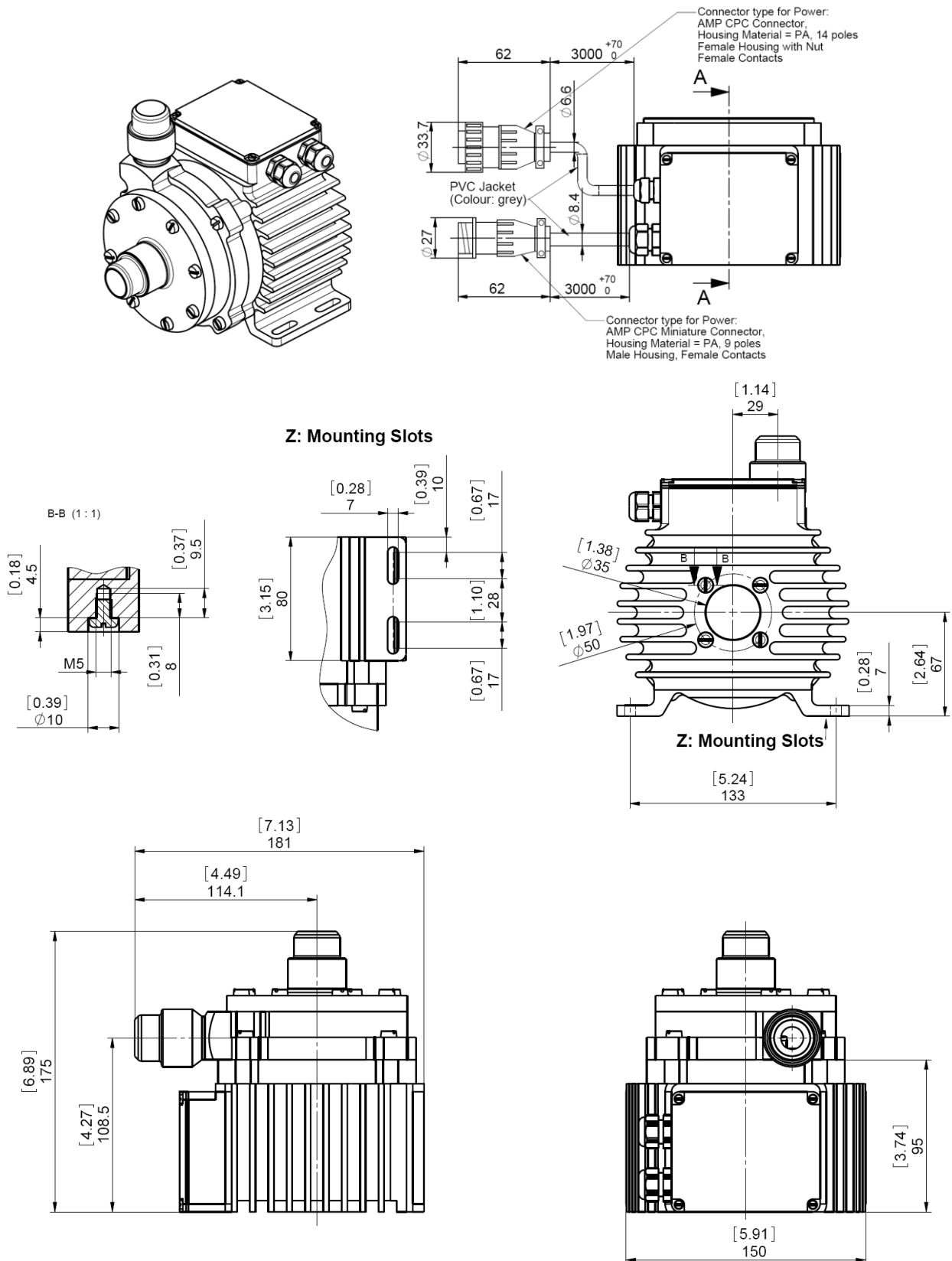


Figure 9: Basic dimensions (in mm and [inch]) of motor LPM-600.2 with pump head LPP-600.1
(for other configurations refer to according drawings)

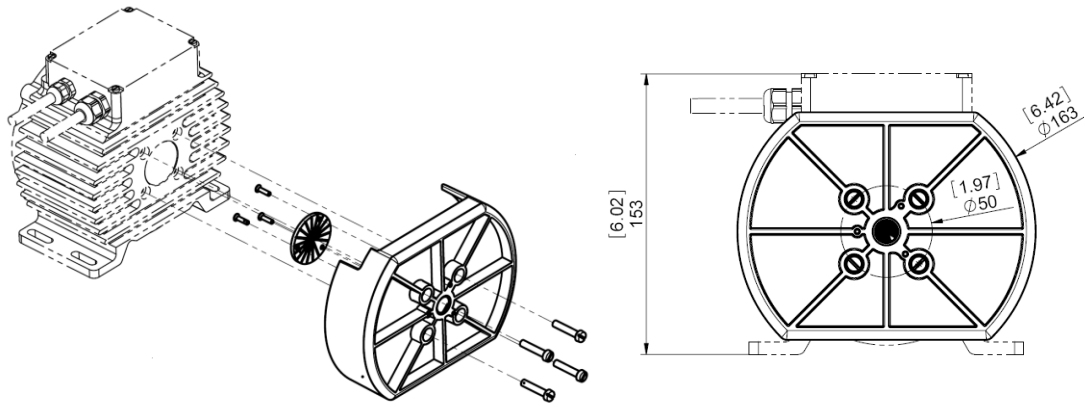


Figure 10: Basic dimensions (in mm and [inch]) motor LPM-600 with Air Cooling Module ACM-600.2

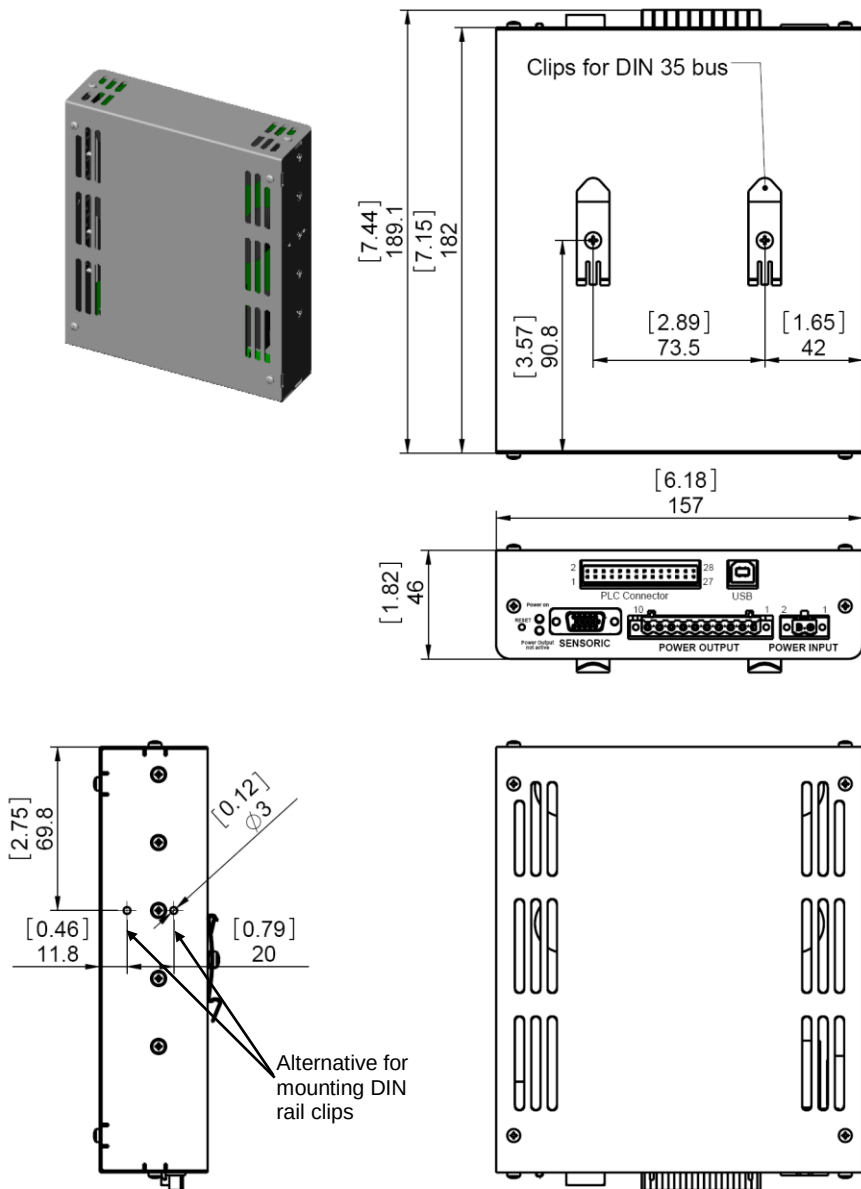


Figure 11: Basic dimensions (in mm) of controller LPC-600.2
(same basic dimensions for LPC-600.1)

3 Engineering Information

3.1 Sealing and Material Concept

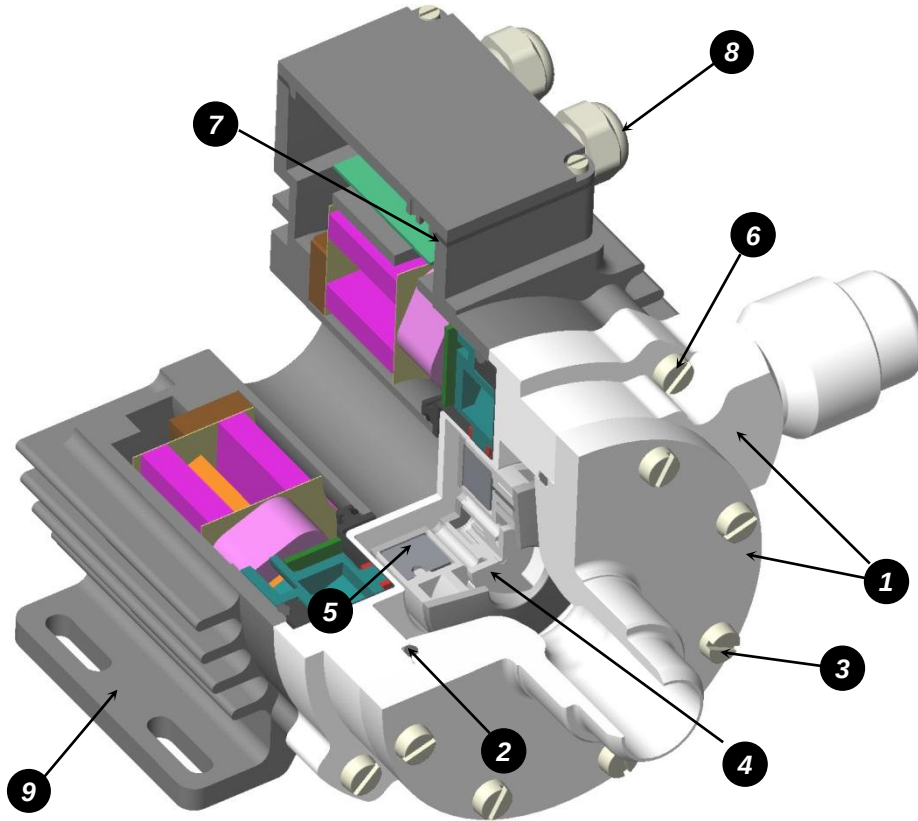


Figure 12: Sealing and material concept

System Component	Item		Materials
	No	Description	
Pump head <i>LPP-600.1</i>	1	Pump housing (lid and bottom)	PTFE
	2	Static sealing O-ring of pump casing	Kalrez
	3	8 screws for pump housing	PVDF
	4	Impeller <i>LPI-600.2</i>	PFA
	5	Rotor magnet	NdFe (rare-earth material)
	6	4 screws for pump/motor mounting	PVDF
Motor <i>LPM-600.2</i>	7	Flat gasket for motor housing	FPM (= FKM)
	8	Cable bushing	PVDF, cable jacket is FEP
	9	Motor housing	ETFE coating, waterproof (IP-67) Coils and electromagnetic circuit potted with an epoxy compound (UL94 V0).

Table 6: Materials used in the LPM-600.2 motor and LPP-600.1 pump head

3.2 Power Consumption

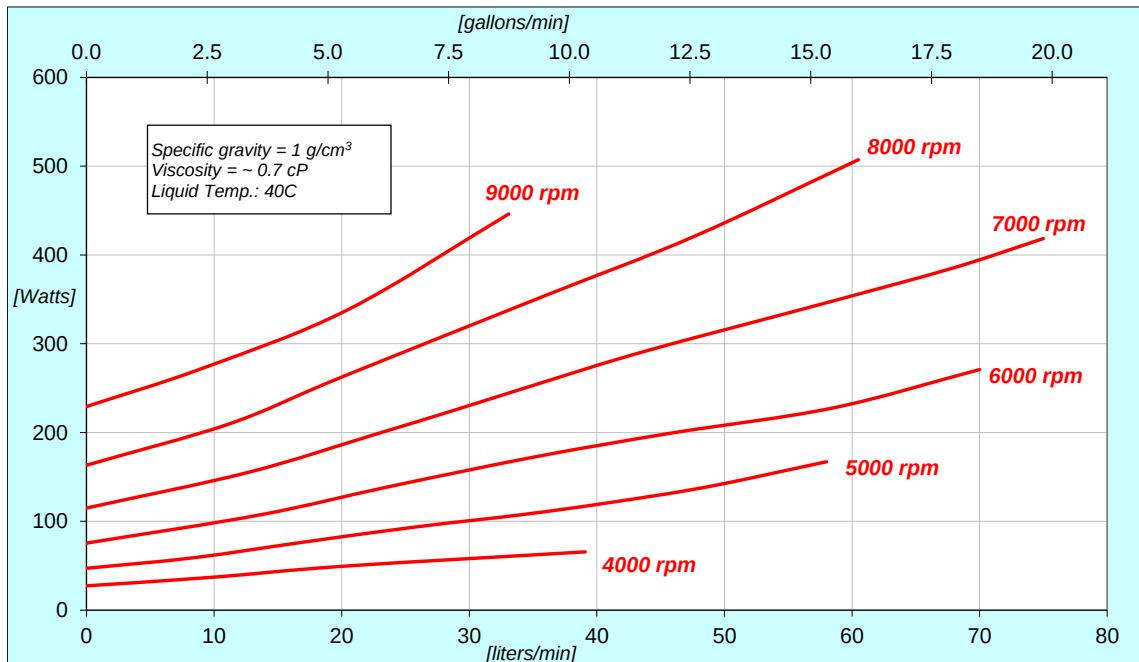


Figure 13: Electrical power consumption
(Controller LPC-600.x with pump head LPP-600.1)

3.3 Temperature Monitoring

To avoid overheating of the system, the controller and motor temperatures are monitored. If the controller temperature exceeds 70°C (158°F) or the motor temperature 90°C (194°F) for a duration of more than 10 minutes, the system goes into an error state and the pump stops. At 80°C (176°F) controller temperature or 100°C (212°F) motor temperature, the system stops immediately.

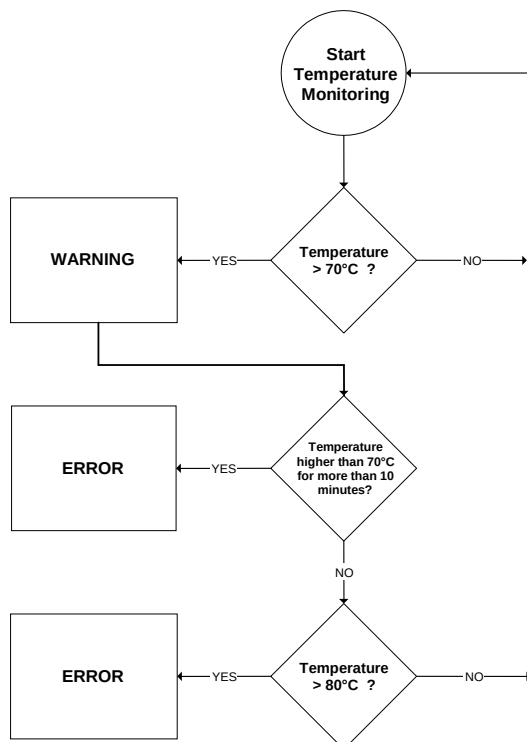


Figure 14: Controller temp. monitoring

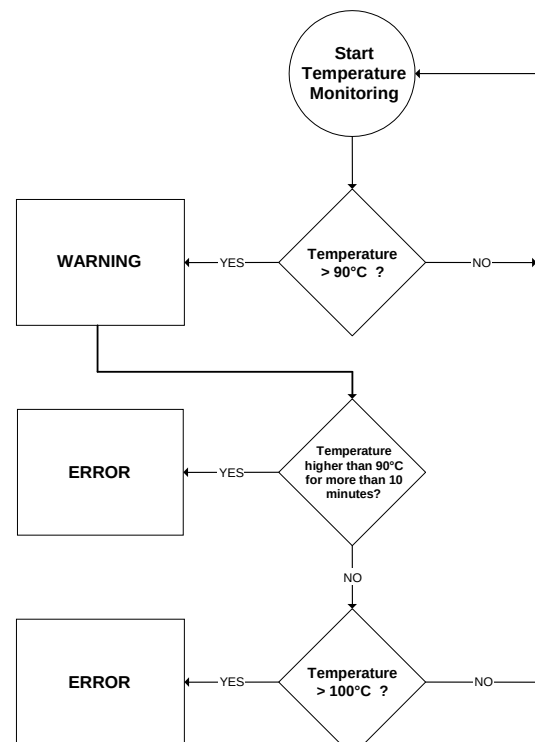


Figure 15: Motor temperature monitoring

3.4 Thermal Management

3.4.1 Motor Temperature

The motor temperature depends on the ambient and liquid temperature, as well as on the hydraulic operation point. *Figure 16*, *Figure 17* and *Figure 19* illustrate the temperature characteristics of the motor depending on these parameters. For higher liquid temperatures and hydraulic operating points active cooling is recommended for example with the air cooling module ACM-600.2 (see *Figure 18*).

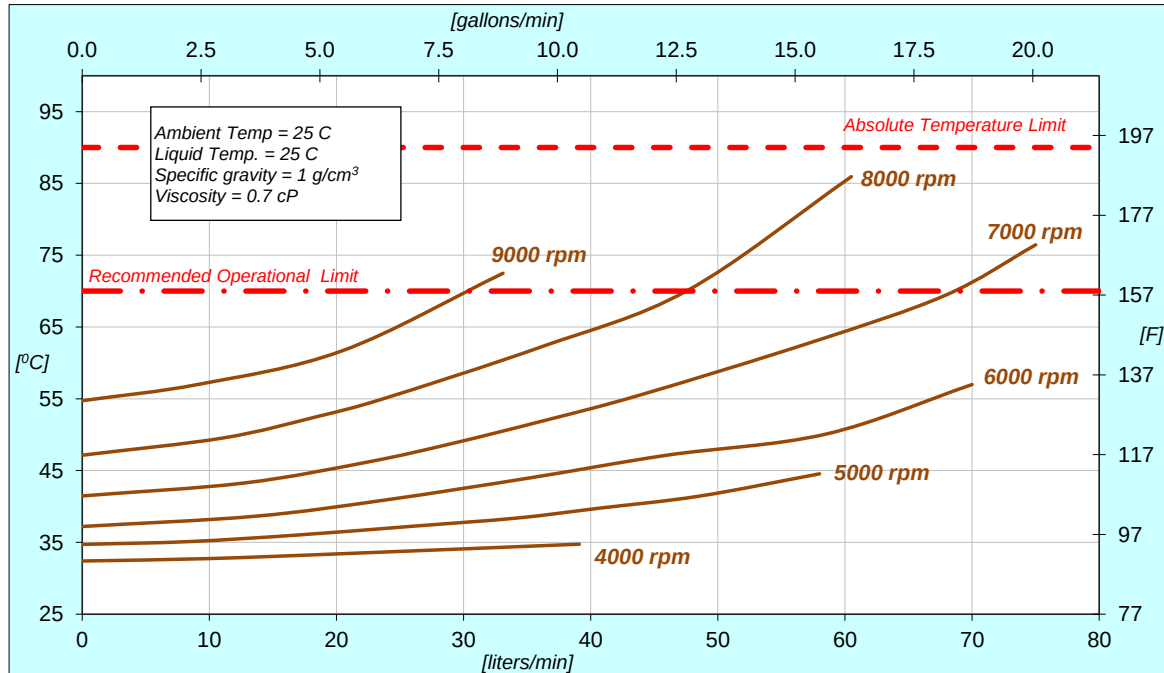


Figure 16: Temperature curves for the LPM-600.2 motor @ 25 °C liquid temperature
(Pumping with pump head LPP-600.1, temperature is measured inside of the motor, contact temperature of surface is below this temperature)

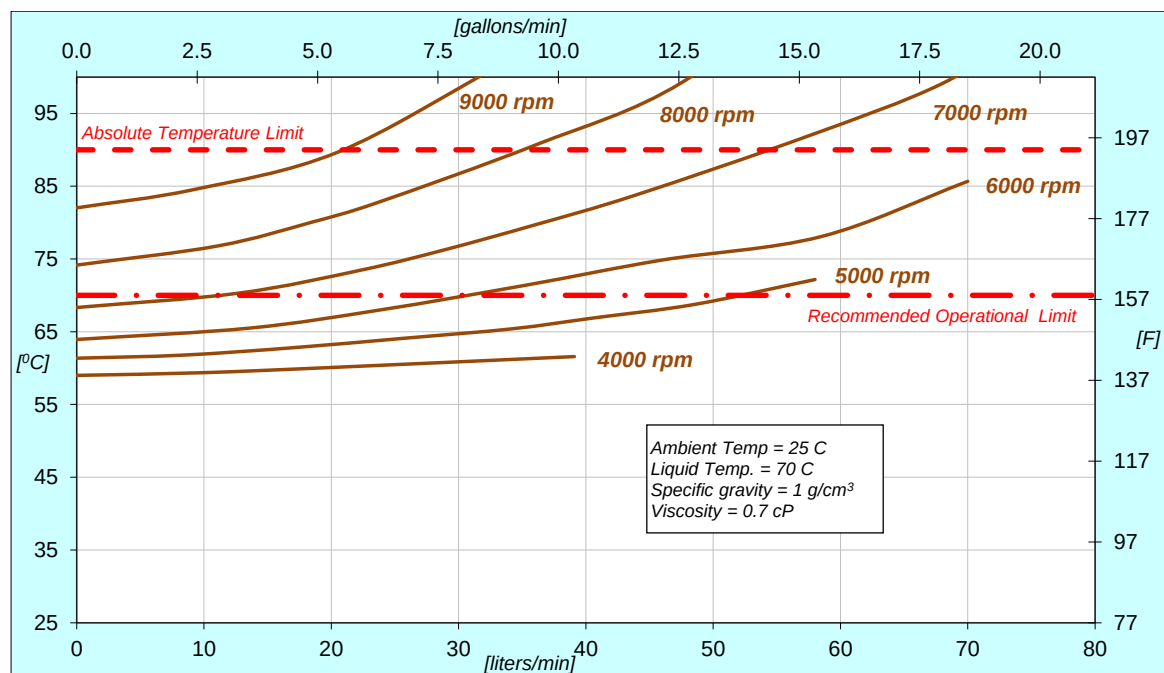


Figure 17: Temperature curves for the LPM-600.2 motor @ 70 °C liquid temperature
(Pumping with pump head LPP-600.1, temperature is measured inside of the motor, contact temperature is below this temperature)

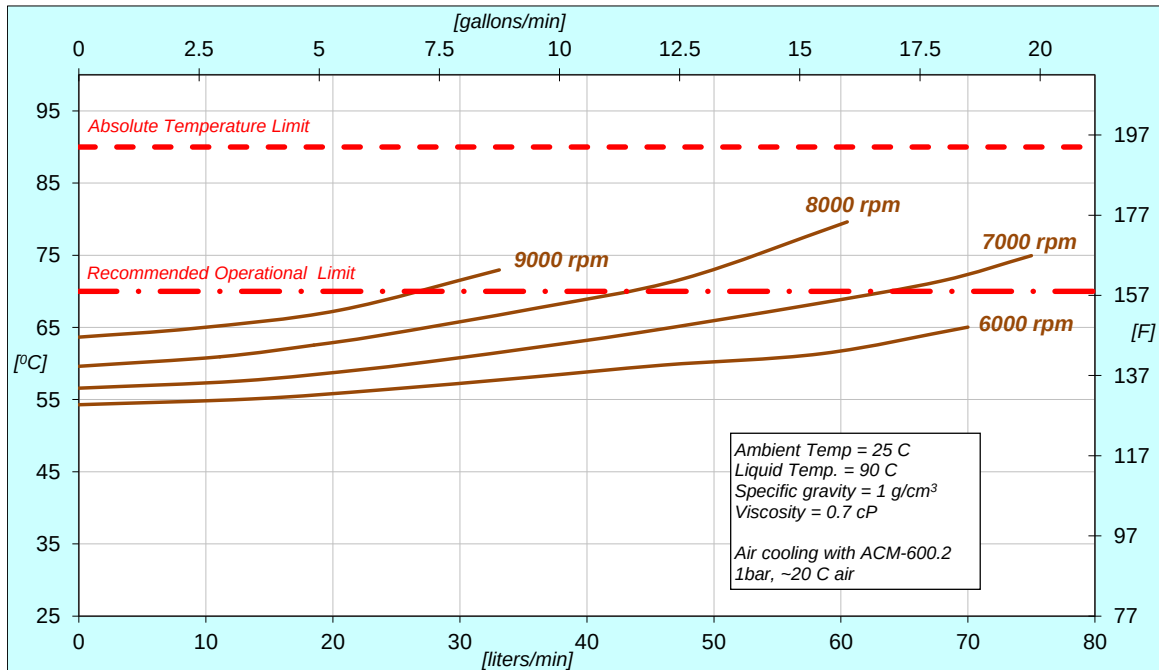


Figure 18: Temperature curves of motor LPM-600.2 with air cooling module ACM-600.2
(Pump head LPP-600.1, liquid. temp = 90 °C, Air Cooling Module ACM-600.2 with 1bar air at 25 °C)

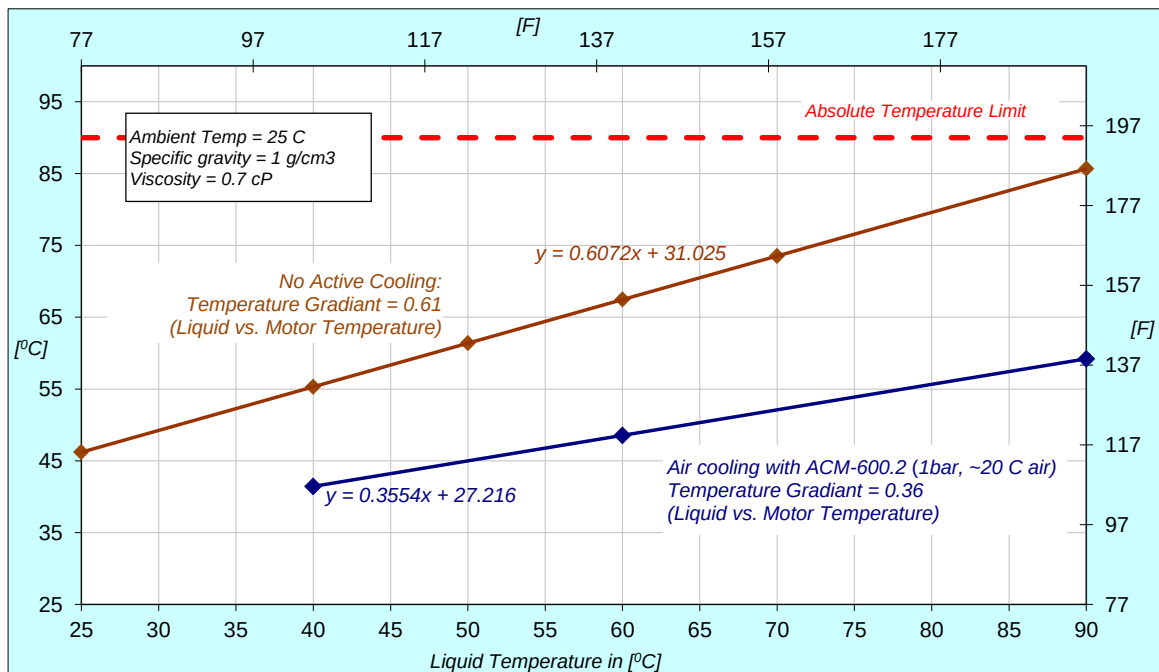


Figure 19: Influence of liquid temperature on motor temperature
(Measurement at 7000 rpm 23 lpm but gradients are representative for other operational points)

The above curves are measurements of the motor temperature at certain liquid and ambient temperatures. Equation (Eq. 1) shows how to calculate the motor temperature for other liquid and ambient temperatures based on these curves.

$$T_M(T_L, T_A) \approx \underbrace{T_M(T_L = 25^\circ\text{C}, T_A = 25^\circ\text{C})}_{\text{see Figure 16}} + (T_L - 25^\circ\text{C}) \cdot \underbrace{tg_{LM}}_{\text{see Figure 19}} + (T_A - 25^\circ\text{C}) \quad (\text{Eq. 1})$$

T_M = Motor temperature

T_L = Liquid temperature

T_A = Ambient temperature

tg_{LM} = Temperature gradient liquid/motor

In order to account for thermal variations (like ambient temperature, closed chemical cabinets or corners without ventilations) and to not significantly reduce the MTBF of the motor it is recommended to keep about 20 °C safety distance to the absolute thermal limit of the motor (90 °C) when designing the thermal concept of the pump system.

3.4.2 Controller Temperature

Depending on the ambient temperature and the placement of the controller additional cooling may be required (see Figure 20). To improve cooling of the controller, place the device into a moving air stream. If the controller is mounted in a compact area or adjacent to additional heat sources (e.g. a 2nd controller) ensure that there is sufficient ventilation.

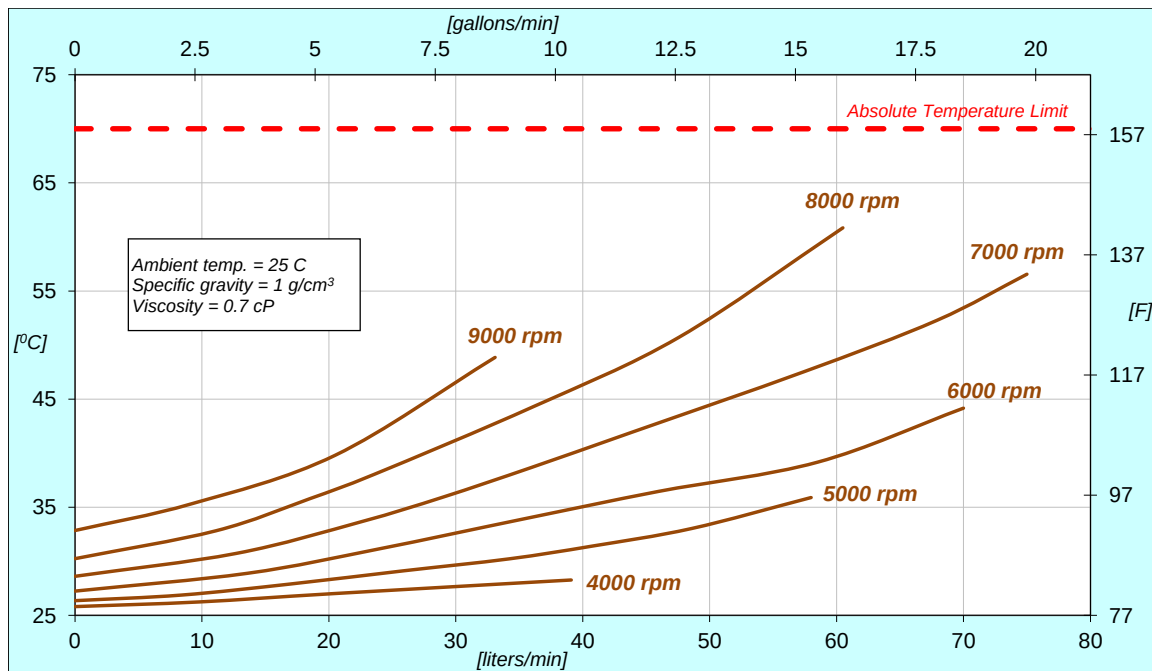


Figure 20: Temperature curves of controller LPC-600 vs. flow and speed
(for pumping with pump head LPP-600.1 and motor LPM-600.2)

The above curves are measurements of the controller temperature at 25 °C ambient. Equation (Eq. 2) shows how to calculate the controller temperature for at other ambient temperatures based on this curve.

$$T_C(T_A) \approx \underbrace{T_C(T_A = 25^\circ\text{C})}_{\text{see Figure 20}} + (T_A - 25^\circ\text{C}) \quad (\text{Eq. 2})$$

T_C = Controller temperature

T_A = Ambient temperature

4 Installation

4.1 Electrical Installation of Controller

4.1.1 Overview

The LPC-600 controllers have signal processor controlled power converters with four switched inverters for the drive and the bearing windings of the motor. The signal processor allows precise control of pump speed and impeller position. *Figure 21* shows the interfaces of the standalone controller LPC-600.1 with stand-alone and minimal PLC functions and *Figure 22* the interfaces of the controller LPC-600.2 with extended PLC functions and USB interface for communication.

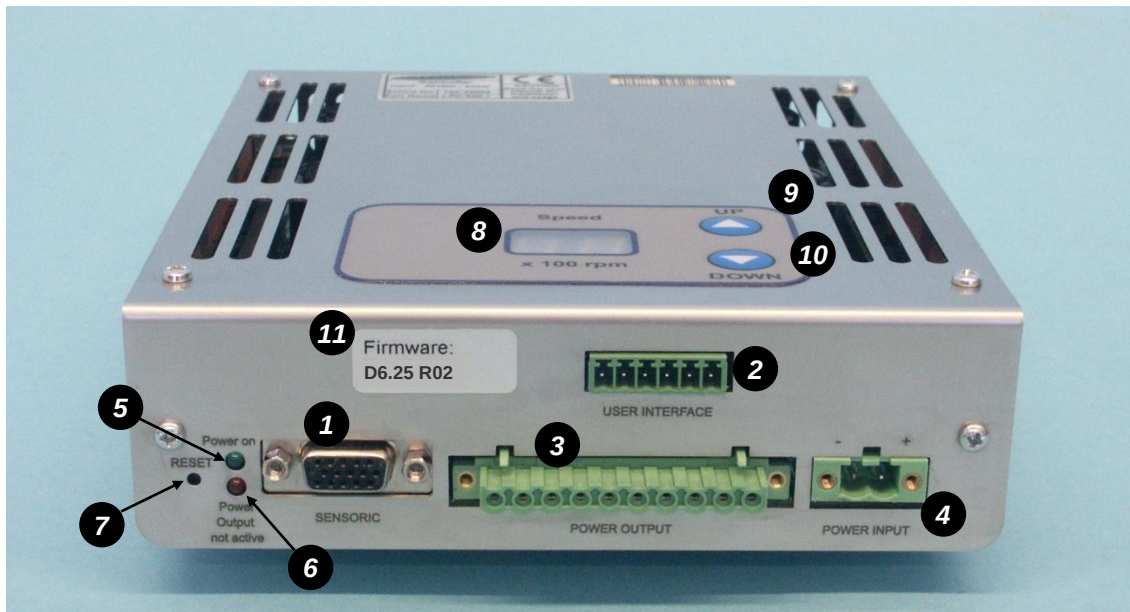


Figure 21: Overview of the controller LPC-600.1 for standalone operation

	Interface (as labeled)	Description
1	"SENSORIC"	Position, field and temperature sensor signals from motor Torque spec. for tightening of connector screws: Min. = 0.4, Max. = 0.6 Nm
2	"USER INTERFACE"	1 Digital Input - Galvanic isolation with optocoupler - Switching voltage / current: minimal 10 V / 7 mA, Typical 24 V / 16 mA, maximal 30 V / 20 mA, (Possible is 5V) - Input resistance: $R_{IN} = 2.2 \text{ k}\Omega$
		1 Digital Output - Galvanic isolation with relay - Relay: 1A / 30VDC, 0.3A / 125 VAC
		1 Analoge Input - Analog current input: 4 – 20 mA - 450 Ohm shunt input
3	"POWER OUTPUT"	Drive and bearing currents of the motor Torque spec. for tightening of connector screws on motor side: Min. = 0.5 Nm, Max. = 0.6 Nm
4	"POWER INPUT"	DC power input Torque spec. for tightening of connector screws on motor side: Min. = 0.5 Nm, Max. = 0.6 Nm
5	"Power on" Green LED	LED is on if supply voltage of signal electronics is present.
6	"Power Output not active" Red LED	Red LED is off if the switched output stage of the controller is enabled. If the LED is on, the bearing and drive coils of the motor carry no current.
7	"RESET" Button	Reset button of the controller stage. The button is sunk mounted and can be activated for example with a small screw driver.
8	2-Digit Display "Speed"	Rotational speed display in 100rpm
9	"UP" Button	Button for speed increasing
10	"DOWN" Button	Button for speed decreasing
11	"Firmware" Label	Firmware version and revision number

Table 7: Description of interfaces of LPC-600.1 controller

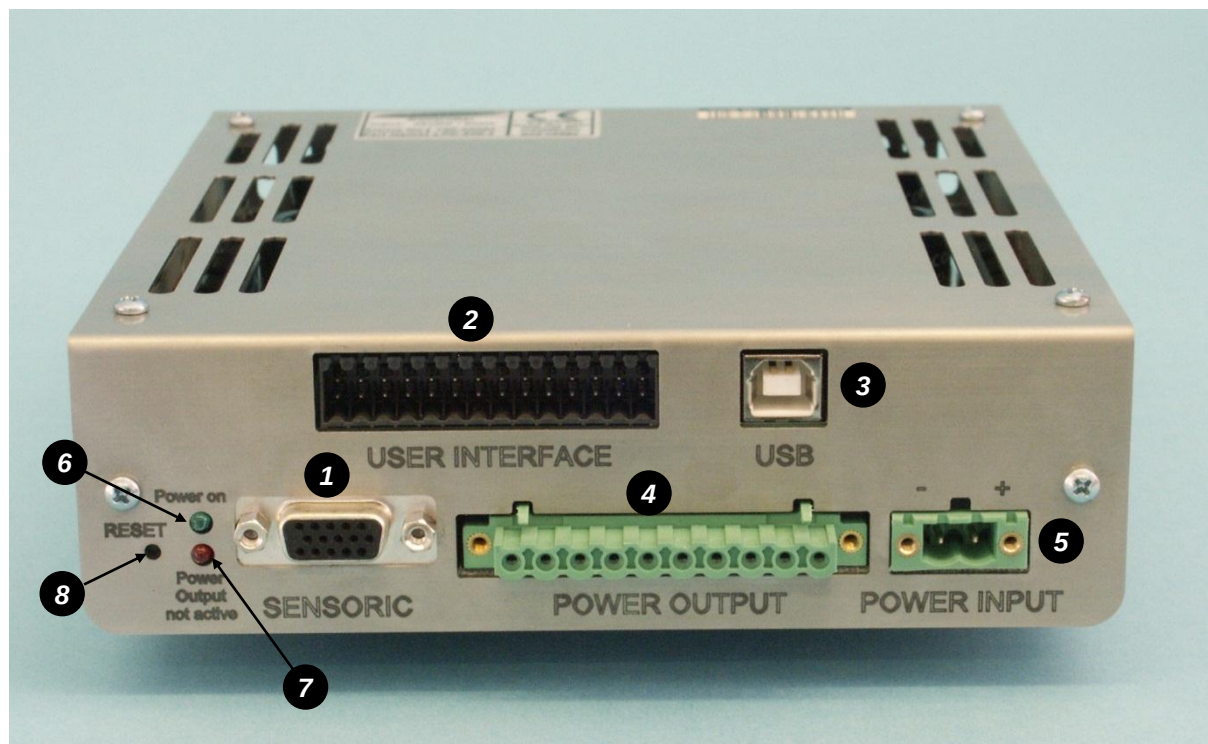


Figure 22: Overview of the controller LPC-600.2 for extended operation

	Interface (as labeled)	Description
1	"SENSORIC"	Position, field and temperature sensor signals from motor. Torque spec. for tightening of connector screws: Min. = 0.4, Max. = 0.6 Nm
2	"USER INTERFACE"	2 Analog Input - Analog current input: 4 – 20 mA - 450 Ohm shunt input
		2 Analog Input - Analog voltage input 0 – 10 V - Direct connection, no galvanic isolation - 7.8 kΩ input resistance
		2 Analog Output - Analog voltage output: 0 – 10 V - Direct connection, no galvanic isolation - Max. Output current: 2mA
		4 Digital Input - Galvanic isolation with optocoupler - Switching voltage / current: minimal 10 V / 7 mA, Typical 24 V / 16 mA, maximal 30 V / 20 mA - Input resistance: $R_{IN} = 2.2 \text{ k}\Omega$
		4 Digital Output - Galvanic isolation with relay - Relay: 1A / 30VDC, 0.3A / 125 VAC
3	"USB"	USB interface
4	"POWER OUTPUT"	Drive and bearing currents of the motor. Torque spec. for tightening of connector screws on motor side: Min. = 0.5 Nm, Max. = 0.6 Nm
5	"POWER INPUT"	DC power input Torque spec. for tightening of connector screws on motor side: Min. = 0.5 Nm, Max. = 0.6 Nm
6	"Power on" Green LED	LED is on if supply voltage of signal electronics is present.
7	"Power Output not active" Red LED	LED is off if the switched output stage of the controller is enabled. If the LED is on, the bearing and drive coils of the motor carry no current.
8	"RESET" Button	Reset button of the controller stage

Table 8: Description of interfaces of LPC-600.2 controller

4.1.2 General Installation Instructions

	 WARNING	
	Hazardous voltage may be present. <i>Always isolate the electrical power supply before making or changing connections to the unit.</i> <i>In case of the usage of an inadequate AC/DC power supply, mains voltages may be present (even if the system is designed for 48VDC). The usage of a galvanic separated power supply, which is certified by a 3rd party (UL or CE), is highly recommended.</i>	

	 WARNING
	Hazardous voltage may be present. <i>The controller housing must be properly grounded and placed in a spill protected environment. Use one of the DIN-rail screws on the back side of the controller housing.</i> <i>Do not use different and longer screws, which may result in short-circuit within the controller.</i>

1. The controller casing must be grounded. The screws of the DIN-rail bracket can be used for grounding.
2. Connect the two motor connectors ("Power Output" and "Sensoric") to the controller.
3. Connect the controller type specific connectors: see *Section 4.1.3* for standalone operation with the *LPC-600.1*, *Section 4.1.4* for extended operation with *LPC-600.1* and *Section 4.1.5* for extended operation with *LPC-600.2*
4. The pump system requires 48 VDC supply voltage at a maximum power of 600 W. Depending on the hydraulic operational point power supplies with smaller power or bigger supplies to supply several pump systems simultaneously may be used. *Figure 13* shows the power consumption depending on the pressure and flow rate. Contact Levitronix for consulting and support on the power supply solution.
5. Connect the DC power supply connector with the cable (included in the controller delivery). Make sure that the polarity is correct (see *Figure 21* and *Figure 22*) and that AC/DC power supply is off.
6. To secure the connectors, tighten all retaining screws according to the torque specifications in *Table 7* for *LPC-600.1* and *Table 8* for *LPC-600.2* controller.

4.1.3 Electrical Installation of Controller *LPC-600.1* for Standalone Operation

For standalone operation the *LPC-600.1* is disabled when power is turned on. It can be enabled manually by using the “UP” button on the display. If the controller shall be enabled automatically, when power is applied the “ENABLE” pin on the “USER INTERFACE” connector (see *Table 9*) has to be active (typically 24V).

4.1.4 Electrical Installation of Controller *LPC-600.1* for Extended Operation

If the *LPC-600.1* shall to be controlled with external signals the “USER INTERFACE” can be used with the PIN designations described in *Table 9*.

Pin Name	Connector Pin Number	Designation	Levels	Note
Analog In, (Signal)	5	<i>Reference Speed</i>	4..20 mA = 0..10000 rpm -> Speed Limit = 9000 rpm $\approx$ 18.4 mA -> Cut-off (min.) speed = 300 rpm	Direct connection, no protection. Galvanic isolation on the user side is required.
Ground Analog In	6			
Digital In, (Signal)	3	<i>Enable</i>	24 V $\Rightarrow$ active 0 V $\Rightarrow$ not active	Is needed to enable the system with an external signal.
Ground Digital In	4			
Digital Out	1	<i>Status</i>	Relay closed $\Rightarrow$ active, system on Relay open $\Rightarrow$ not active, system off	This signal indicates if the system is active.
Ground Digital Out	2			

Table 9: Description of „USER INTERFACE“ connector

(Description is for firmware D6.25 with Revision > 01, for other configurations refer to alternate firmware documentation)

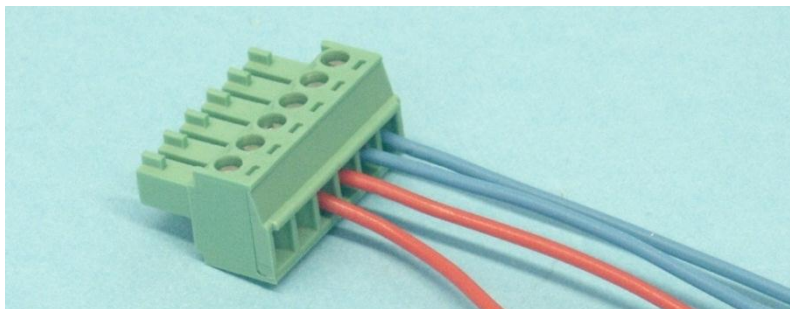


Figure 23: „USER INTERFACE“ connector

- Delivered with controller *LPC-600.1*
- Supplier: PTR Messtechnik GmbH, Germany
- Connector Type: AK1550/06-3.81-GREEN

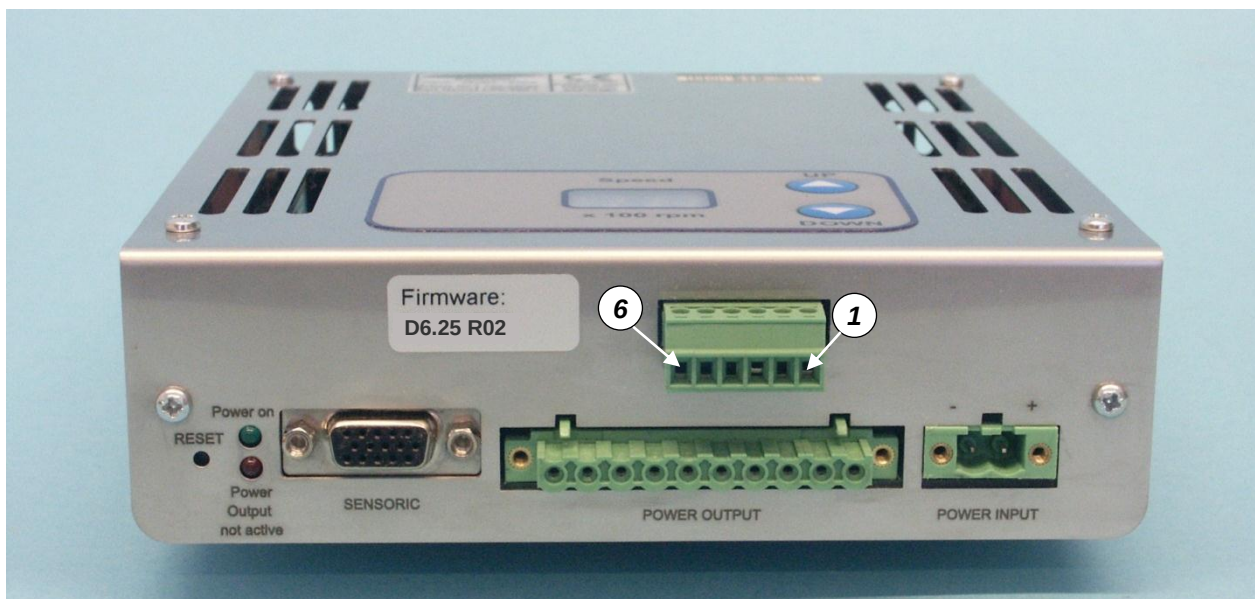


Figure 24: Mounted “USER INTERFACE” connector and Pin numbering

4.1.5 Installation of PLC Interface for Extended Controller LPC-600.2

To operate the pump system with a PLC, a minimum set of two digital inputs and one analog input is needed. The digital and analog outputs can be used to monitor the pump status and operating parameters.

CAUTION

The analog inputs and outputs are not galvanic isolated from the controller electronics. To avoid ground loops and malfunctions, use floating analog signals.

1. Detach the PLC connector from the controller
2. Connect the designated wires of a cable the pins of the detached connector according to *Table 10*. Assignment and functions of the I/Os can be changed with the controller firmware version (refer to according firmware documentation).
3. Connect the PLC connector (*Figure 25*) to the controller.

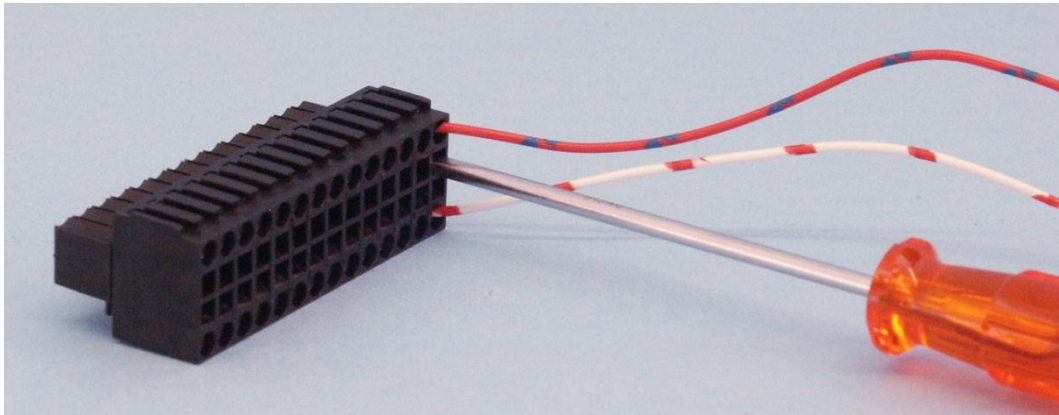


Figure 25: PLC connector
 - Delivered with controller LPC-600.2
 - Supplier: Weidmüller
 - Connector Type: B2L 3.5/28 SN BK BX

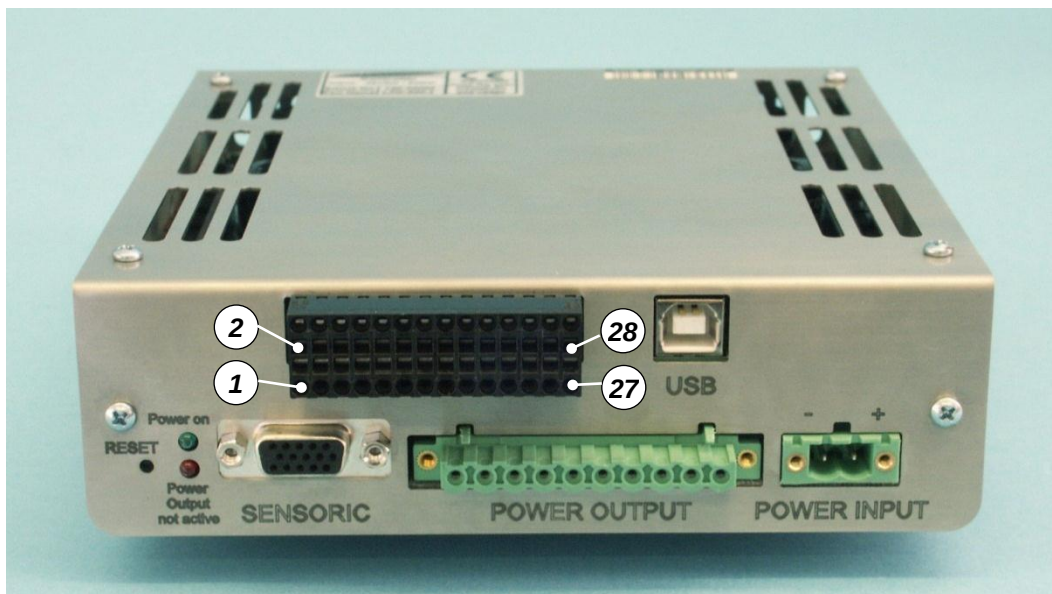


Figure 26: Mounted PLC connector and Pin numbering

Pin Name	Connect. Pin #	Designation	Levels	Note
Analog In1, (Signal)	18	Ref Value (Current Input)	4..20 mA = 0..10000 rpm (<i>speed mode</i>) -> Speed Limit = 9000 rpm $\approx$ 18.4mA -> Cut-off (min.) speed = 300 rpm 4..20 mA = 0..100% (<i>process mode</i>)	- Grounds are internally connected - Direct connection, no protection. Galvanic isolation on the user side is required. - Default input settings: Current inputs selected. Voltage input can be selected with EEPROM–editor in Service Software (consult detailed firmware specification Service Software Manual with Doc.# PL-2034-00)
Ground Analog In1	17			
Analog In2, (Signal)	20	Actual Process Control Value (Current Input)	4..20 mA = 0..100%	
Ground Analog In2	19			
Analog In3, (Signal)	22	Ref Value (Voltage Input)	0..10 V = 0..10000 rpm -> Speed Limit = 9000 rpm $\approx$ 9 V -> Cut-off (min.) speed = 300 rpm 0..10 V mA = 0..100% (<i>process mode</i>)	
Ground Analog In3	21			
Analog In4, (Signal)	24	Actual Process Control Value (Voltage Input)	0..10 V = 0..100 %	
Ground Analog In4	23			
Analog Out1, (Signal)	26	Actual Speed	0.5 V = 0..10000 rpm	Direct connection, no protection. Galvanic isolation on the user side is required. 5V is given by firmware, hardware allows up to 10V output voltage.
Analog Out2, (Signal)	28	Actual Process Control Value	0.5 V = 0..100%	
Com. Ground Analog Out	25, 27	--	--	
Digital In1, (Signal)	2	Reset	24 V $\Rightarrow$ active 0 V $\Rightarrow$ not active	Resets error state
Ground Digital In1	1			
Digital In2, (Signal)	4	Process mode	24 V $\Rightarrow$ active 0 V $\Rightarrow$ not active	Switches between process mode and speed mode
Ground Digital In2	3			
Digital In3, (Signal)	6	Enable	24 V $\Rightarrow$ active, system on 0 V $\Rightarrow$ not active, system off	The Enable signal switches the pump system on and off.
Ground Digital In3	5			
Digital In4, (Signal)	8	Not used	--	--
Ground Digital In4	7			
Digital Out1	10	Status	Relay closed $\Rightarrow$ active, system on Relay open $\Rightarrow$ not active, system off	This signal indicates the state of the pump system.
Ground Digital Out1	9			
Digital Out2	12	Error	Relay closed $\Rightarrow$ not active, system on Relay open $\Rightarrow$ active, system off	When active, the system drives the impeller to zero rpm and shuts down. With a reset pulse the system can be re-initialized.
Ground Digital Out2	11			
Digital Out3	14	Warning	Relay closed $\Rightarrow$ not active, system o.k. Relay open $\Rightarrow$ active, system not o.k.	The warning signal indicates if a system fault has been detected. The warning signal indicates a system fault but the system does not shut down
Ground Digital Out3	13			
Digital Out4	16	Default Setting: Trend Warning	Relay closed $\Rightarrow$ warning active Relay open $\Rightarrow$ warning not active	Default setting: Relay closed if trend warning is active. Can be changed in EEPROM with Service Software
Ground Digital Out4	15	Option: Priming Valve Signal	Relay closed $\Rightarrow$ Priming valve active Relay open $\Rightarrow$ Priming valve inactive	Can be used to control a priming valve for priming of the pump. Feature can be activated and configured with Service Software.

Table 10: Signals of the PLC connector for standard firmware D6.48

- For other configurations of PLC Inputs and Outputs refer to alternate firmware documentation
- Configurations can be done with Levitronix Service Software (see Manual with Doc.# PL-2034-00)

4.2 Mechanical Installation of the Pump/Motor

- The motor can be fixed with four screws on the motor feet (see *Figure 9*)
- As an alternative the motor can be mounted with four screws on the (see *Figure 9*) backside
- The motor can be mounted in either the horizontal or vertical position
- Each motor is identified with a unique serial number. This serial number consists of a series of 6 digits where the 5th and the 6th digit representing the manufacturing year.

4.3 Installation of ATEX Motors

	! WARNING <i>Motors for ATEX applications. Only specific types of motors LPM-600 are classified for the use in ATEX Zone 2 classified locations. Refer to the corresponding section in the manual.</i>	
---	---	---

An ATEX conform solution is needed for the motor cables to leave the ATEX area (see *Figure 5*). One option is an ATEX certified cable system as listed in *Table 4* and shown in *Figure 2*.

A protective earth wire shall be attached to the ATEX specific motor housings by using one of the four M5 threads on the backside of the motor.

- Remove one of the four M5 screws on the backside of the motor
- Use a crimp-type end together with a spacer sleeve to connect a earth wire
- Attach the grounding wire with a M5 stainless steel screw to the motor according to *Figure 27*



Figure 27: Attachment of a protective earth wire to the backside of the motor

4.4 Mechanical Installation of the Controller

	! WARNING
	Hazardous voltage may be present. <i>In order to avoiding fluid spills shorting mains or other voltages within the controller, place the controller in a spill protected environment (for example protected electronic cabinets).</i> <i>If explosive flammable gases are present, place the controller in an explosion-proof cabinet.</i>

CAUTION
<i>Make sure the controller is mounted in a position that allows free air circulation around the controller. A minimum distance of 10cm (4") to other objects above or below the controller casing is recommended.</i>

- Use the Din-Rail bracket to mount the controller.
- If no forced air-cooling is used, mount the controller in upright position.
- The Din-Rail brackets can also be mounted on the controller side according to *Figure 11*

CAUTION
<i>Use only 3,5 x 6,5mm self-tapping screw for the fixation of the Din-Rail brackets. The controller may be damaged if other type or too long screws types are used!</i>

5 Operation

5.1 System Operation with LPC-600.1 (Stand-Alone Version)

5.1.1 State Diagram of LPC-600.1

The controller *LPC-600.1* allows stand-alone operation with manual speed setting ("Button Control Mode") as well as extended operation with analogue speed setting (Analog Control Mode). *Figure 28* shows the state diagram which can be controlled with the manual buttons and the signals on the "USER INTERFACE" connector. The operation mode can be chosen by pressing the "UP" and "DOWN" buttons simultaneously during 5 seconds. For the standard firmware D6.25 default setting ex factory is "Button Control Mode".

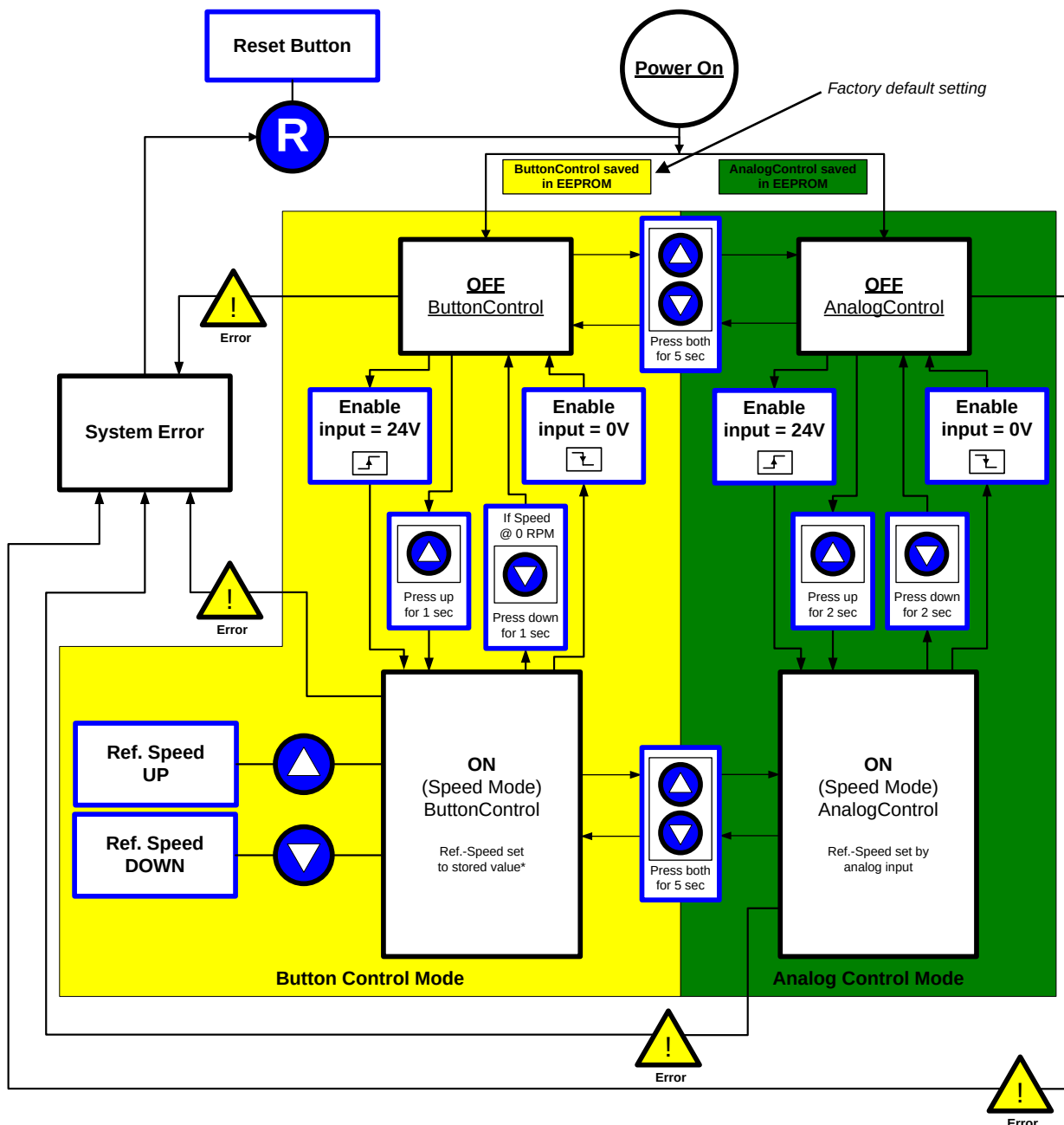


Figure 28: State diagram for operation with LPC-600.1 controller

(Description is for firmware D6.25 with Revision > 01, for other configurations refer to alternate firmware documentation)

5.1.2 Standalone Operation (Button Control Mode)

- When applying power the system defaults into the “*Button Control Mode*” and goes into the status “*OFF Button Control*” according to *Figure 28*. Levitation is disabled and the display indicates “*OF*”.
- Levitation can be enabled by pressing the “*UP*” button during 1 second (display shortly indicates “*ON*”) or by activating (typically 24V) the “*ENABLE*” pin on the “*USER INTERFACE*” connector (see *Table 9*). The system goes then into the status “*ON Button Control*” and is running at the speed, which is stored in the EEPROM.
- The speed can be changed by pressing accordingly the “*UP*” and “*DOWN*” buttons. As long as the digits on the display are blinking the set speed is shown. As soon as blinking stops the actual speed is shown and the set-speed is stored in the EEPROM of the controller.
- The system can be disabled by pressing the “*DOWN*” button until 0 rpm is achieved. Pressing further 1 second the “*DOWN*” button the system disables levitation and shows “*OF*” on the display. The system can also be disabled by deactivating (0 V) the “*ENABLE*” pin on the “*USER INTERFACE*” connector (see *Table 9*). Before disabling the system the speed is automatically reduced to 0 rpm and the impeller is properly touched down without grinding the wall.
- In case of an error the “*RESET*” button (see *Figure 21*)
-) can be used to restart the system or the power can be switched off and on.
- For error analysis the codes described in *Table 11* are displayed (blinking between “*Er*” and the according code number).
- If the system shall be enabled automatically, when power is applied the “*ENABLE*” pin on the “*USER INTERFACE*” connector (see *Table 9*) has to be active (typically 24V). When switching on the power the system is running with the stored speed.
- For monitoring purposes a digital output on the “*USER INTERFACE*” connector (see *Table 9*) indicates the status of the system. When the impeller is rotating the digital output “*Status*” turns active.

The digital input “*ENABLE*” is normally edge-triggered to allow control by digital input and buttons in order to enable/disable systems simultaneously. An exception is, when the system is powered up or a system reset occurs: than the system checks the level of the digital input and switches to the desired state. Hence on startup a high level of the digital input “*ENABLE*” is sufficient to switch on the system and a transition from low to high is not required.

5.1.3 Extended Operation (“Analog Control Mode”)

- In order to be able to control the pump with external signals (PLC) the mode “*Analog Control Mode*” has to be set with the display buttons. The “*UP*” and “*DOWN*” buttons have to be pressed simultaneously during 5 seconds. The display should feedback the change by blinking between the stored speed value and “*An*”. The chosen mode is then stored in the EEPROM of the controller. After startup the system returns to the operation mode selected previously.
- The system and levitation can be enabled/disabled with the digital input on the “*USER INTERFACE*” connector (see *Table 9*). When disabling the running system the speed is automatically reduced to 0 rpm and the impeller is properly touched down without grinding the wall. The display is blinking between “*An*” and “*OF*”. Alternatively “*UP*” button is also able to switch on system, with “*DOWN*” button the system can be switched off.
- The speed can be set with an analog signal on the “*USER INTERFACE*” connector according to *Table 9*. It is strongly recommended to use galvanic separated signal values.
- For monitoring purposes a digital output “*Status*” on the “*USER INTERFACE*” connector (see *Table 9*) indicates the status of the system. When the impeller is rotating the digital output “*Status*” turns active.

5.1.4 Error Display on the Integrated Panel

Error Source	Errors	Error Code on Display
Motor	No Motor	Er 01
Motor	Motor cable (power wires) not connected to controller	Er 02
Motor	Motor cable (sensor wires) not connected to controller	Er 03
Motor	No Rotor	Er 04
Controller	Short circuit	Er 05
Controller	Over current in the bearing coils	Er 06
Controller	Over current in the drive coils	Er 07
Controller	DC-Link voltage out of range (Voltage range for monitoring: 18 – 54 VDC) If the voltage is out of extended range the system starts to reduce the speed and an error is generated. When reaching 0 rpm and the voltage is still out of extended range the system is disabled.	Er 08
Controller	Communication problems EEPROM Controller	Er 09
Motor	Communication problems EEPROM Motor	Er 10
Controller	Controller temp. over 80 °C or more than 10 minutes above 70 °C	Er 11
Motor	Motor temp. over 100 °C or more than 10 minutes above 90 °C	Er 12
Pump	Dry running of pump circuit: -> Pump keeps running on reduced speed (5000 rpm) -> The system accelerates to the original speed value when the pump is refilled with liquid. -> Note that the speed is only reduced during dry running if the pump speed was $\geq$ 6000 rpm.	<i>Blinking dots on display</i>

Table 11: Errors and warnings with indication on display of LPC-600.1

- In case of an error the system can only be restarted with a reset or a power supply restart
- Standard firmware is D6.25
- For other configurations of error codes refer to alternate controller or firmware documentation

5.2 System Operation with Controller LPC-600.2 (PLC version)

5.2.1 State Diagram of the PLC Interface

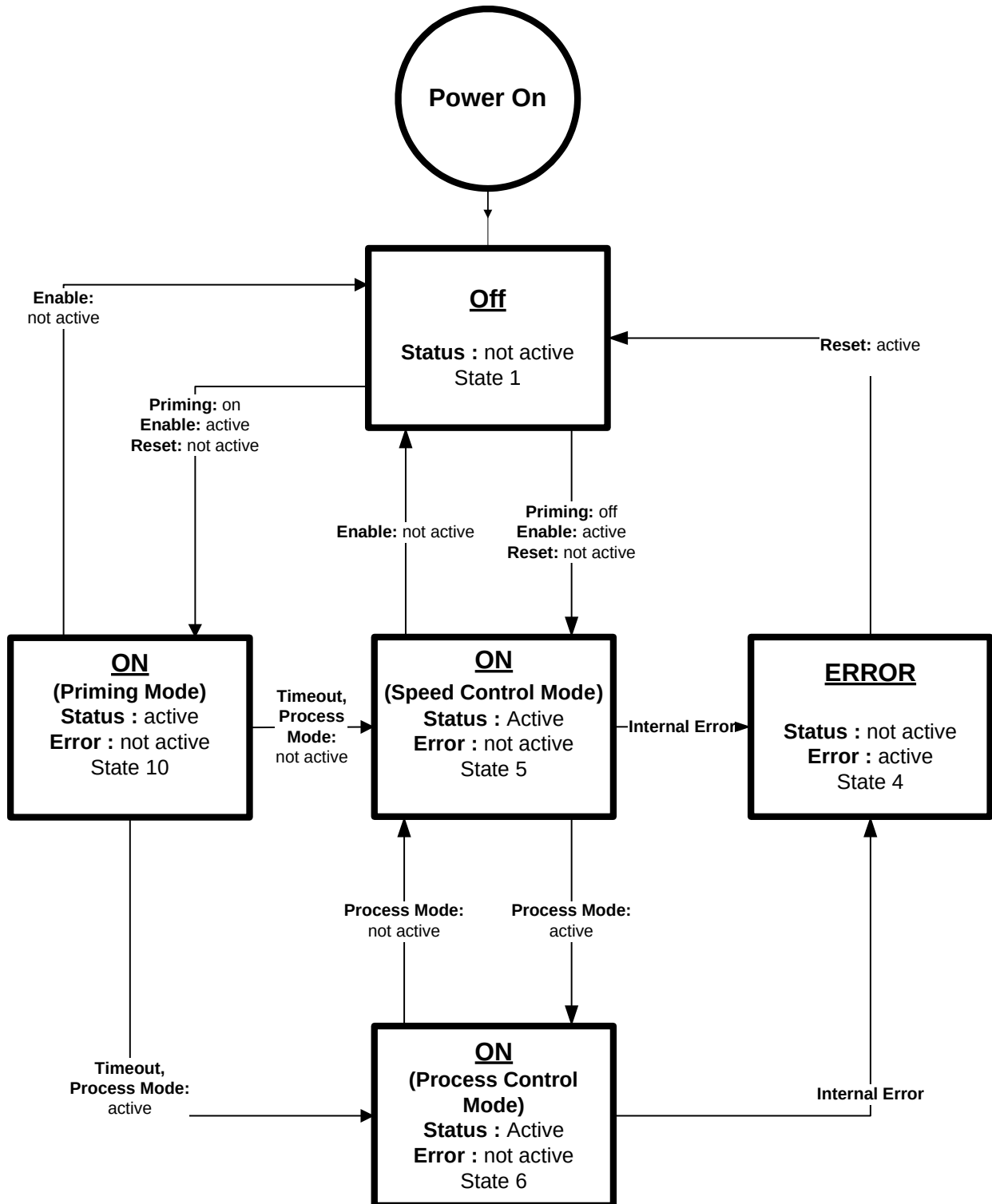


Figure 29: PLC interface state diagram for standard firmware D6.48
(For other configurations refer to alternate firmware documentation)

State “Off”:

The pump system is switched off and the motor has no power. In this state, Levitronix Service Software has full control.

State “ON” (speed control mode):

The pump system is switched ON and the impeller is rotating with the referenced speed. The motor has electrical power when in this state.

State “ON” (process control mode):

The pump system is switched ON and the impeller is rotating with the priming speed. The motor has electrical power when in this state. In this state, the Manual User Interface LUI is disabled. This mode can only be accessed by activating priming feature within EEPROM-editor in *Levitronix® Service Software*. Priming speed and Timeout-time can also be configured within EEPROM-editor.

State “Error”:

If an error according to *Table 12*. occurs in the pump system, the system defaults to the *Error* state. The designated digital output on the PLC Interface is activated. The pump system is switched OFF. By activating the “Reset” input the system gets back to the “Off” state.

Error Source	Errors	Effect on Designated Digital Output of the PLC
Motor	No rotor	Error = relay open
Motor	Temperature over 100°C	Error = relay open
Motor	Temp. was higher than 90°C for more than 10 minutes.	Error = relay open
Motor	Temperature more than 90°C	Warning = relay open
Motor	No motor temperature signal	Warning = relay open
Motor	Motor power cable not connected with controller	Error = relay open
Motor	Motor sensor cable not connected with controller	Error = relay open
Controller	Over-current	Error = relay open
Controller	Power channel interrupted	Error = relay open
Controller	Temperature over 80°C	Error = relay open
Controller	Temp. was higher than 70°C for more than 10 minutes.	Error = relay open
Controller	DC link (supply voltage) out of range (< 18 or > 54 V DC) If the voltage is out of range the system starts to reduce the speed and a warning is generated. When reaching 0 rpm and the voltage is still out of range the system is disabled and an error is generated. In case the voltage is again within the range during speed reduction the system switches to normal operation and no Error is generated.	Error = relay open
Controller	Temperature over 70°C	Warning = relay open
Controller	Dry Running Detection -> Pump keeps running on reduced speed (5000 rpm) -> The system accelerates to the set speed value when the pump is refilled with liquid -> Note that the speed is only reduced during dry running if the pump speed was ≥ 6000 rpm	Warning = relay open
Controller	Trend warning (actual speed too high)	Warning = relay open

Table 12: Errors and warnings with indication on PLC interface for standard firmware D6.48
(For other configurations refer to alternate firmware documentation)

5.3 System Operation for ATEX Applications

Specific precautions may be considered while using the pump system in potential explosive gas atmospheres according to ATEX category 3G/3D (Zone 2 and 22).

The user shall prevent priming issues during normal pump operation. Especially precautions have to be considered during installing and maintenance operations to prevent the occurrence of combustible atmospheres.

The user shall prevent electrostatic charging of the system at cleaning processes by using dry cleaning cloth. User shall use wet cleaning rags to avoid issues with charging at a cleaning process.

CAUTION		
<i>Precautions have to be considered to prevent priming issues during installation operation and maintenance of the pump head / motor.</i>		

	⚠ WARNING	
	Operational Temperature T5 Maximum allowed pump liquid temperature is 90°C / 194°F for the use in ATEX Zone2 applications.	

	⚠ WARNING	
	Do not operate the pump against closed valves Refer to the corresponding section in the manual.	

6 Inspection and Maintenance

6.1 Replacement Interval of the Impeller

The impeller has a limited lifetime depending on the chemical type, concentration and temperature of the fluid which is pumped. Therefore a preventive periodical exchange of the impeller is recommended. Contact the *Levitronix Technical Service Department* (see *Section 6.2*) for further information on replacement times.

6.2 Impeller Replacement Procedure

6.2.1 Preparation

Before starting the impeller replacement procedure the parts and tools illustrated in *Figure 30* and *Figure 31* should be prepared. Impeller exchange kits, which contain this parts and tools are available at Levitronix (see *Table 4*). Please verify that you have the right types of impellers, O-rings and screws.

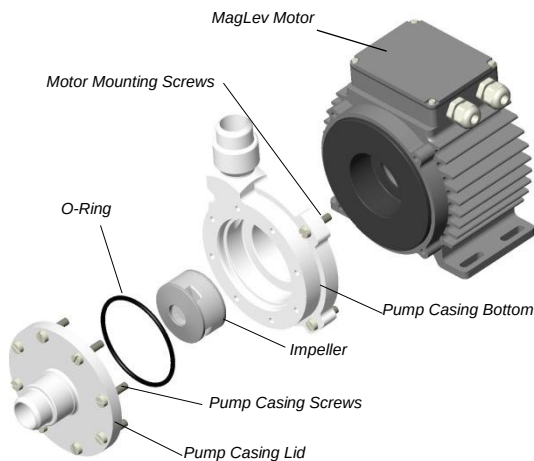


Figure 30: Explosion view of pumphead with motor

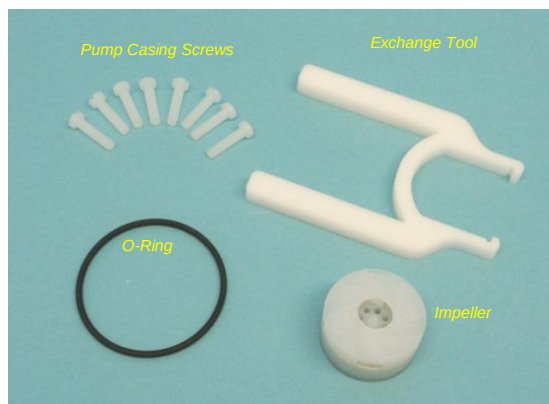


Figure 31: Components for impeller replacement
(The 4 M6x25 screws for motor mounting are not on the picture)

The following warnings and cautions should be read carefully before starting the replacement of the impeller.

! WARNING				
				
The impeller could splash TOXIC or CORROSIVE CHEMICALS because of the strong magnetic forces. Flush the pump housing before opening it.				

! WARNING				
				
HARMFUL CHEMICALS may be present. Skin contact and toxic gases may be hazardous to your health. Wear safety gloves and other appropriate safety equipment.				

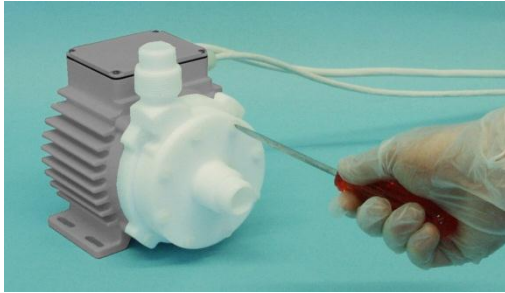
! CAUTION				
				
The rotating impeller could cause injury. Do not run the pump system when opening the pump head.				

CAUTION				
				
Pay attention to the magnetic forces when handling the impeller. The attraction of magnetic parts and particles should be avoided in order to keep the impeller and the pump head clean and free of contamination.				

6.2.2 Instructions for Replacement

1. Power down the pump system and remove the AC power. If necessary, allow the housing to cool down to a workable temperature.

2. Unscrew the top of the pump head and remove it along with the sealing ring.

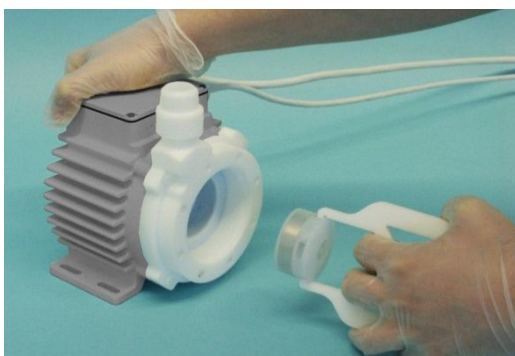


3. Remove the impeller with the *Impeller Exchange Tool*. Hook the claws of the *Impeller Exchange Tool* into two opposing orifices of the impeller.



4. Inspect the wet area of the pump head carefully. In case of material damage, also replace the pump casing.

5. Place the new impeller into the pump casing using the *Impeller Exchange Tool*.



6. If necessary, remove the existing *O-Ring* and gently press the new *O-Ring* into the lid of the pump casing.

CAUTION

Use the correct *O-Ring* type for your process. If necessary, consult the Levitronix Technical Service Department.

Do NOT twist or roll the *O-Ring* as this may cause leaking to occur.

7. Press the lid with the *O-ring* flush into the bottom of the pump casing.



8. Carefully tighten the 8 screws. The screws should not be used to press the lid with the *O-ring* into the bottom of the pump casing. Do not apply too much torque. The torque specifications are:

Maximum torque for pump screws
PVDF M6: 50 Ncm

9. Start up the system and check if the impeller is rotating properly and the pump head doesn't leak.

10. If the pump head leaks, check and make sure the lid and the *O-Ring* are properly pressed into the bottom of the pump casing. It may be necessary to change the *O-Ring* if it has been damaged.

7 Troubleshooting

7.1 Troubleshooting for Operation with Controller *LPC-600.1*

For troubleshooting and failure analysis with the stand-alone controller *LPC-600.1* the following procedure is recommended:

- Check the status of the LEDs. The specific LEDs are described in *Table 7*
- Use the ERROR codes on the display. The specific error codes are described in *Table 11*
- A digital output on the “USER INTERFACE” connector (“Status”) indicates if the system is active. However, the source of an error cannot be identified by this signal

7.2 Troubleshooting for Operation with Controller *LPC-600.2*

The integrated *PLC* provides a Warning and an Error signals according to *Table 12*. However, the source of error cannot be identified by these signals.

For more detailed analysis the *Levitronix® Service Software* can be used with a PC and a USB interface to the controller.

7.3 Troubleshooting with Service Software

The *Levitronix® Service Software* allows communication with the pump system in connection with a PC and a USB interface. The software can be used for performing detailed troubleshooting. For usage of the Service Software refer to the *Service Software User Manual* (Document #: *PL-2034-00*), which is available in the download section on the Levitronix Web-page or contact the *Levitronix® Technical Service Department* (see under *Section 8*).

Note: the Service Software can not be used with the standalone controller *LPC-600.1*.

8 Technical Support

For troubleshooting, support and detailed technical information contact *Levitronix® Technical Service Department*:

Levitronix®
Technical Service Department
Technoparkstr. 1
CH-8005 Zurich
Switzerland

Phone for US:	888-569 07 18
Phone for outside US:	+1 888-569 07 18
E-Mail:	support@levitronix.com

9 Appendix

9.1 Regulatory Status

9.1.1 CE Marking

The *Bearingless Pump System BPS-600*, in his various configurations, is in conformity with the essential requirements of the *EMC Directive 2004/108/EC* and the *Machinery Directive 2006/42/EC*. The following particular harmonized standards of the *EMC Directive 2004/108/EC* are tested and confirmed at a certified laboratory (Hochschule für Technik, Switzerland, Test Laboratory, CH-8005 Zurich, Switzerland, Swiss Certification No.: STS 404):

EN55011	Limits and methods of measurement of electromagnetic disturbance characteristic of industrial, scientific and medical (ISM) radiofrequency equipment.
EN61000-4-2	ESD
EN61000-4-3	Radiated RF Immunity
EN61000-4-4	Fast transient / Burst
EN61000-4-5	Surge
EN61000-4-6	Immunity to conducted disturbances, induced by RF
ENV 50204	Radiated RF Immunity
EN61000-6-2	Generic standards, Immunity for industrial environments
EN61000-6-4	Generic standards, Emission standard for industrial environments

The following particular standards of the *Machinery Directive* are followed for the design, validation and risk assessment:

EN809	Pumps for Fluids: basic requirements are followed.
EN12162	Procedure for hydrostatic pressure testing in fluid pumps: max. pressure testing.
ISO12100	Safety for machinery – principles for risk assessments.

9.1.2 ATEX Marking

Specific motors together with the pump head of the *BPS-600* pump system are in conformity with the requirements of the *Directive 94/9/EC*. The following standards are tested and confirmed at a certified laboratory (Electrosuisse, Switzerland, CH-8320 Fehraltorf, Swiss testing No. STS 001).

EN60079-0	Electrical apparatus for explosive gas atmospheres. General requirements.
EN60079-15	Electrical Apparatus for explosive gas atmospheres. Construction, test and marking of type of protection, "n" electrical apparatus.
EN 60079-31	Equipment dust ignition protection by enclosure
EN 13463-1	Non-electrical equipment for use in potentially explosive atmospheres – Part 1: Basic method and requirements
EN13463-5	Non-electrical equipment intended for use in potentially explosive atmospheres – Part 5: Protection by constructional safety 'c'

The Levitronix Ex motors are marked clearly and in accordance to the ATEX Directive. The protection EX nA means non sparking electrical apparatus.


 II 3G Ex c nAc IIC T5
 II 3D Ex c tc IIIC T100°C IP67

Classification:	Category 3GD (Zone 2 for Gas and Zone 22 for Dust)
Explosion Groups:	Group IIA: Propane (IPA), Methane, Aceton, Acetaldehyde Group IIB: Ethylene, Ethylenglycol Group IIC: Acetylene, Hydrogen (not carbon disulphide) Group IIIC: conductive dust
Thermal Classification:	Thermal classification of motor is T5 (< 100 °C = 212 °F) for maximum full-load operating temperature at a maximum liquid temperature of 60°C / 140 °F.
UL Correspondence:	ATEX listing corresponds to UL hazardous location Class 1 Division 2.

9.2 Symbols and Signal Words

Symbol / Signal Word	Description	Type	Source
DANGER	Indication of an imminently hazardous situation that, if not avoided, will result in death or severe injury. Limited to the most extreme situation	Signal word	SEMI S1-0701
WARNING	Indication of a potentially hazardous situation which, if not avoided, could result in death or severe injury.	Signal word	SEMI S1-0701
CAUTION	Indication of potentially hazardous situations which, if not avoided, could result in moderate or minor injury. Also alert against unsafe practice. Without safety alert indication of hazardous situation which, if not avoided, could result in property damage.	Signal word	SEMI S1-0701
	Safety alert for "Warning" and "Caution"	Safety alert	SEMI S1-0701
	Safety alert for "Danger"	Safety alert	SEMI S1-0701
	Caution (refer to accompanying documents) (is used on article labels for reference to manual)	Refer to manual	ISO 3864
	Toxic material, poison	Hazard identification	IEC 61310
	Corrosive material, corrosion	Hazard identification	IEC 61310
	Cut/sever hand, sharp object	Hazard identification	ANSI Z535.3
	Strong magnetic field	Hazard identification	SEMI S1-0701
	Danger: electricity, electrical hazard	Hazard identification	IEC 61310, ISO 3864
	Wear safety gloves	Hazard avoidance Mandatory action	IEC 61310
	Wear face shield	Hazard avoidance Mandatory action	SEMI S1-0701
	Unplug power line	Hazard avoidance Mandatory action	SEMI S1-0701
	No pacemakers	Hazard avoidance Prohibition	SEMI S1-0701
	ATEX Logo	Used for hazard identificat. in warnings	--

Table 13: Safety symbols and signal words

G Glossary

This glossary defines all terms and abbreviations used in this publication that may be new or unfamiliar. For a listing of commonly used computer and data processing terms, see a dictionary written expressly for the compilation of computer terms, such as the *IBM Dictionary of Computing*. For a listing of commonly used electrical and electronic terms, see a dictionary written expressly for the compilation of electrical and electronic terminology, such as the *New IEEE Standard Dictionary of Electrical and Electronic Terms*. For a comprehensive listing of commonly used semiconductor terms, see the *SEMI International Standards Program* handbook.

The following cross-references are used in this glossary:

Contrast with This refers to a term that has an opposed or substantively different meaning.

See This refers the reader to a multiple-word terms in which this term appears.

See also This refers the reader to a term that has a related, but not synonymous, meaning.

Synonym for This indicates that the term has the same meaning as a preferred term, which is defined in the glossary.

A

affected individual Any person who routinely operates machinery or equipment that is being serviced under lockout or tagout procedures or who works in an area where such servicing is performed. These are typically machine operators, laboratory workers, or other non-maintenance personnel.

AIC (ampere interrupting capacity) The fault current that a device (usually a fuse or a circuit breaker) is capable of breaking without damage.

authorized individual A person trained in lockout/tagout procedures who is involved in any such covered activity.

B

backflow The flow of liquid in the direction opposite its intended direction.

C

cabinet interlock A safety device that places the equipment in a safe state when a door is opened.

cabinet leak The error state created by liquid collecting inside the cabinet.

chemical delivery module (CDM) A machine that transfers and distributes pressurized chemical to the chemical distribution system.

Chemical System Shutdown (CSS) The signal that causes the chemical distribution system to shut a valve that is immediately upstream of the point that generated the alarm. This system alarm will not shut down a device, but will isolate points of the chemical distribution system that are included in, or are downstream of, the point that generated the CSS alarm. This alarm can be generated by, for example, the system operator pressing a CSS pushbutton on a valve box, or a leak interlock actuation within the chemical distribution system.

clean dry air (CDA) Fabrication plant air that is void of moisture and other contaminants. Used to activate valves and pumps.

corrosive A corrosive chemical (e.g., sulfuric acid, hydrofluoric acid). It can be an acid, a base or an oxidizing agent.

D

deionized (DI) water Extremely pure water that has all ions removed.

disable The opposite of enable. To take out of service or turn off.

door interlock A safety device that places the equipment in a safe state when a door is opened.

download The process of loading the operating software from one computing device to another.

drum quick-connect The physical connection between the chemical supply drum and the chemical delivery tubing.

drum-connect verification A device used in drum quick-connect fittings that verifies a proper connection.

E

enable To turn on or make functional.

emergency machine off (EMO) A control switch or circuit that, when activated, cuts all power to a device or series of related devices and places them in a known safe state.

emergency power off (EPO) A control switch or circuit that, when activated, cuts all power to a device or series of related devices and places them in a unknown or unsafe state. Pressure trapped in the system can create an unsafe condition if maintenance procedures are performed.

energy isolating device A device such as manually operated circuit breakers, line valves, electrical disconnect switches, or other devices designed to prevent the transmission or release of energy.

energy source Electrical, mechanical, chemical, heat (thermal), pneumatic, noise or other energy source which, if released or contacted, may directly or indirectly cause injury to an individual.

F

fab Fabrication plant. The facility that manufactures or fabricates semiconductor wafers.

fault state The condition of equipment when an error is present.

L

light-emitting diode (LED) A lamp, typically red, that indicates the state of a device or electrical signal.

lockout device A device that has a lock (such as a padlock) and is used to hold an energy isolating device in place.

lockout/tagout (LOTO) The process of identifying, shutting off, or otherwise removing an energy source, and then placing a lockout and a tagout device on the energy isolating device.

M

main valve box (MVB) The valve box that is used as the first branch point in the chemical distribution system.

N

National Electric Code (NEC) The standards organization that is responsible for the wiring of facilities and structures.

National Electrical Manufacturers Association (NEMA) The standards organization that is responsible for environmental ratings of enclosures.

National Fire Protection Agency (NFPA) The standards organization has the responsibility of overlapping U.L. for panel construction standards.

nitrogen purge *See Purge.*

O

offline An inactive or not ready state.

online An active or ready state.

P

Plumbed, process-grade isopropyl alcohol (IPA) Commercially acceptable grade of alcohol used for cleaning and prewetting filter elements.

PM *See Preventive Maintenance.*

PV Pneumatic valve. Provides isolation between the chemical and the electrical control systems. The PV is the “wetted” component that is controlled by solenoid valve.

power distribution panel (PDP) A device that converts standard electrical voltage (110 or 220 Vac) to 24 Vdc.

power-down The process of removing electrical power from a device.

power-up The process of applying electrical power to a device.

pre-chemical startup The process of preparing equipment for chemical. It consists of checking out all safety features and running the equipment with a safe liquid, typically DI water.

prewetting When referring to filters, the process of readying the filter for chemical by soaking it in alcohol.

preventive maintenance The steps performed on a regular basis to ensure optimum functionality.

programmable logic controller, or PLC A dedicated, industrially hardened computer used to control a device or system. PLCs communicate to non-PLC devices (valves, pumps, sensors, etc.) through inputs and outputs (I/O).

pulse dampener A device that suppresses the surge of chemical from an operating pump.

pump stroke rate The speed at which a pump operates.

purge The process of blowing chemical out of the plumbing lines with pressurized nitrogen.

Q

quick-connect A device that connects the chemical supply plumbing line to the chemical supply drum.

S

SCADA System Control and Data Acquisition. Hardware or software device that provides an interface to other devices.

solvent A flammable chemical.

stabilize/stabilization Recirculate chemical to minimize shedding of particles from a filter.

steady-on A constantly lit light-emitting diode (LED).

stored energy Potential energy that is residual or stored and includes hydraulic or pneumatic pressure, pressure below atmospheric pressure (negative or vacuum pressure), compressed or extended springs, potential energy due to gravity, potential energy stored in flywheels, batteries, capacitors, static electricity, thermal or subzero temperatures or any other pressure buildup.

T

tagout device A tag or other sign that can be secured to an energy-isolating device and is used to indicate the owner of the tag and the operation that is taking place.

(Note: A tagout device shall be used only when an authorized individual has made the determination that a lockout device cannot be used or would not be feasible).

U

U.L. Underwriters Laboratories. A standards agency that governs the safety and construction concerns associated with control panel construction.

V

valve manifold box (VMB) Controls the distribution of the chemical to the POU.

P Prints

This section lists the engineering prints, by model, to be used for reference and troubleshooting the Inline Blender, with Integrated Daytanks. [Table P-1](#) lists the prints, which are included in this section.

Table P-1. Engineering Drawings and Documentation

Drawing No.	Application	Dwg	P/S
P&ID			
514510-001	P&ID, Inline Blender, Integrated DT	X	
Facility and Installation Drawing			
514530-001	Drawing, Facility & Installation, Integrated Blender, AQ	X	
514536-001	Drawing, System, DHF Blender Pair, 1 CDM	X	
Electrical Schematic			
514580-001	Schem., Electrical, DHF Blender	X	

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